Principal Component Analysis (PCA)

A graphical explanation

	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

Let’s start with a simple data set.

	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

We've measured transcription of two genes, gene 1 and gene 2…

	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

…in 6 different mice.

	Sample 1	Sample 2	Sample 3	Sample 4	...
Variable 1	10	11	8	3	...
Variable 2	6	4	5	3	...

…think of the mice as individual samples…

	Sample 1	Sample 2	Sample 3	Sample 4	…
Variable 1	10	11	8	3	…
Variable 2	6	4	5	3	…

…and the genes as variables that we measure for each sample.

	Student 1	Student 2	Student 3	Student 4	...
Math	95	88	93	75	...
Reading	96	79	98	81	...

For example, the samples could be students in high school and the variables could be test scores in math and reading…

	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1

If we only measure 1 gene, we can plot the data on a number line…



	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1

Mice 1, 2 and 3 have relatively high values…



	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1

…and mice 4, 5 and 6 have relatively low values.



	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1

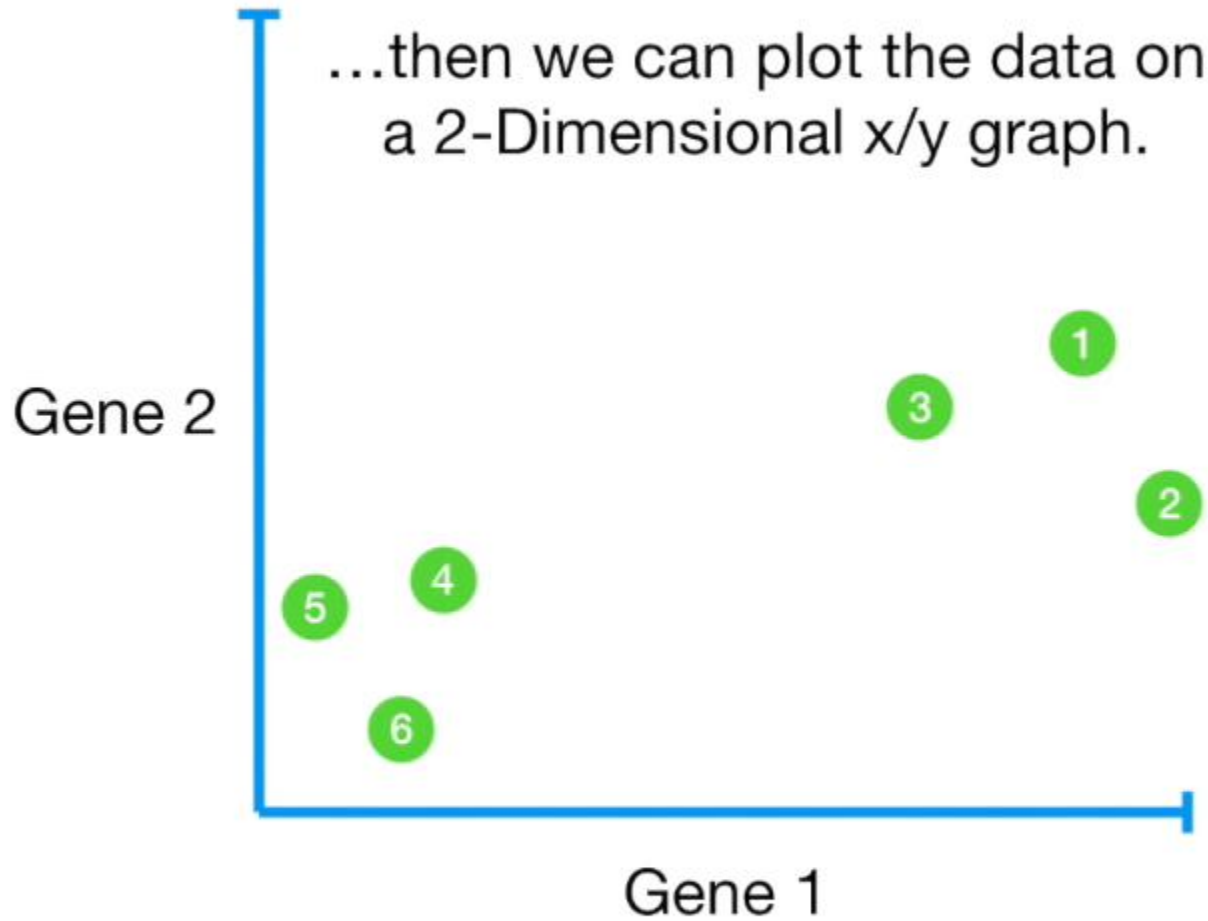
Even though it's a simple graph, it shows us that mice 1, 2 and 3 are more similar to each other than they are to mice 4, 5 6.



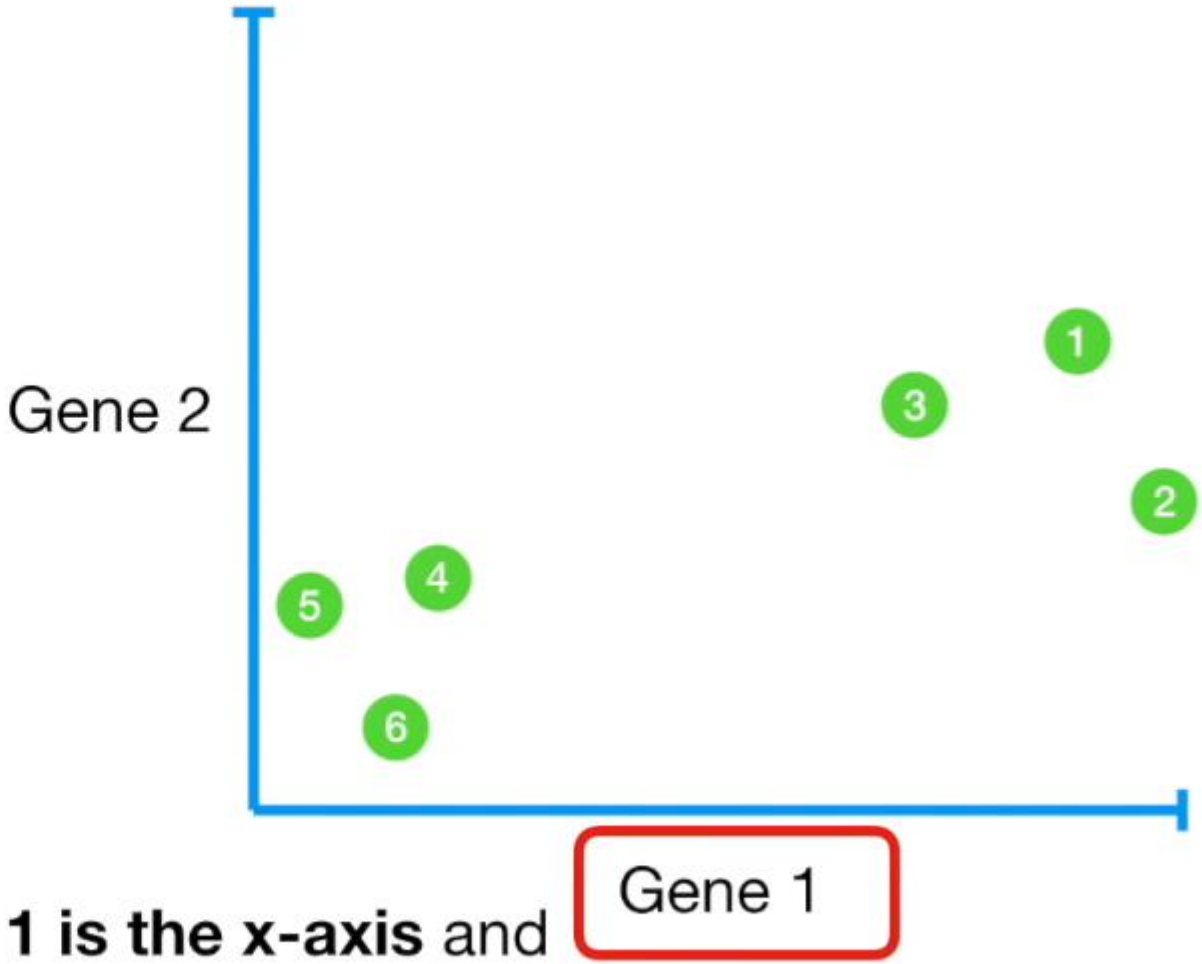
	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

If we measured 2 genes…

	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

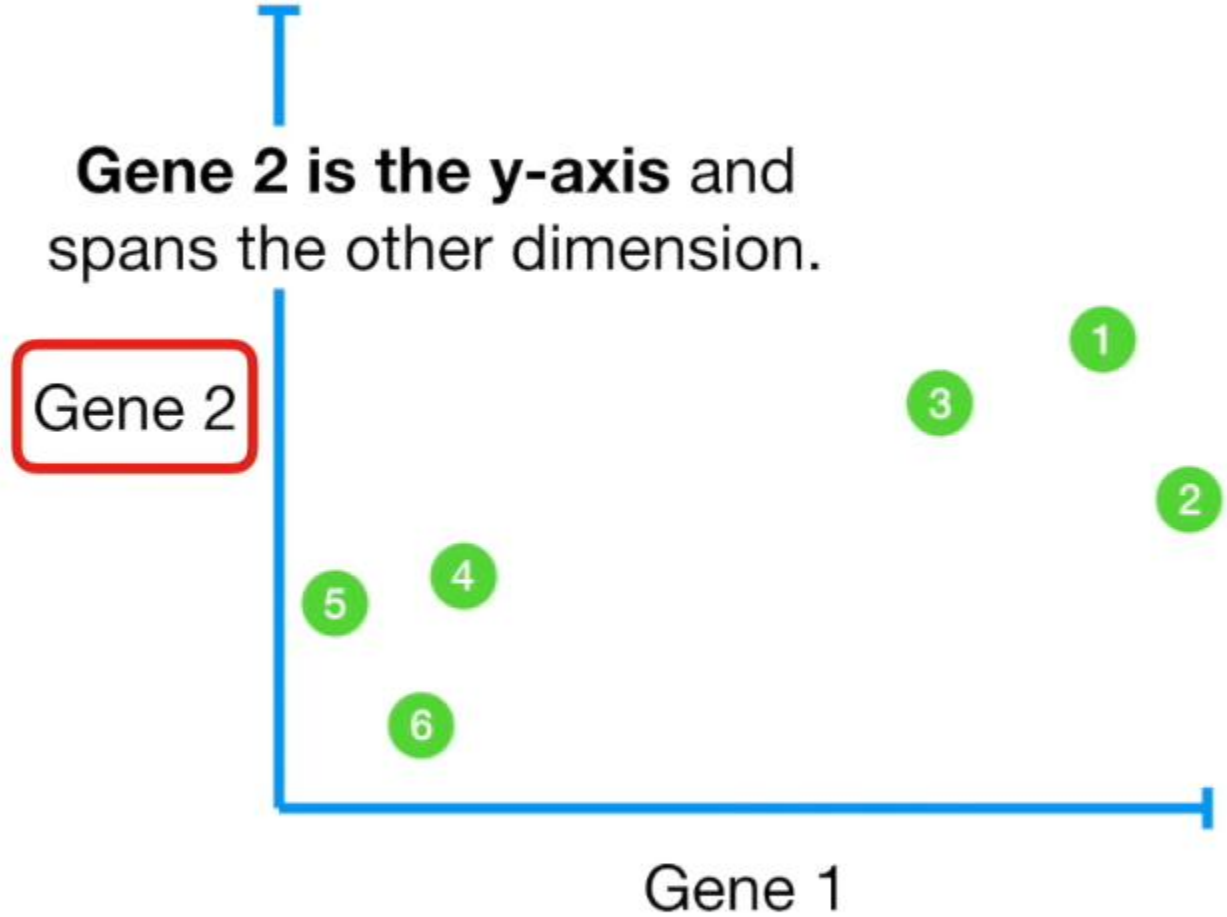


	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

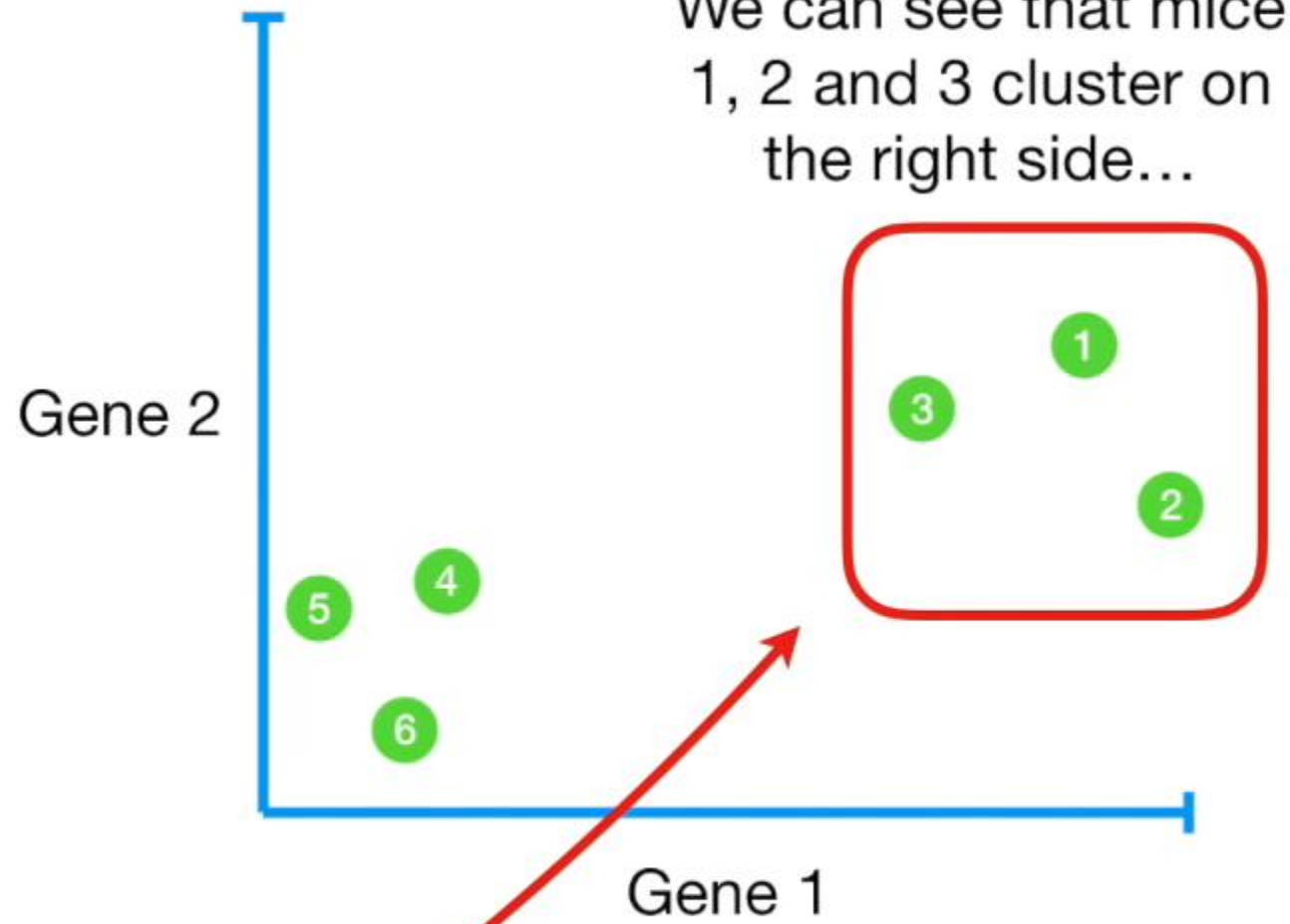


Gene 1 is the x-axis and spans one of the 2 dimensions in this graph.

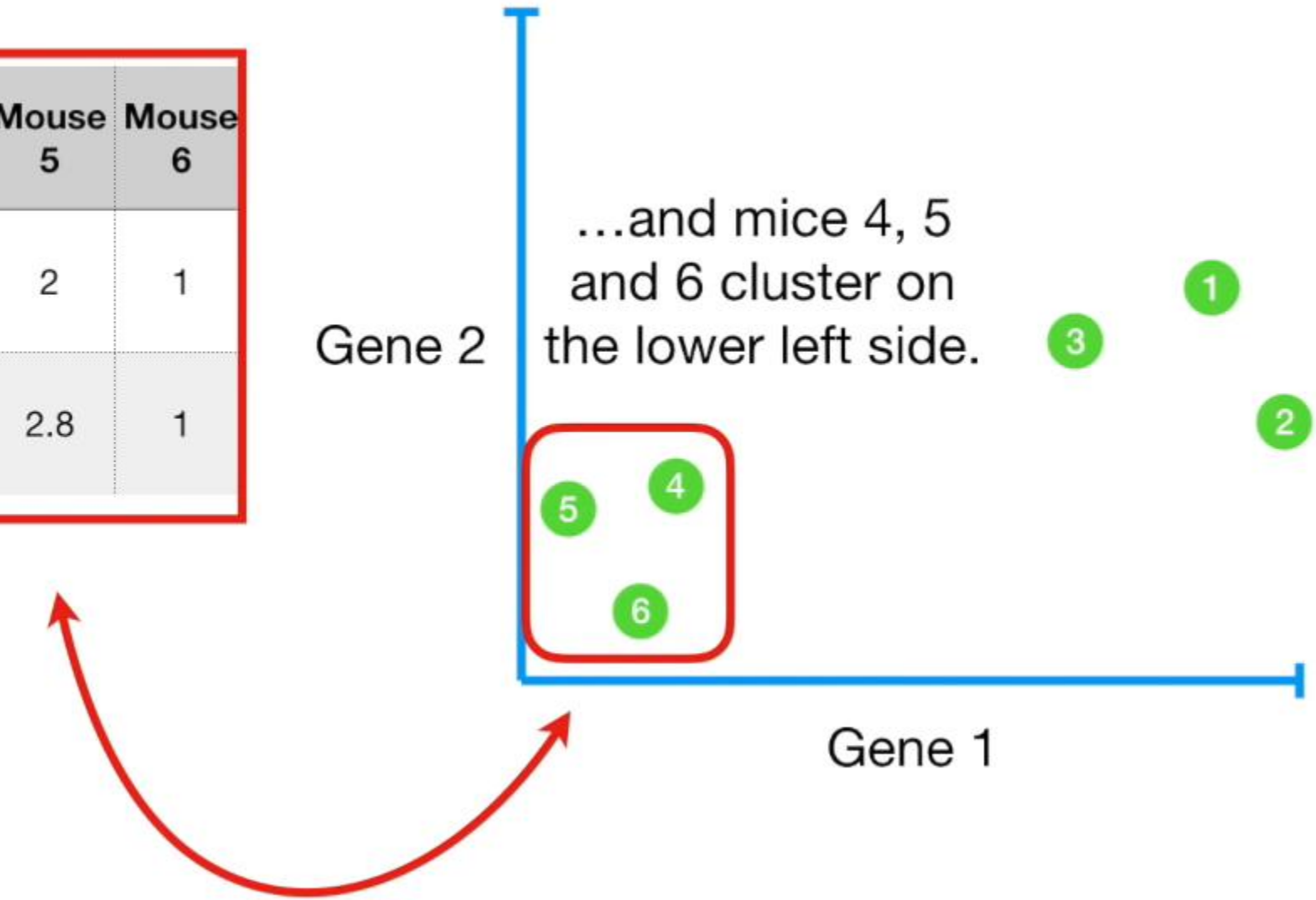
	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1



	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

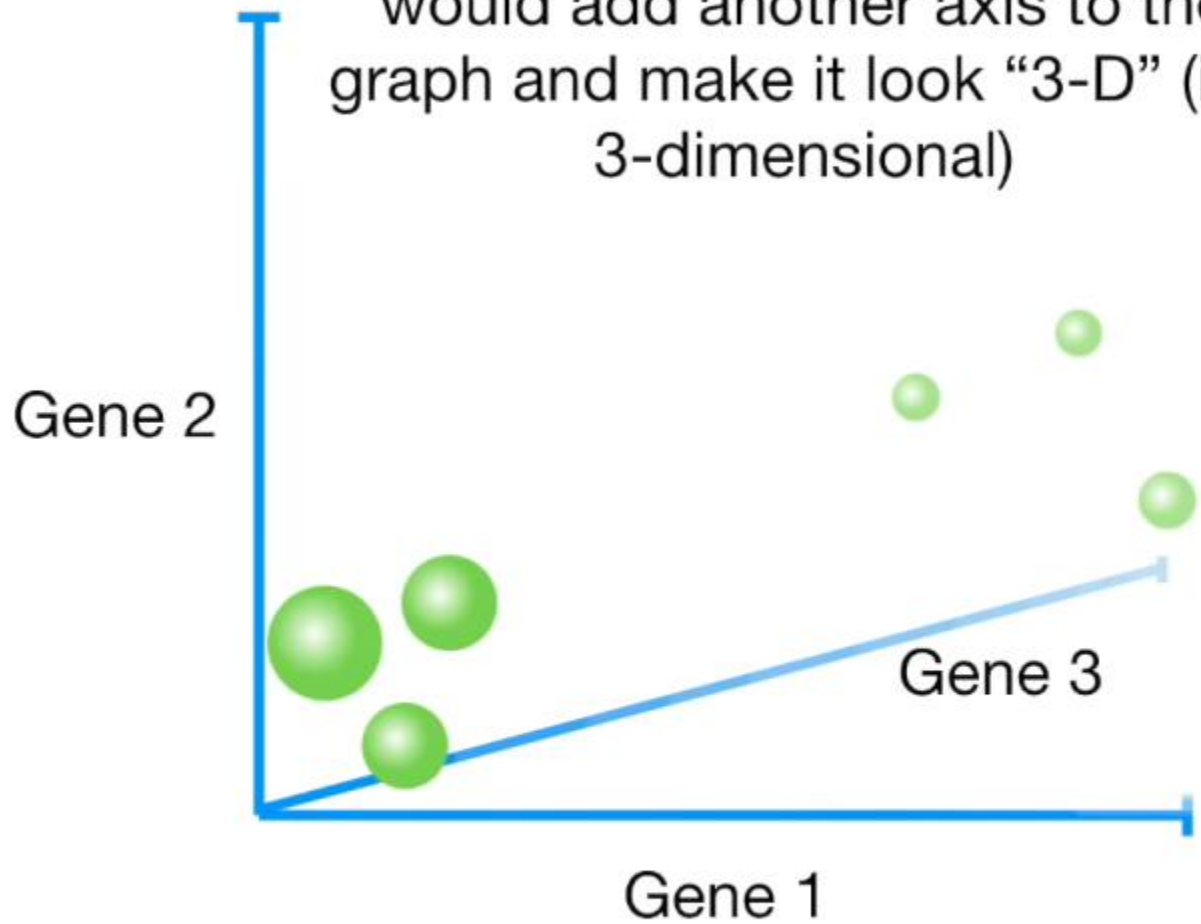


	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

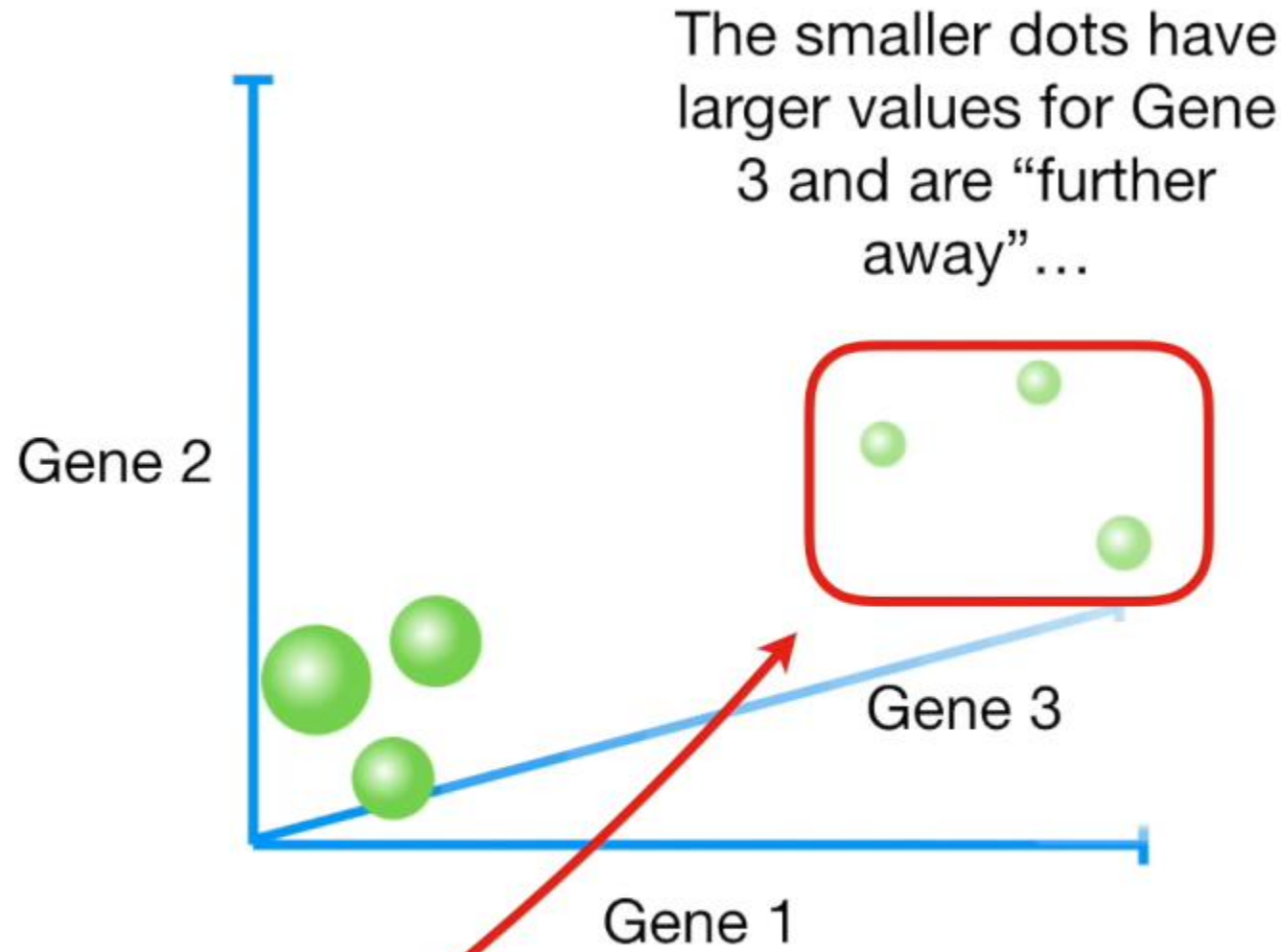


	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1
Gene 3	12	9	10	2.5	1.3	2

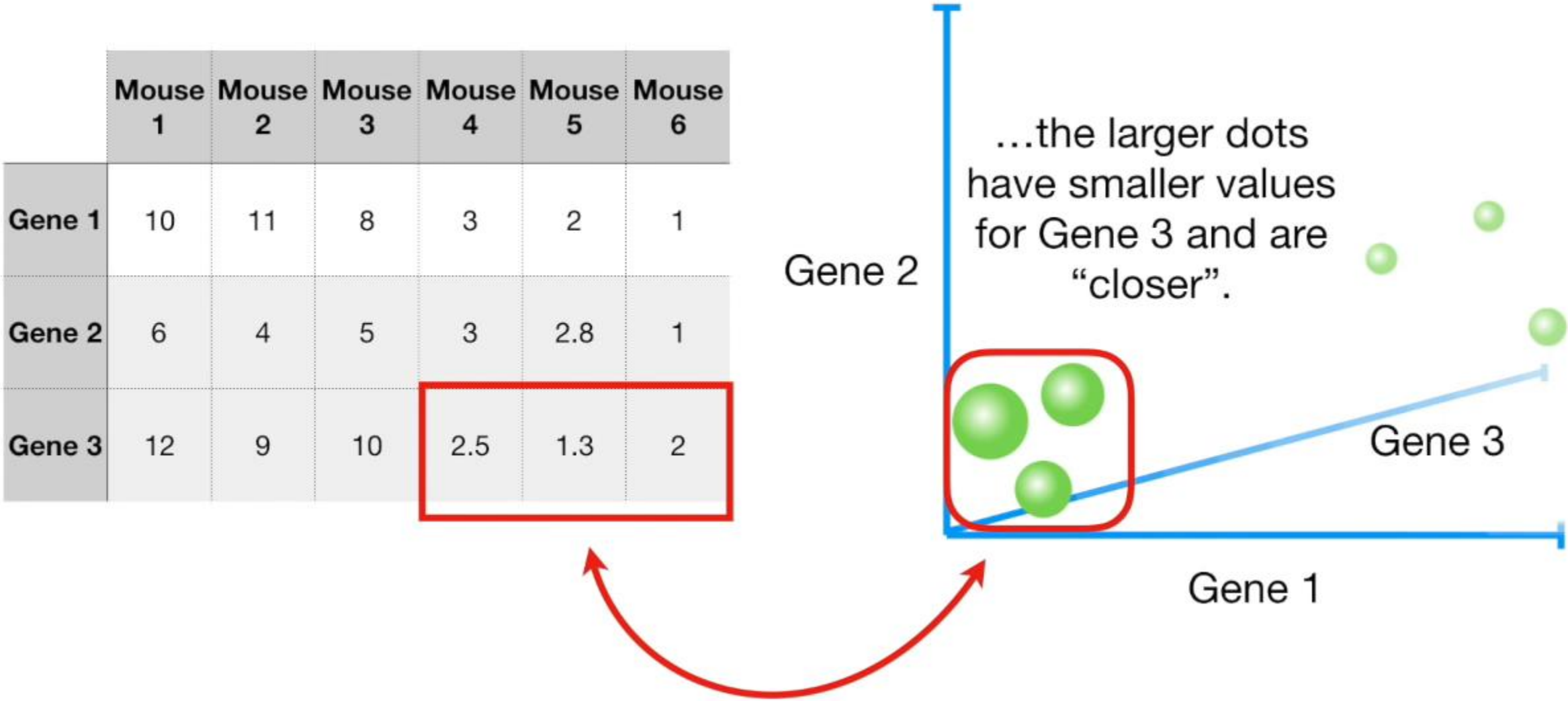
If we measured 3 genes, we would add another axis to the graph and make it look “3-D” (i.e. 3-dimensional)



	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1
Gene 3	12	9	10	2.5	1.3	2



	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1
Gene 3	12	9	10	2.5	1.3	2

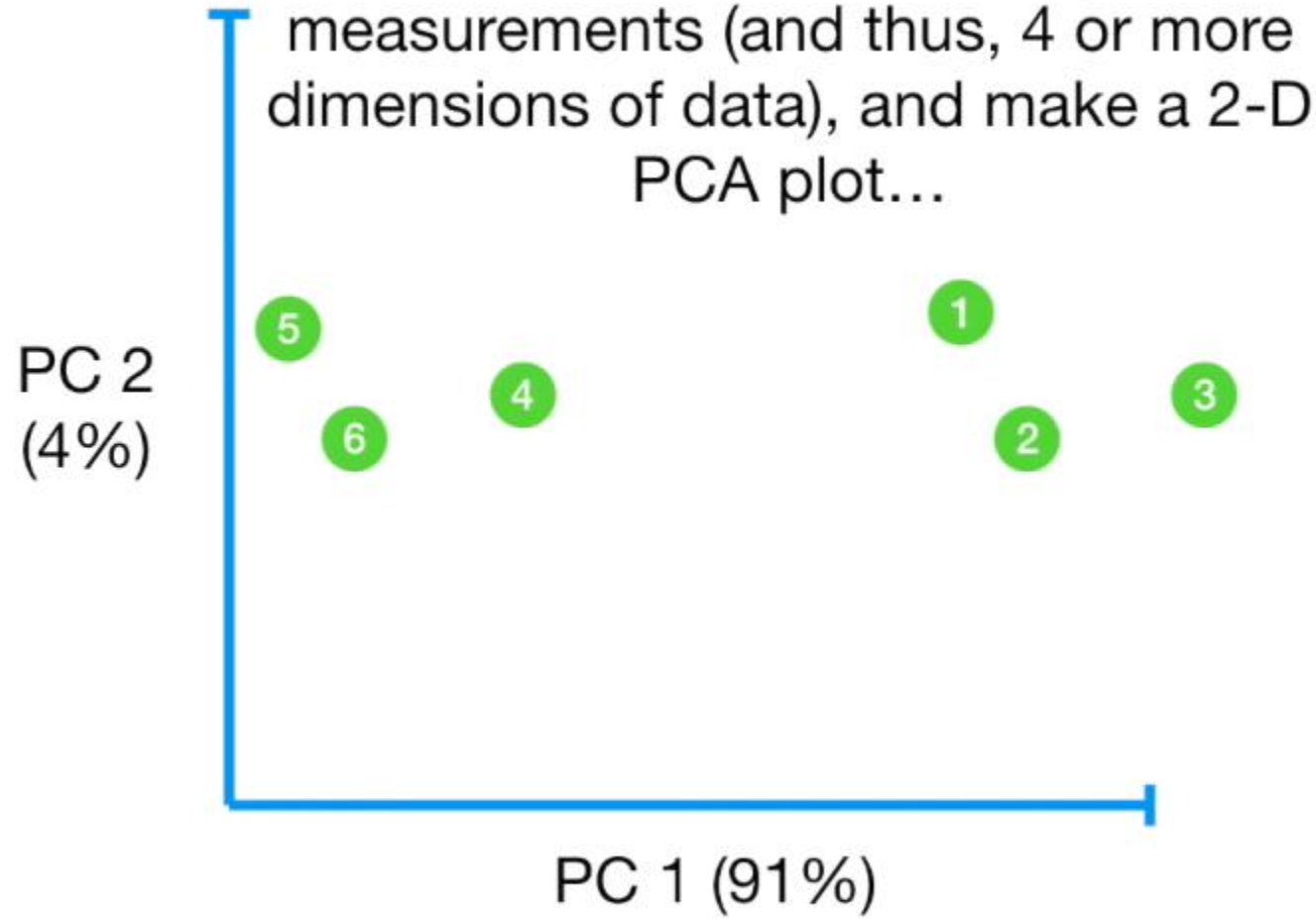


	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1
Gene 3	12	9	10	2.5	1.3	2
Gene 4	5	7	6	2	4	7

If we measured 4 genes, however, we can no longer plot the data - 4 genes require 4 dimensions.

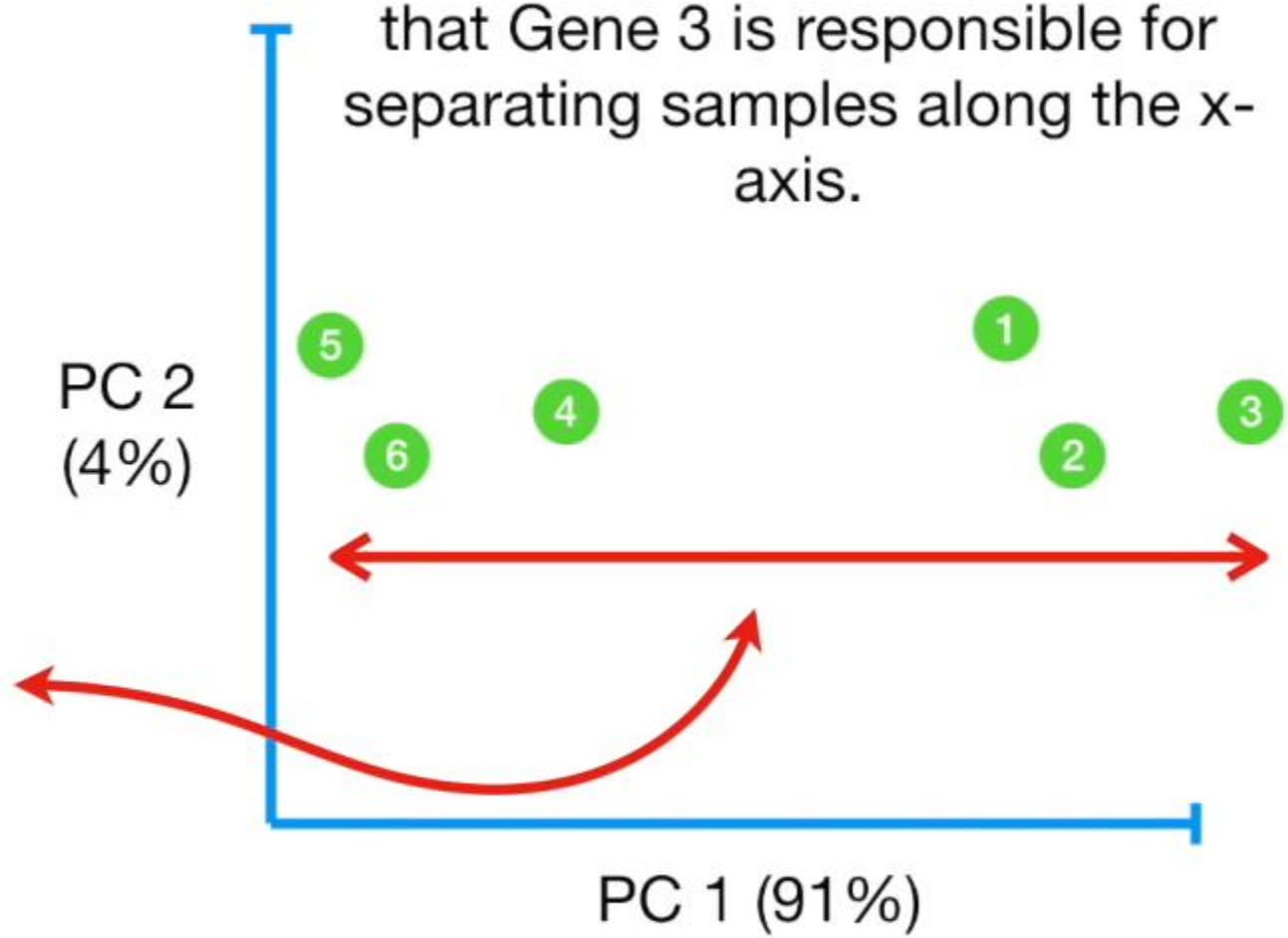
	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1
Gene 3	12	9	10	2.5	1.3	2
Gene 4	5	7	6	2	4	7

So we're going to talk about how PCA can take 4 or more gene measurements (and thus, 4 or more dimensions of data), and make a 2-D PCA plot…



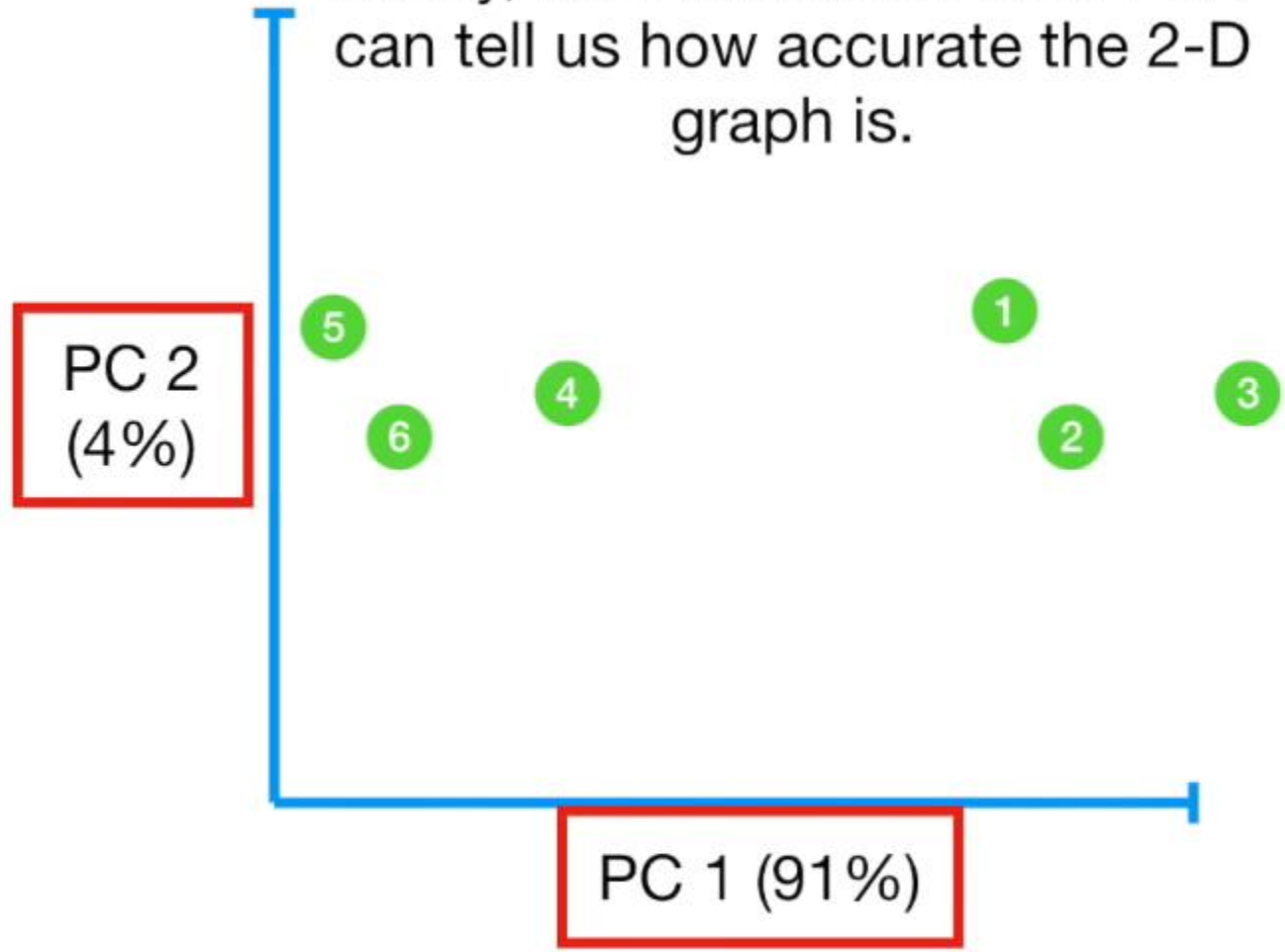
	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1
Gene 3	12	9	10	2.5	1.3	2
Gene 4	5	7	6	2	4	7

For example, PCA might tell us that Gene 3 is responsible for separating samples along the x-axis.



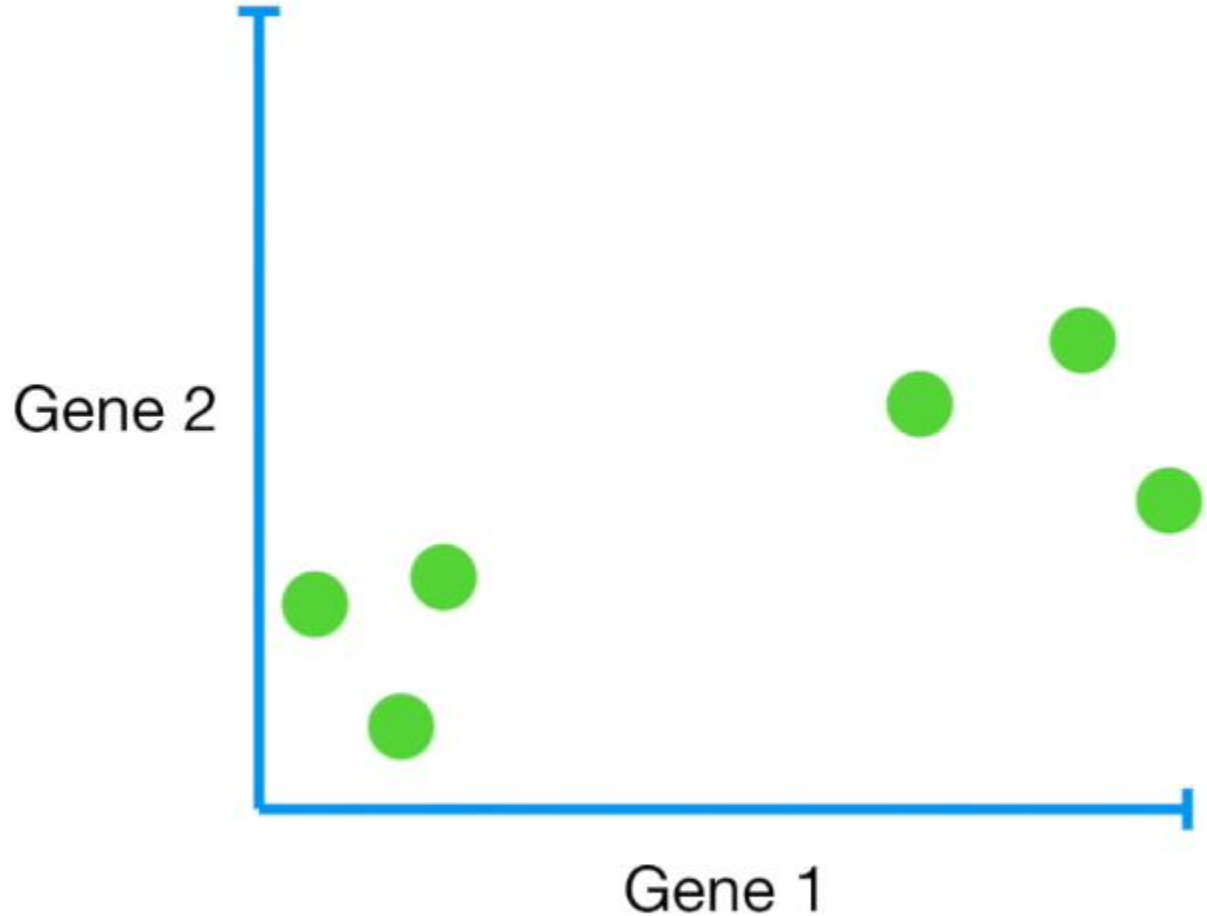
	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1
Gene 3	12	9	10	2.5	1.3	2
Gene 4	5	7	6	2	4	7

Lastly, we'll talk about how PCA can tell us how accurate the 2-D graph is.



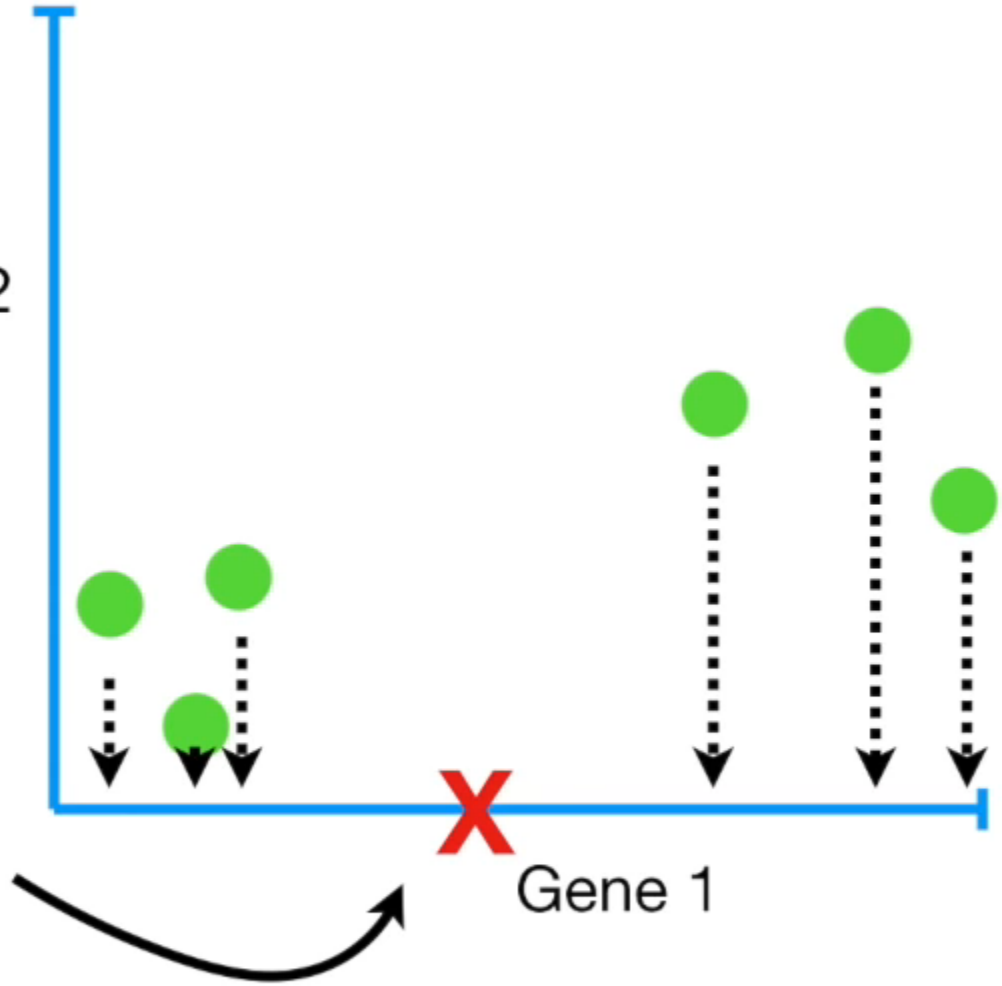
	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

We'll start by plotting the data…

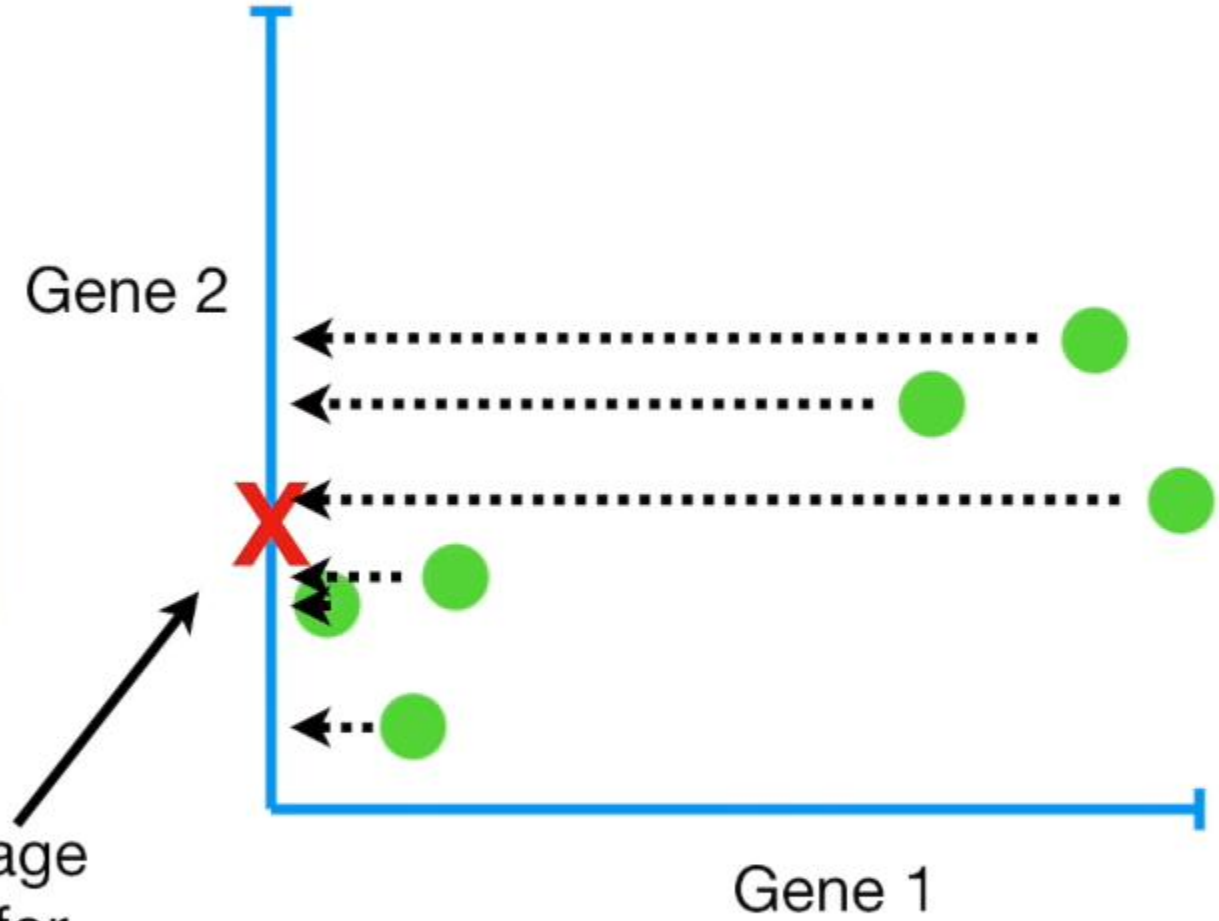


	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

Then we'll calculate the average measurement for Gene 1…

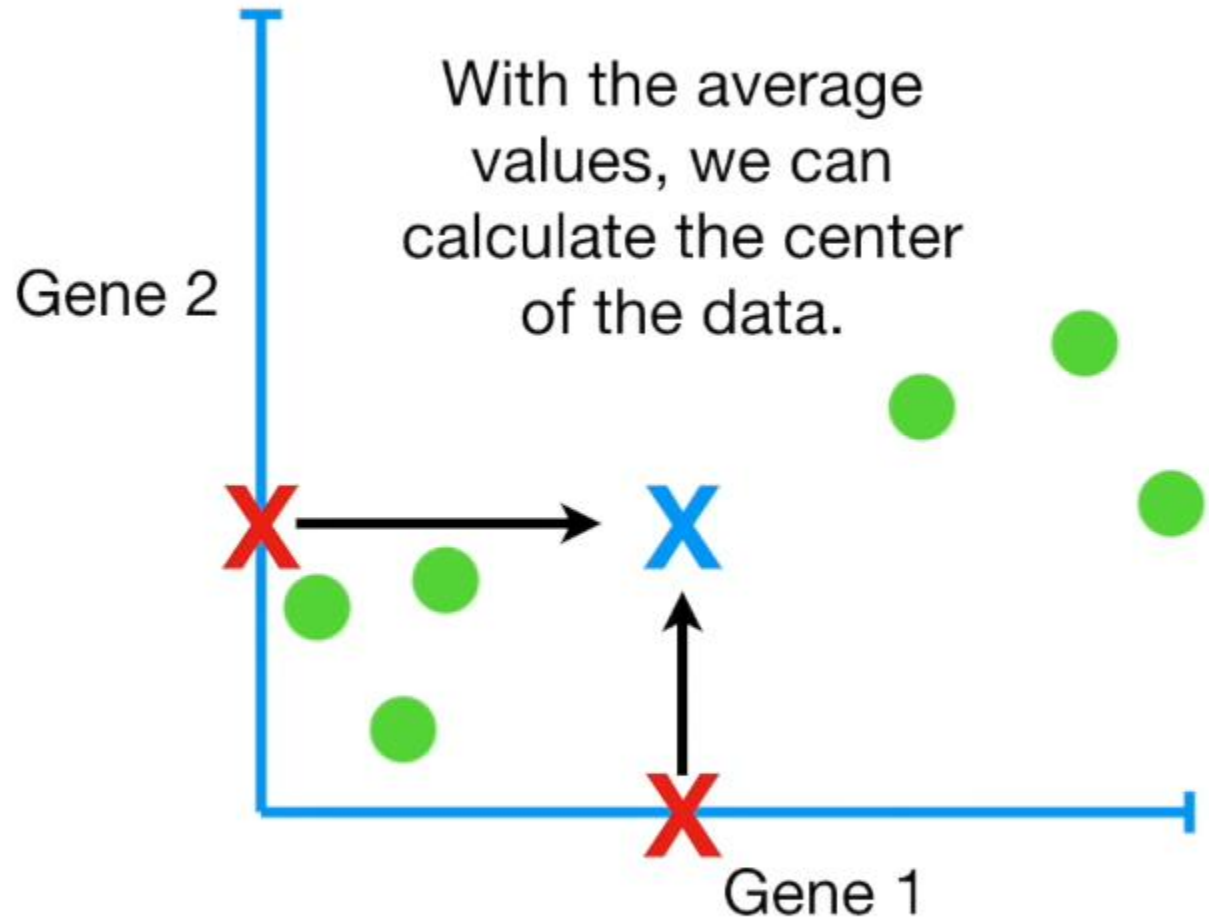


	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1



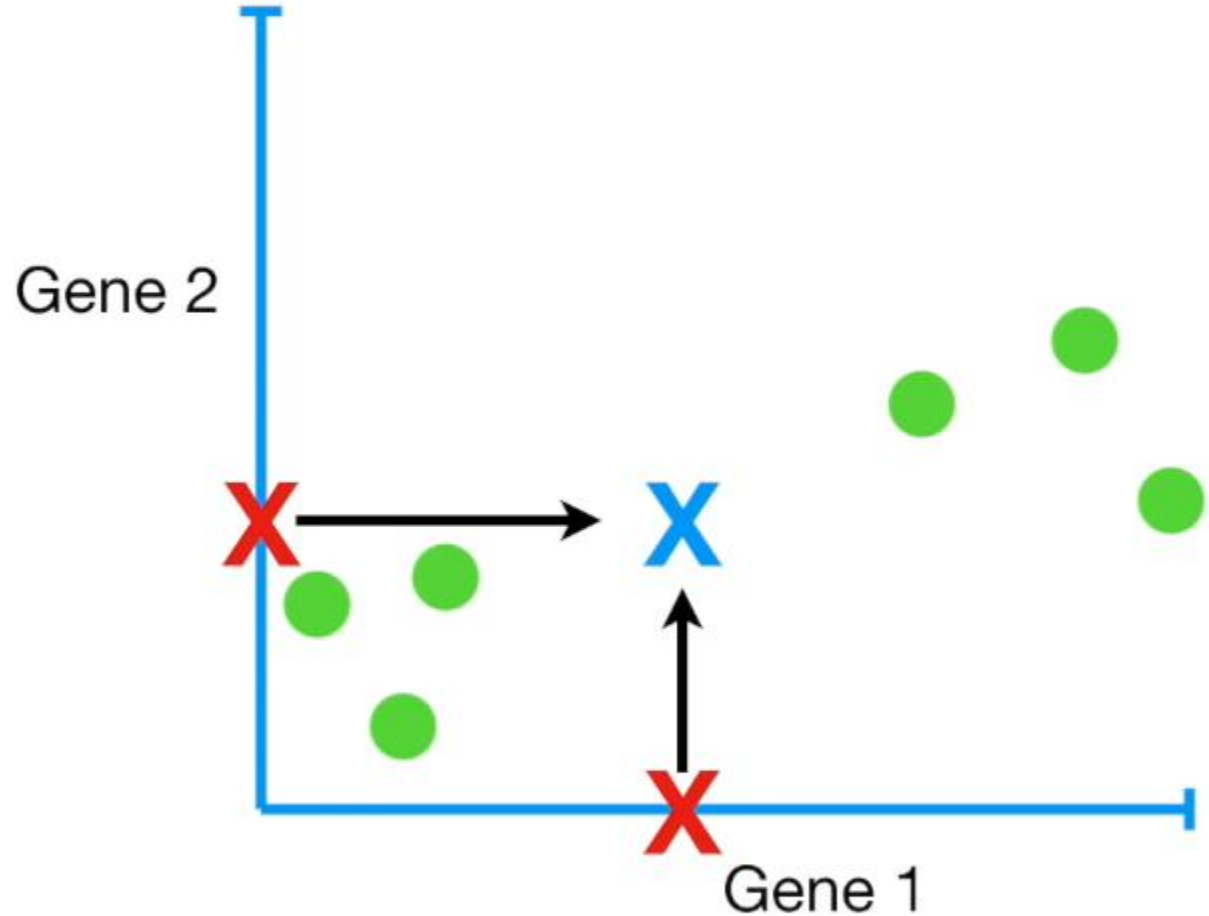
…and the average measurement for Gene 2.

	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

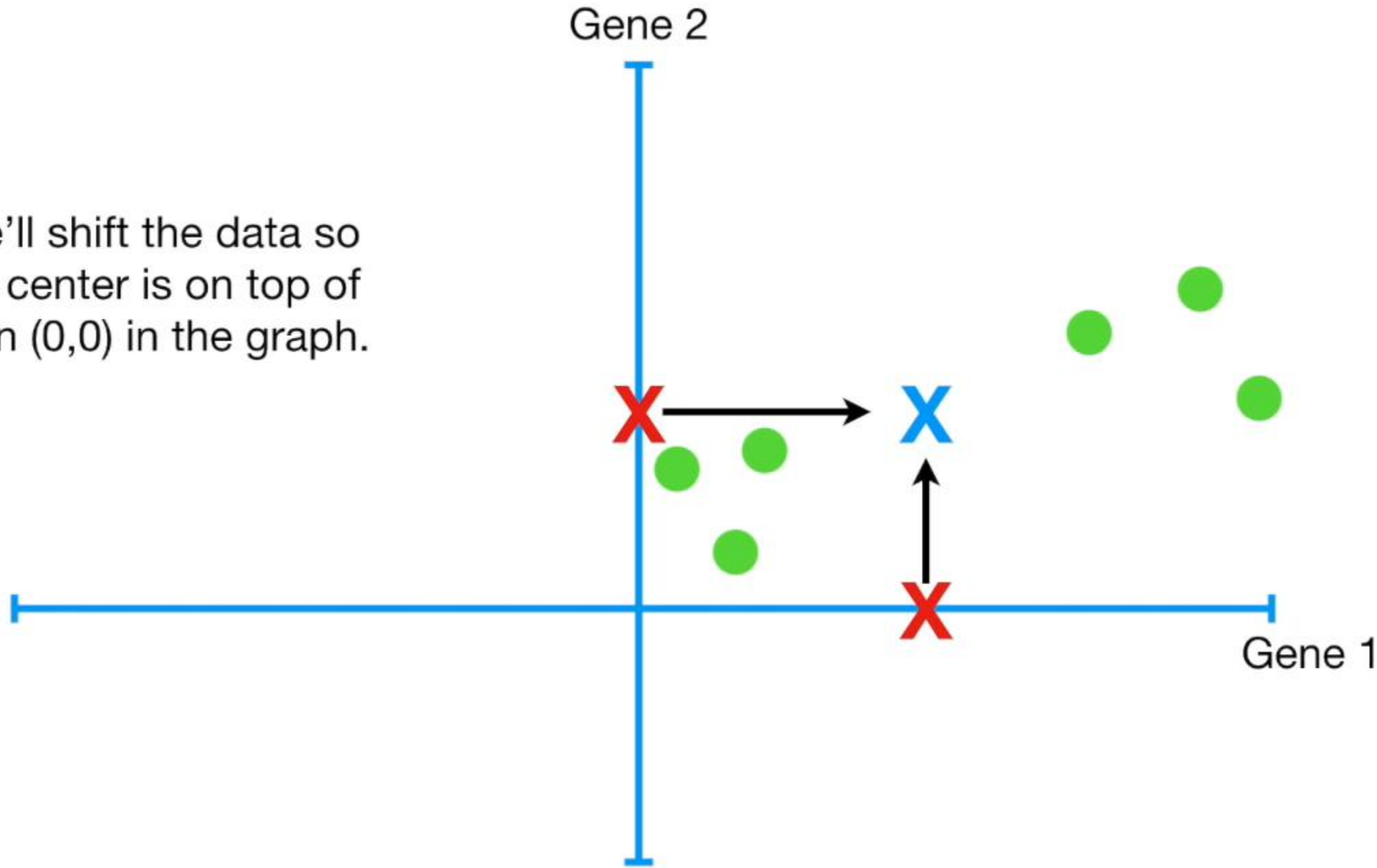


	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1

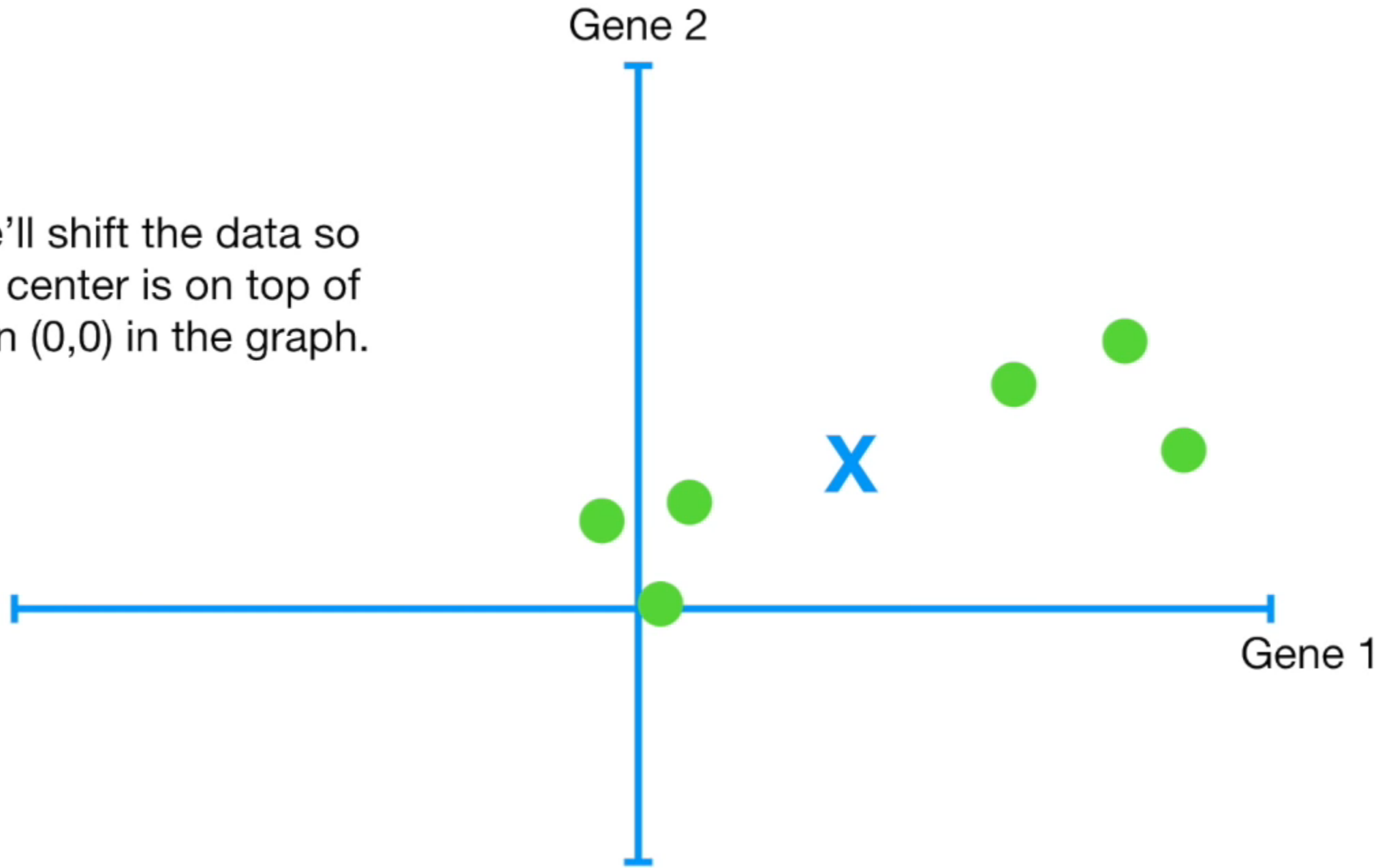
From this point on, we'll focus on what happens in the graph; we no longer need the original data…



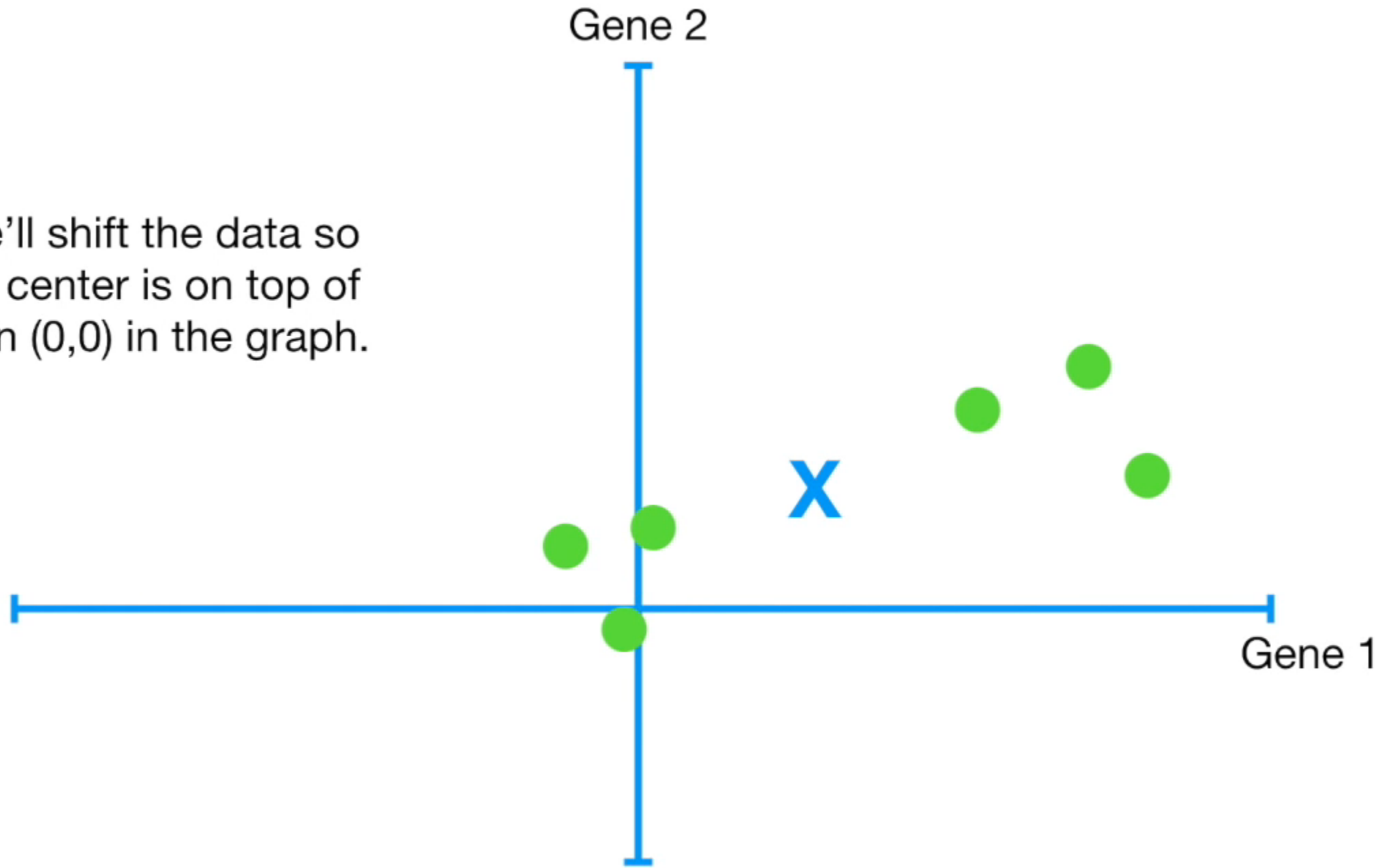
Now we'll shift the data so that the center is on top of the origin (0,0) in the graph.



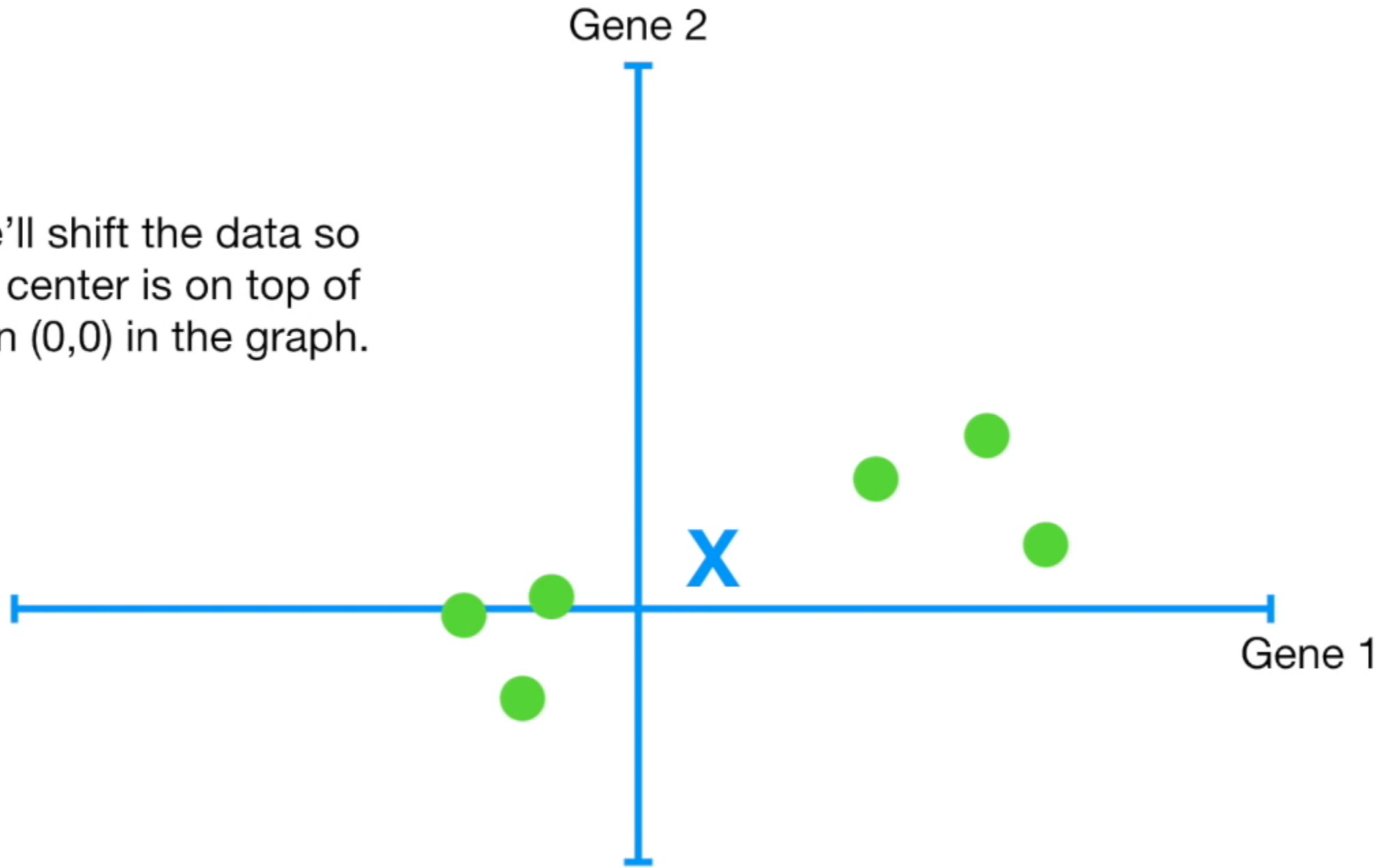
Now we'll shift the data so that the center is on top of the origin (0,0) in the graph.



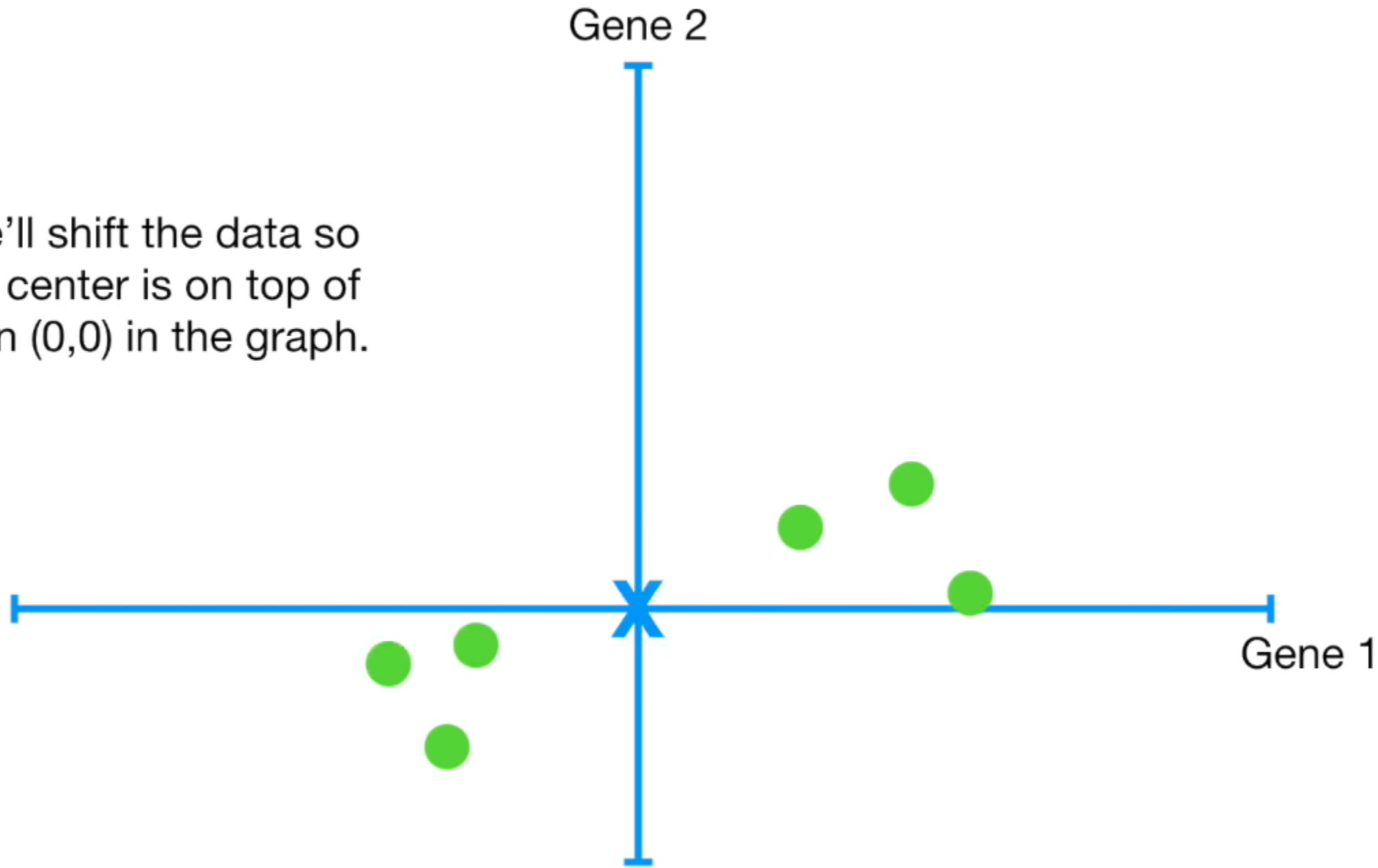
Now we'll shift the data so that the center is on top of the origin (0,0) in the graph.



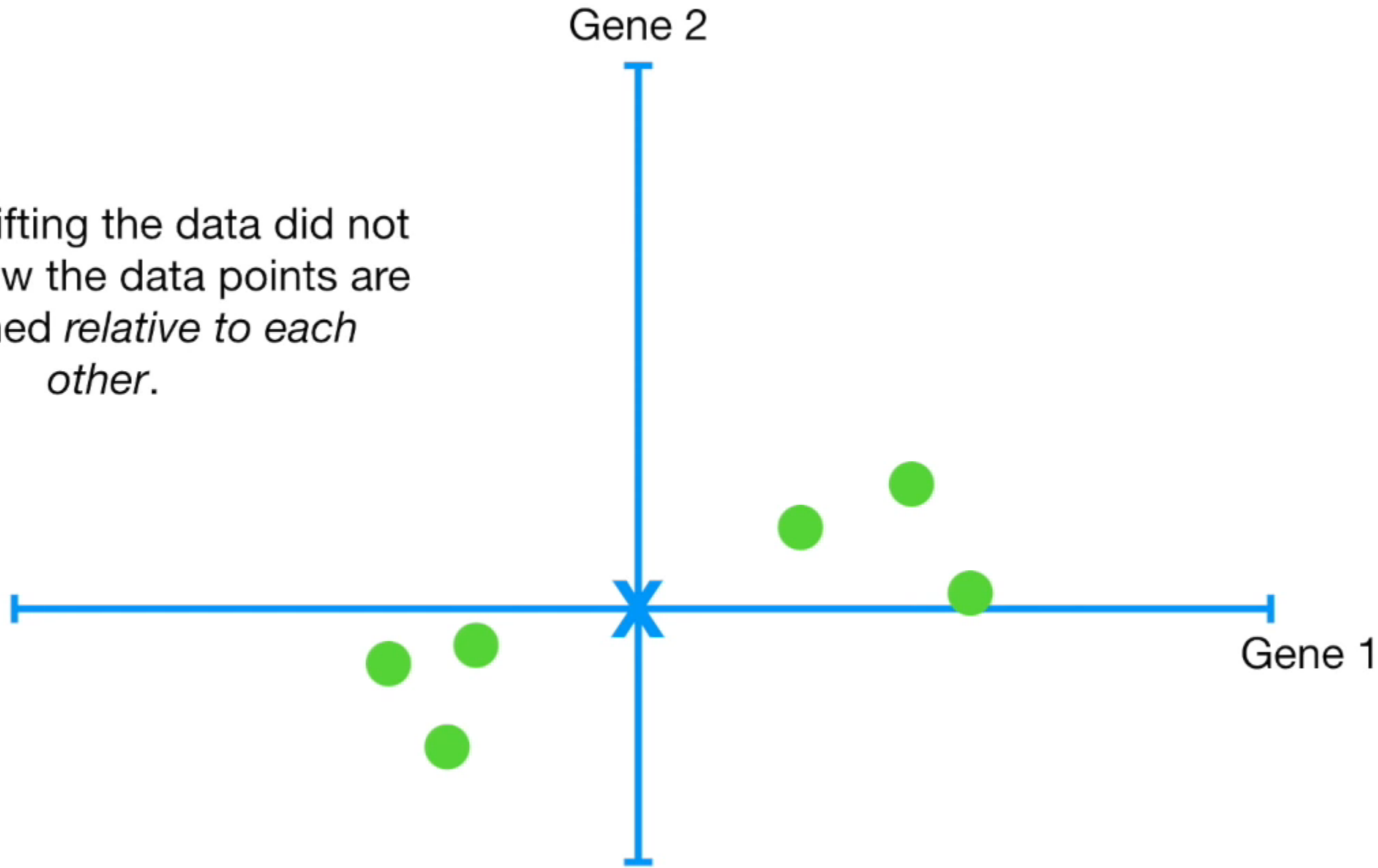
Now we'll shift the data so that the center is on top of the origin (0,0) in the graph.



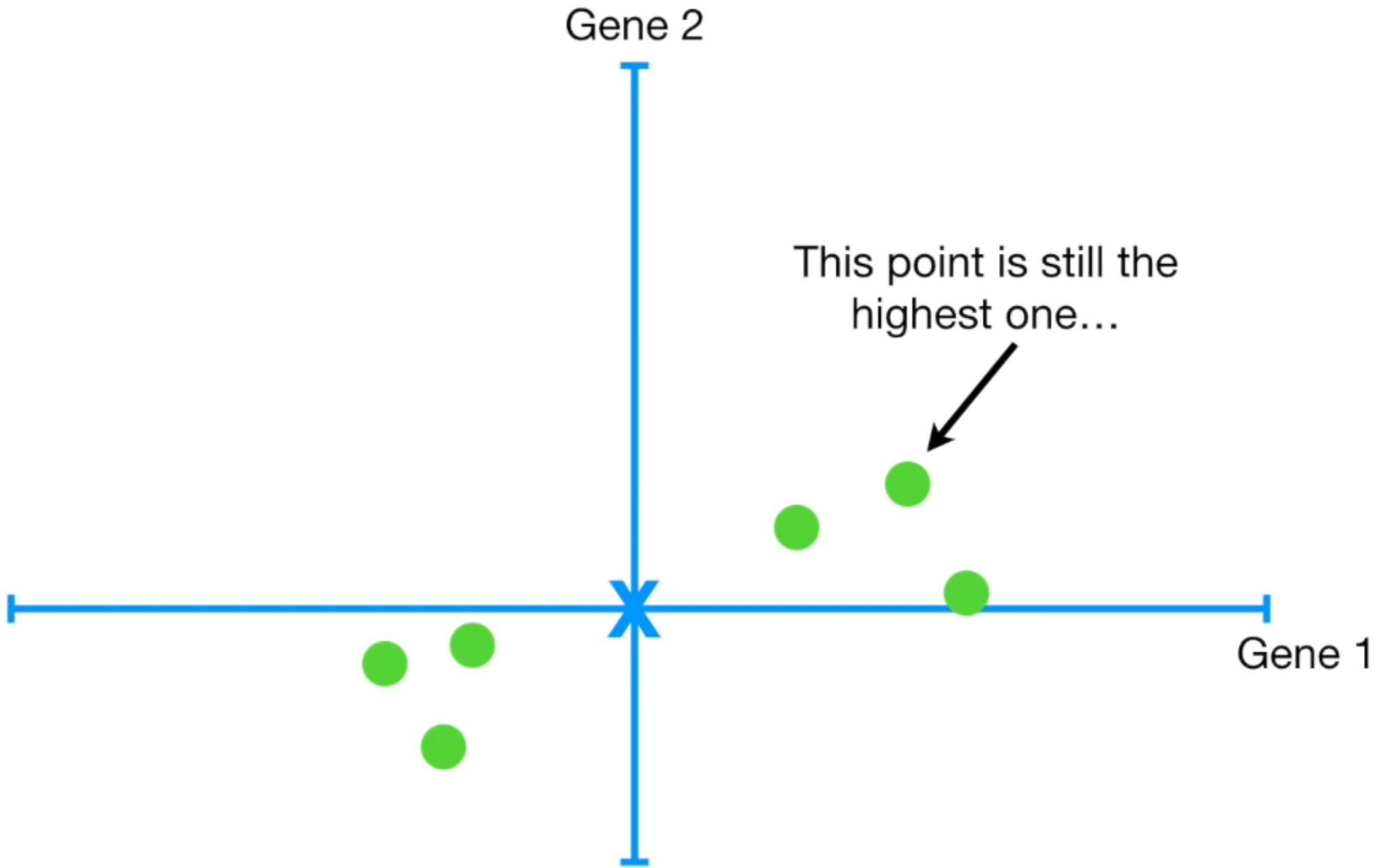
Now we'll shift the data so that the center is on top of the origin (0,0) in the graph.



NOTE: Shifting the data did not change how the data points are positioned *relative to each other*.

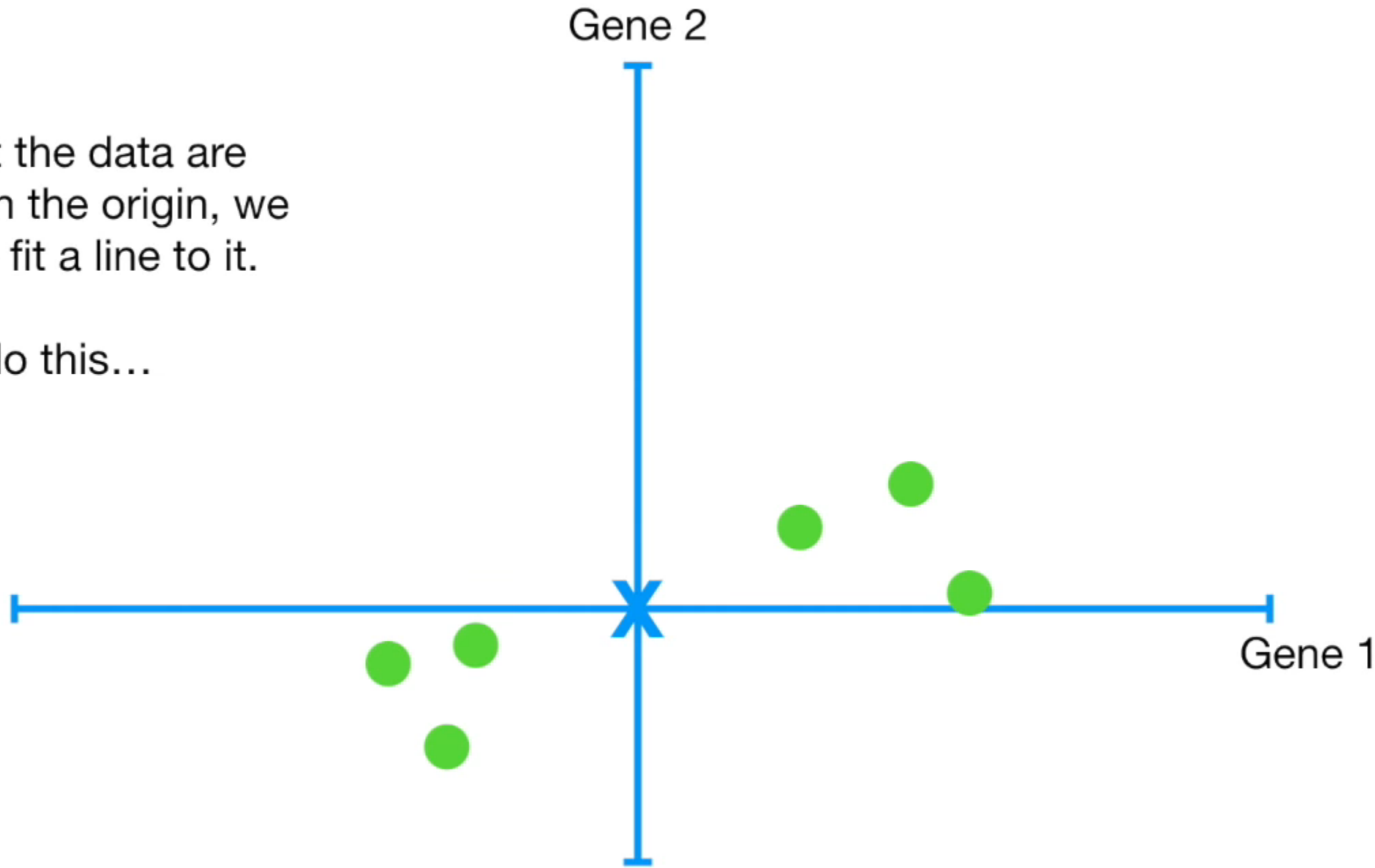


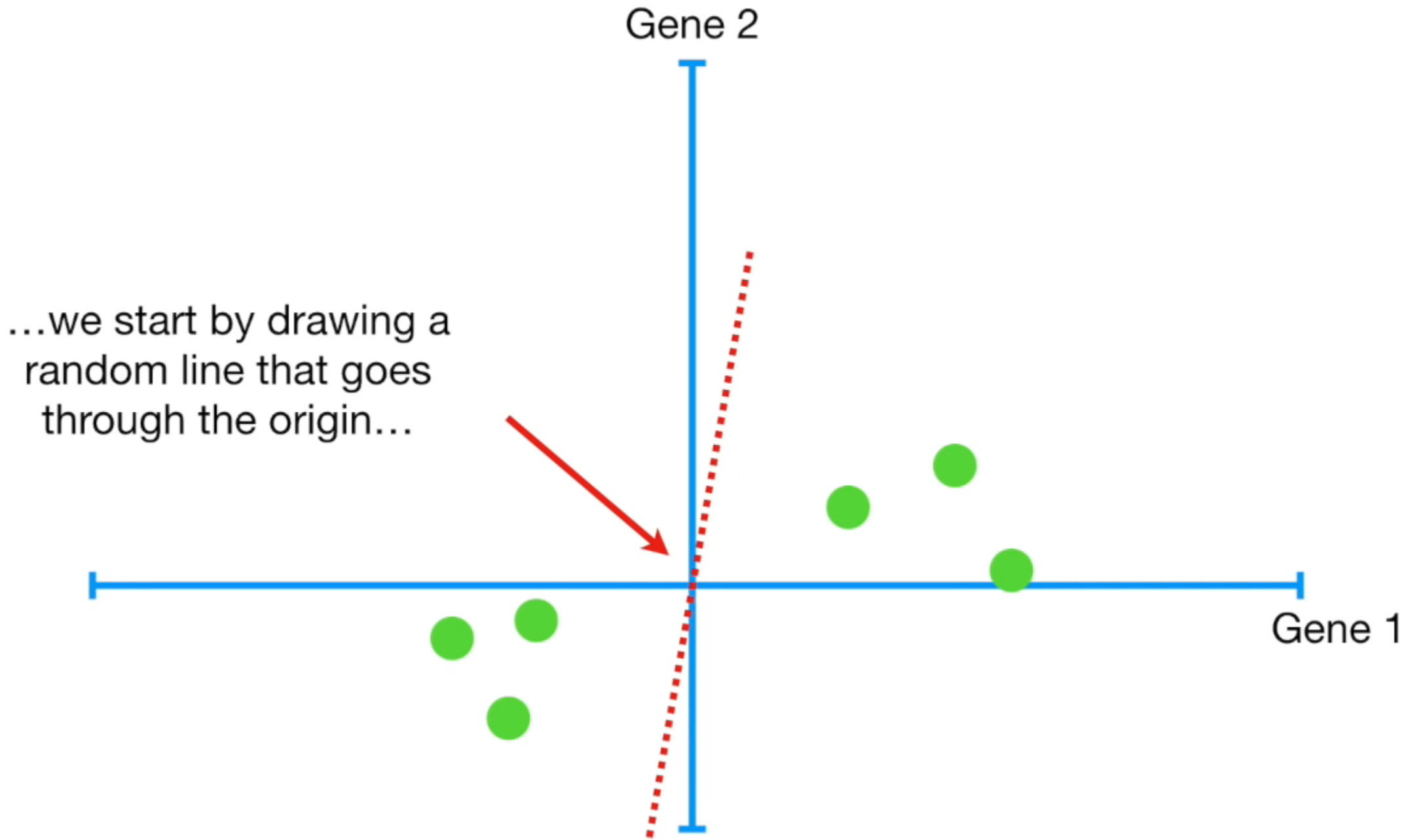
Q: How can we make the data centered around its mean?


Gene 2
This point is still the highest one…
Gene 1

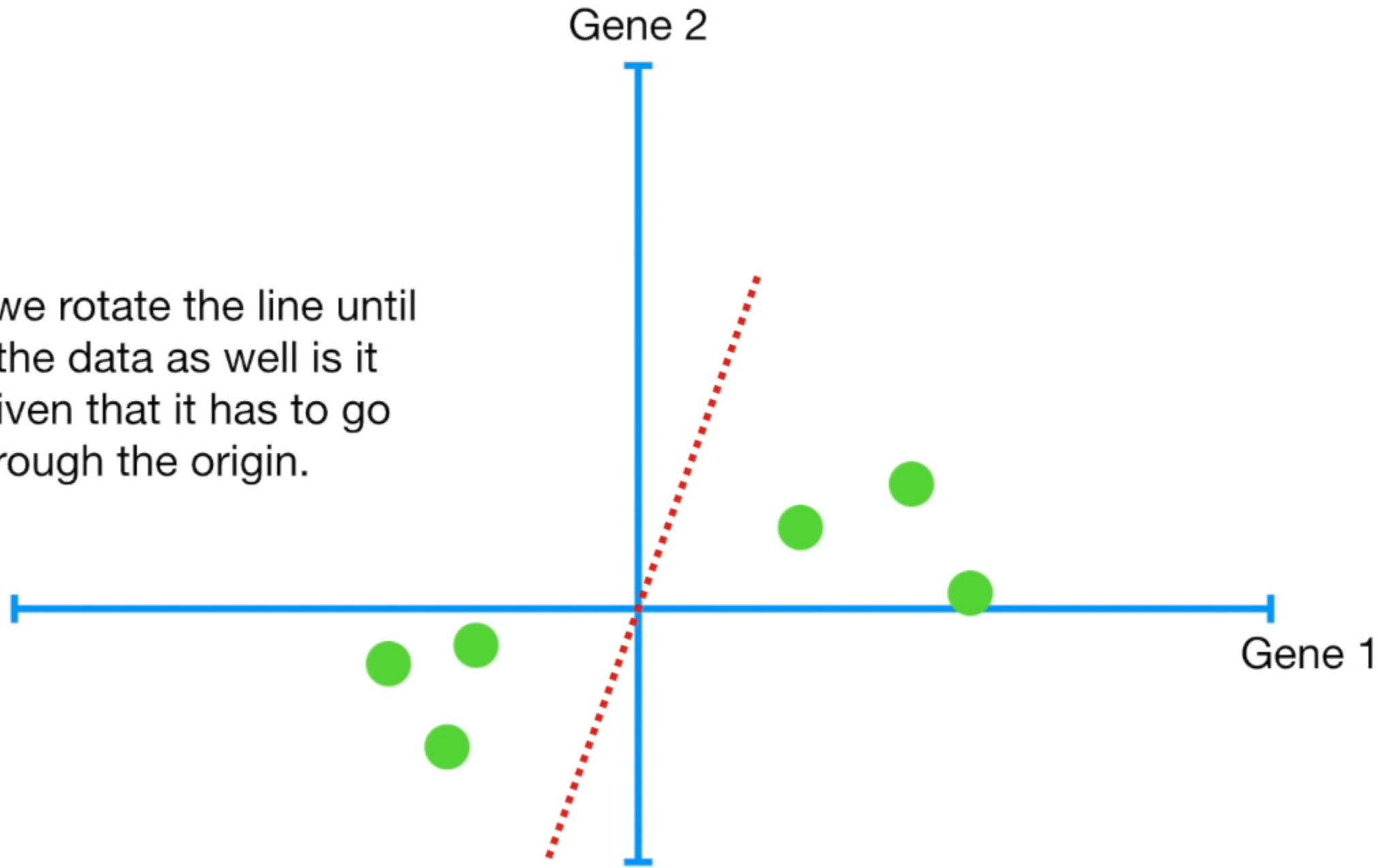
Now that the data are centered on the origin, we can try to fit a line to it.

To do this…

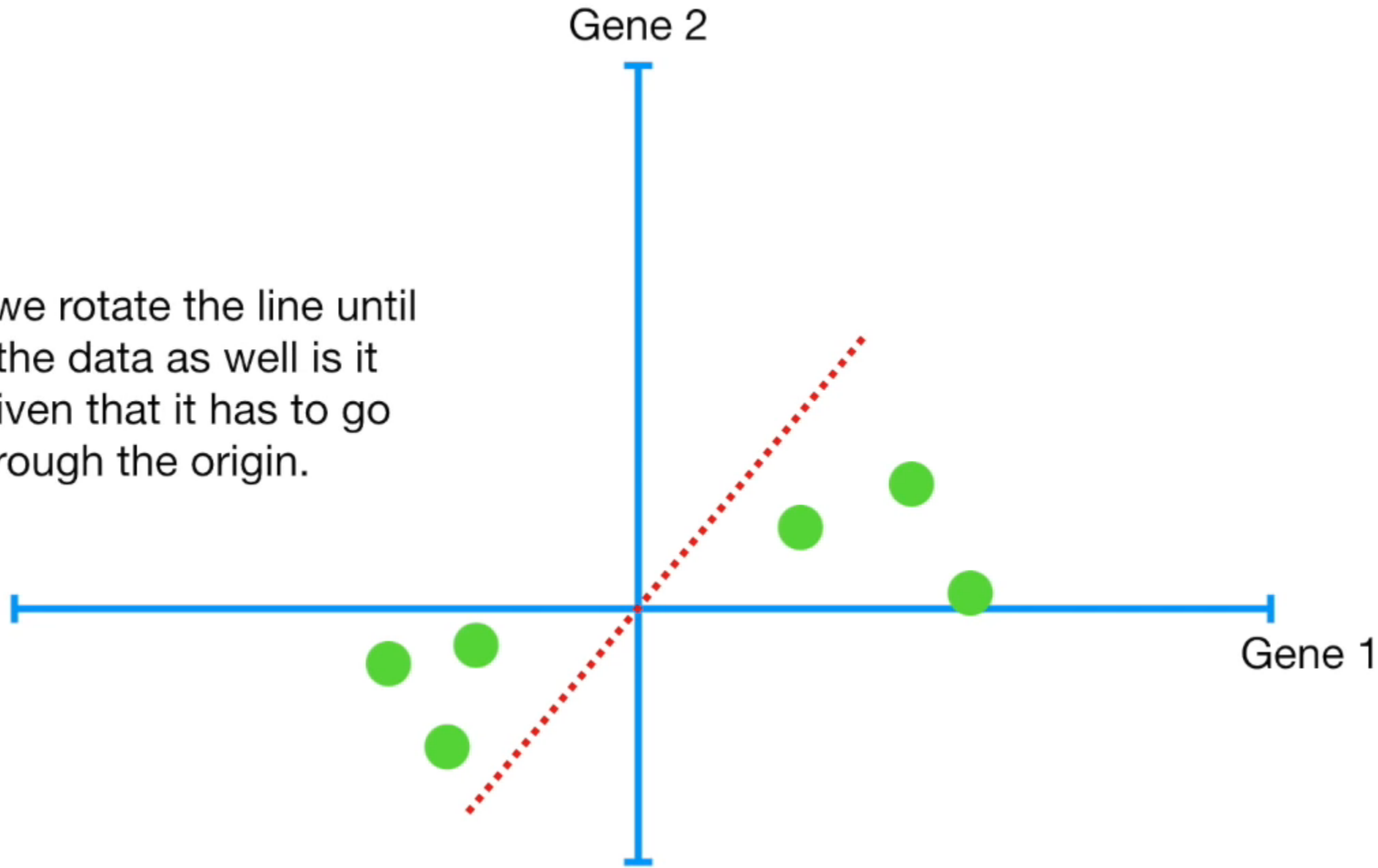



Gene 2
…we start by drawing a random line that goes through the origin…
Gene 1

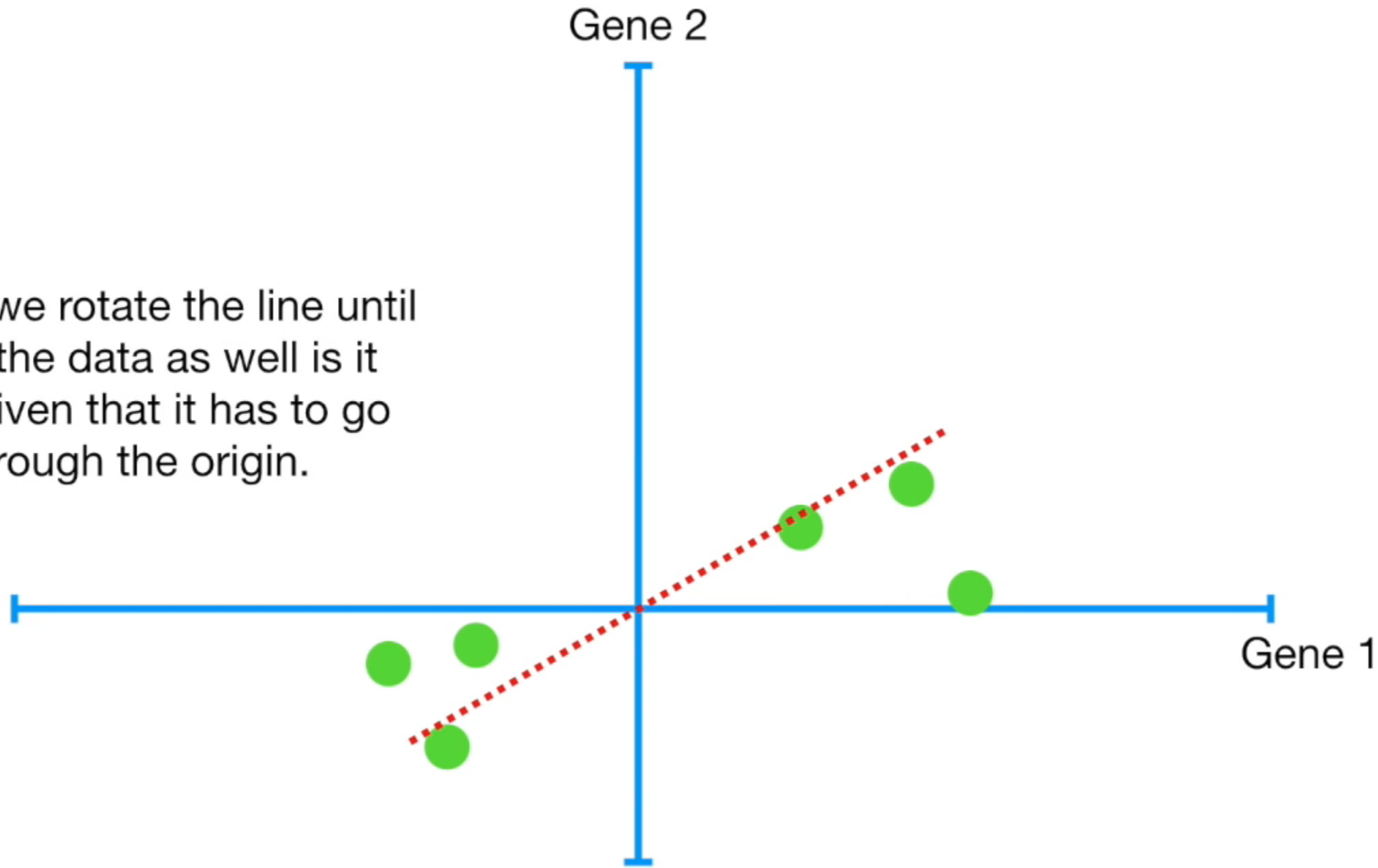
…then we rotate the line until it fits the data as well is it can, given that it has to go through the origin.

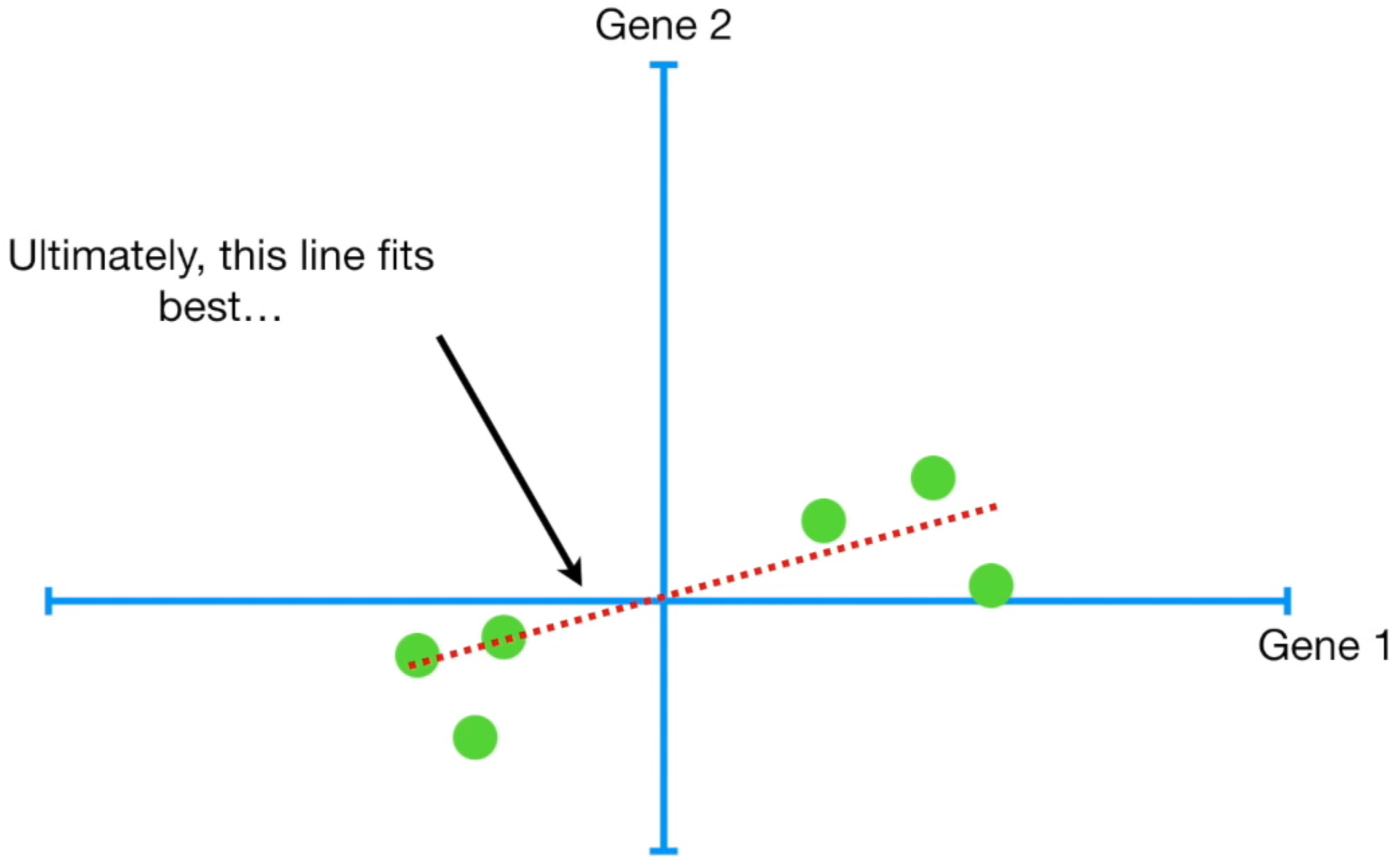


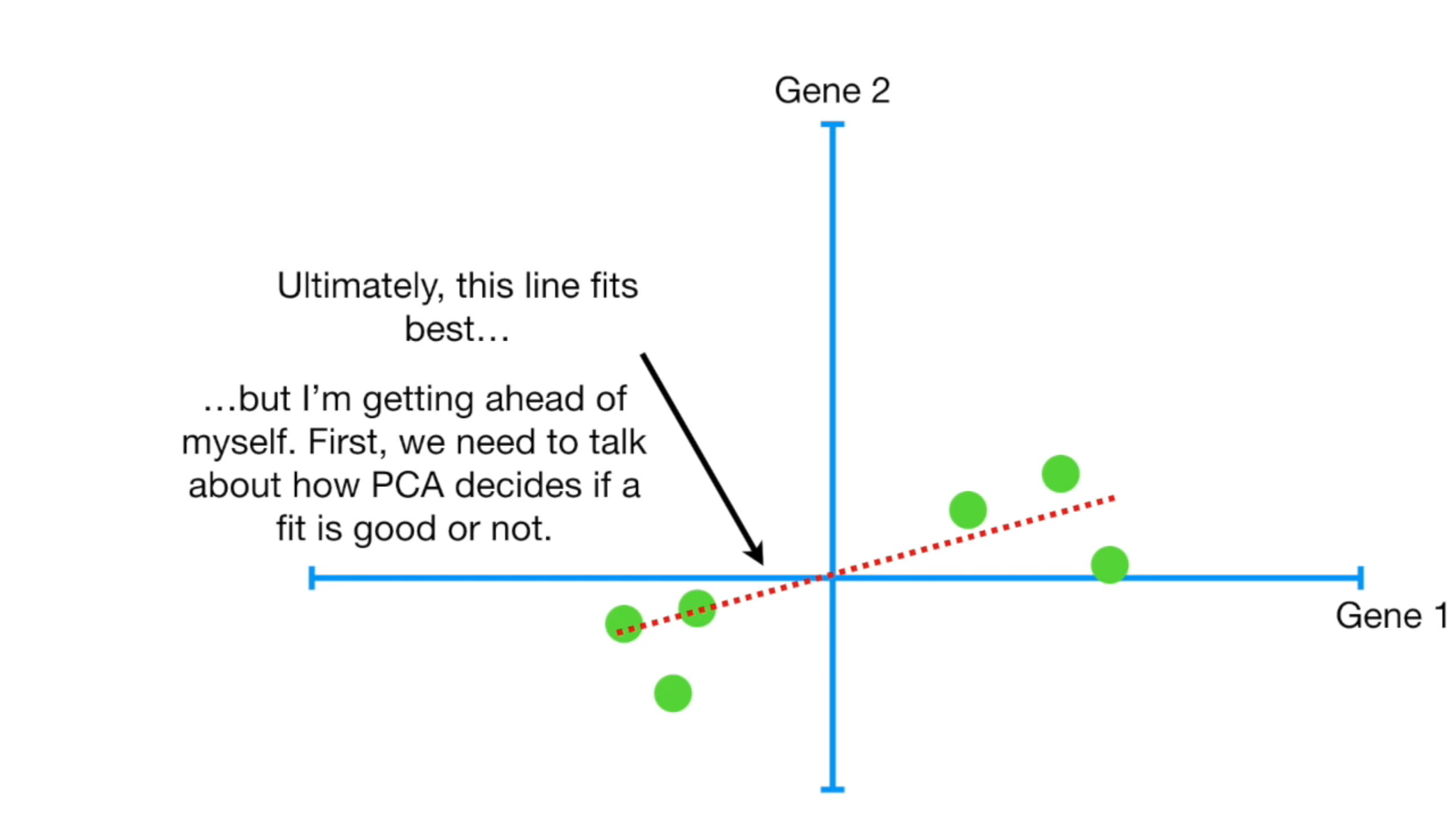
…then we rotate the line until it fits the data as well is it can, given that it has to go through the origin.



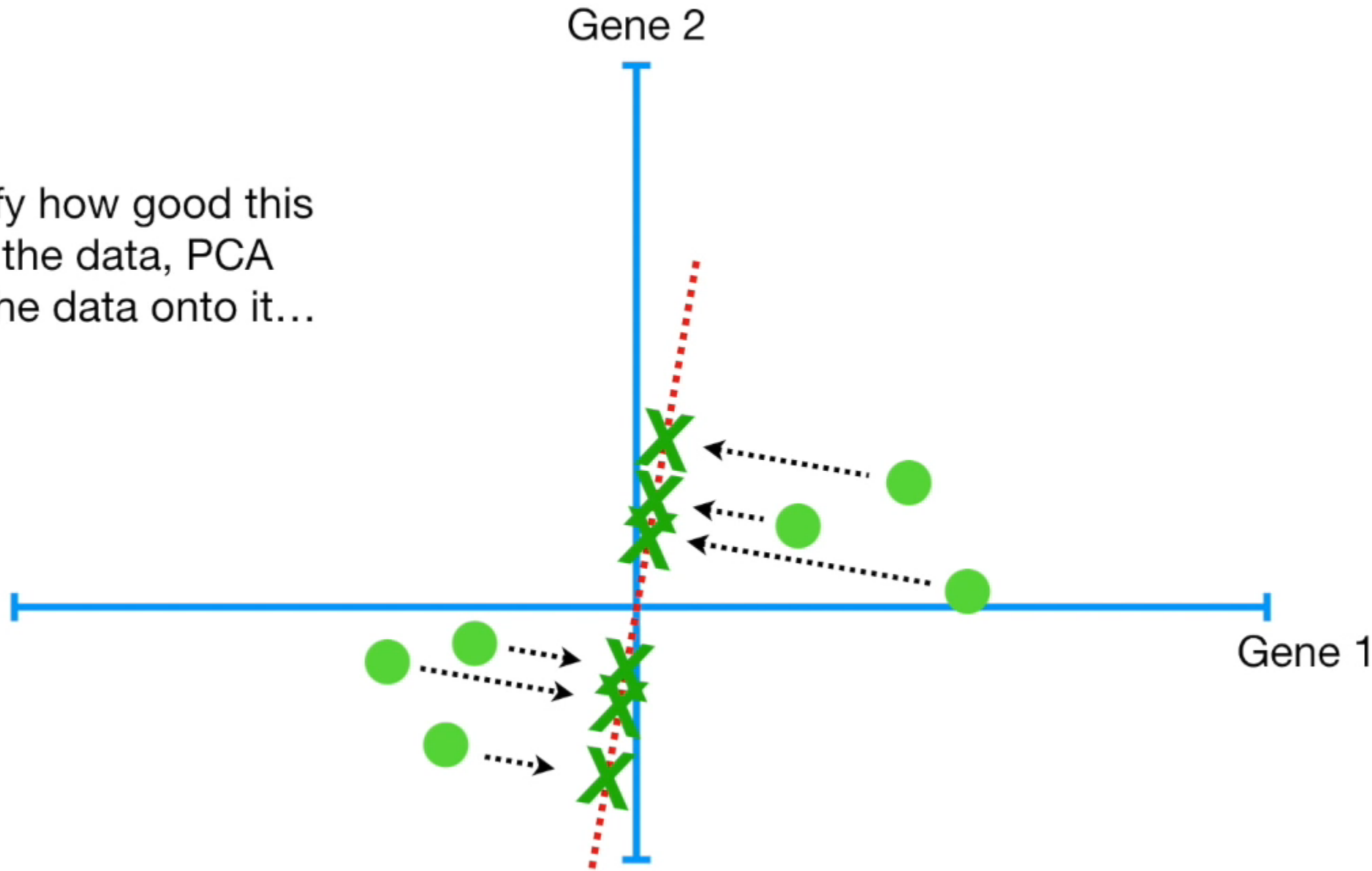
…then we rotate the line until it fits the data as well is it can, given that it has to go through the origin.

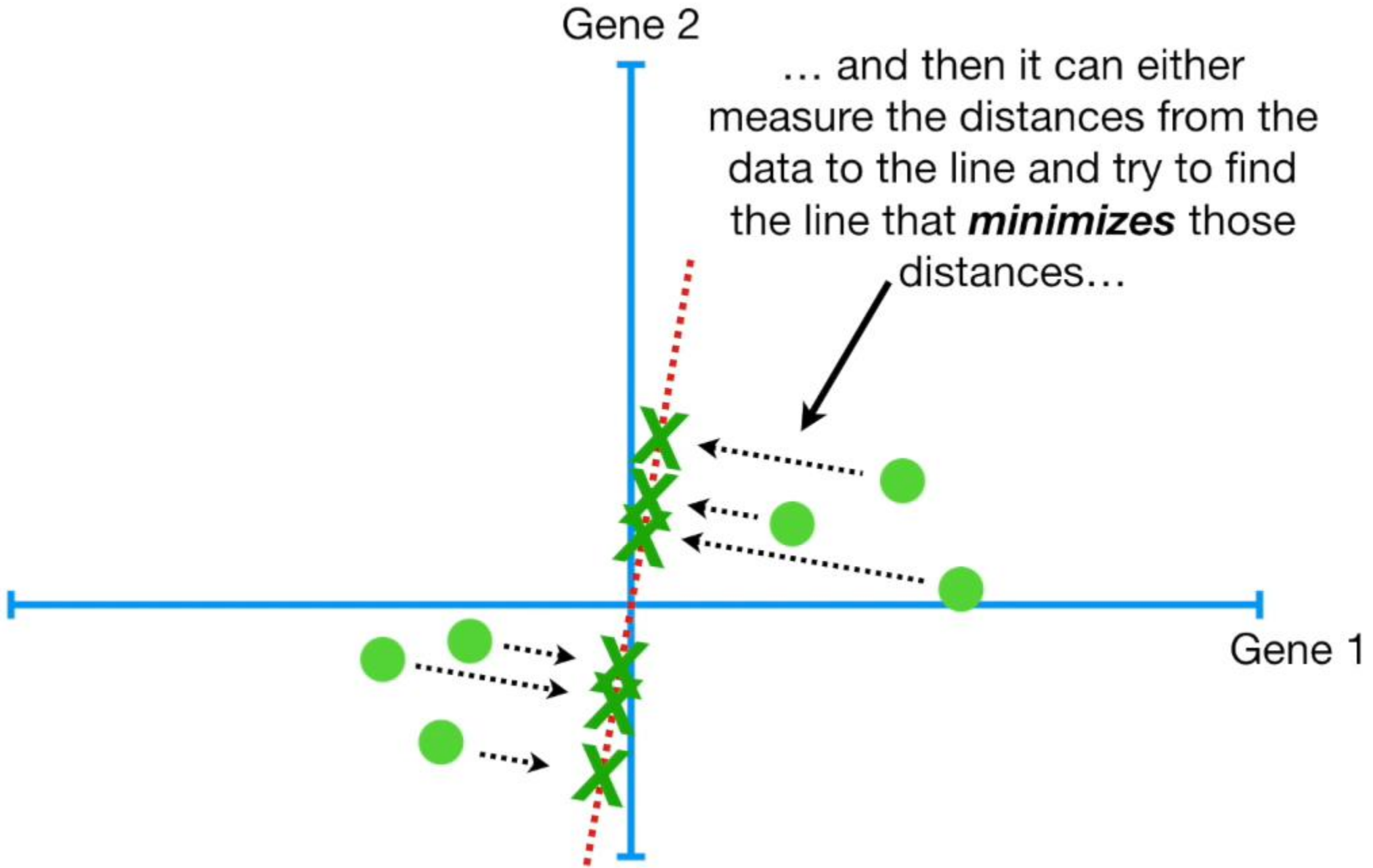



Gene 2
Ultimately, this line fits best…
Gene 1

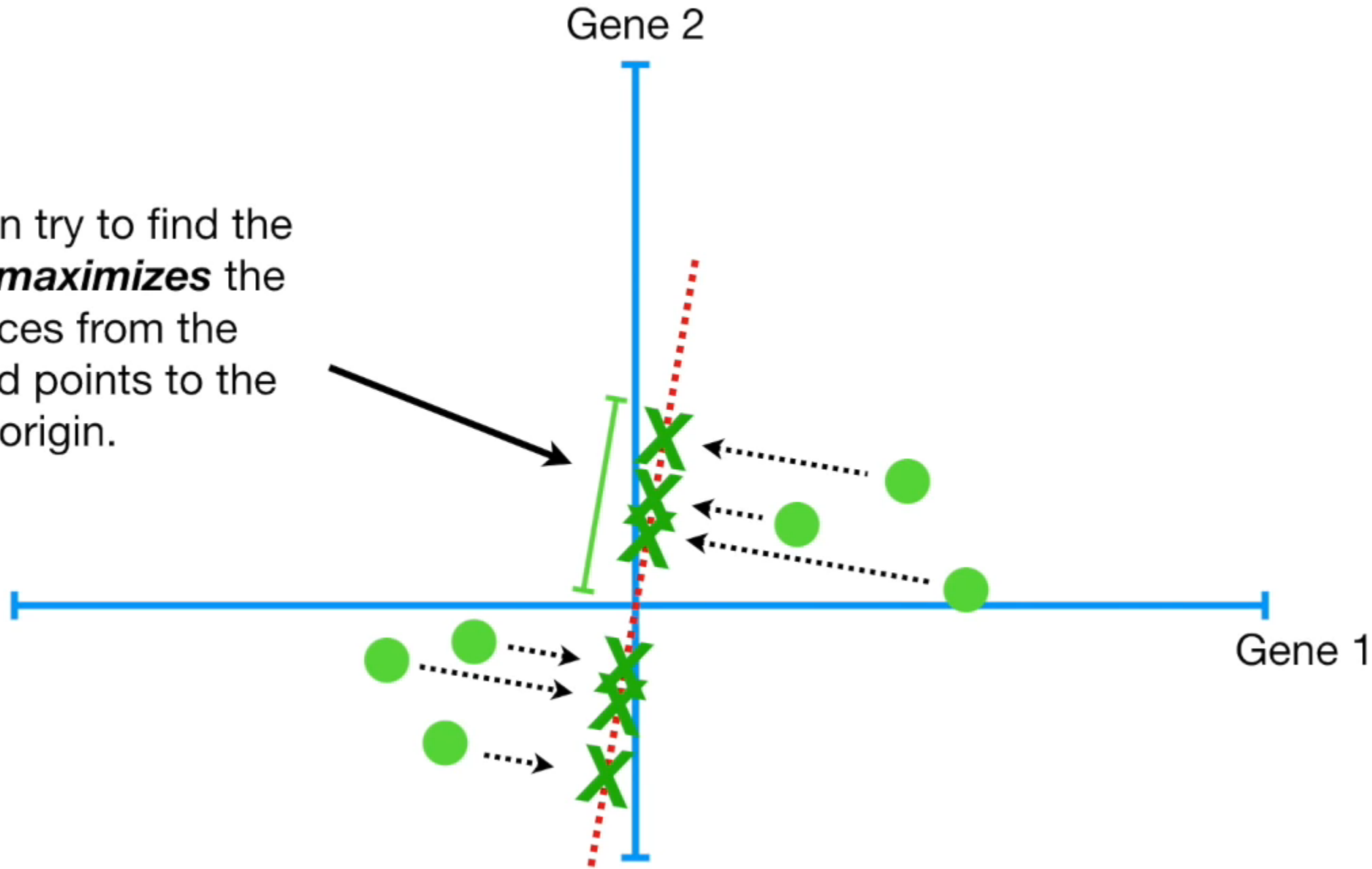

Gene 2
Ultimately, this line fits best…
…but I'm getting ahead of myself. First, we need to talk about how PCA decides if a fit is good or not.
Gene 1

To quantify how good this line fits the data, PCA projects the data onto it…

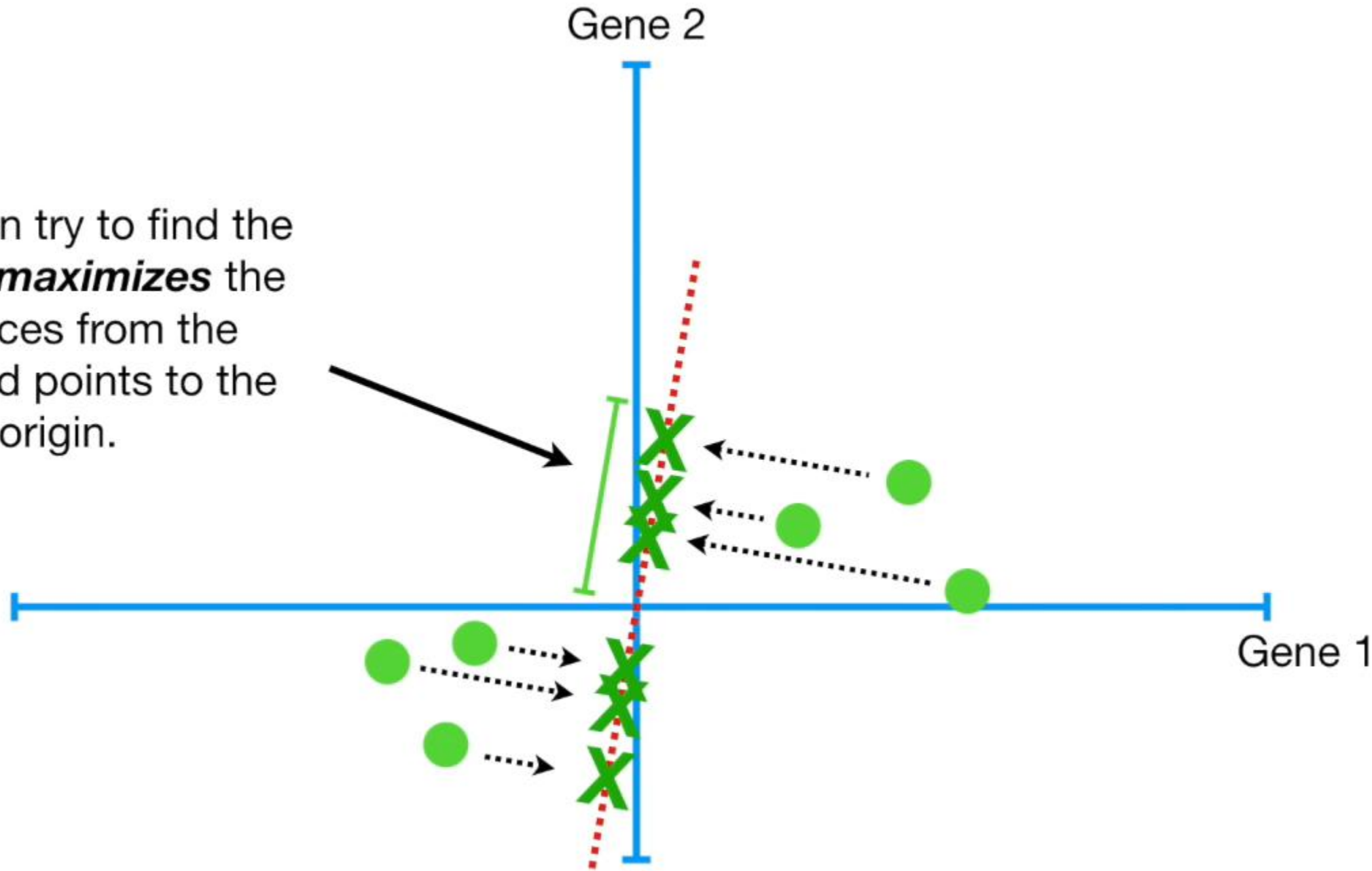



Gene 2
… and then it can either measure the distances from the data to the line and try to find the line that ***minimizes*** those distances…
Gene 1

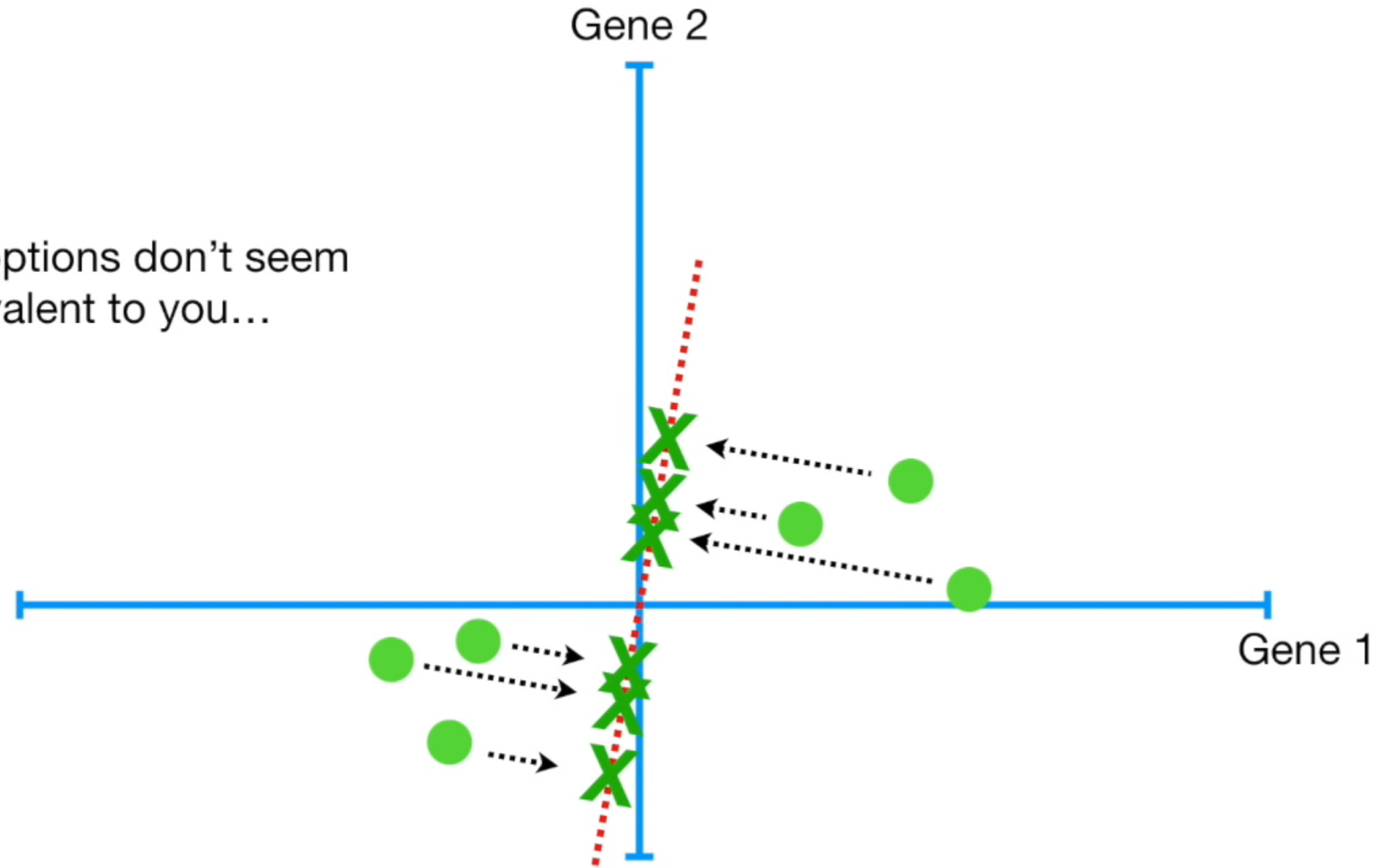
…or it can try to find the line that ***maximizes*** the distances from the projected points to the origin.

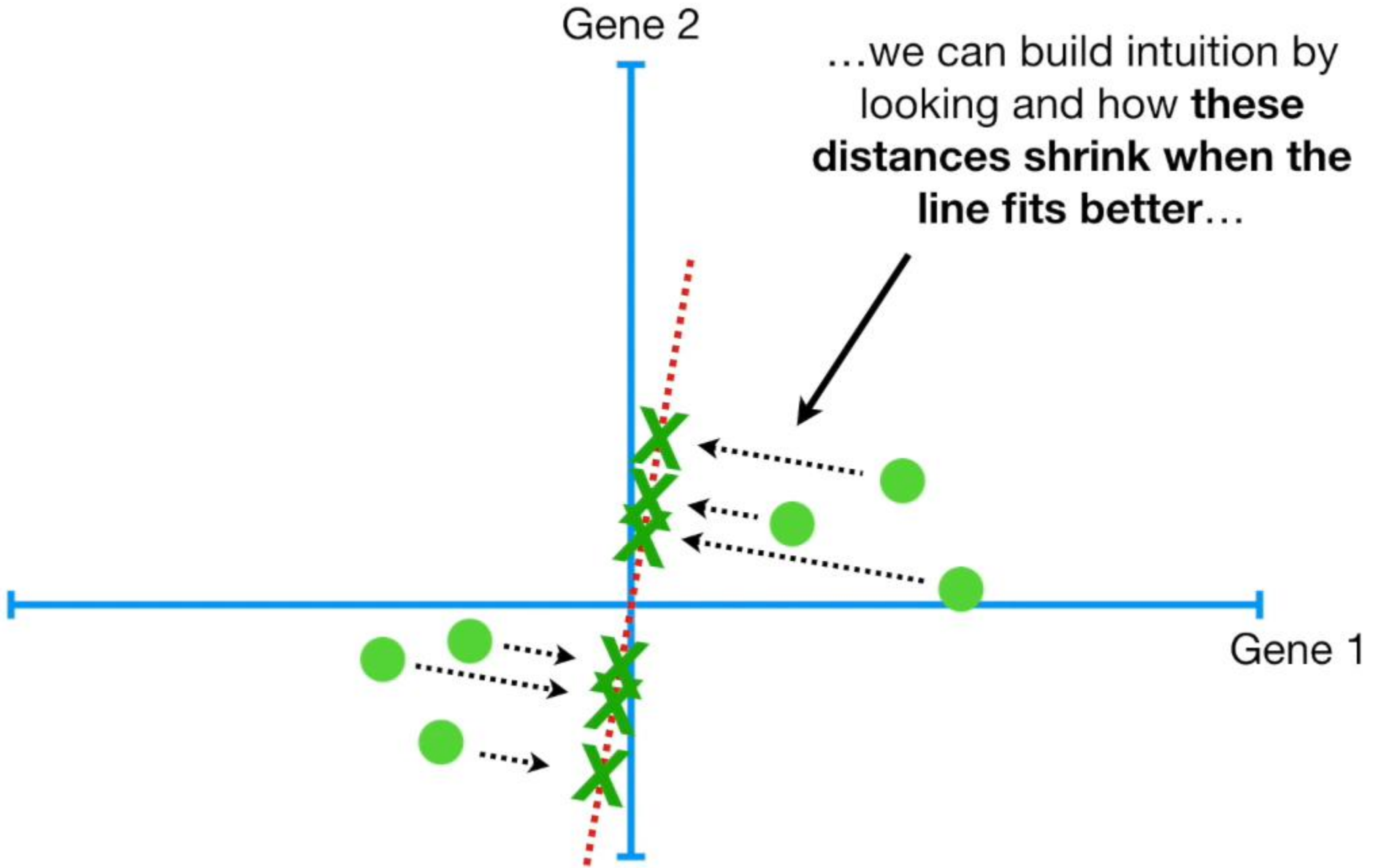


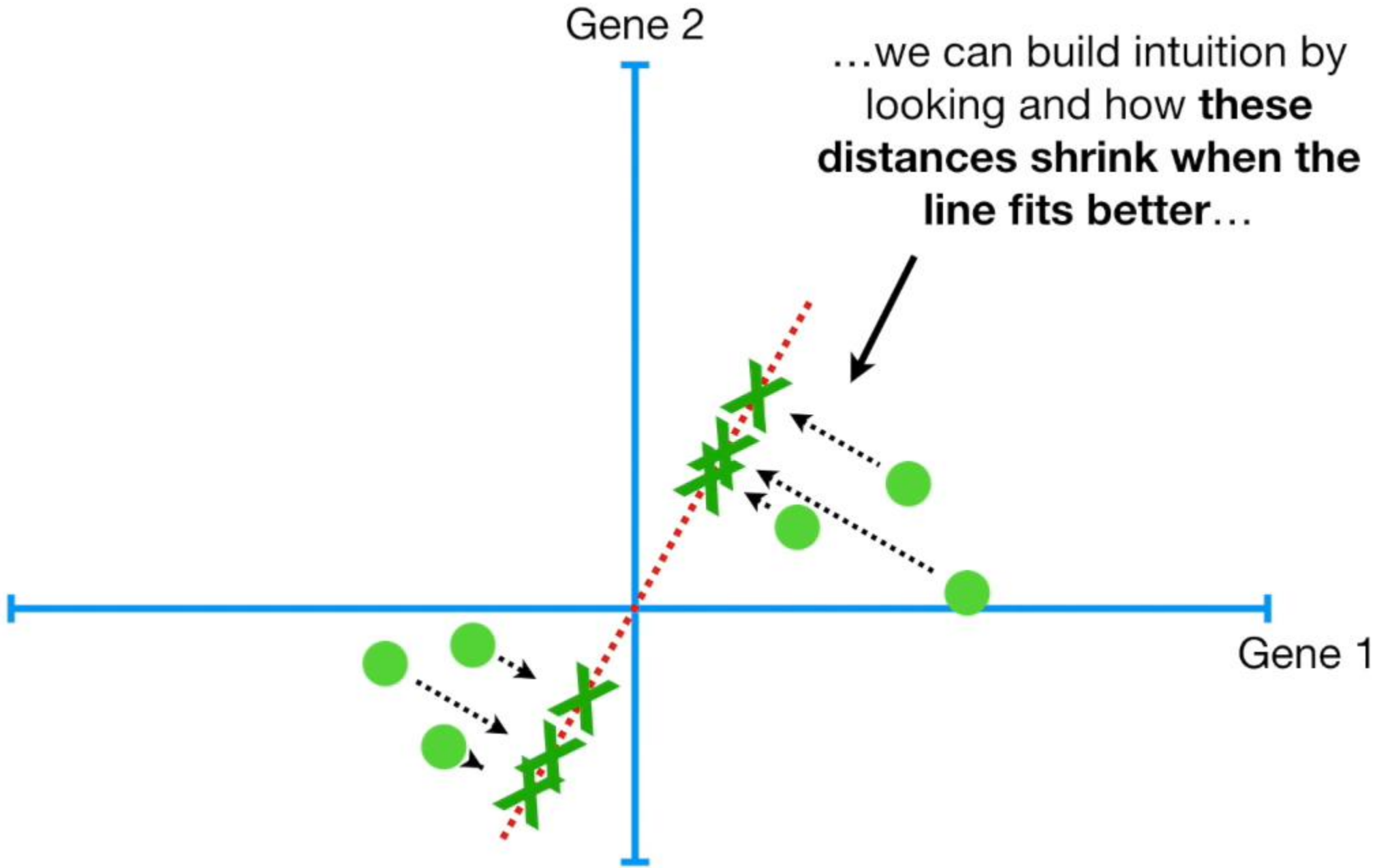
…or it can try to find the line that ***maximizes*** the distances from the projected points to the origin.

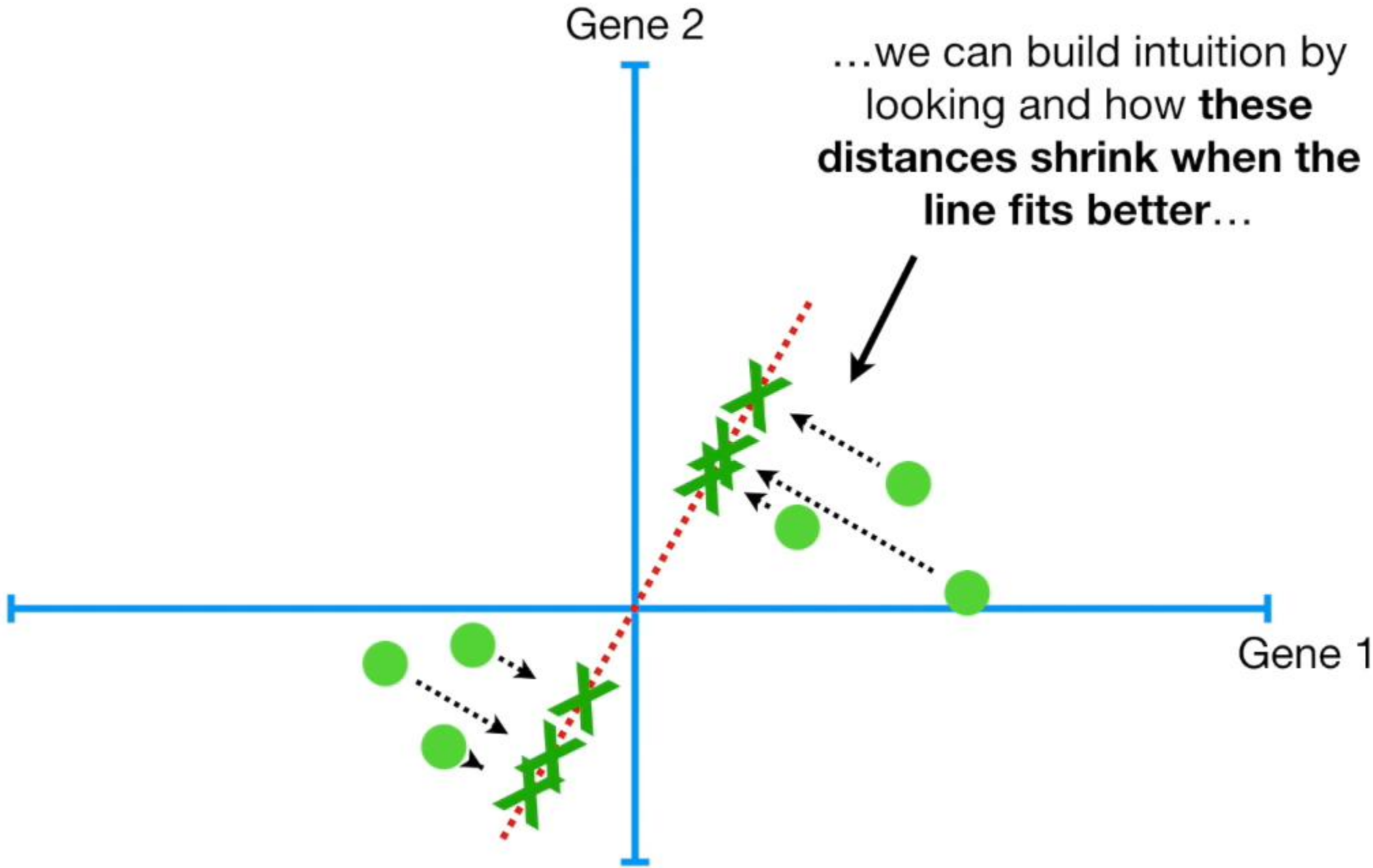


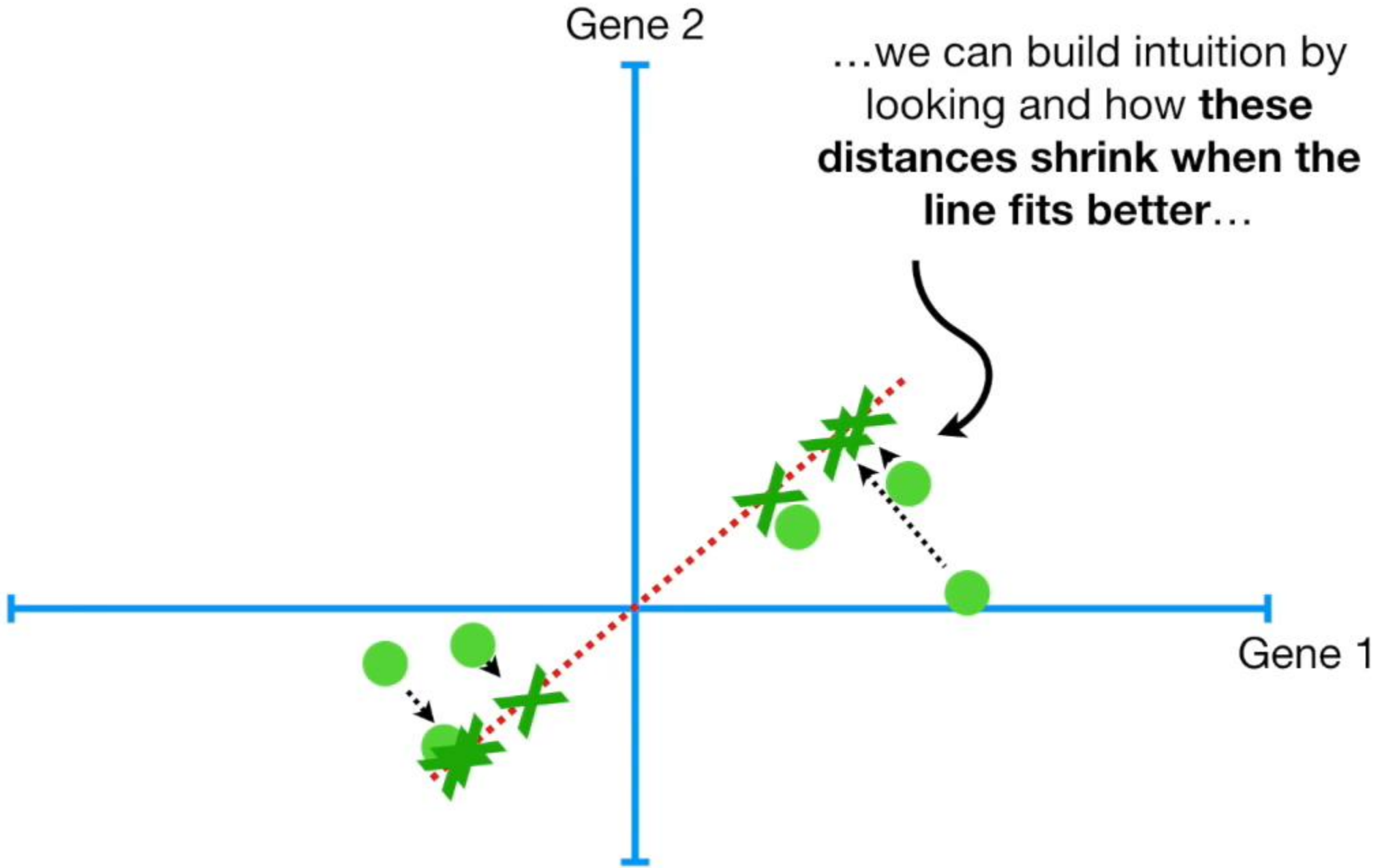
If those options don't seem equivalent to you…

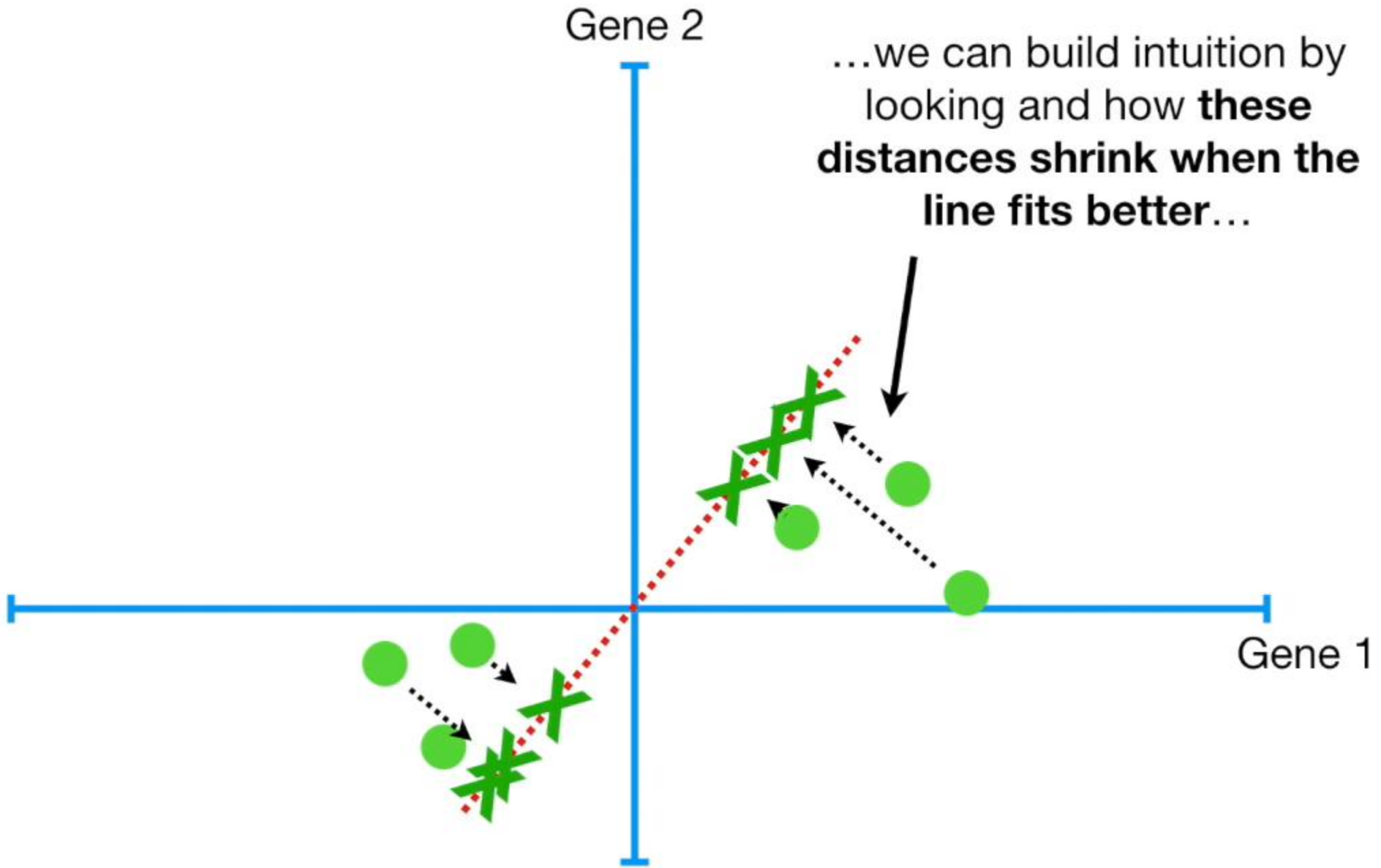


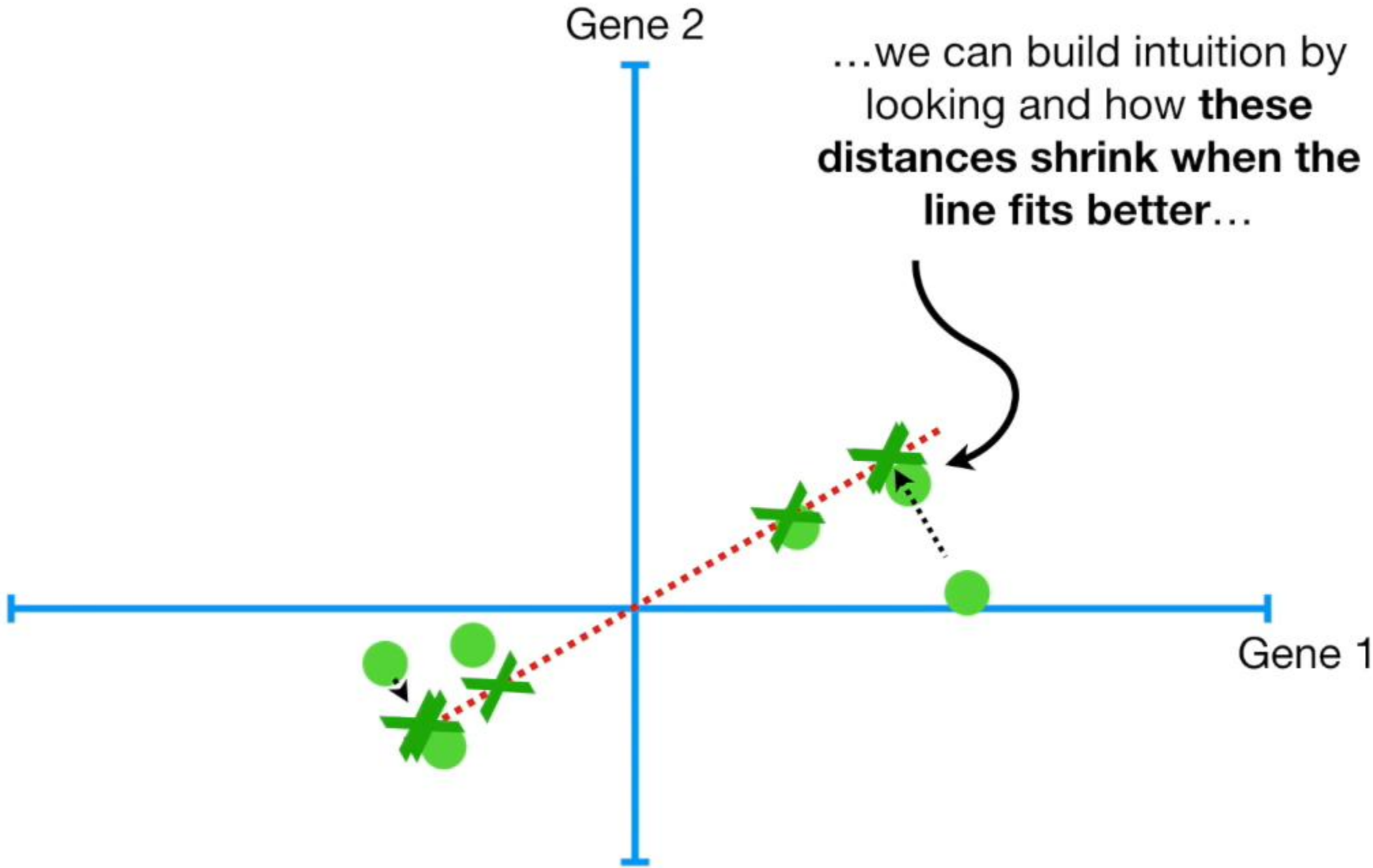

Gene 2
…we can build intuition by looking and how **these distances shrink when the line fits better**…
Gene 1

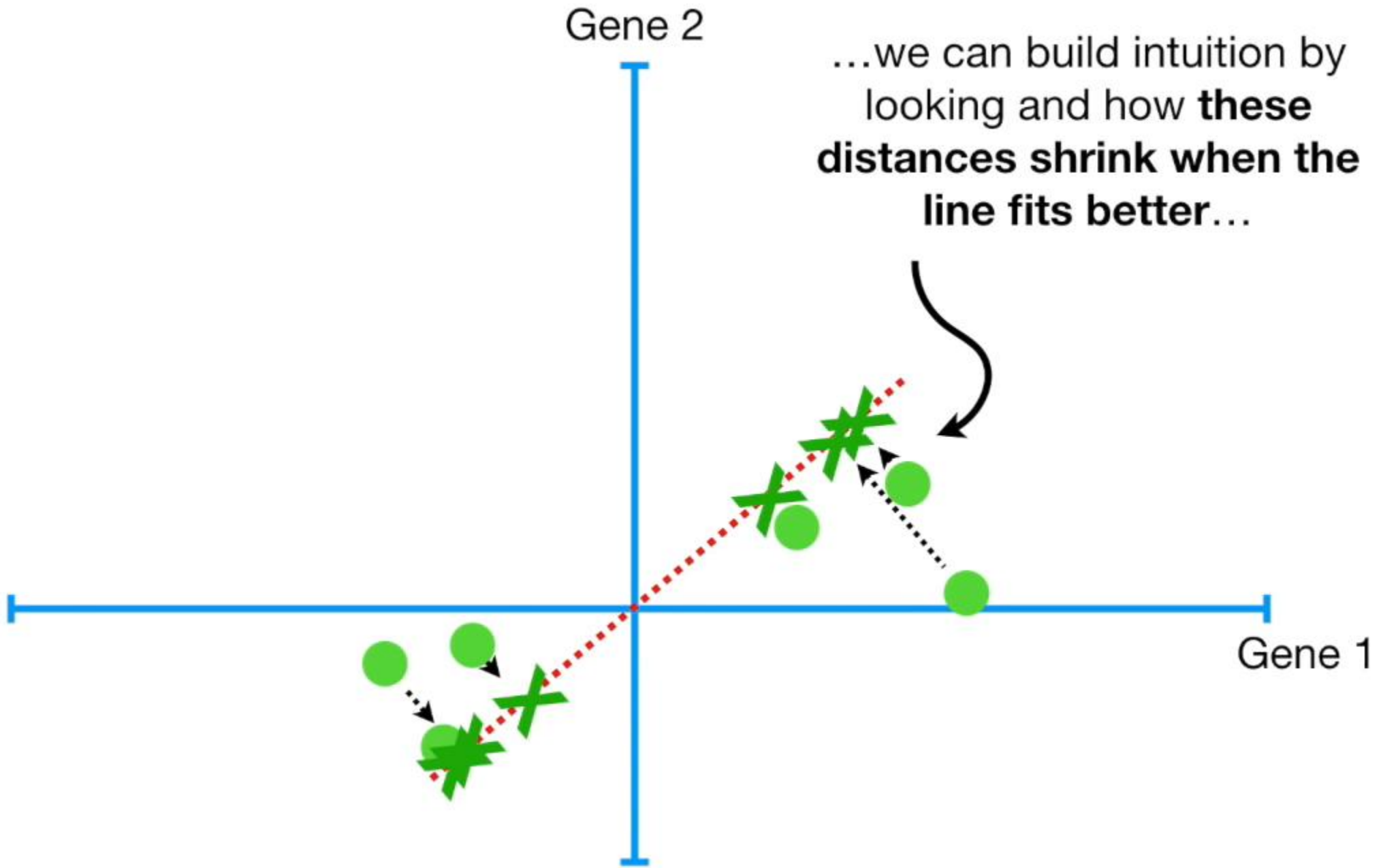

Gene 2
…we can build intuition by looking and how **these distances shrink when the line fits better**…
Gene 1

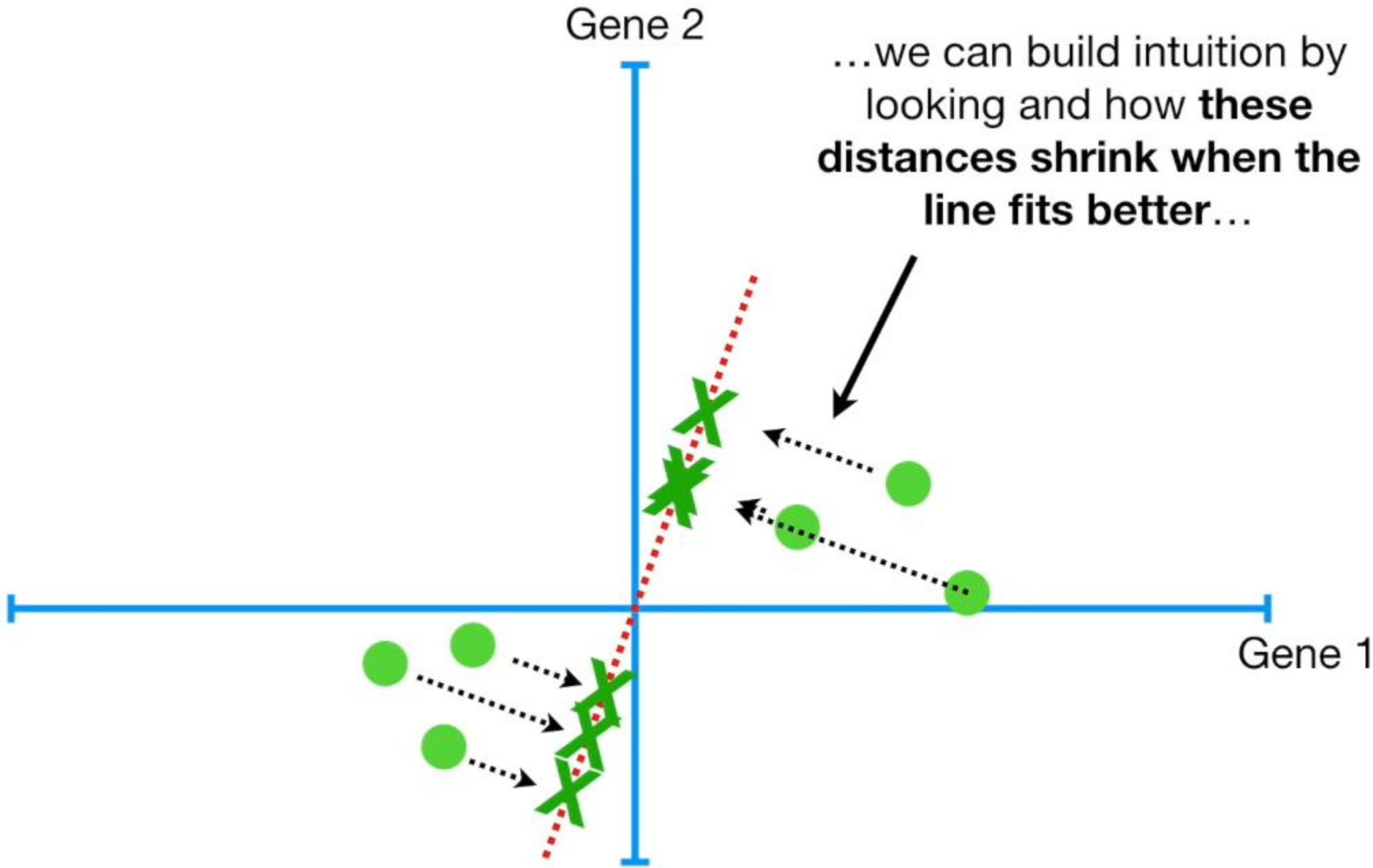

Gene 2
…we can build intuition by looking and how **these distances shrink when the line fits better**…
Gene 1

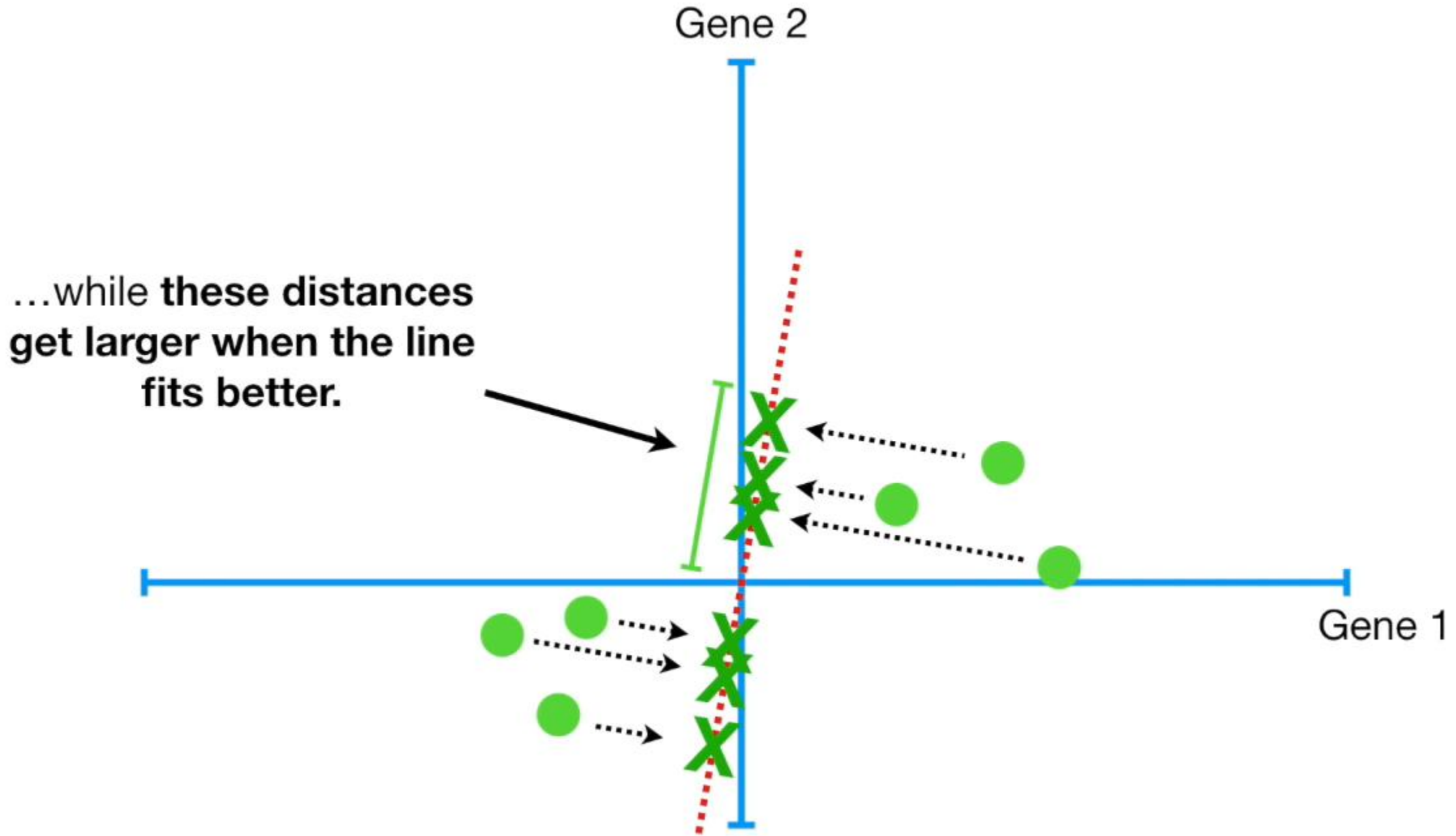

Gene 2
…we can build intuition by looking and how **these distances shrink when the line fits better**…
Gene 1

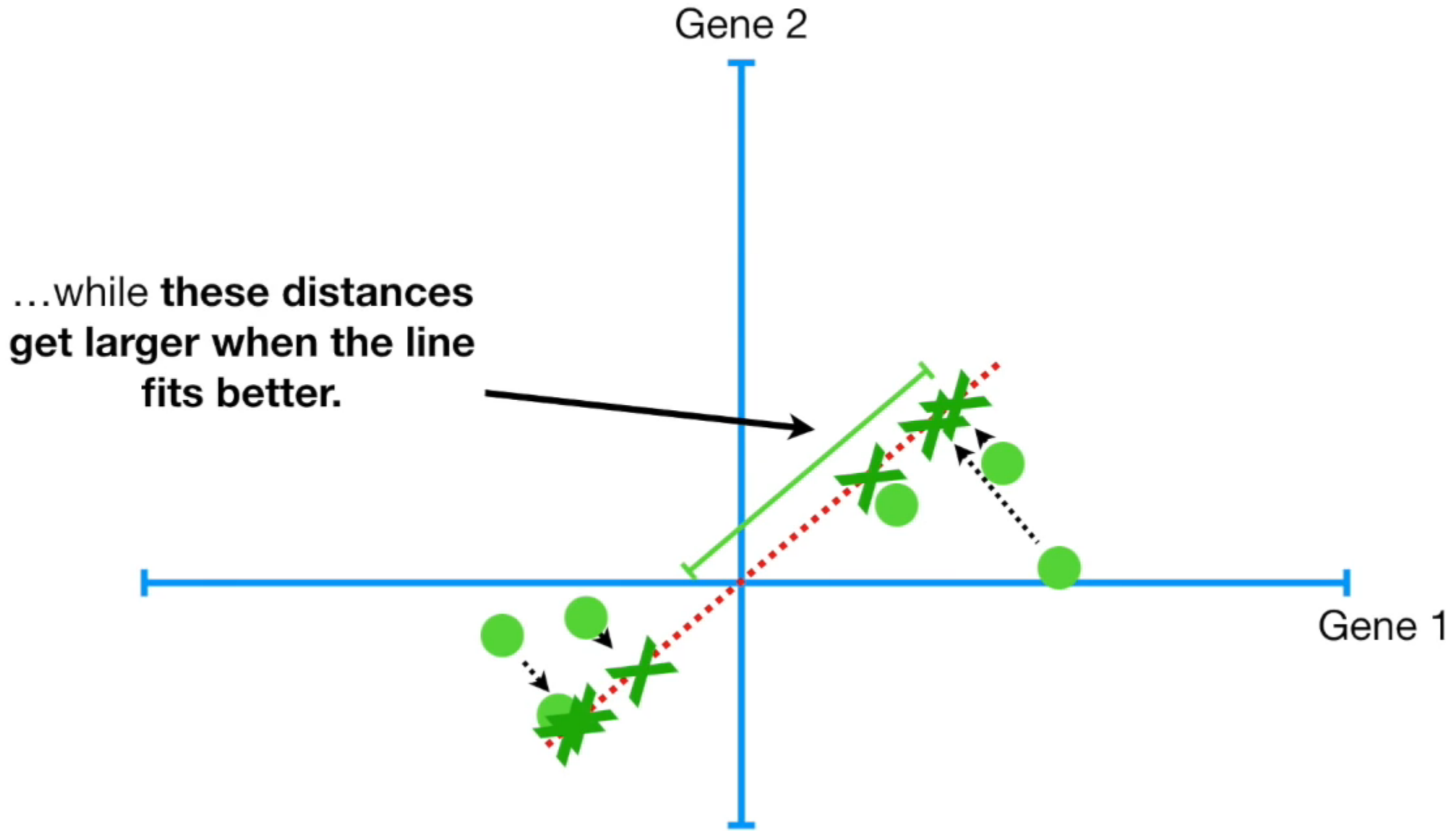

Gene 2
…we can build intuition by looking and how **these distances shrink when the line fits better**…
Gene 1

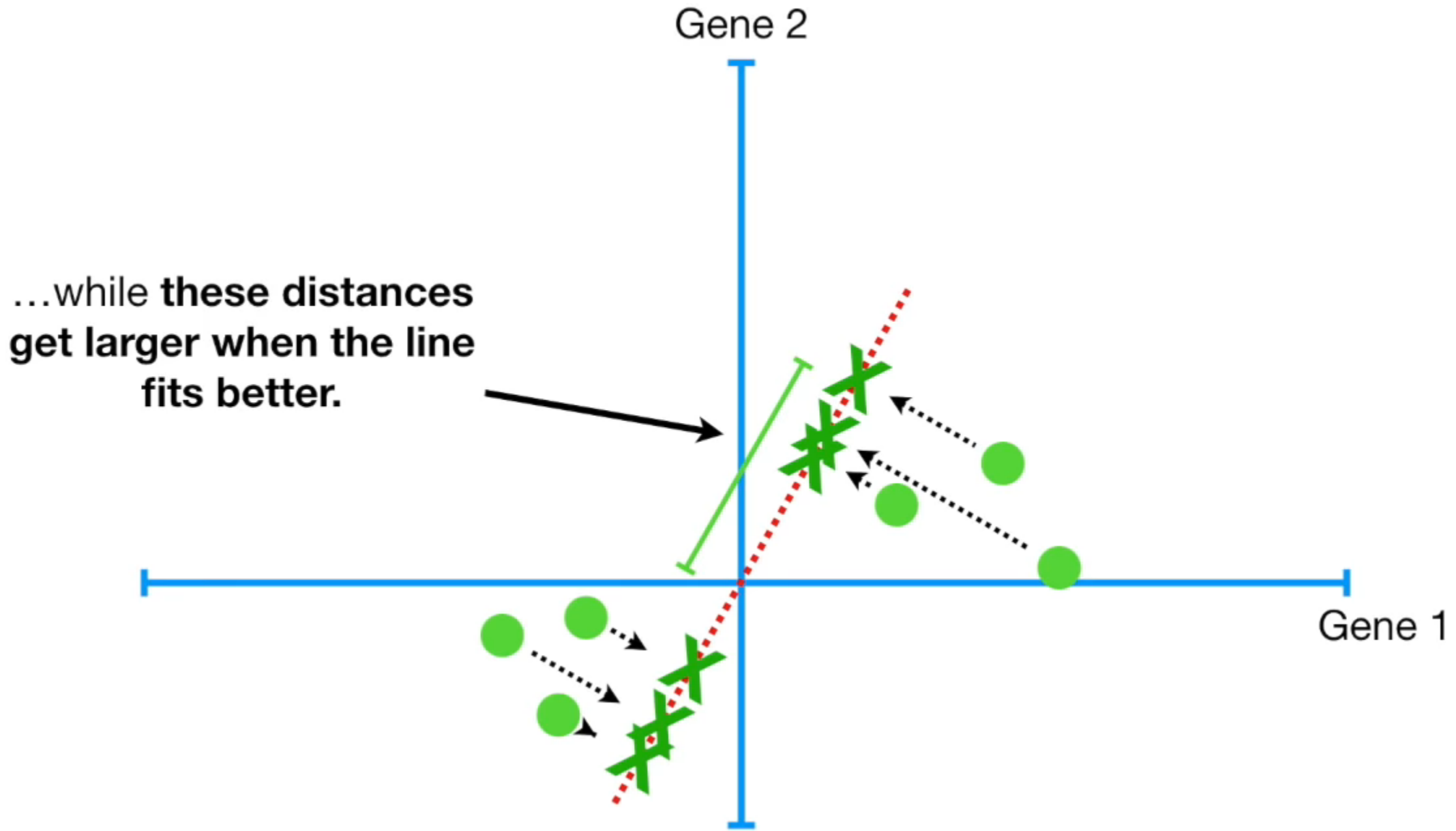

Gene 2
…we can build intuition by looking and how **these distances shrink when the line fits better**…
Gene 1

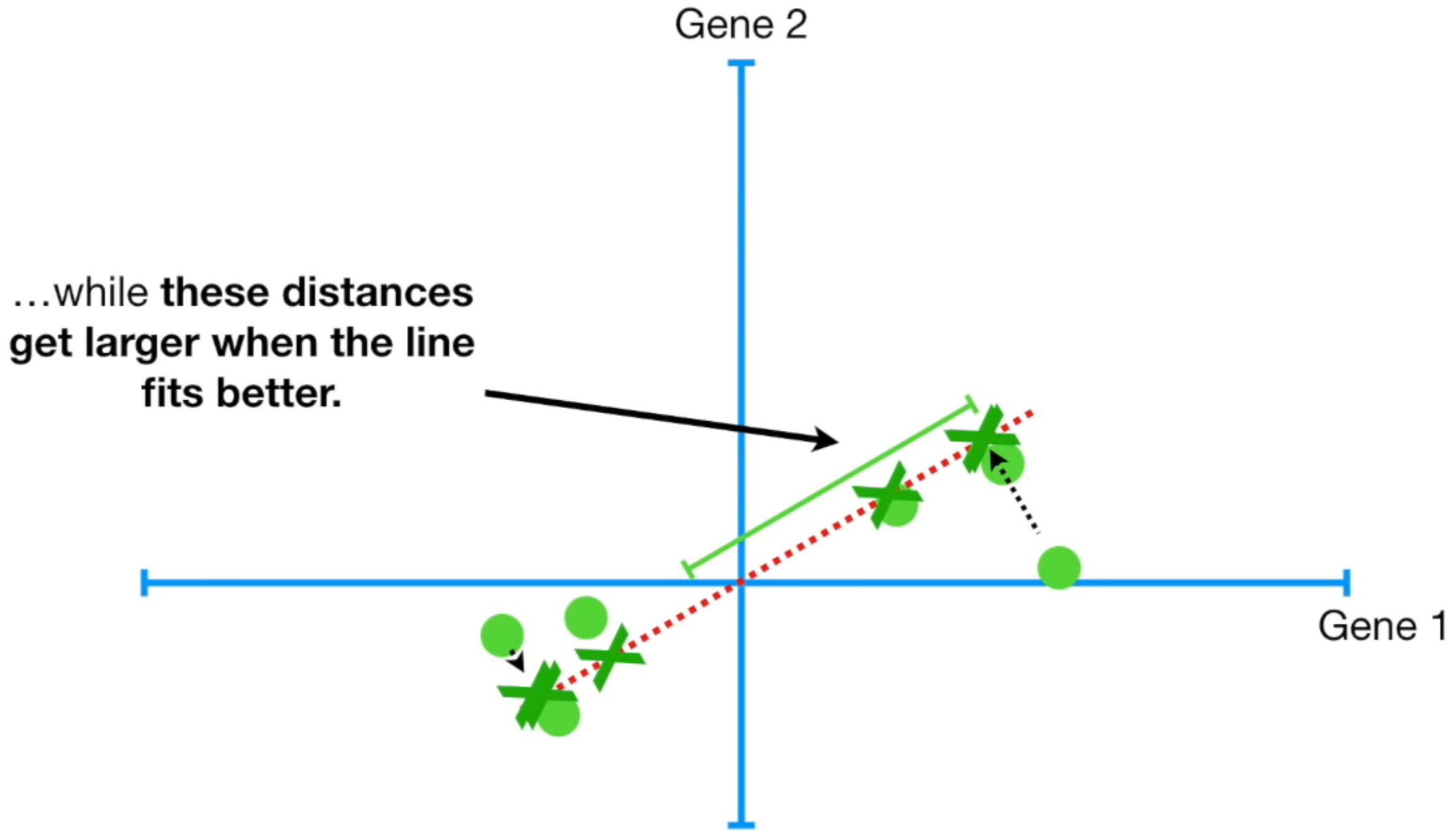

Gene 2
…we can build intuition by looking and how **these distances shrink when the line fits better**…
Gene 1

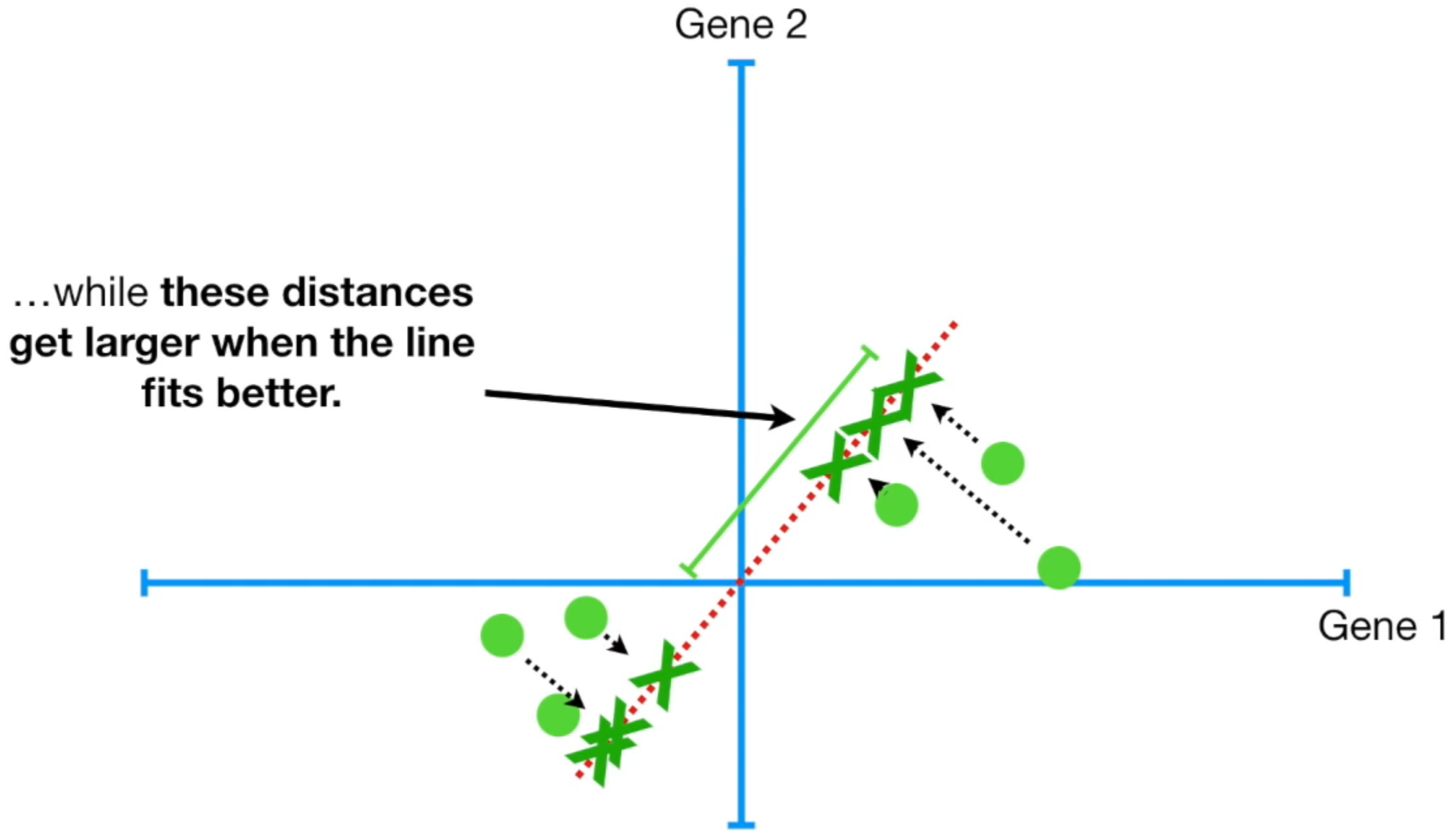

Gene 2
…we can build intuition by looking and how **these distances shrink when the line fits better**…
Gene 1

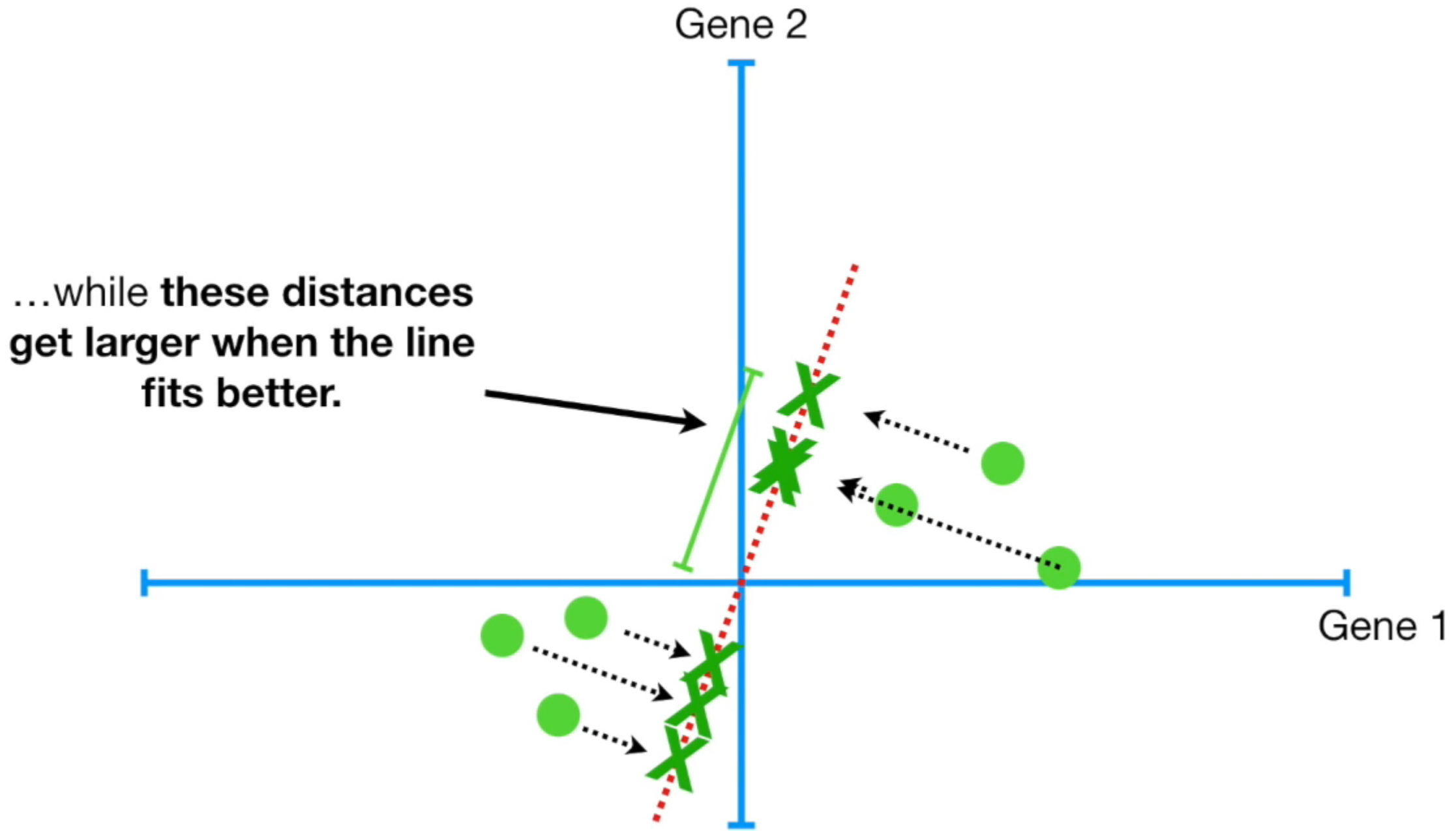

Gene 2
…while **these distances get larger when the line fits better.**
Gene 1

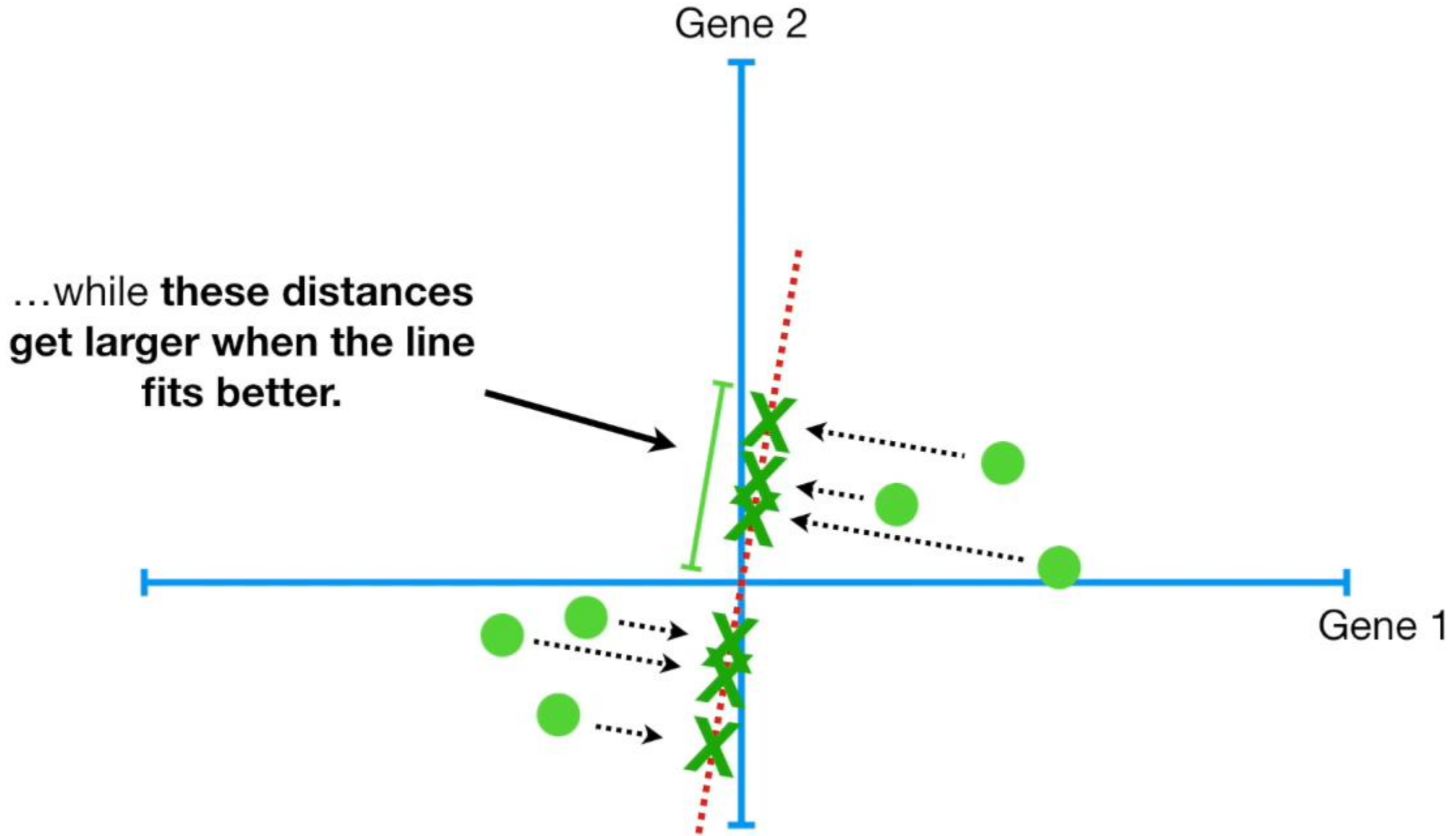

Gene 2
…while **these distances get larger when the line fits better.**
Gene 1

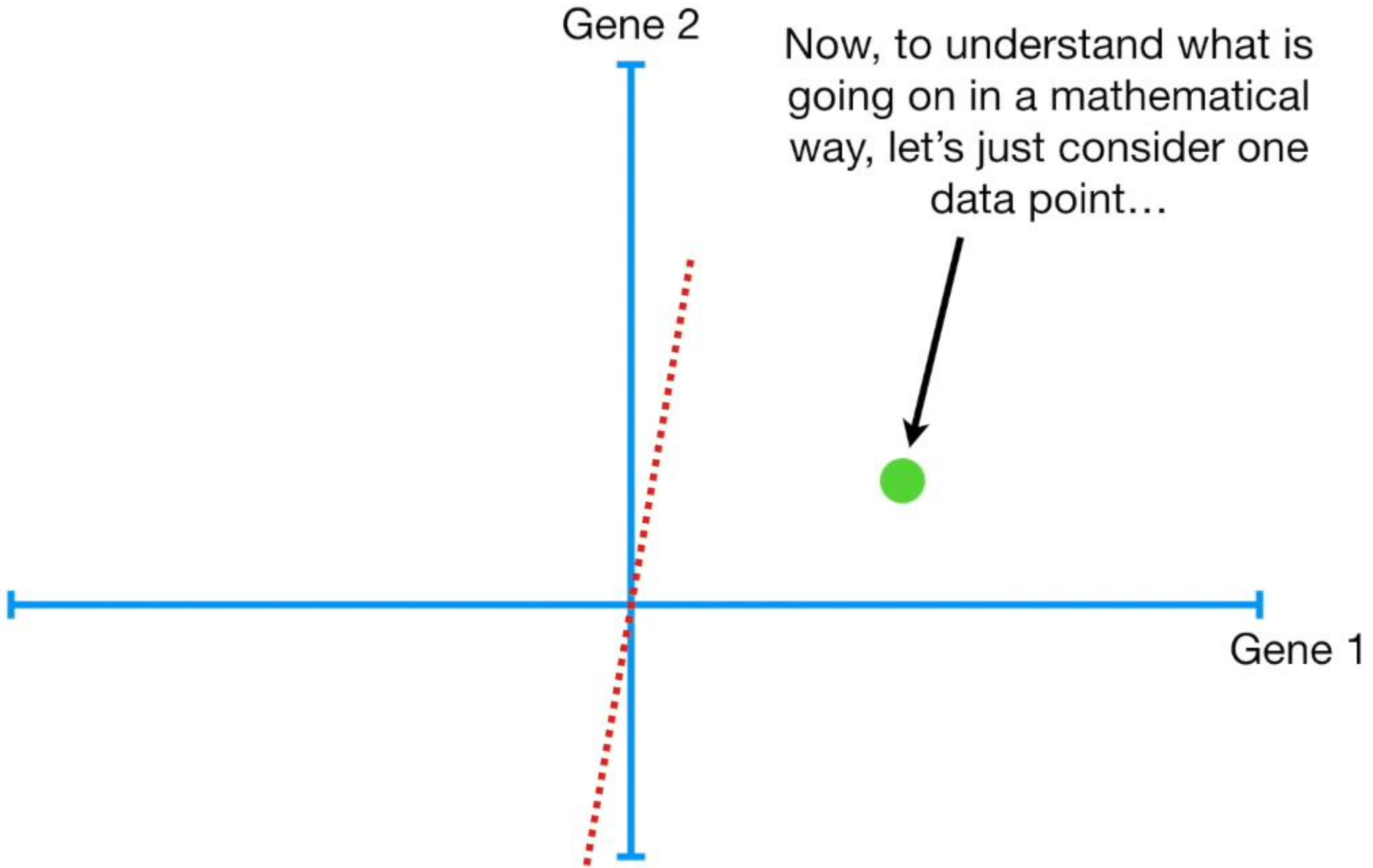

Gene 2
…while **these distances get larger when the line fits better.**
Gene 1

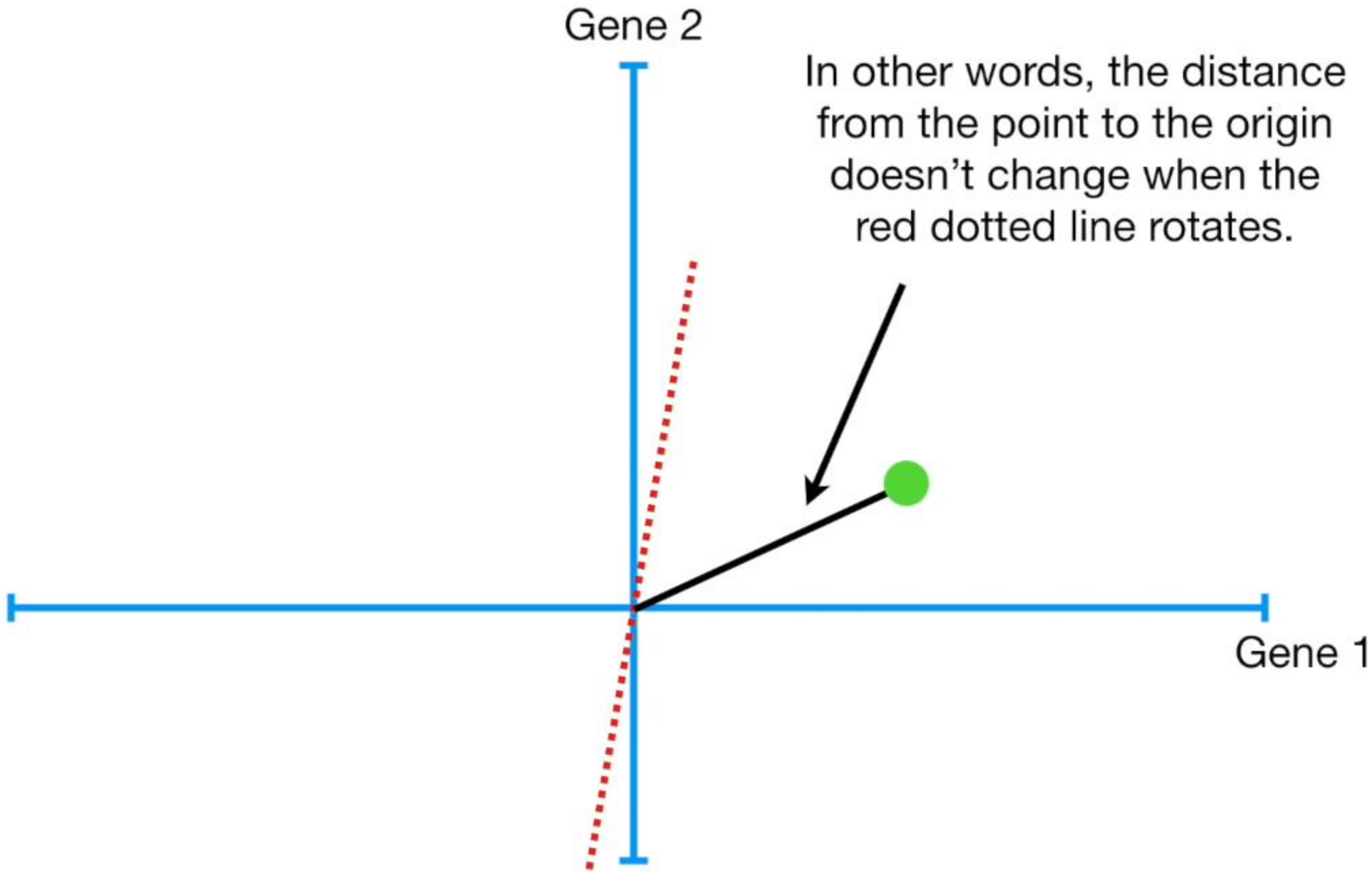

Gene 2
…while these distances get larger when the line fits better.
Gene 1

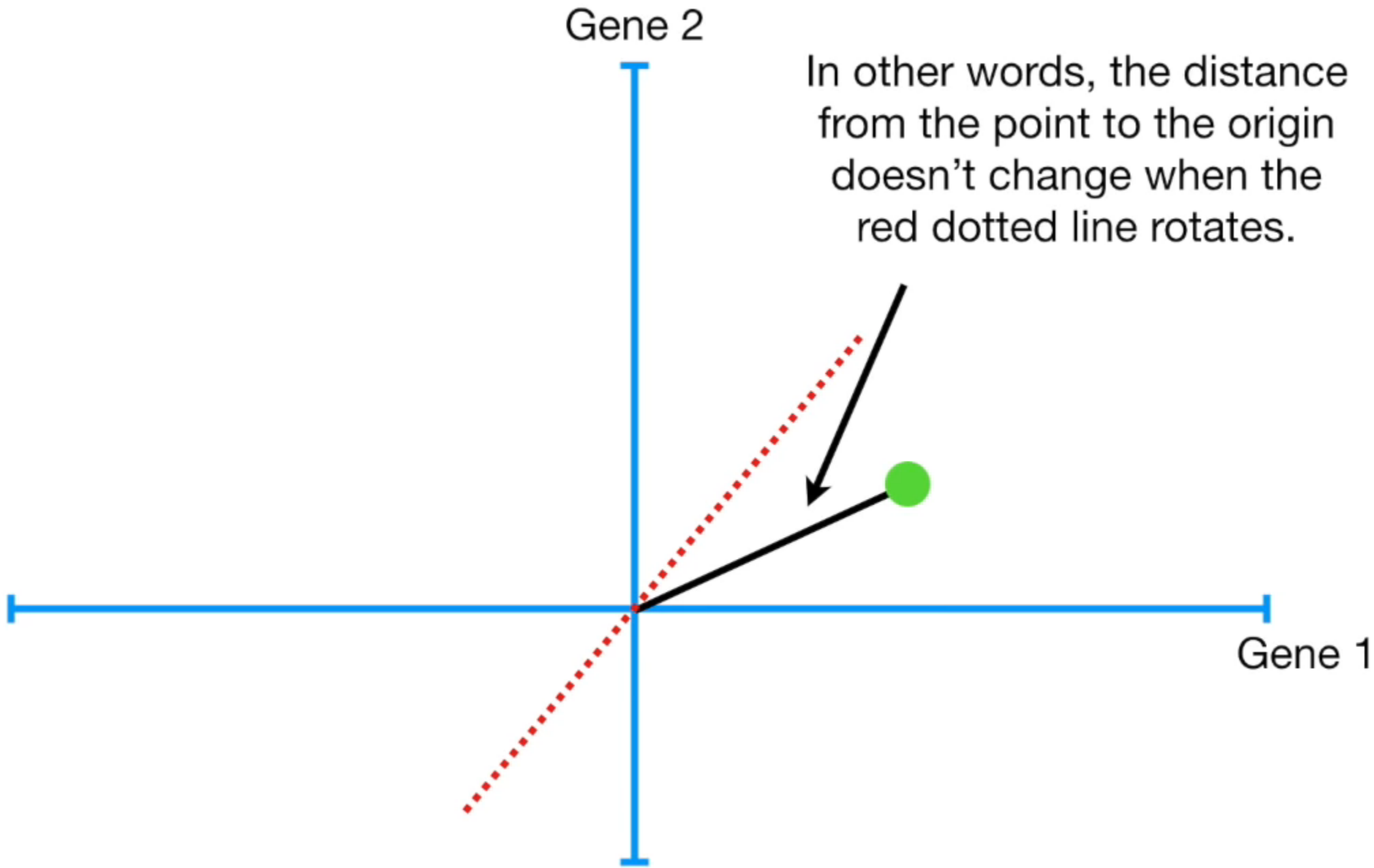

Gene 2
…while **these distances get larger when the line fits better.**
Gene 1

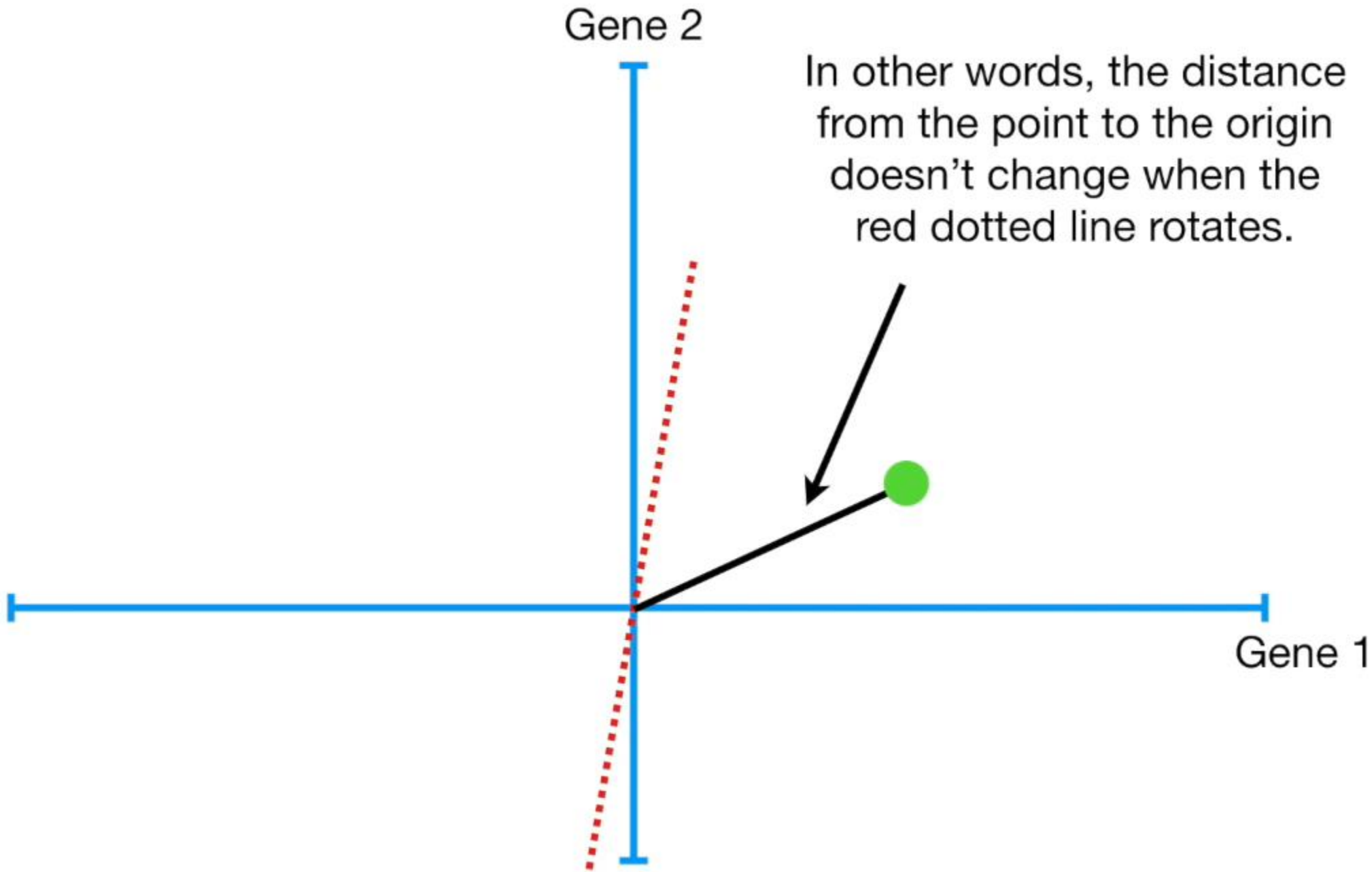

Gene 2
…while **these distances get larger when the line fits better.**
Gene 1

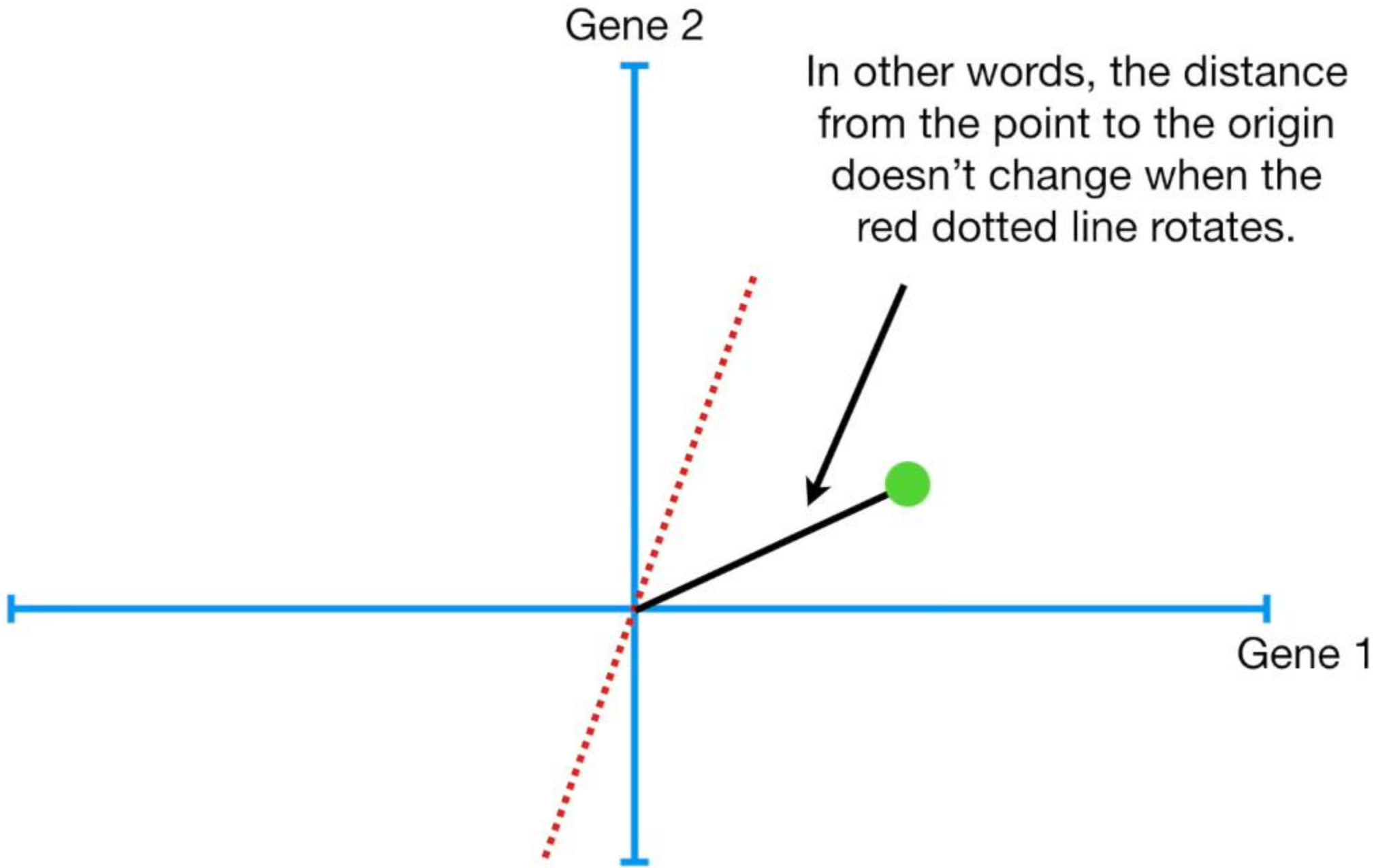

Gene 2
…while **these distances get larger when the line fits better.**
Gene 1

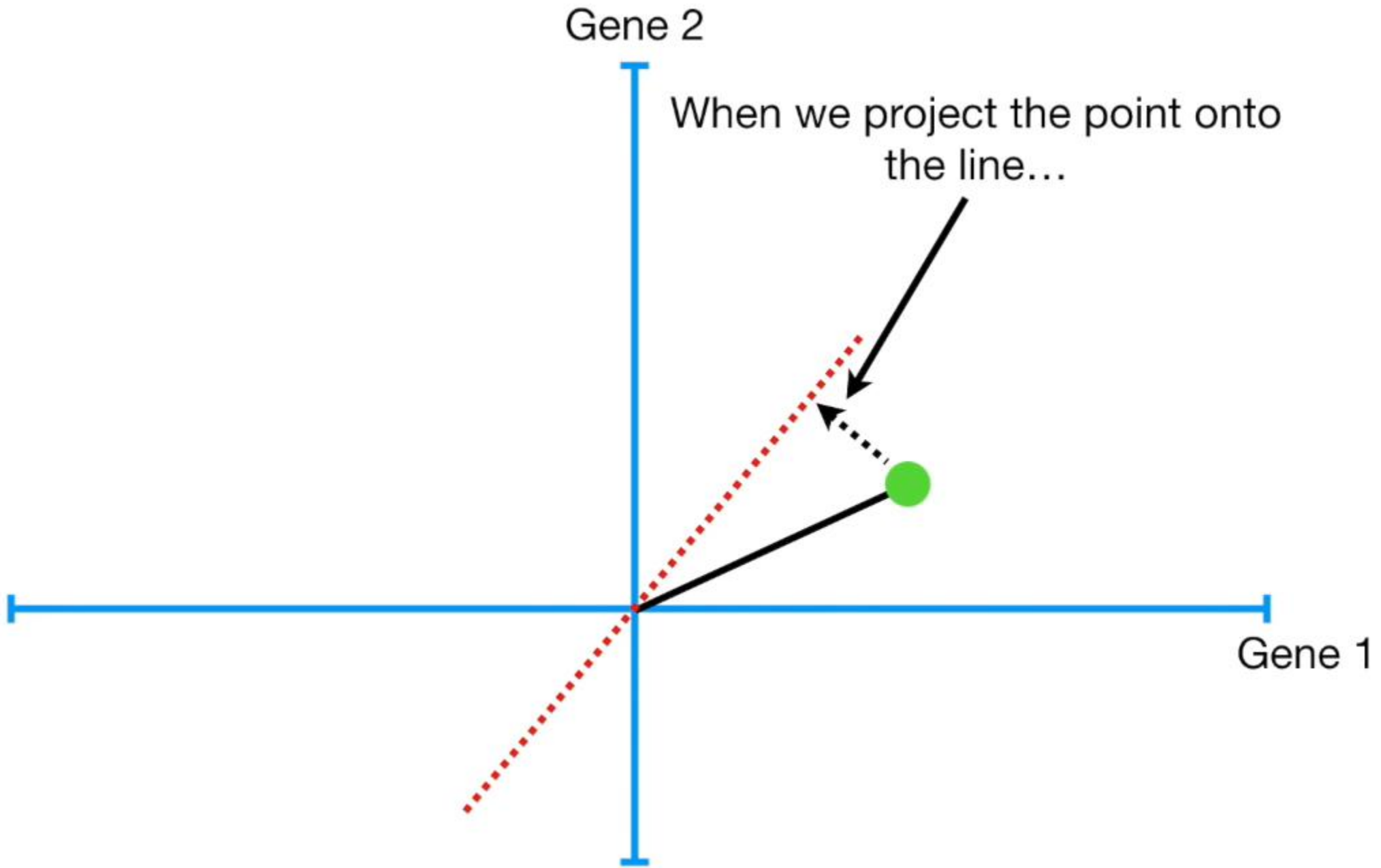

Gene 2
Now, to understand what is going on in a mathematical way, let's just consider one data point…
Gene 1

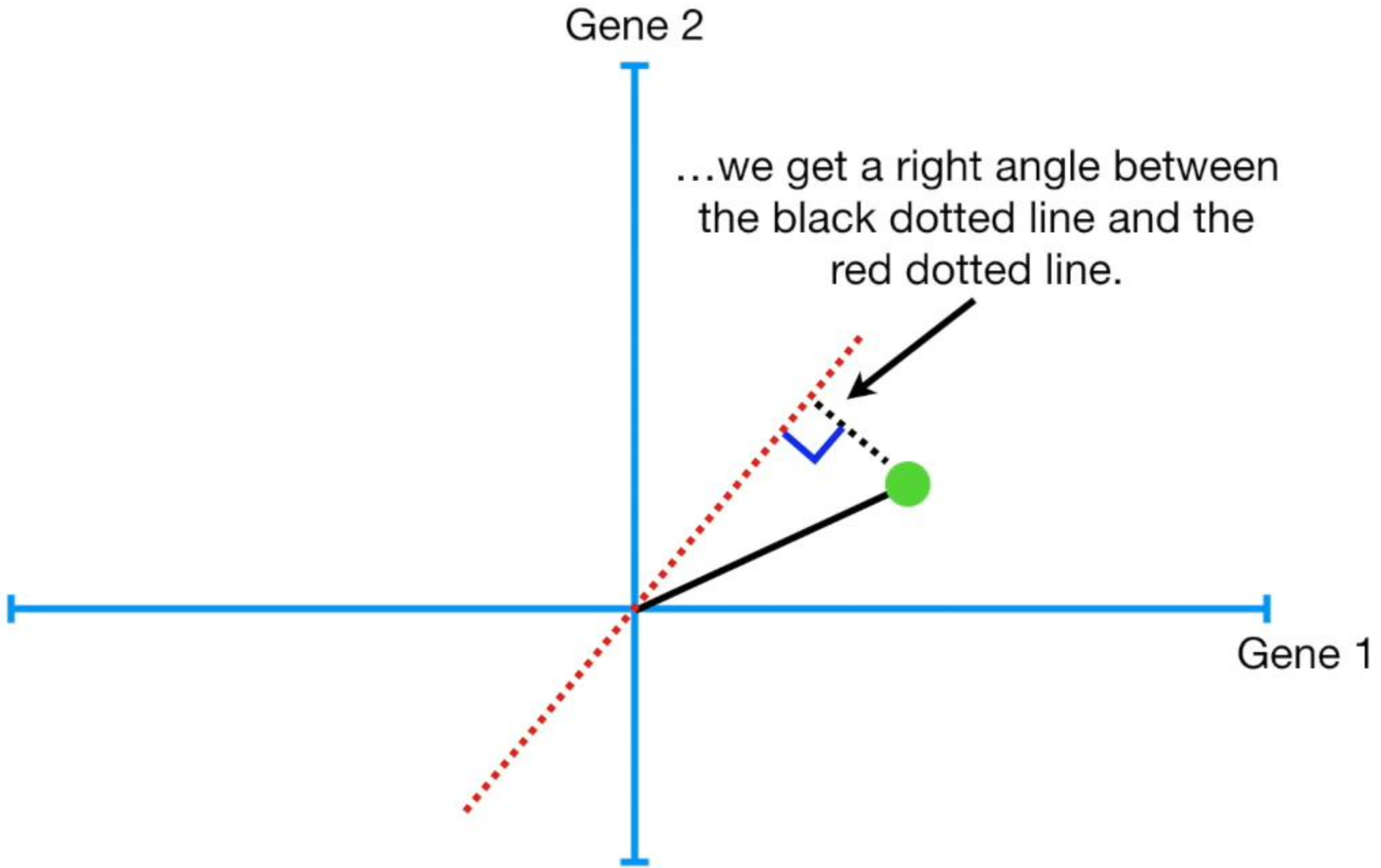

Gene 2
In other words, the distance from the point to the origin doesn't change when the red dotted line rotates.
Gene 1

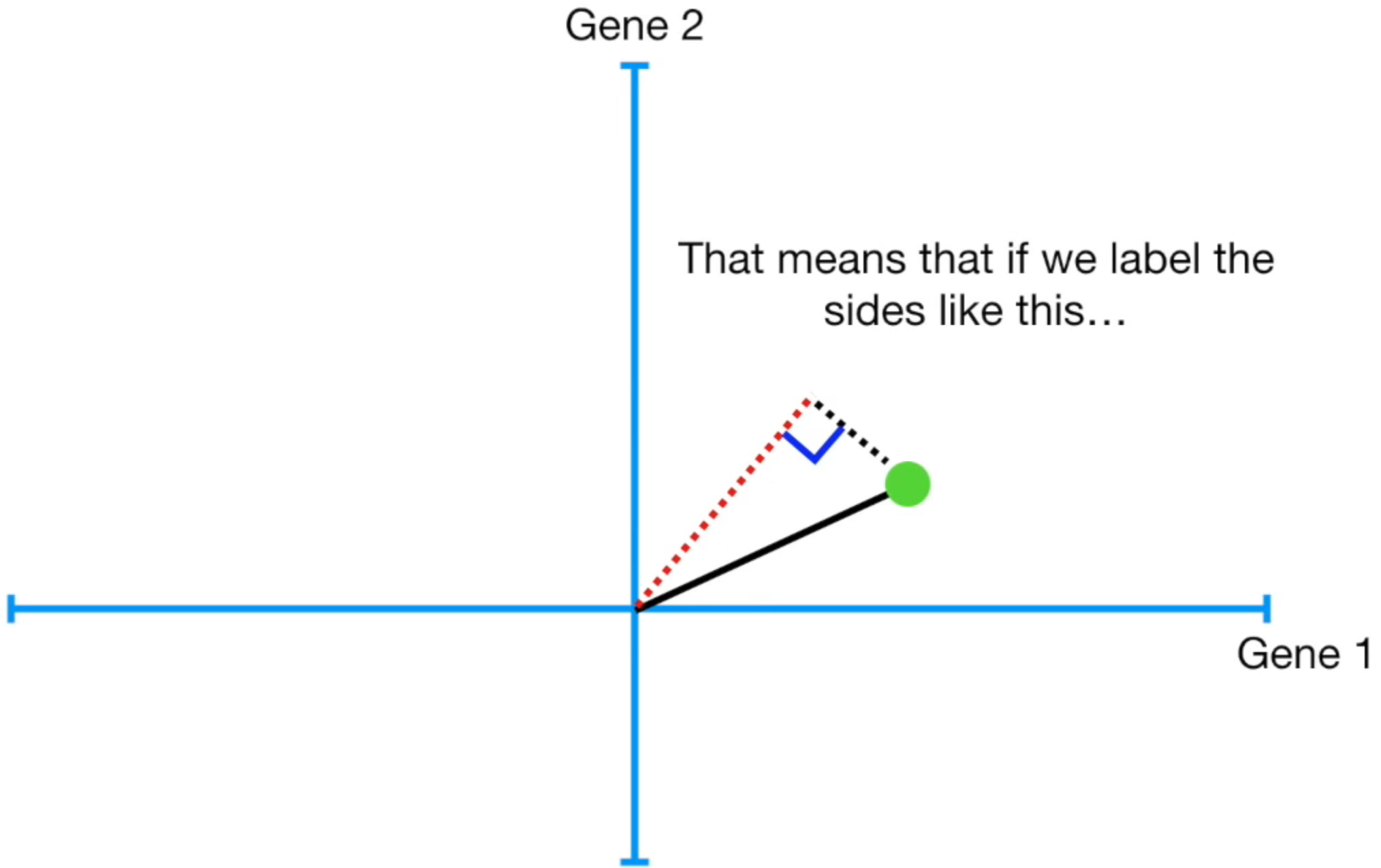

Gene 2
In other words, the distance from the point to the origin doesn't change when the red dotted line rotates.
Gene 1

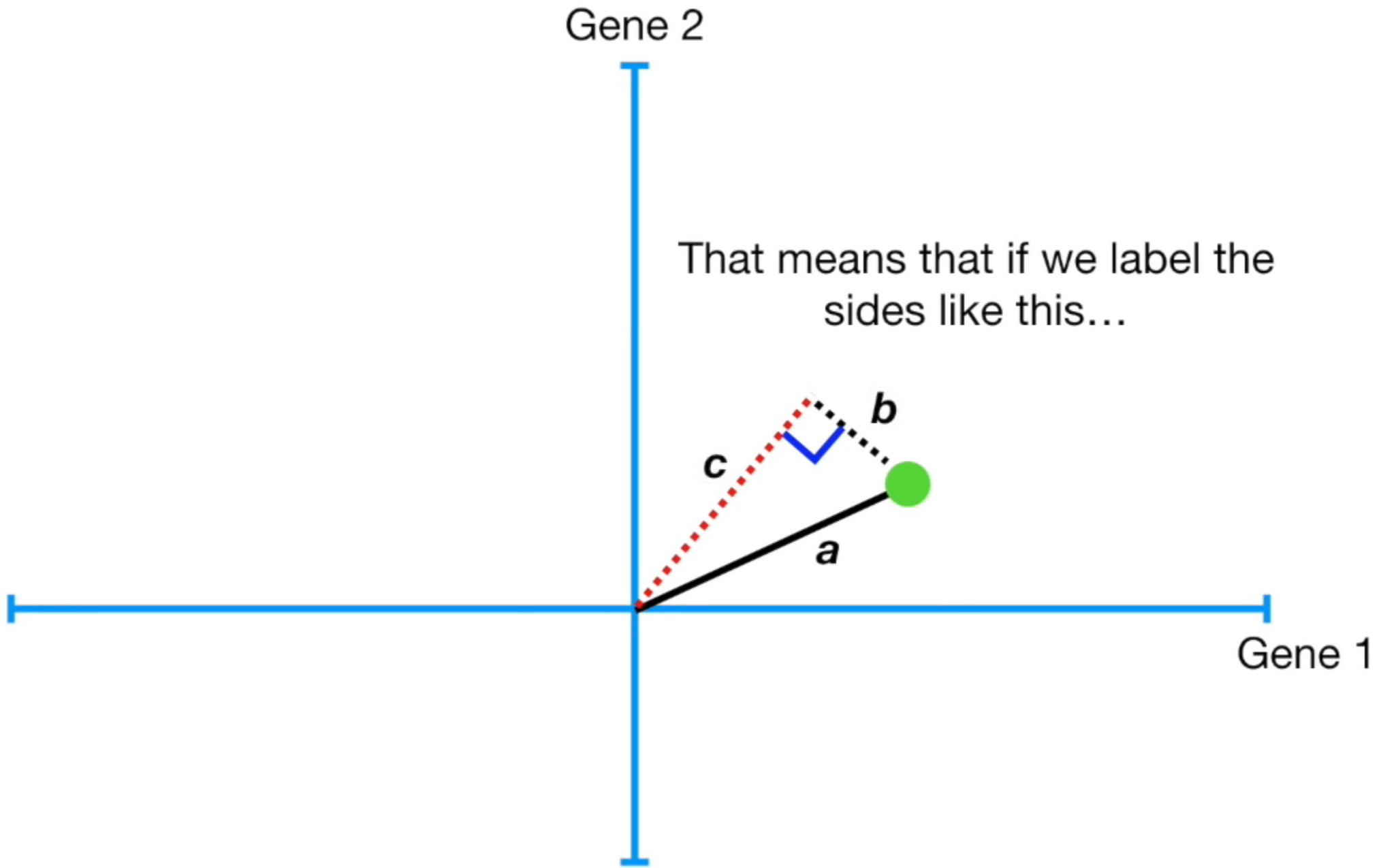

Gene 2
In other words, the distance from the point to the origin doesn't change when the red dotted line rotates.
Gene 1


Gene 2
In other words, the distance from the point to the origin doesn't change when the red dotted line rotates.
Gene 1


Gene 2
When we project the point onto the line…
Gene 1


Gene 2
…we get a right angle between the black dotted line and the red dotted line.
Gene 1

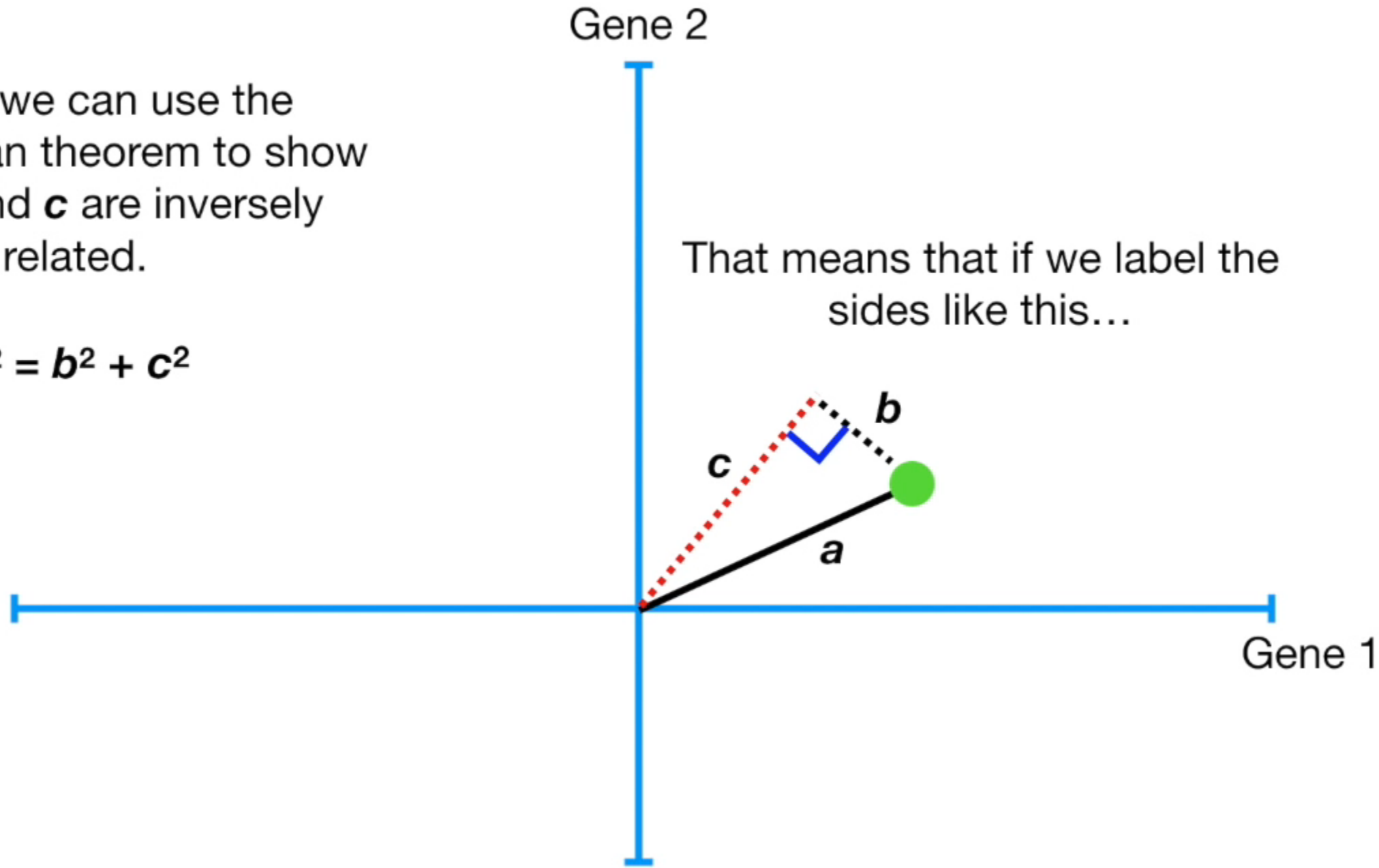

Gene 2
That means that if we label the sides like this…
Gene 1

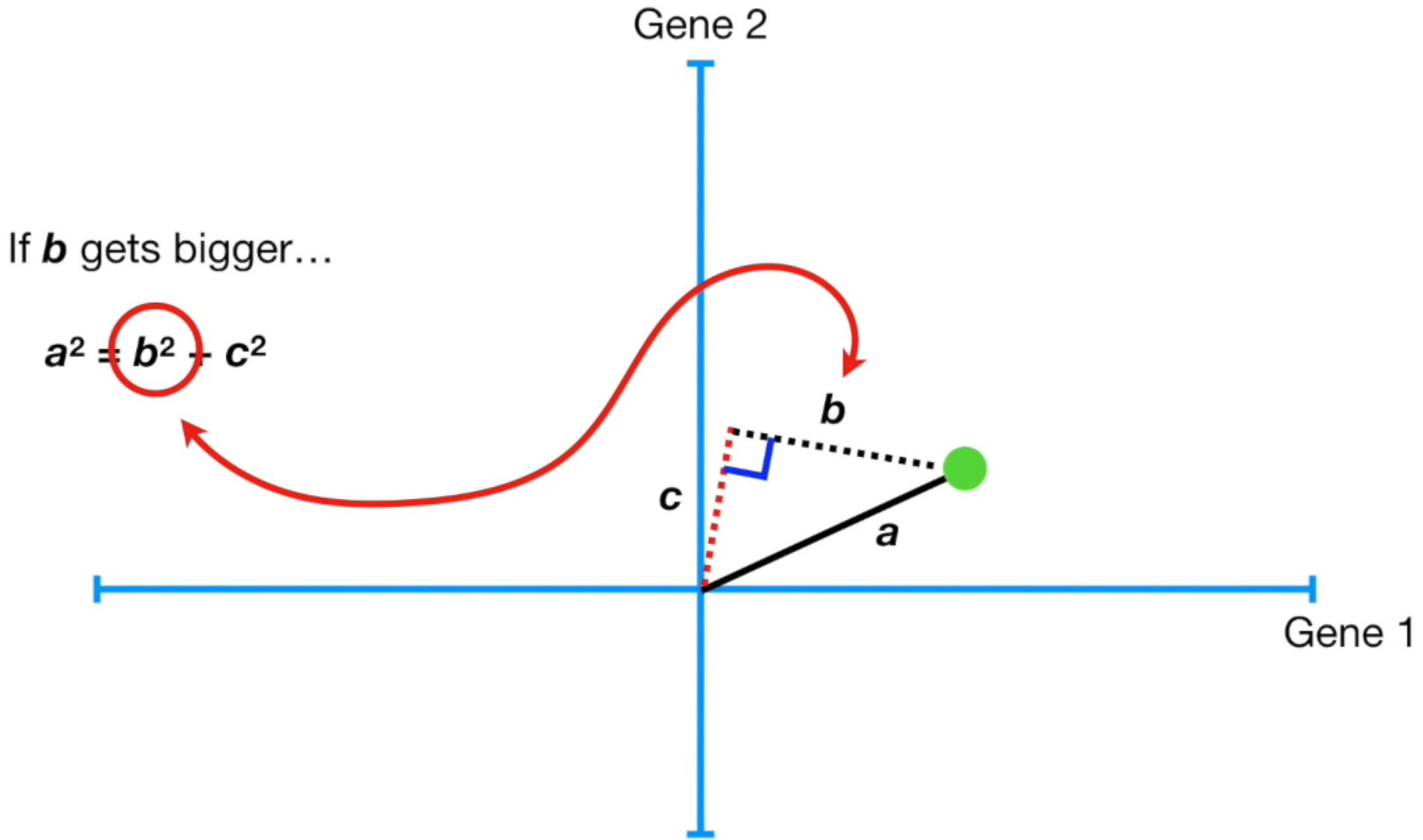

Gene 2
That means that if we label the sides like this…
b
c
a
Gene 1

…then we can use the Pythagorean theorem to show how ***b*** and ***c*** are inversely related.

$$a^2 = b^2 + c^2$$

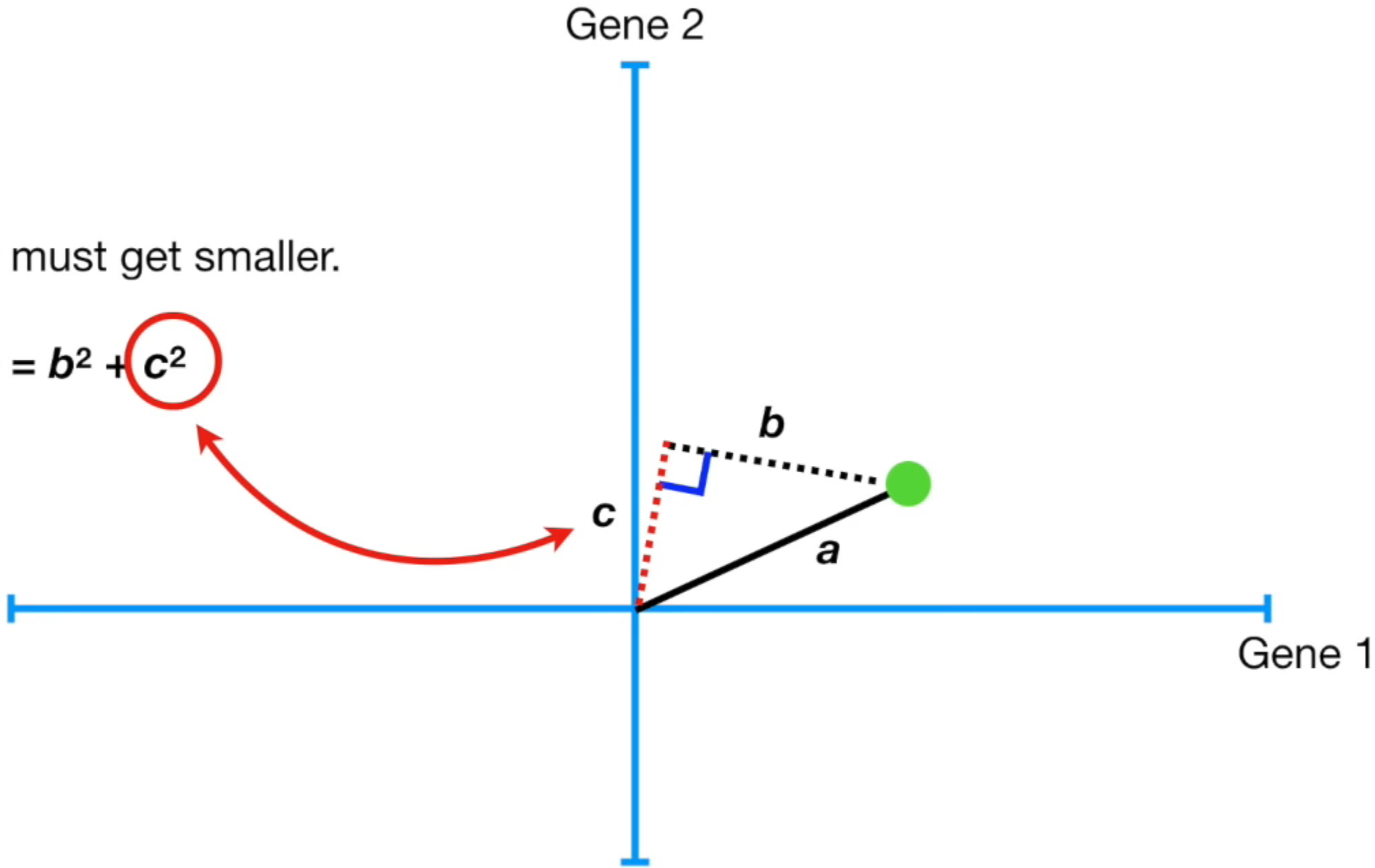
That means that if we label the sides like this…

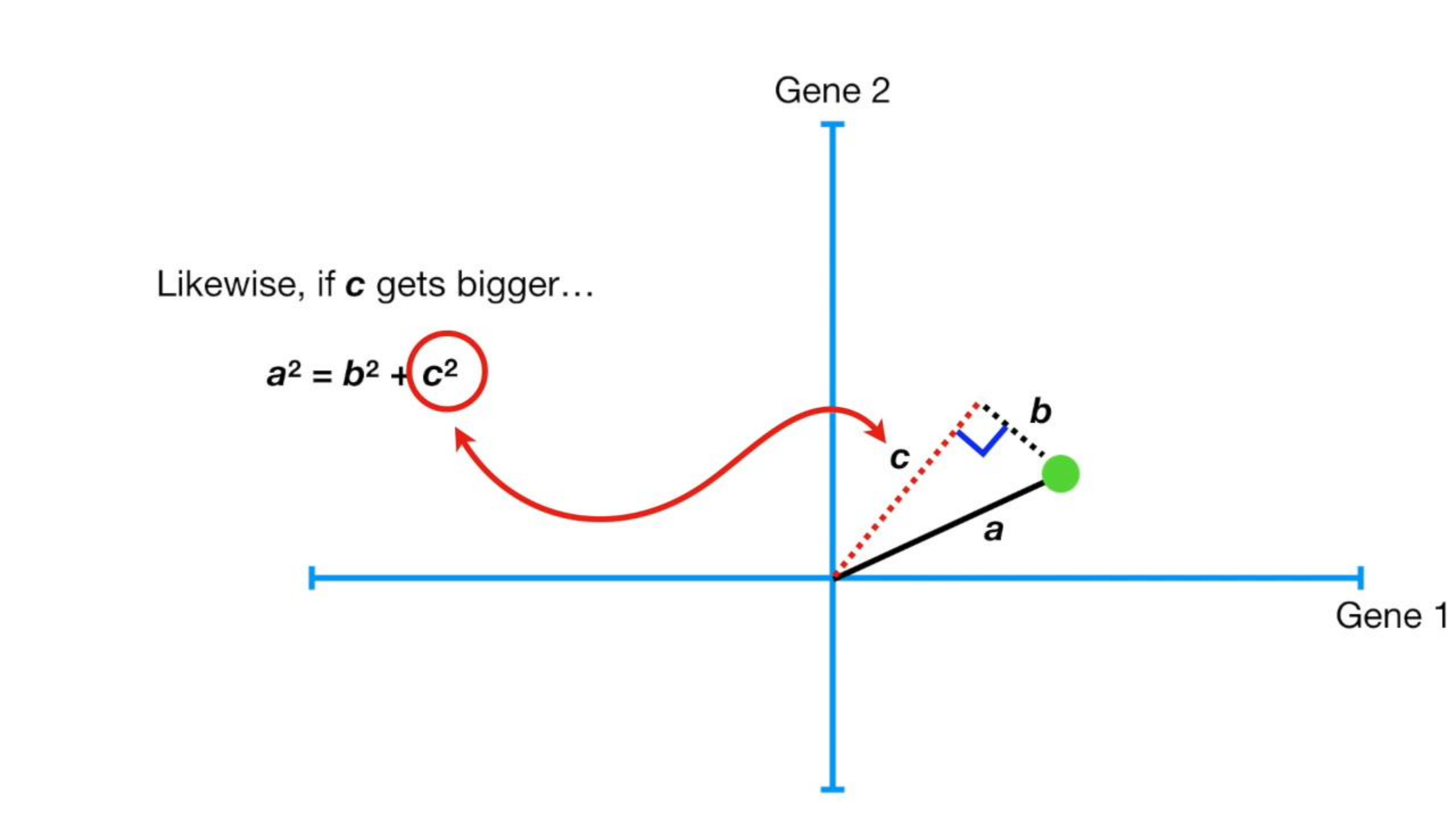


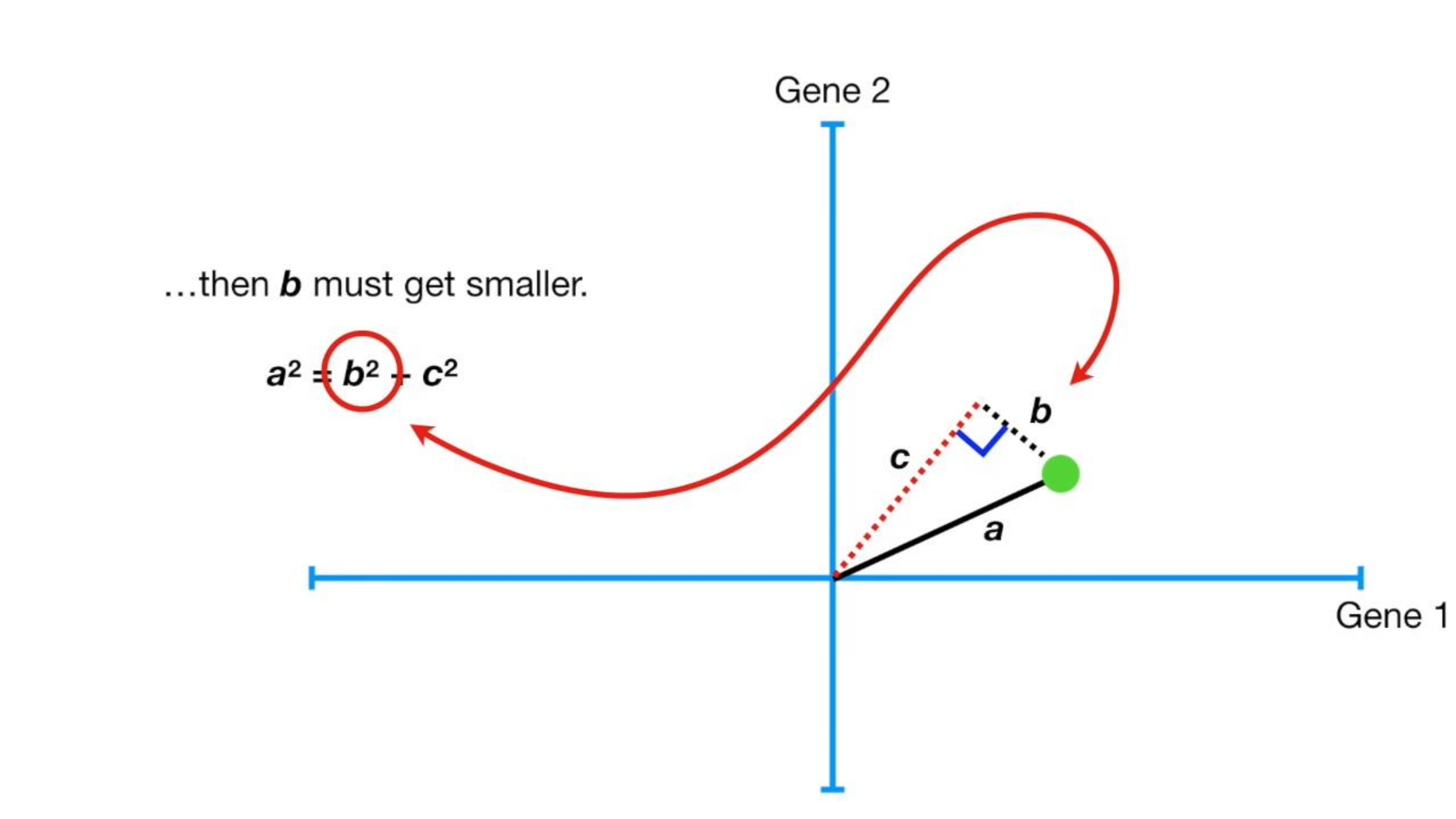

Gene 2
If b gets bigger…
a² = b² + c²
b
c
a
Gene 1

…then ***c*** must get smaller.

$$a^2 = b^2 + c^2$$

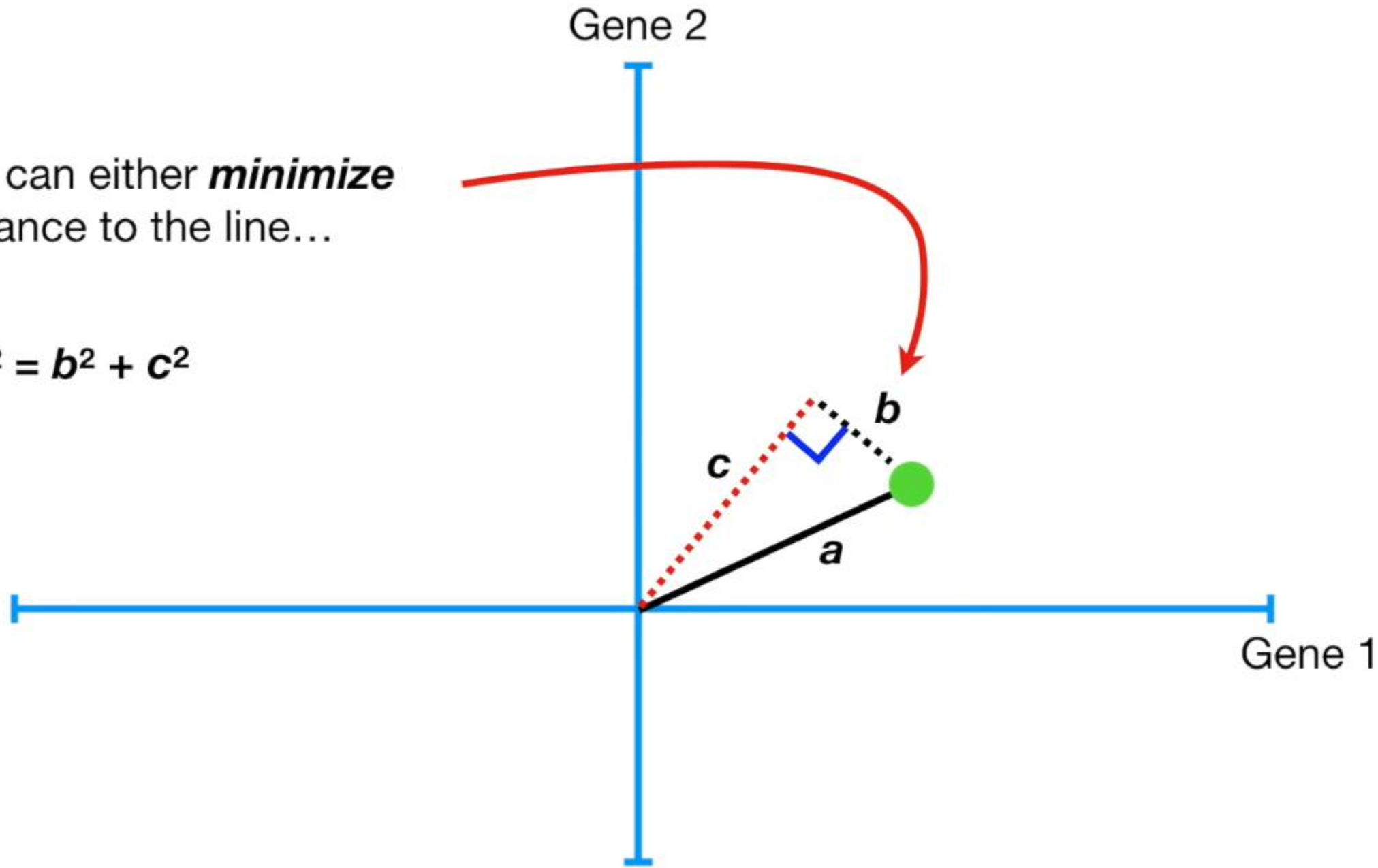



Gene 2
Likewise, if c gets bigger…
$a^2 = b^2 + c^2$
b
c
a
Gene 1


Gene 2
…then b must get smaller.
$a^2 = b^2 + c^2$
b
c
a
Gene 1

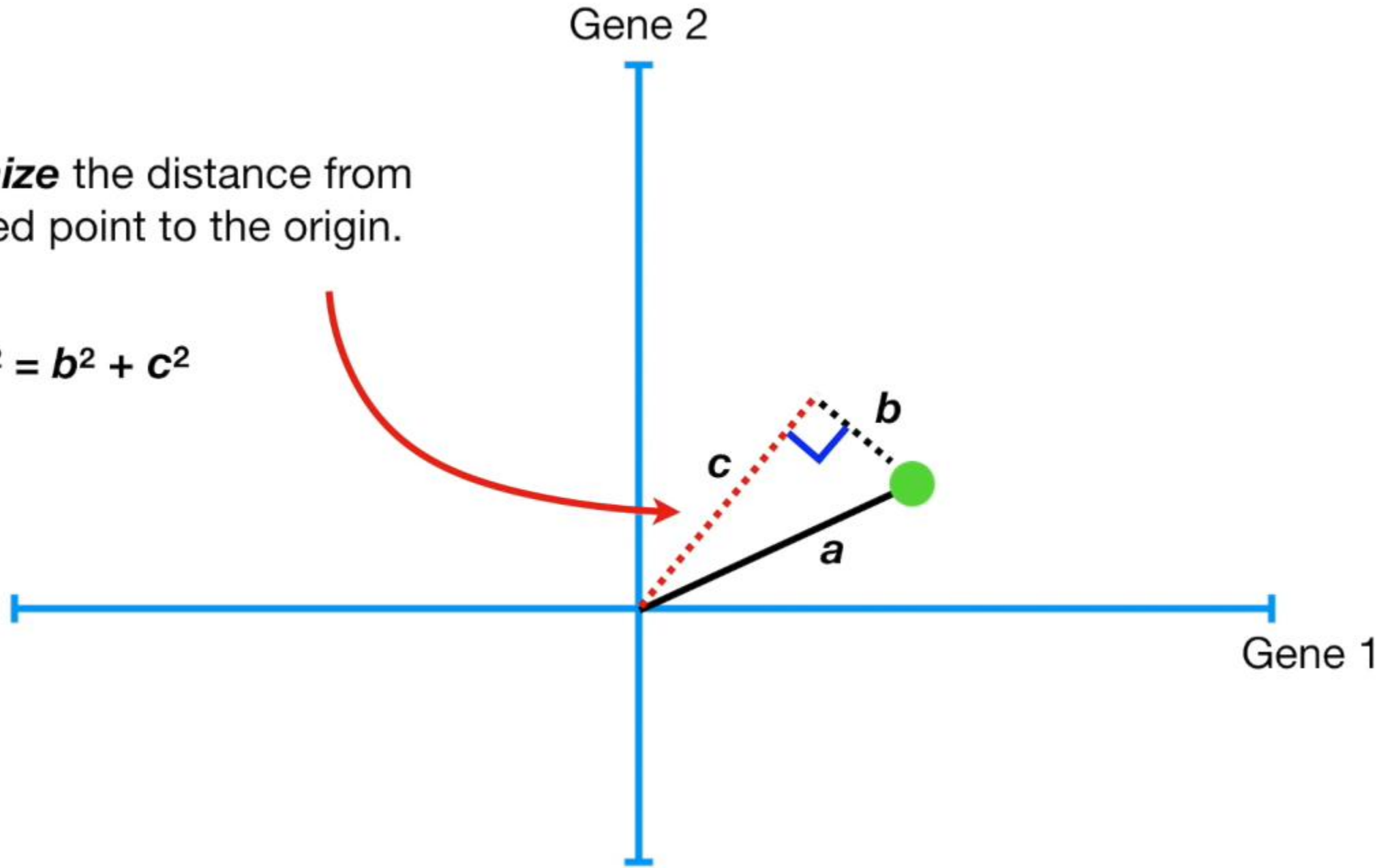
Thus, PCA can either ***minimize*** the distance to the line…

$$a^2 = b^2 + c^2$$

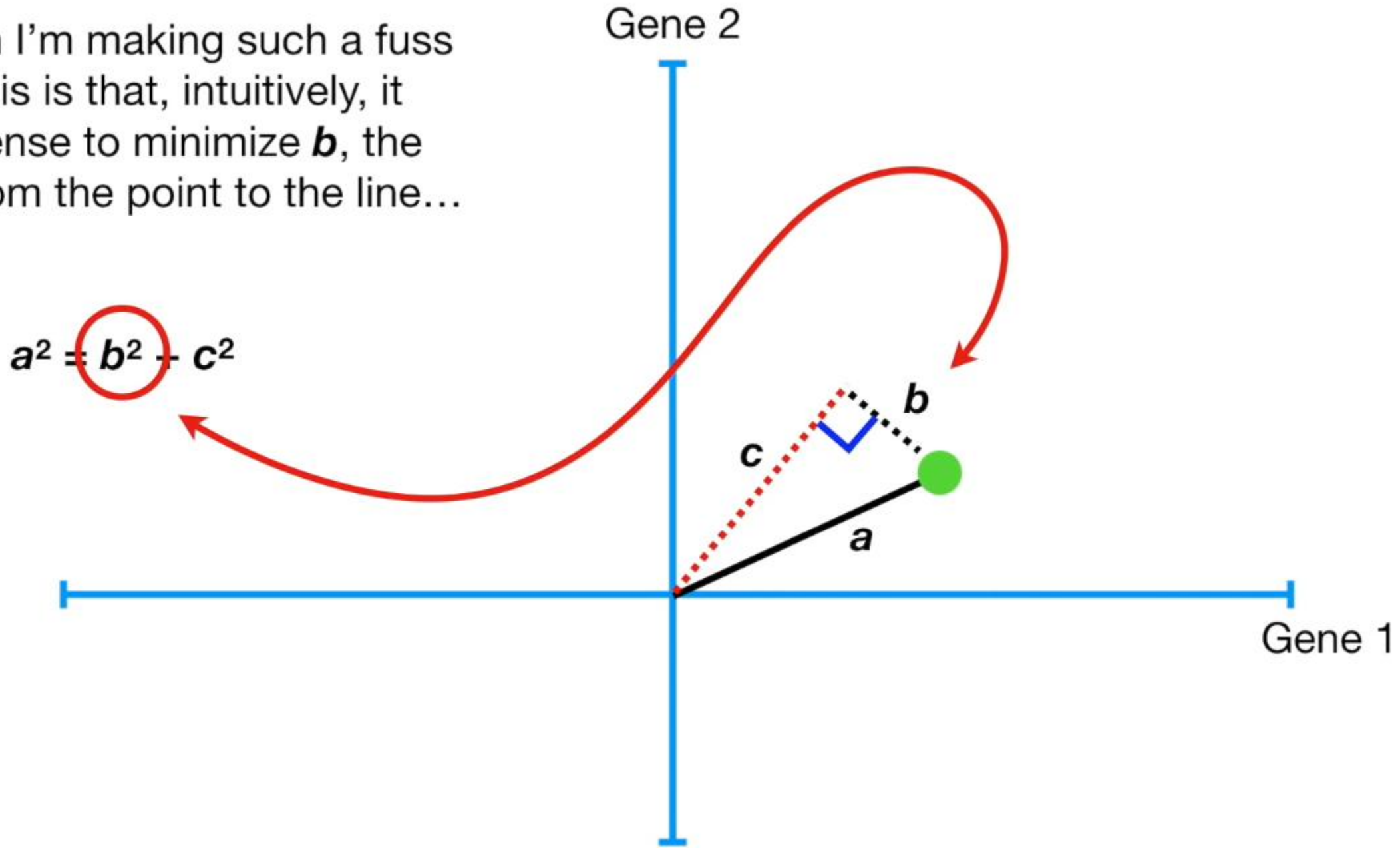


…or ***maximize*** the distance from the projected point to the origin.

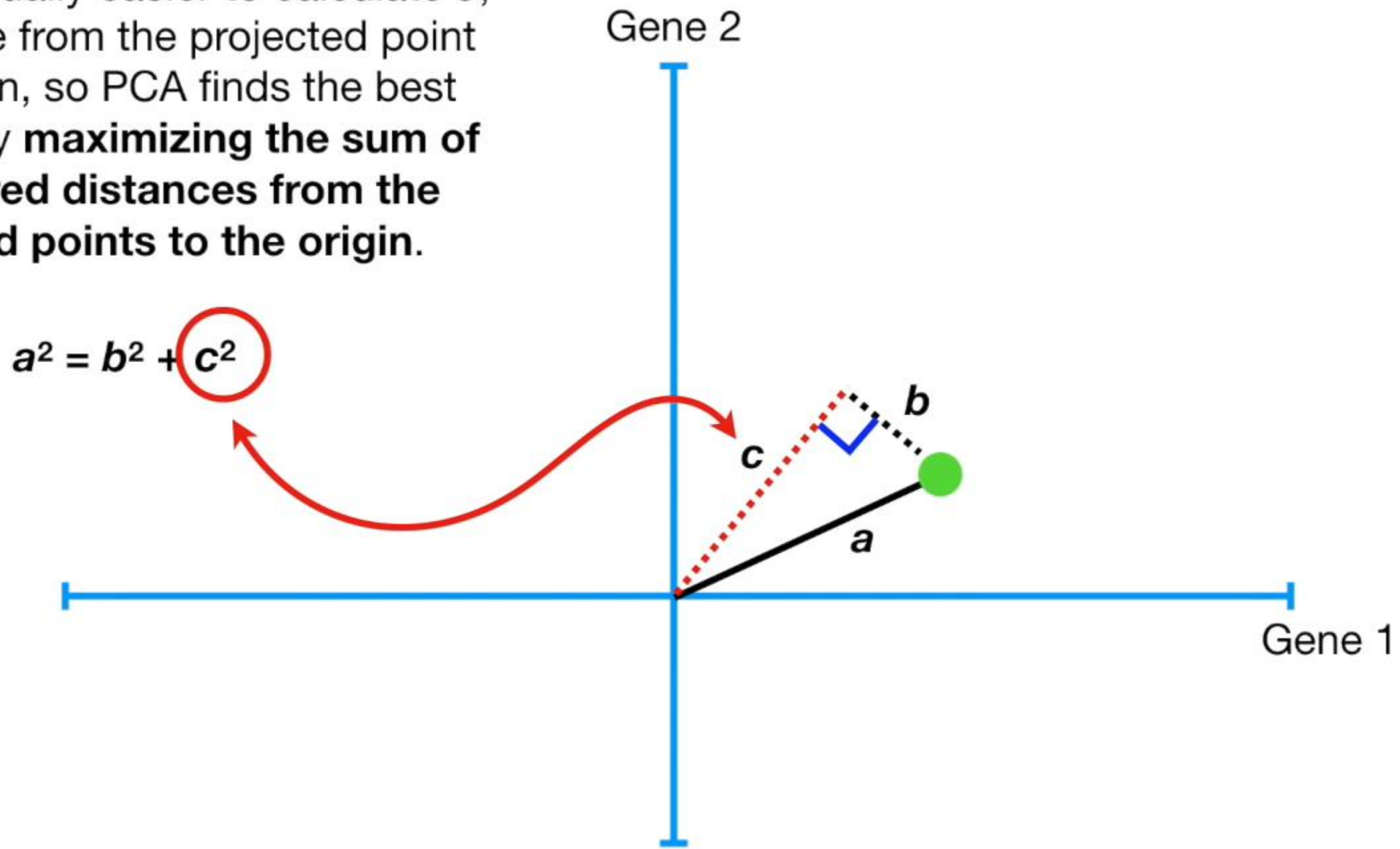
$$a^2 = b^2 + c^2$$

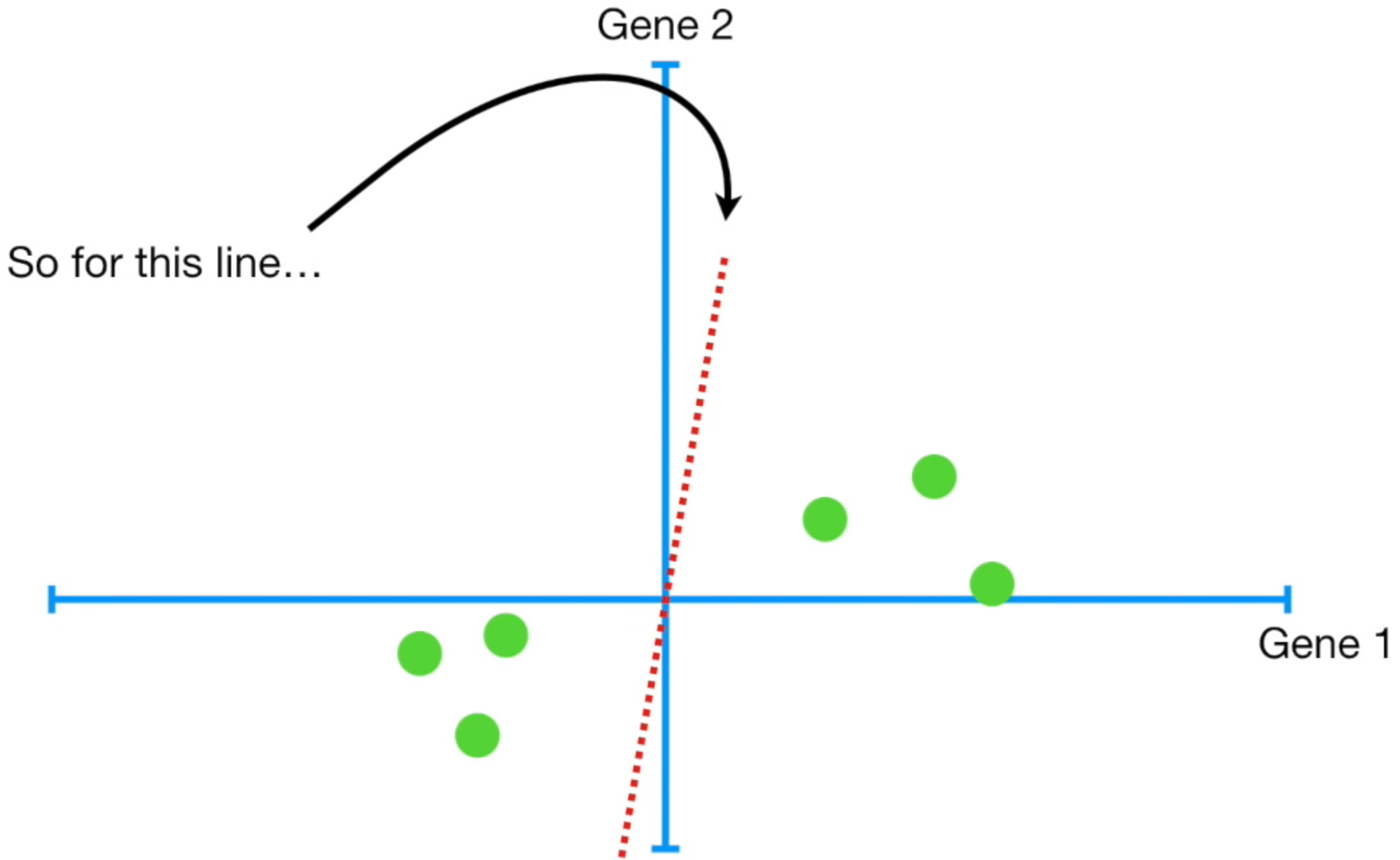


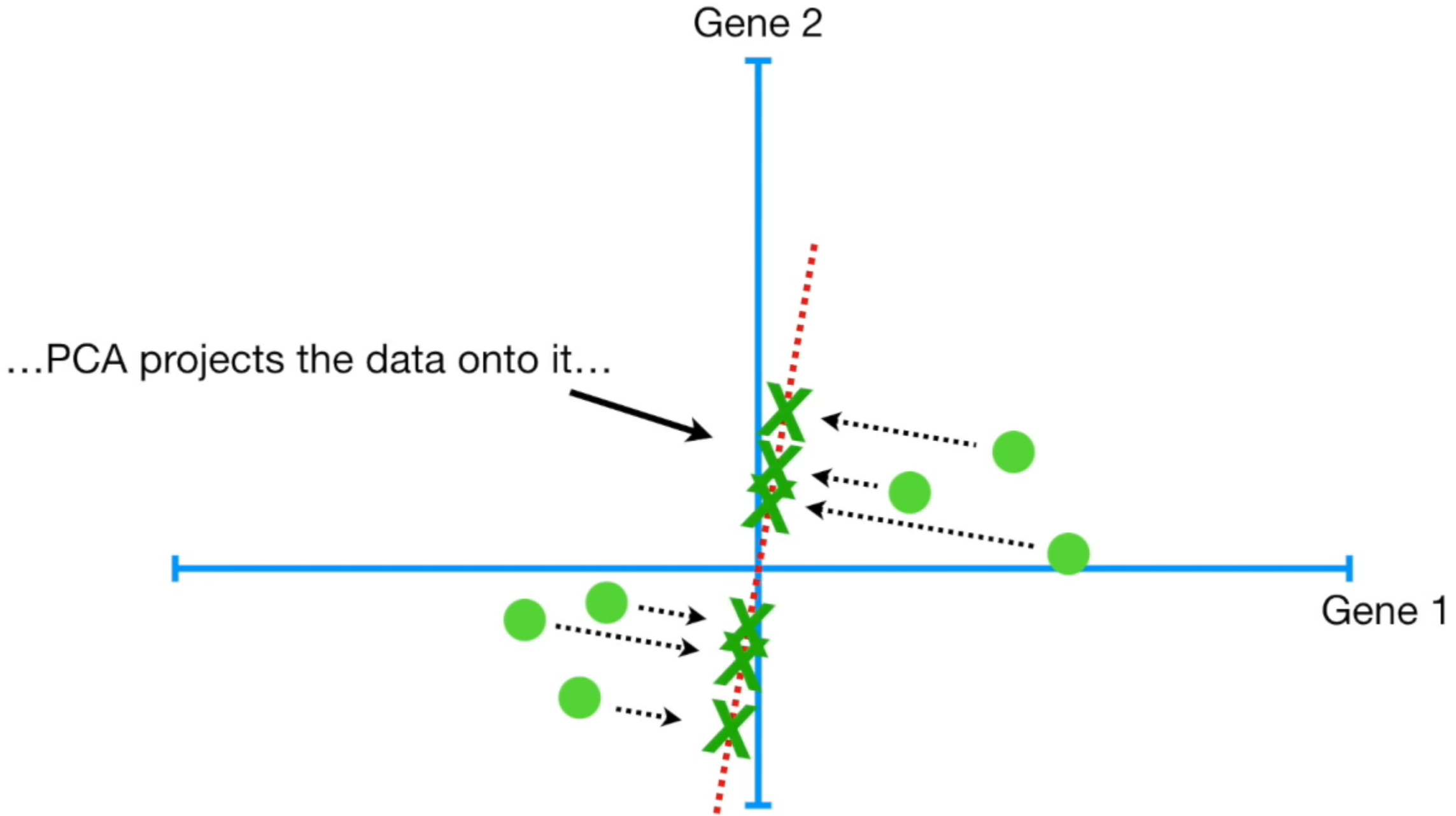
The reason I'm making such a fuss about this is that, intuitively, it makes sense to minimize ***b***, the distance from the point to the line…

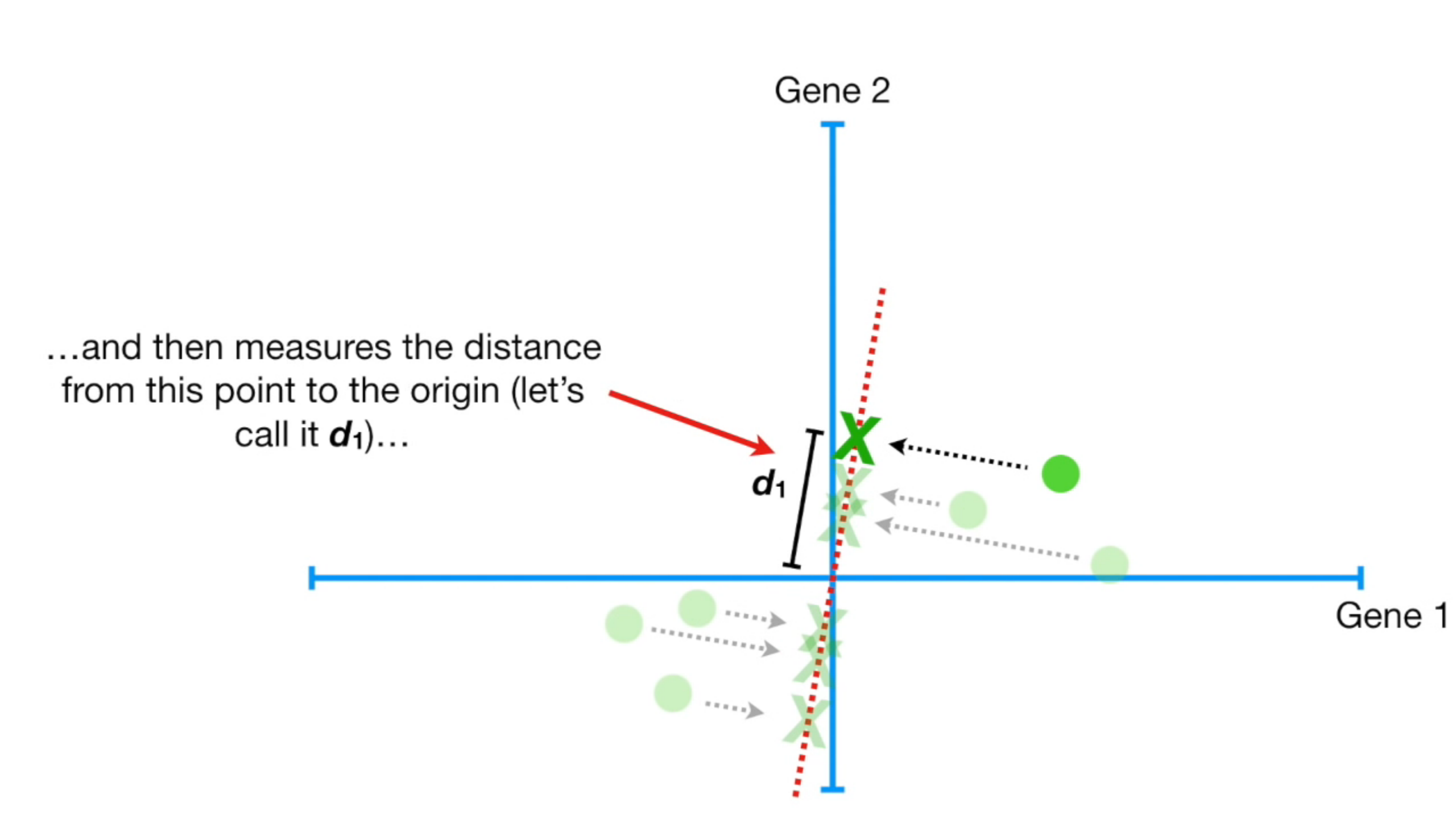


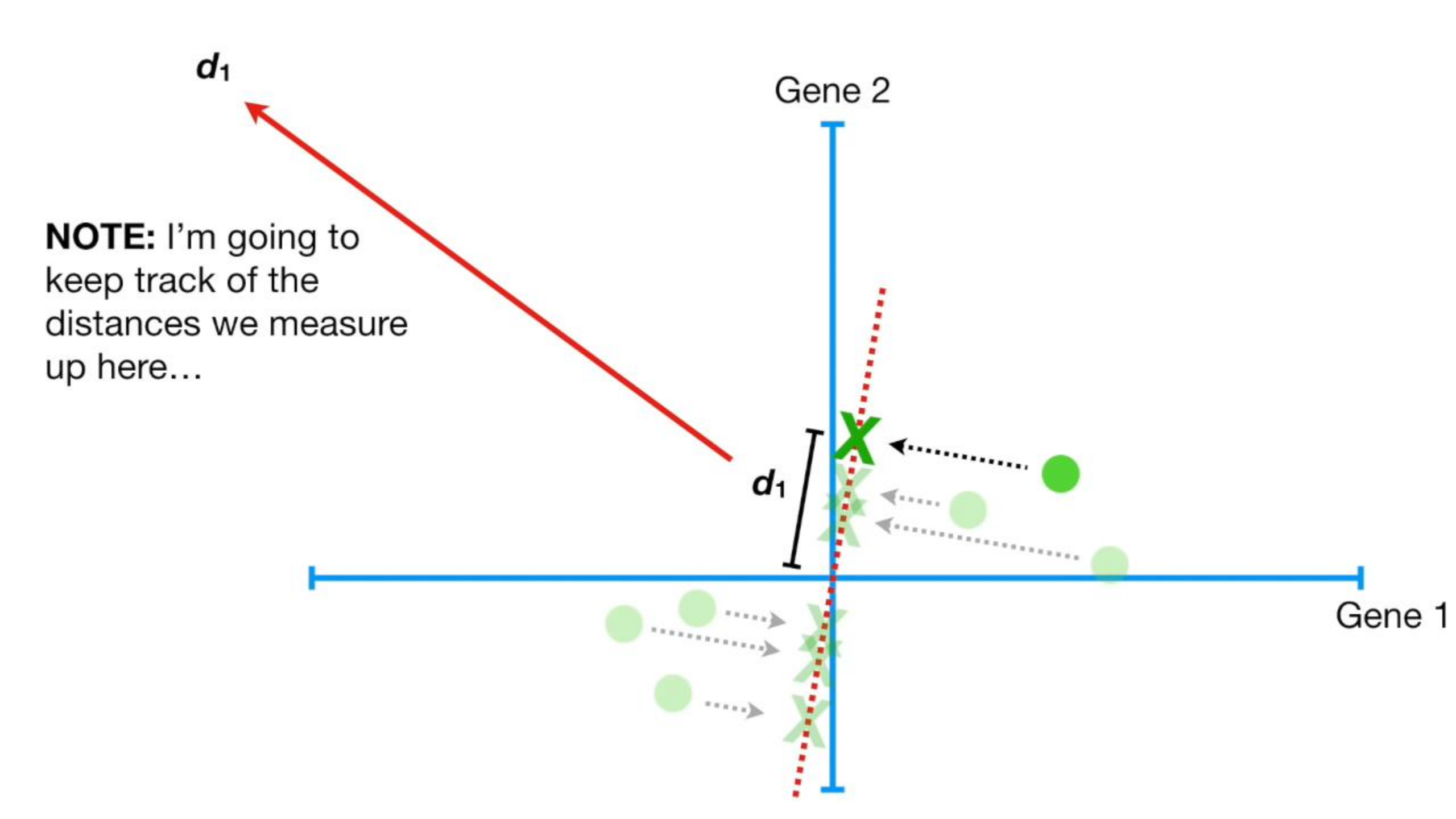
…but it's actually easier to calculate ***c***, the distance from the projected point to the origin, so PCA finds the best fitting line by **maximizing the sum of the squared distances from the projected points to the origin**.




Gene 2
So for this line…
Gene 1

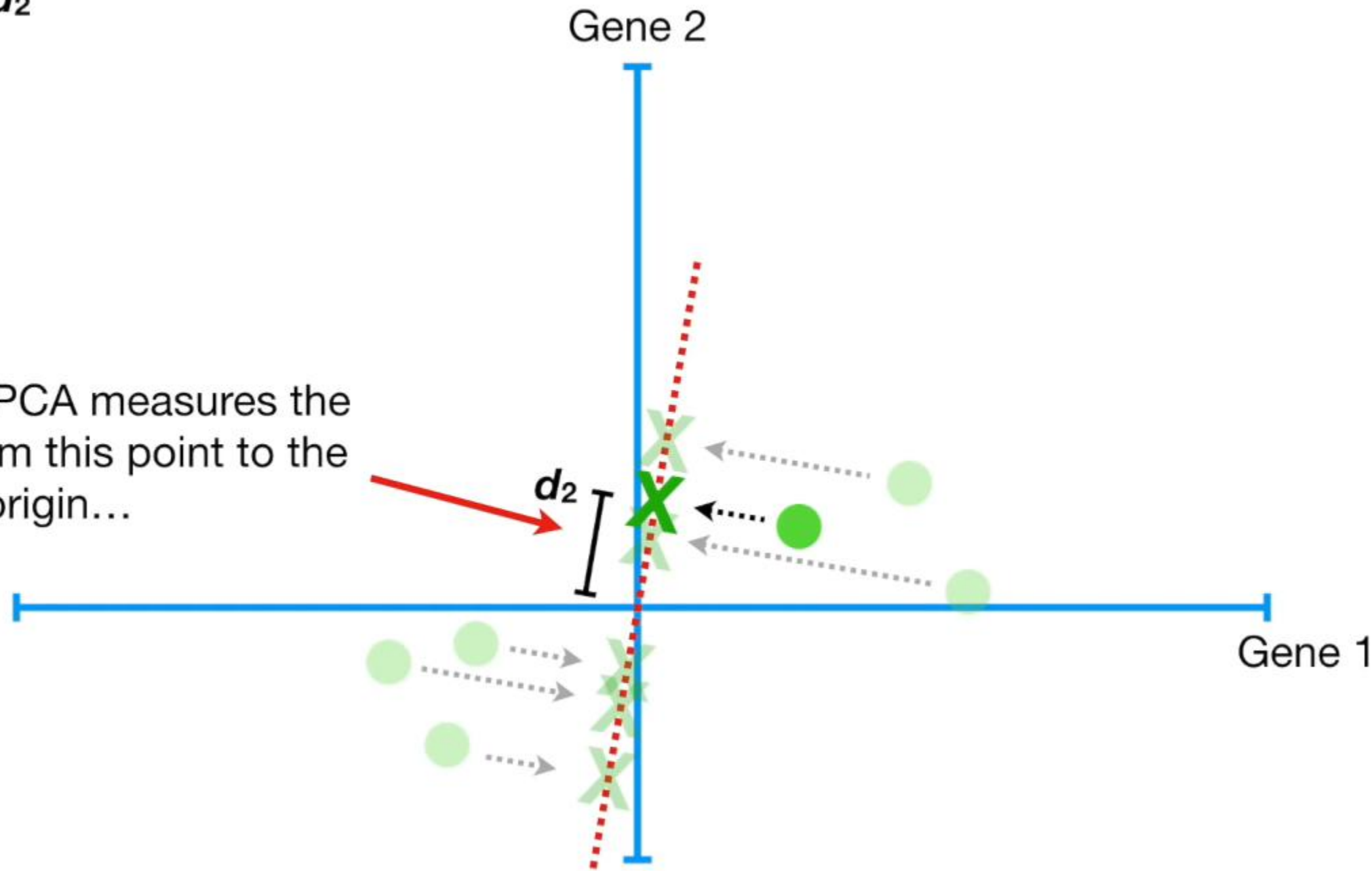

Gene 2
…PCA projects the data onto it…
Gene 1


Gene 2
…and then measures the distance from this point to the origin (let's call it d_1)…
d_1
Gene 1

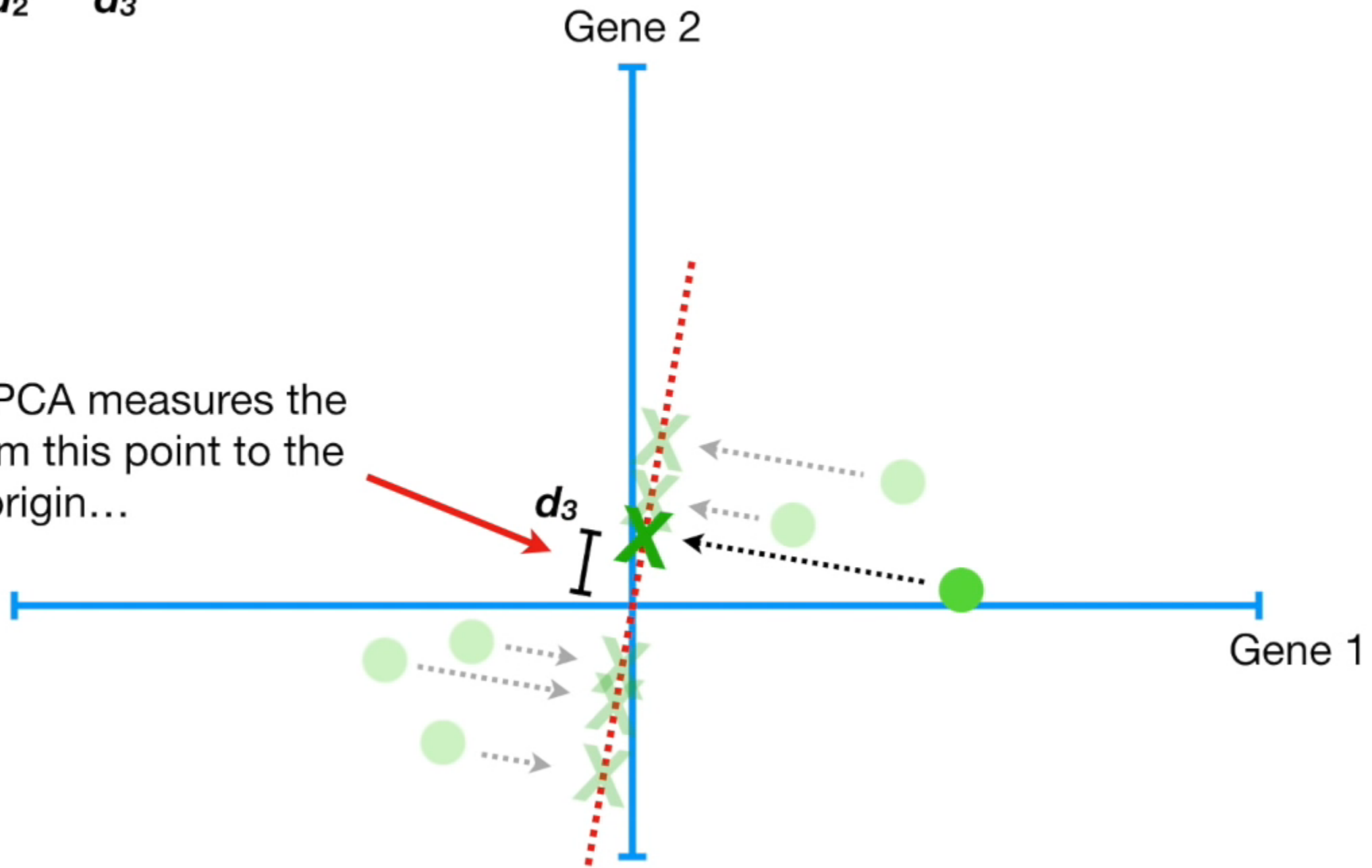

d_1
Gene 2
NOTE: I'm going to keep track of the distances we measure up here…
d_1
Gene 1

d_1 d_2

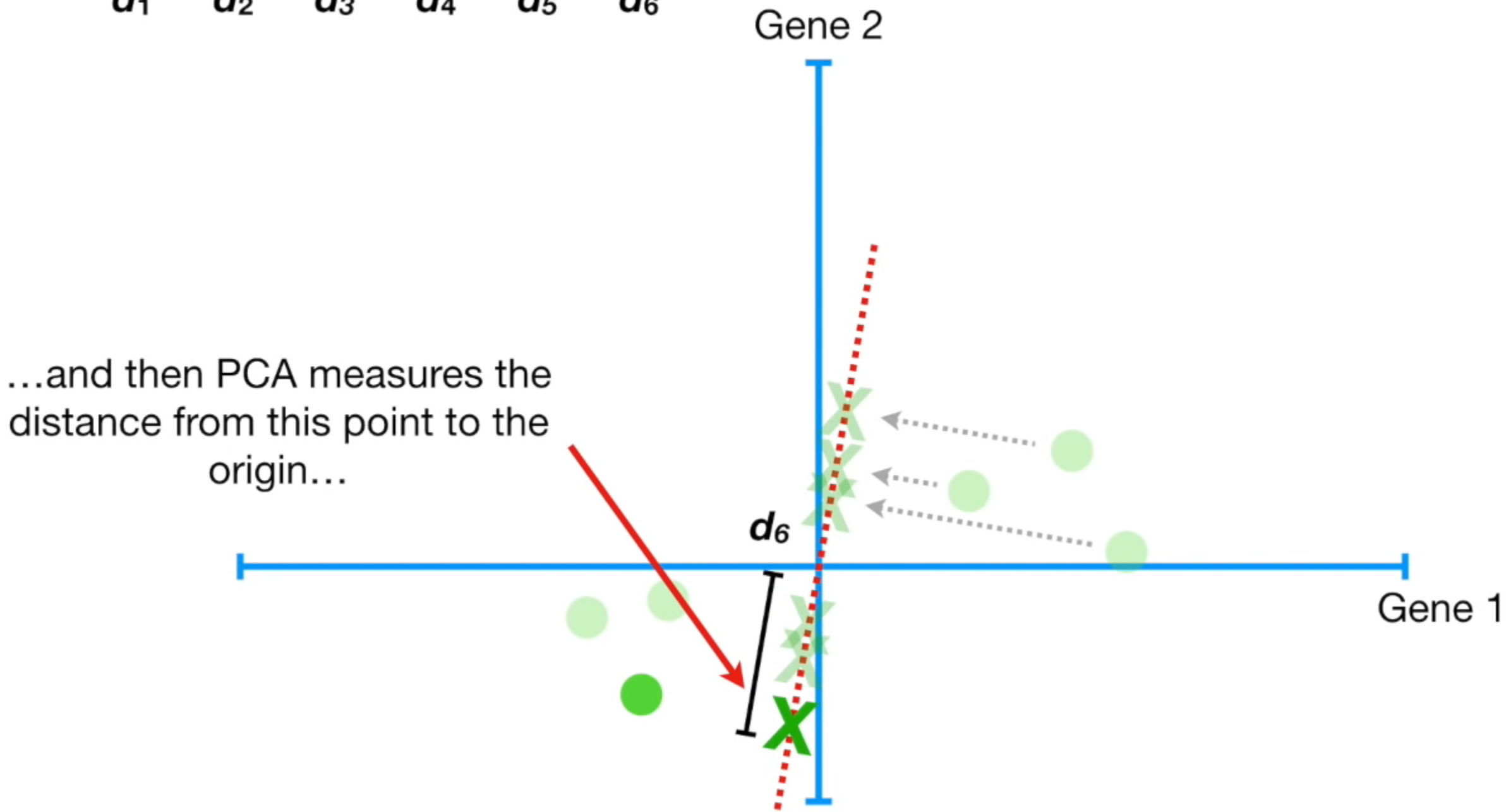
…and then PCA measures the distance from this point to the origin…



d_1 d_2 d_3

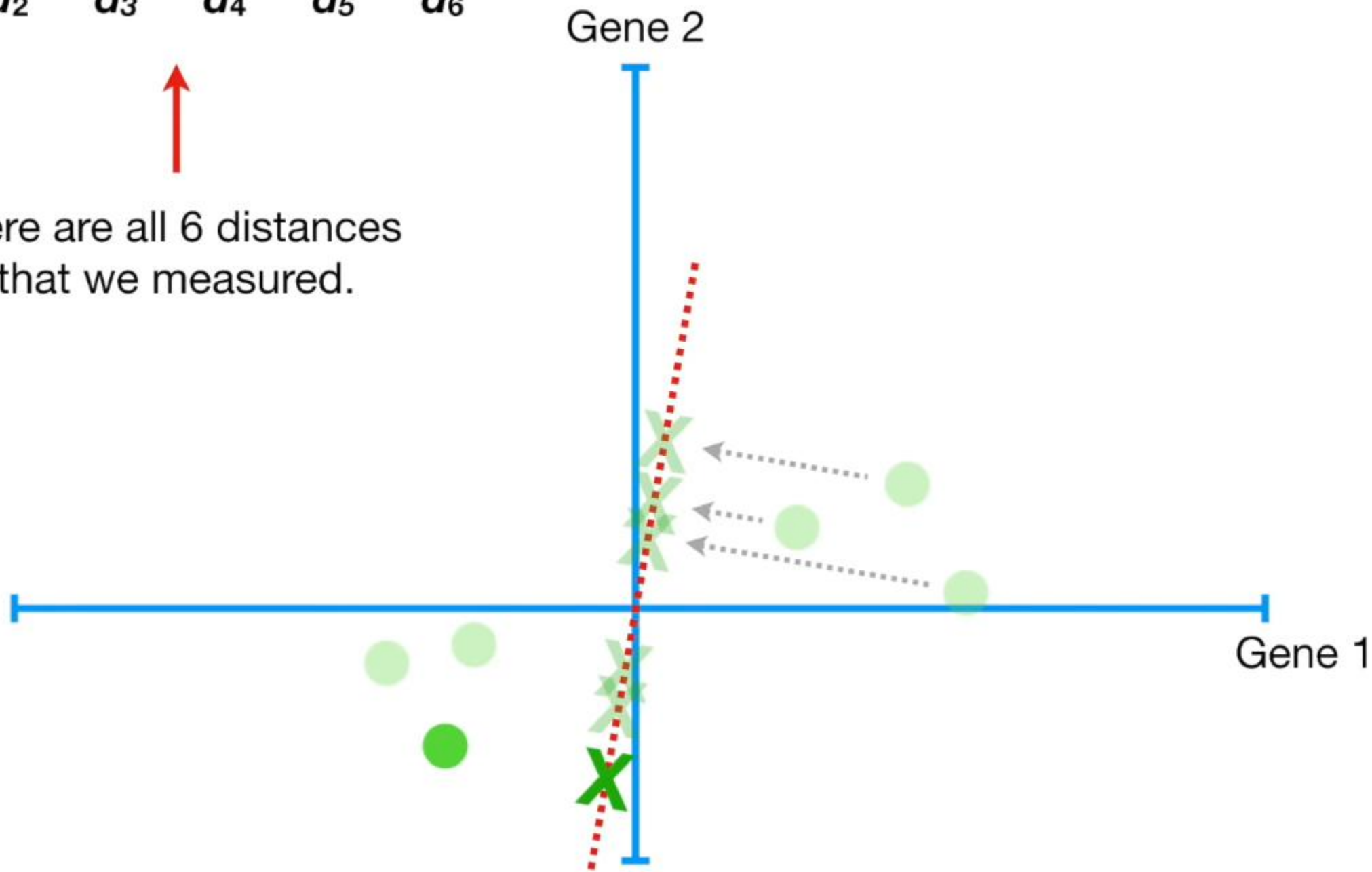

Gene 2
…and then PCA measures the distance from this point to the origin…
d_3
Gene 1

d_1 d_2 d_3 d_4 d_5 d_6



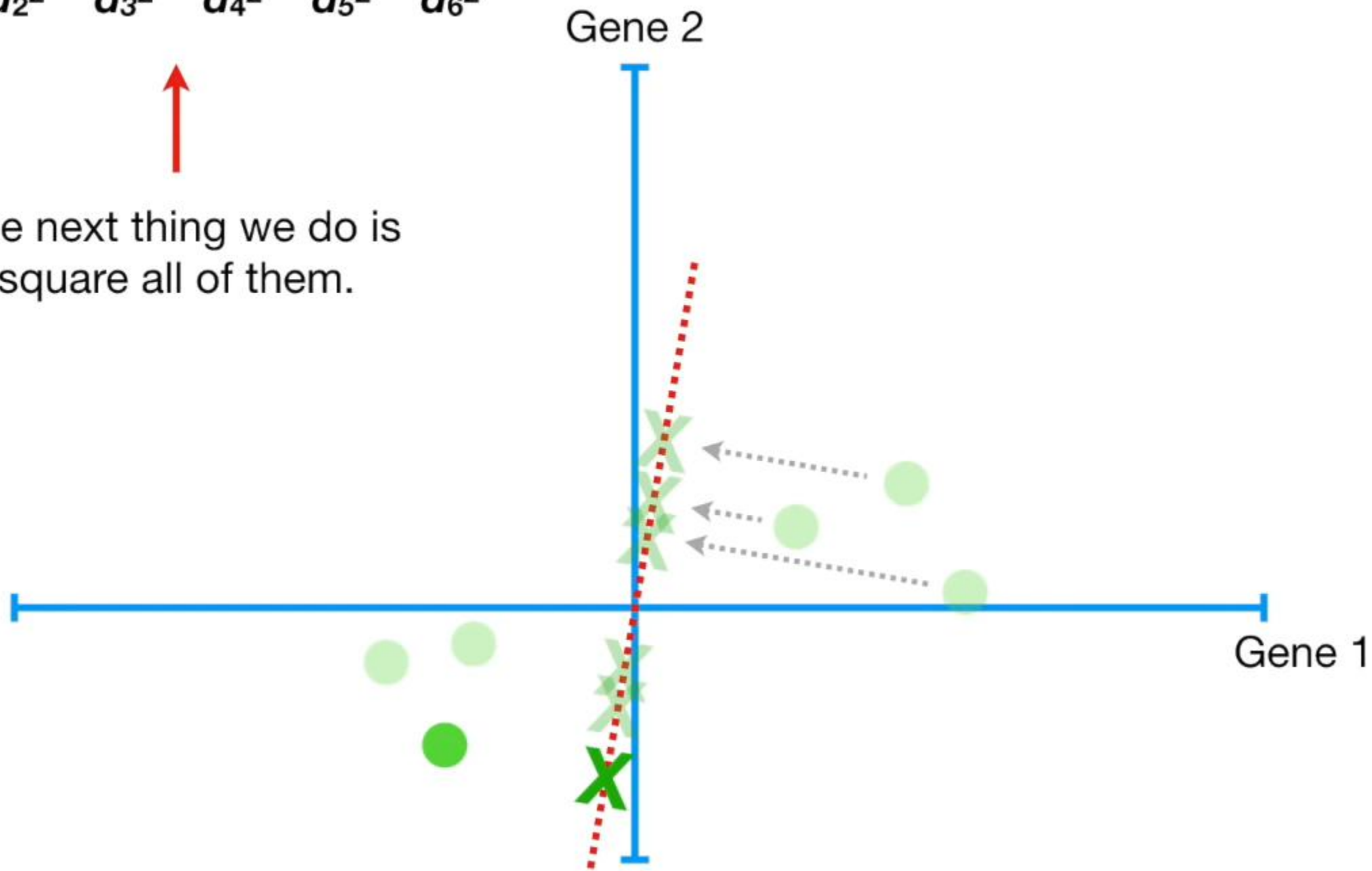
d_1 d_2 d_3 d_4 d_5 d_6

Here are all 6 distances that we measured.



d_1^2 d_2^2 d_3^2 d_4^2 d_5^2 d_6^2

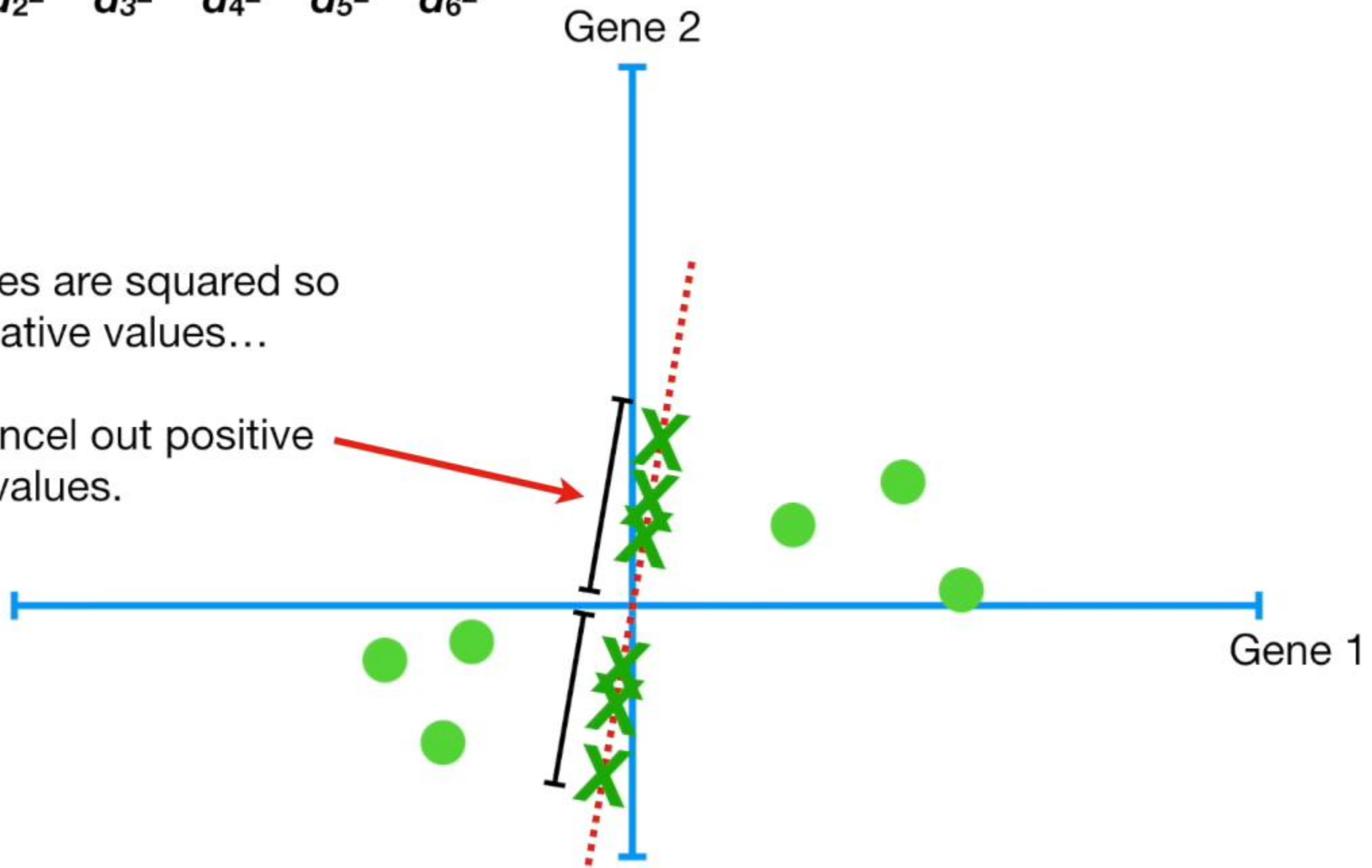
The next thing we do is square all of them.



d_1^2 d_2^2 d_3^2 d_4^2 d_5^2 d_6^2

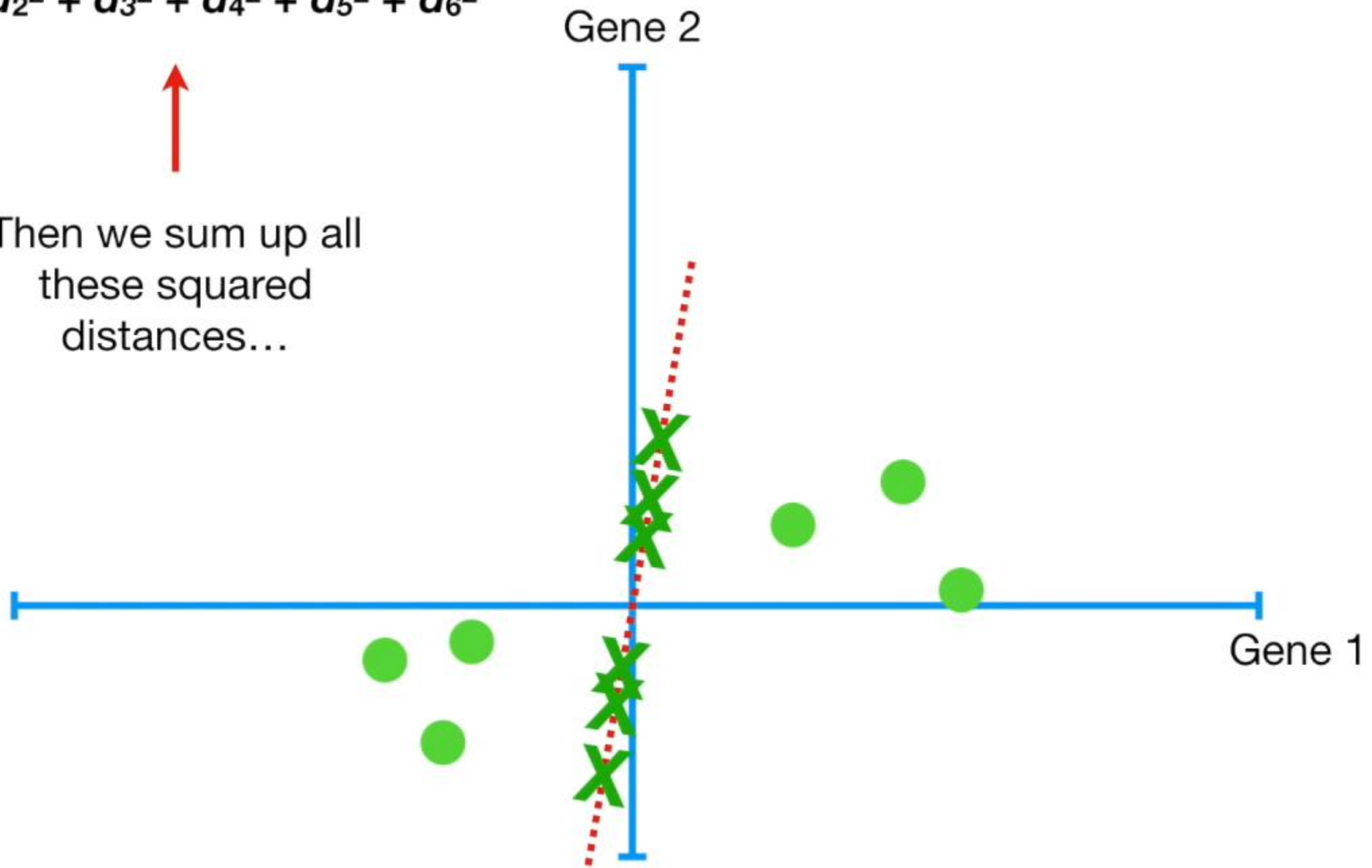
The distances are squared so that negative values…

…don't cancel out positive values.

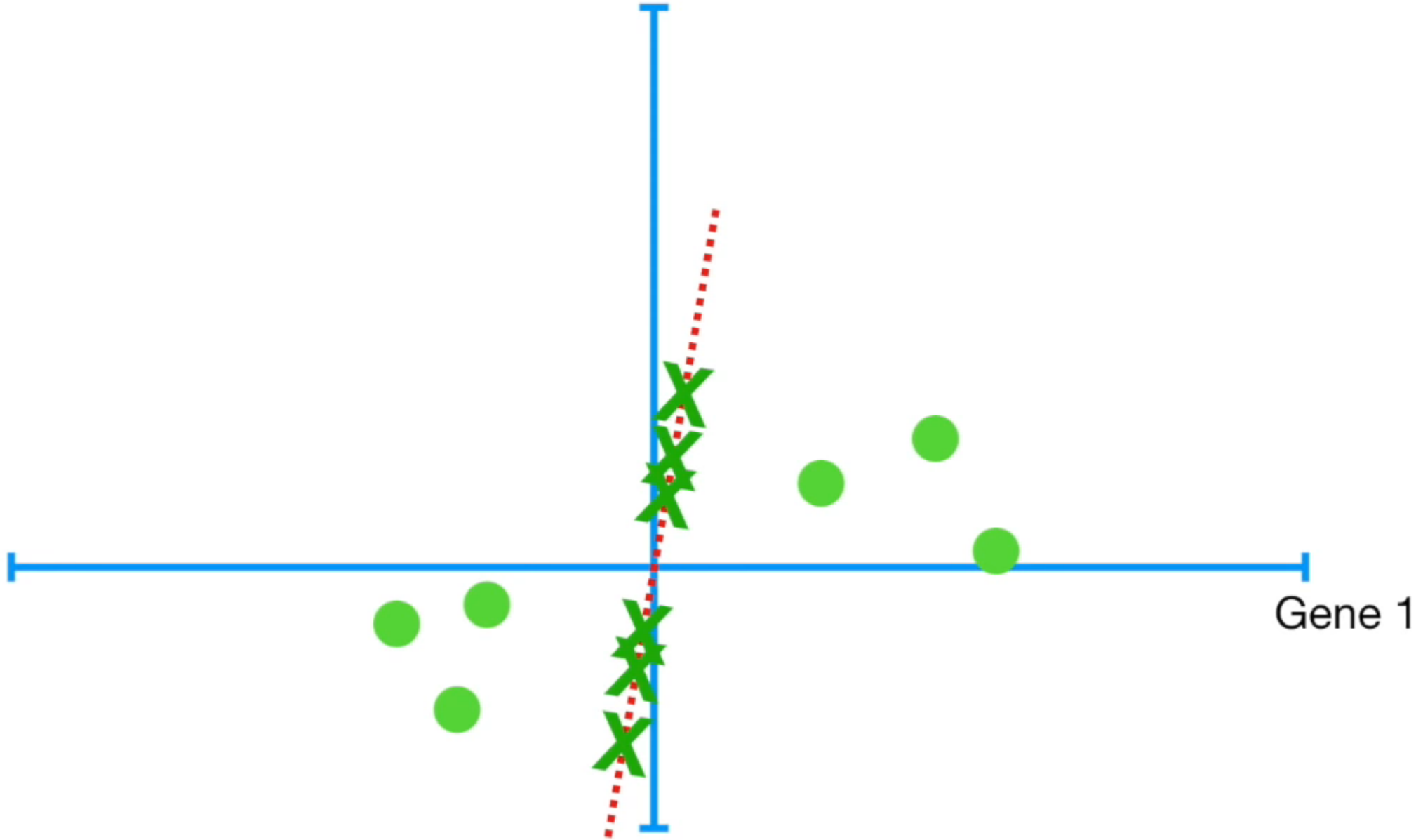


$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$

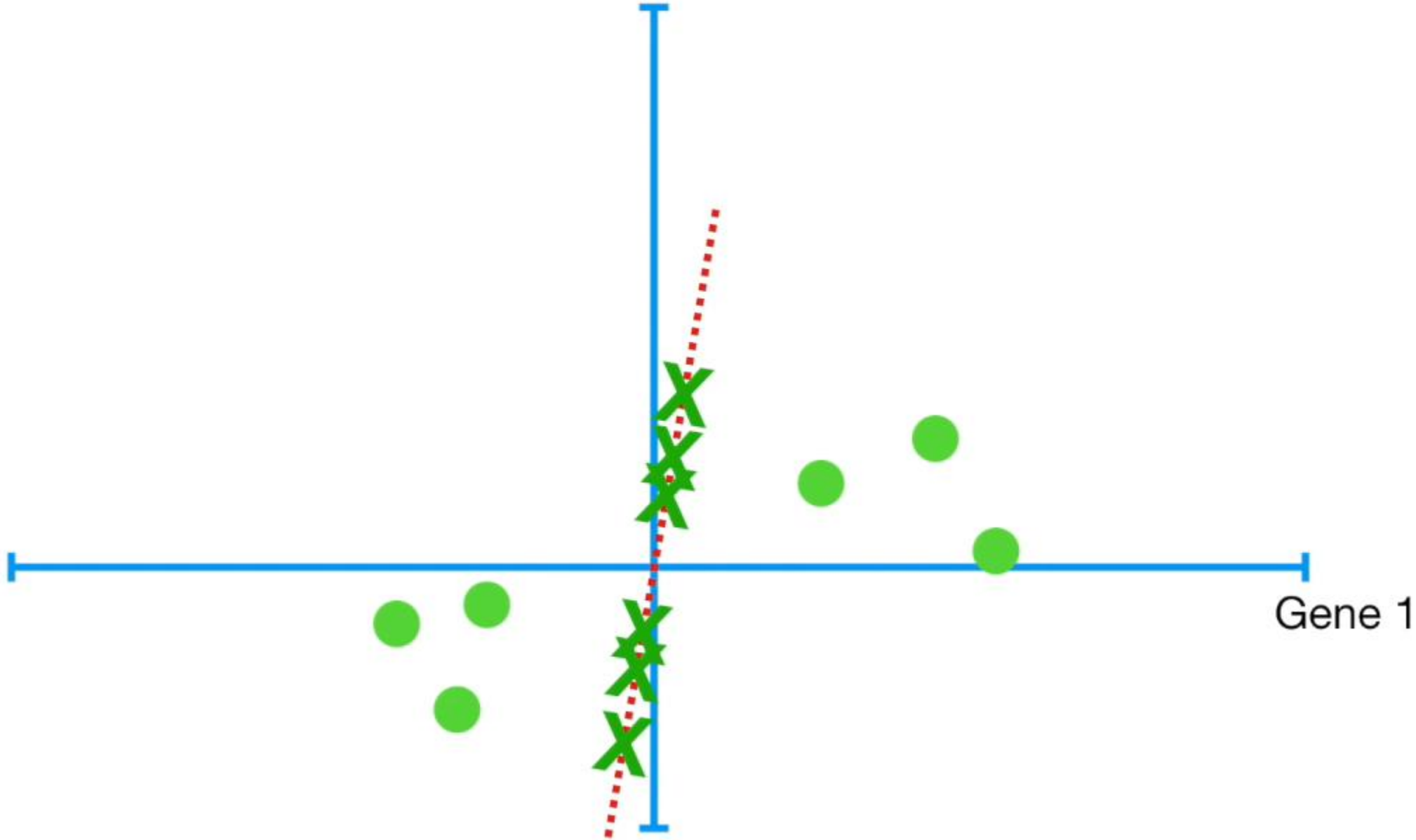
Then we sum up all these squared distances…



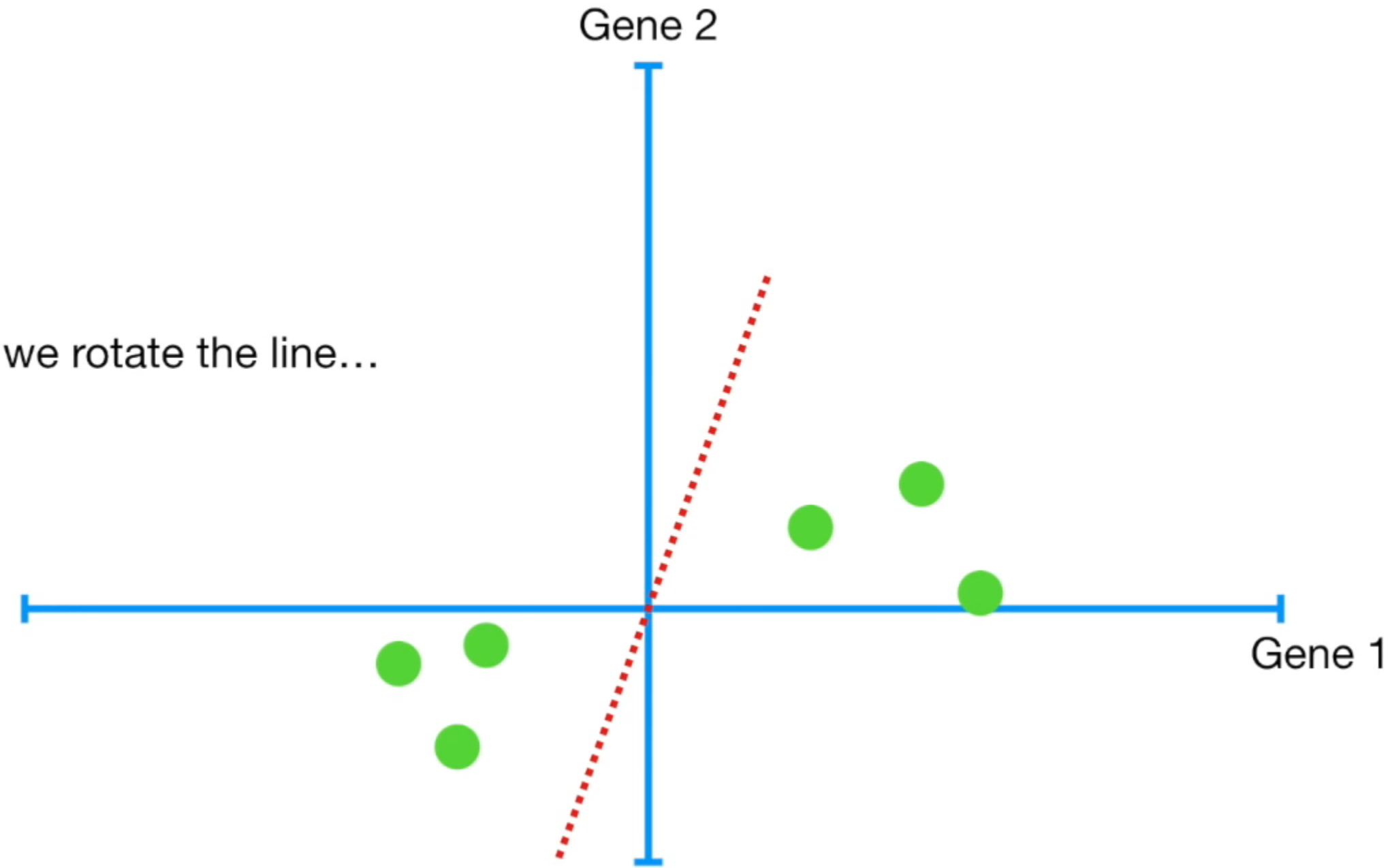
$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances

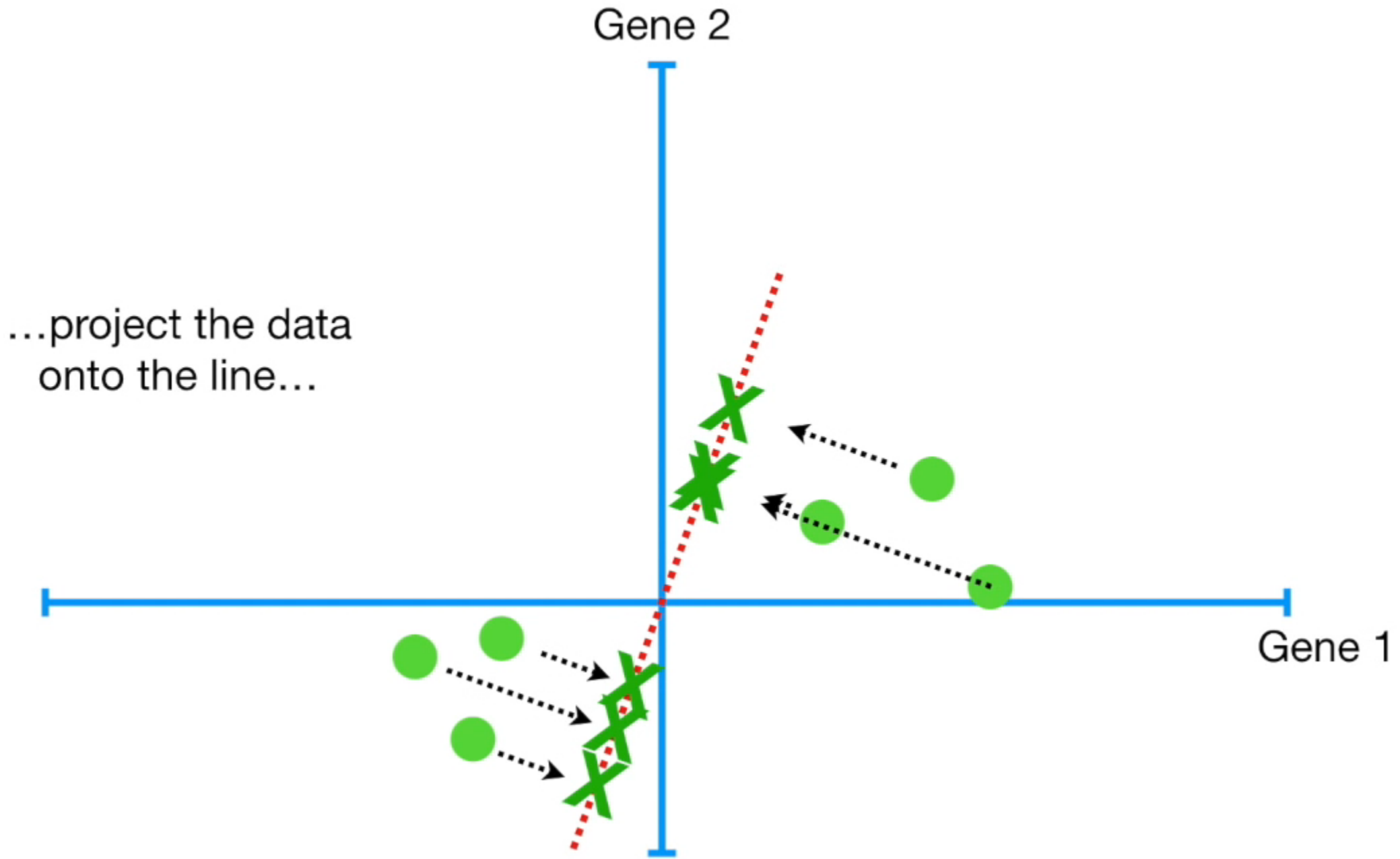


$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)



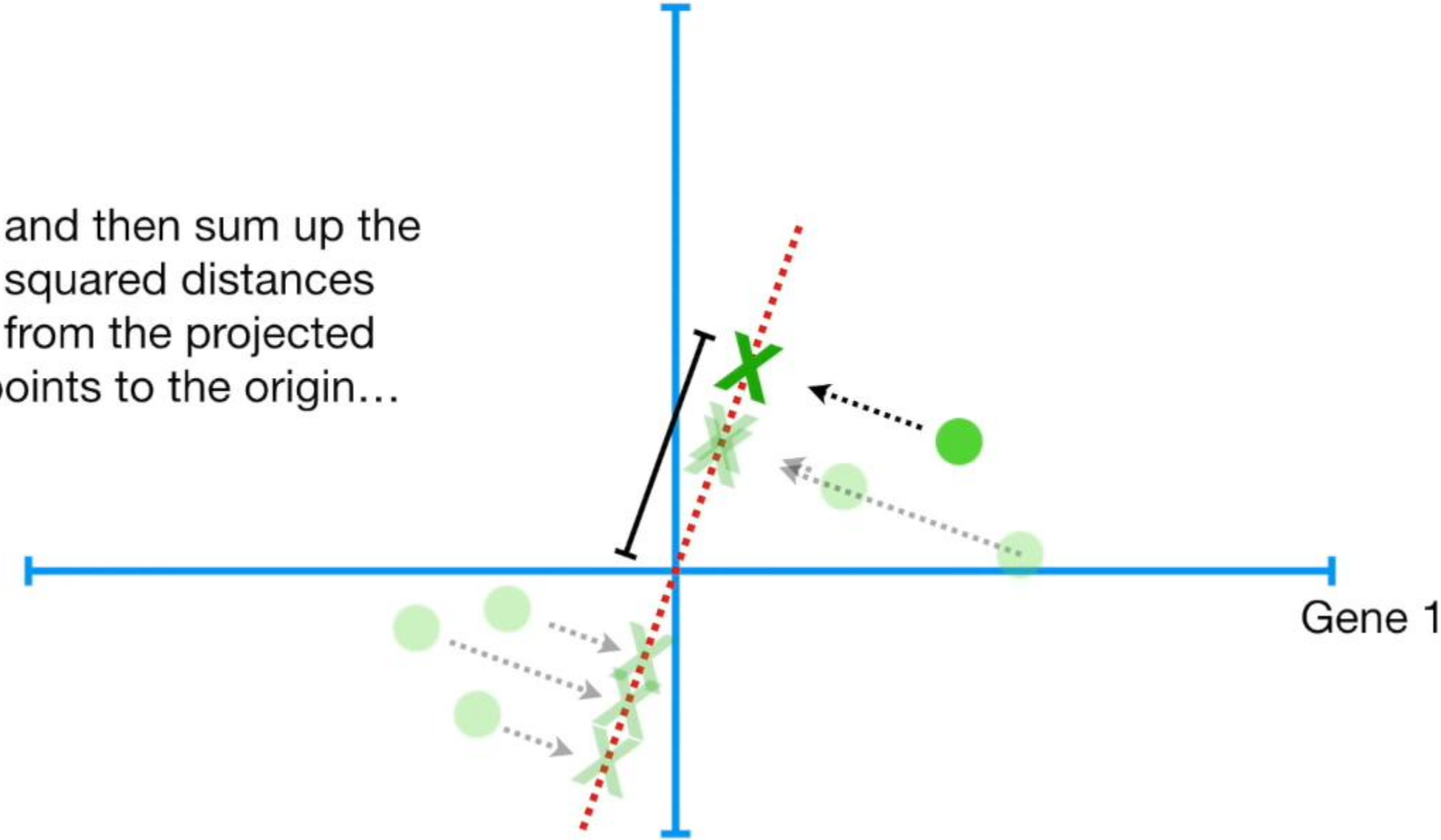
Now we rotate the line…




Gene 2
…project the data
onto the line…
Gene 1

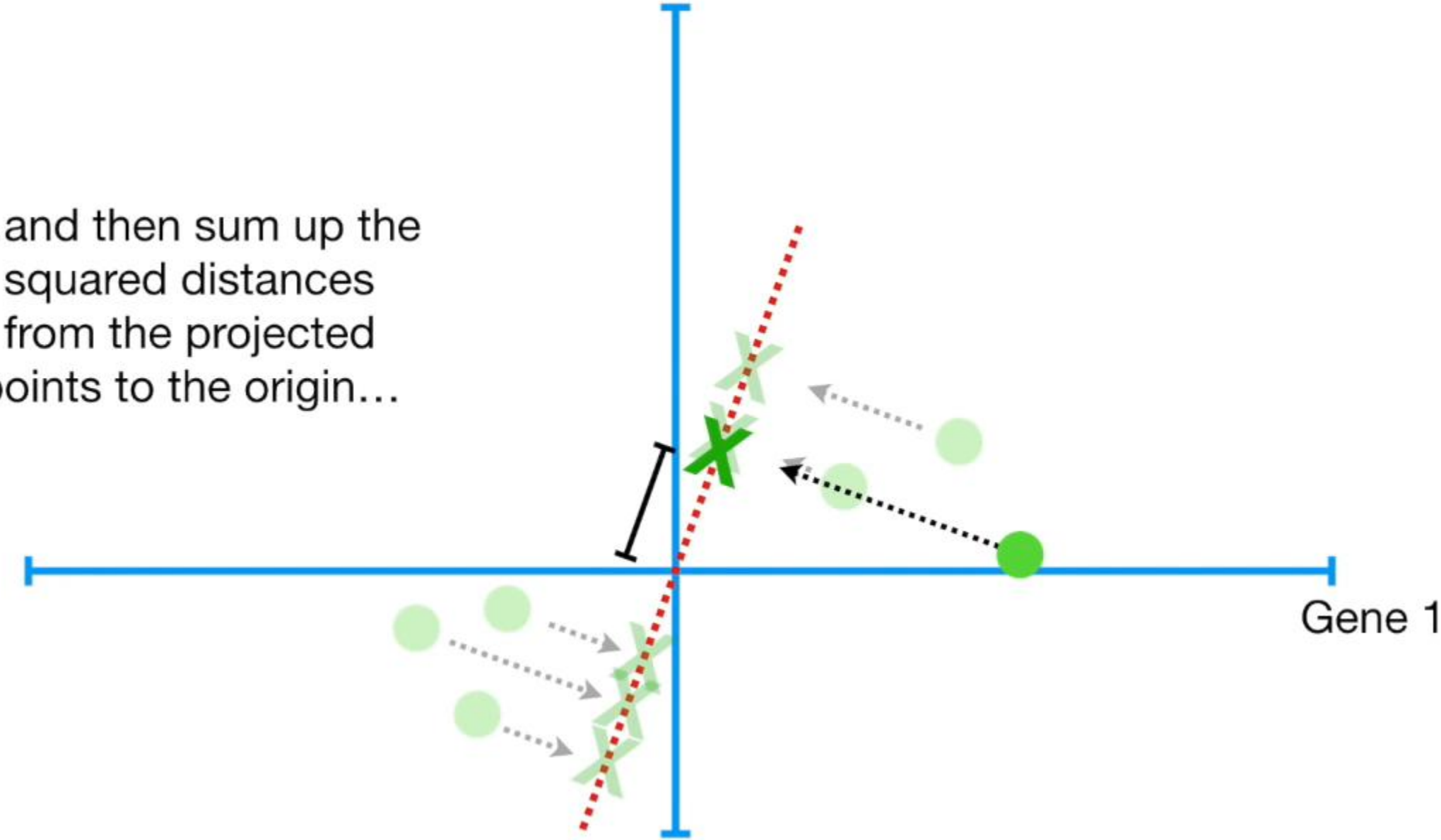
$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

…and then sum up the squared distances from the projected points to the origin…



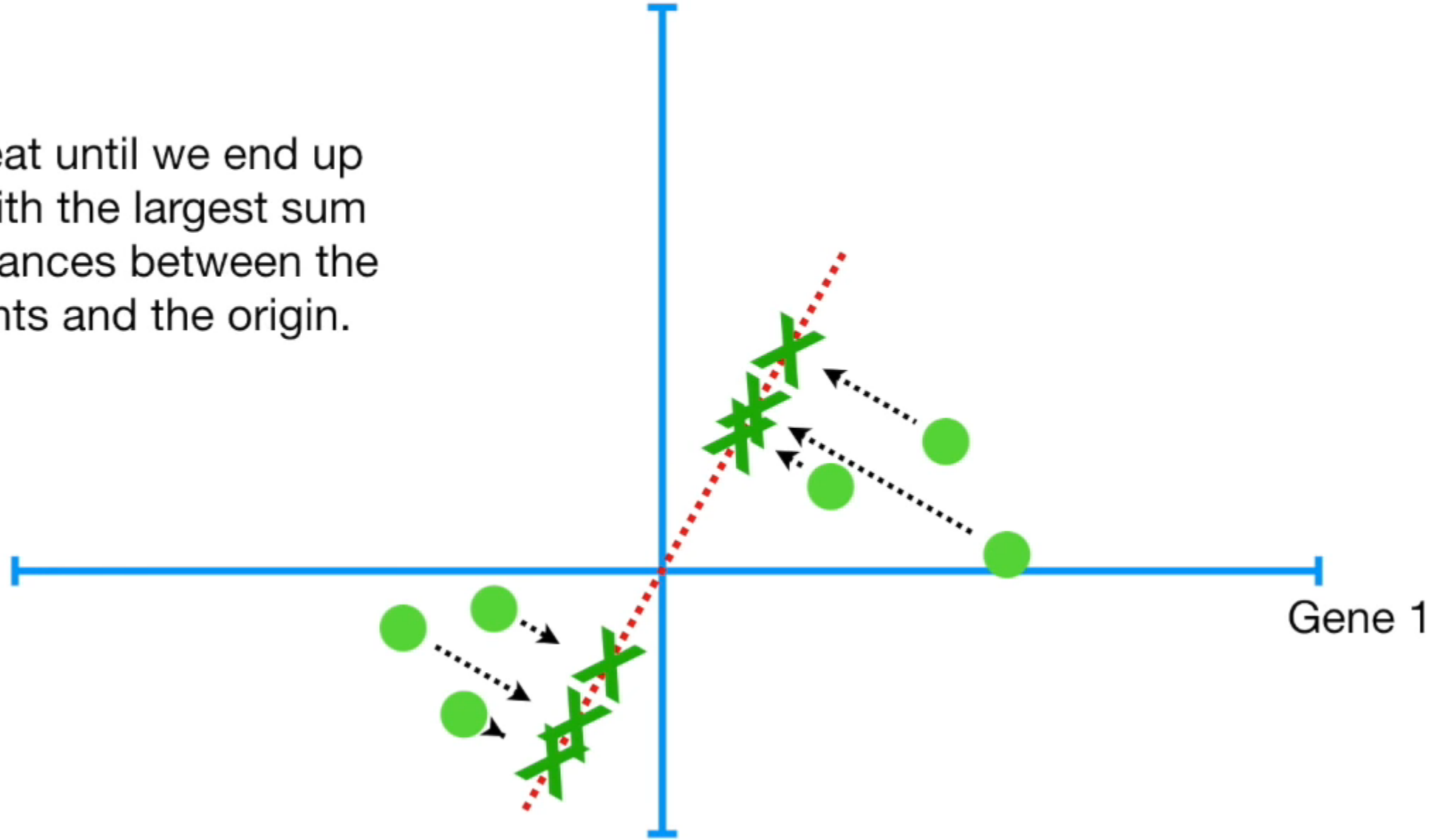
$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

…and then sum up the squared distances from the projected points to the origin…



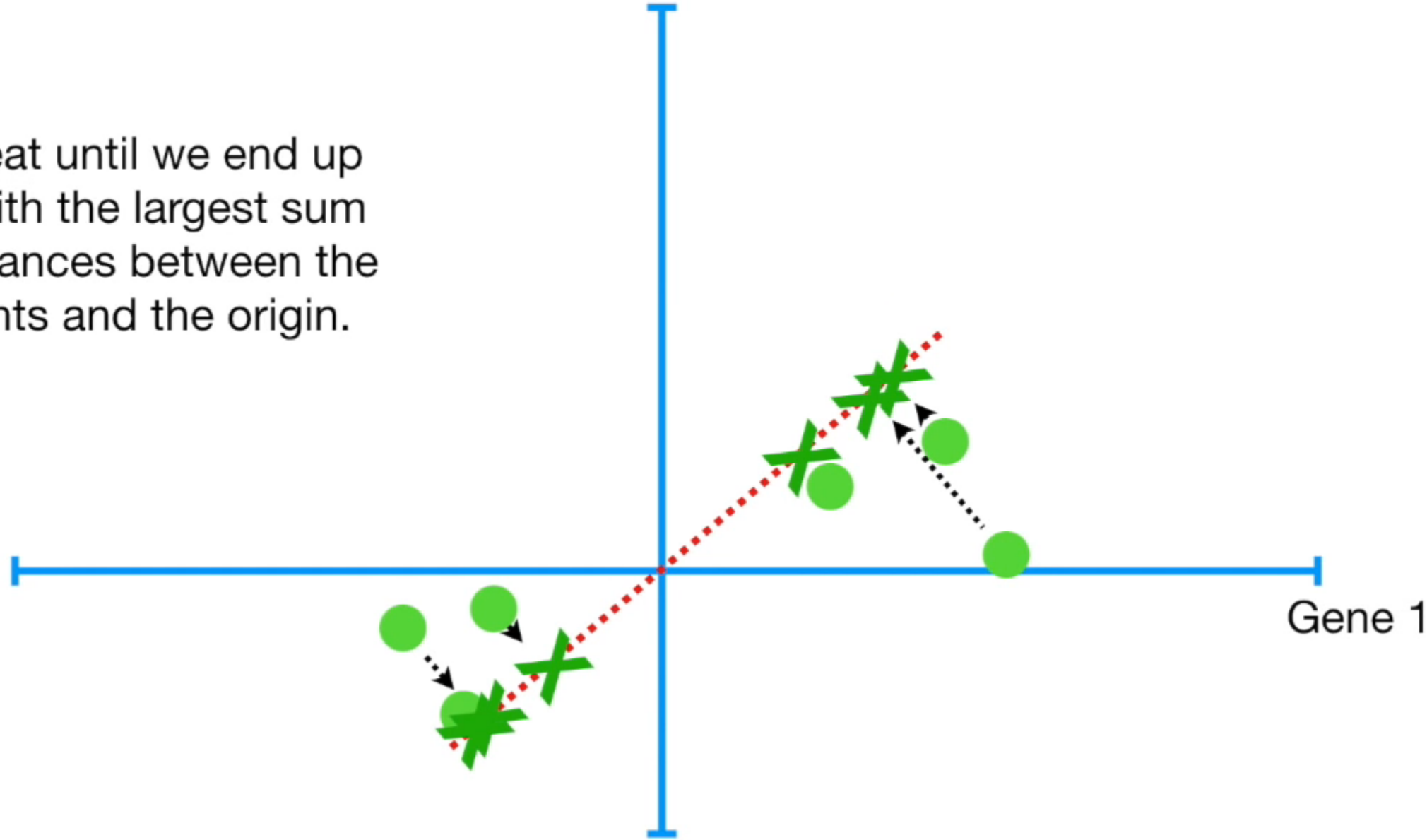
$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

…and we repeat until we end up with the line with the largest sum of squared distances between the projected points and the origin.



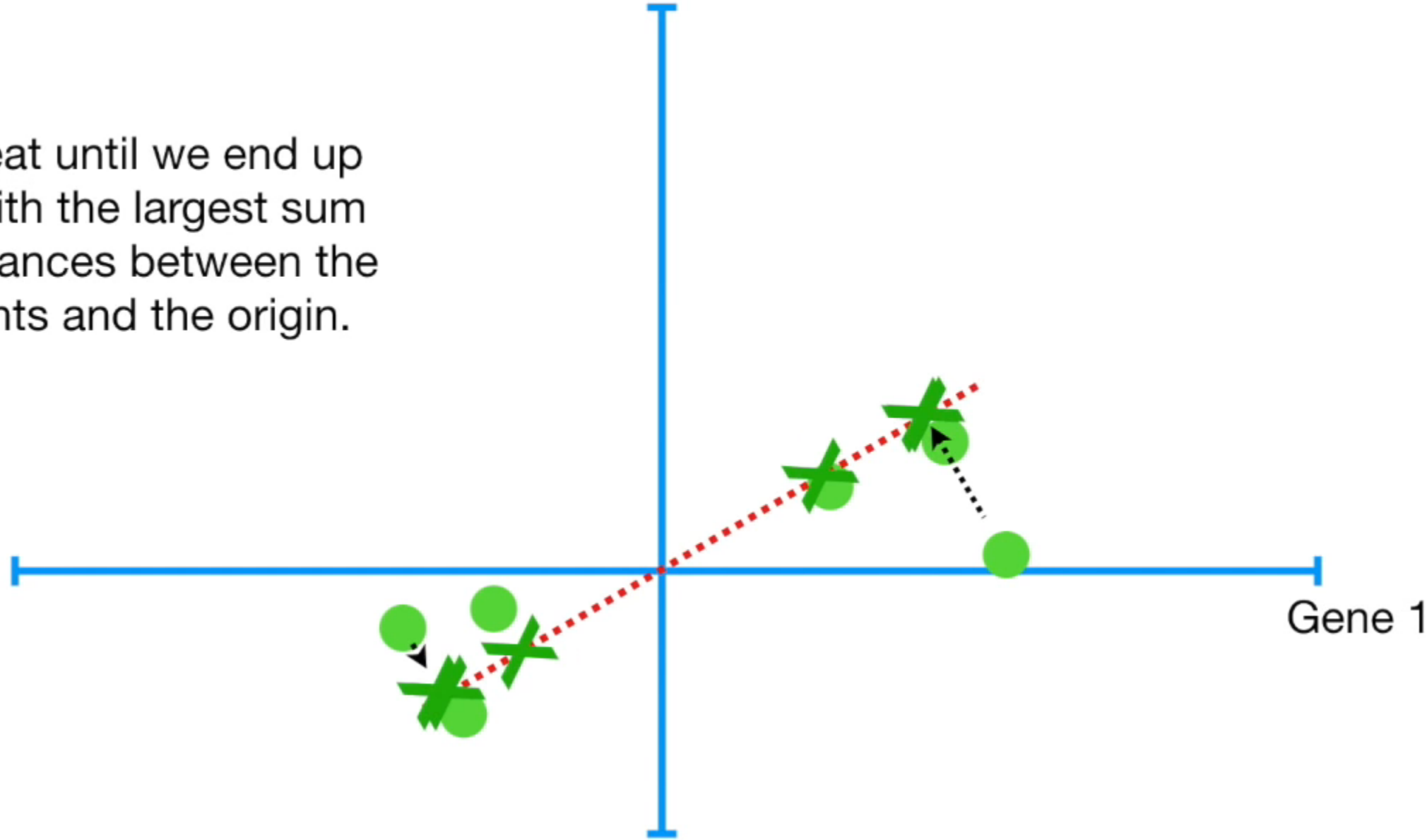
$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

…and we repeat until we end up with the line with the largest sum of squared distances between the projected points and the origin.



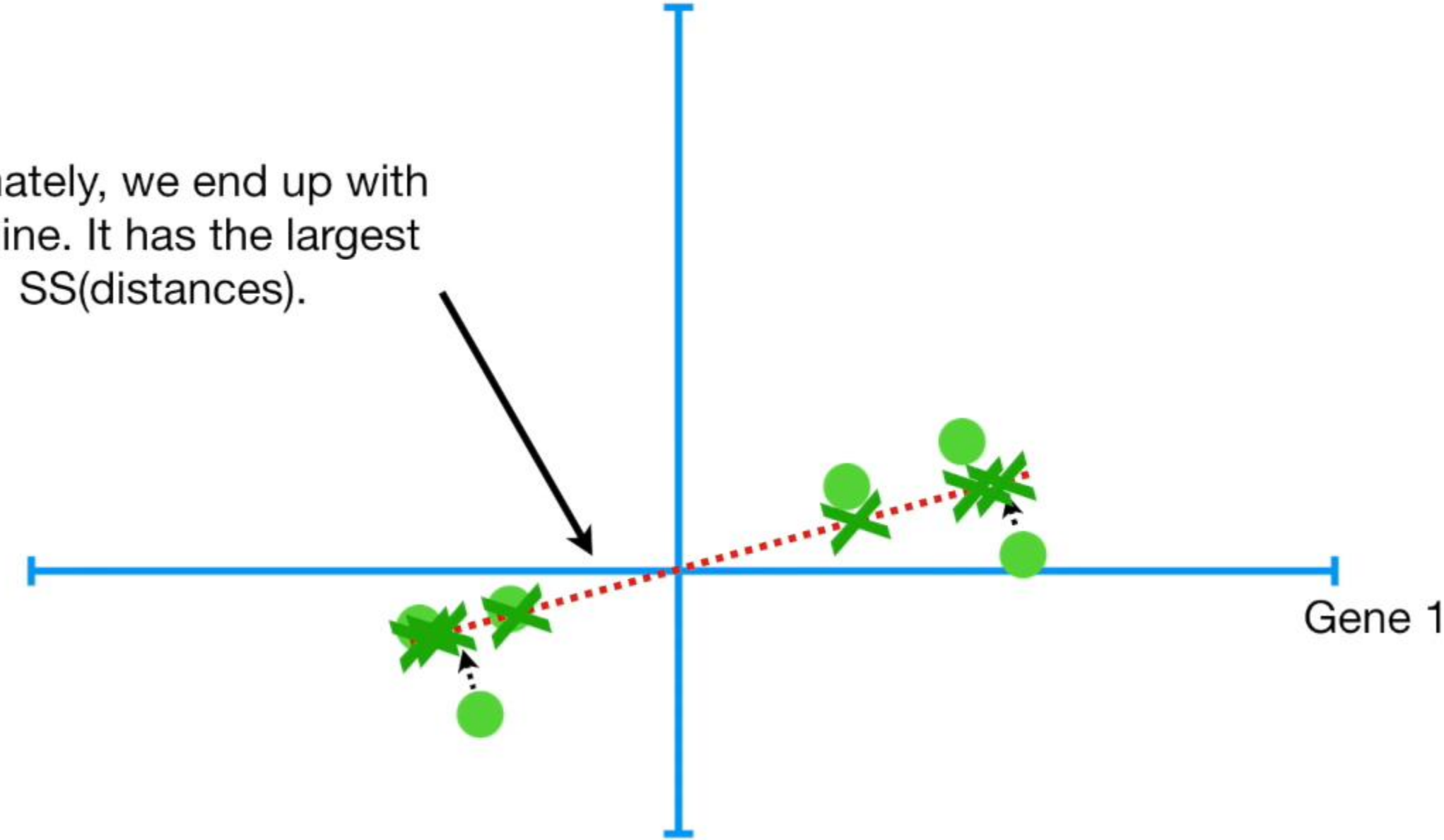
$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

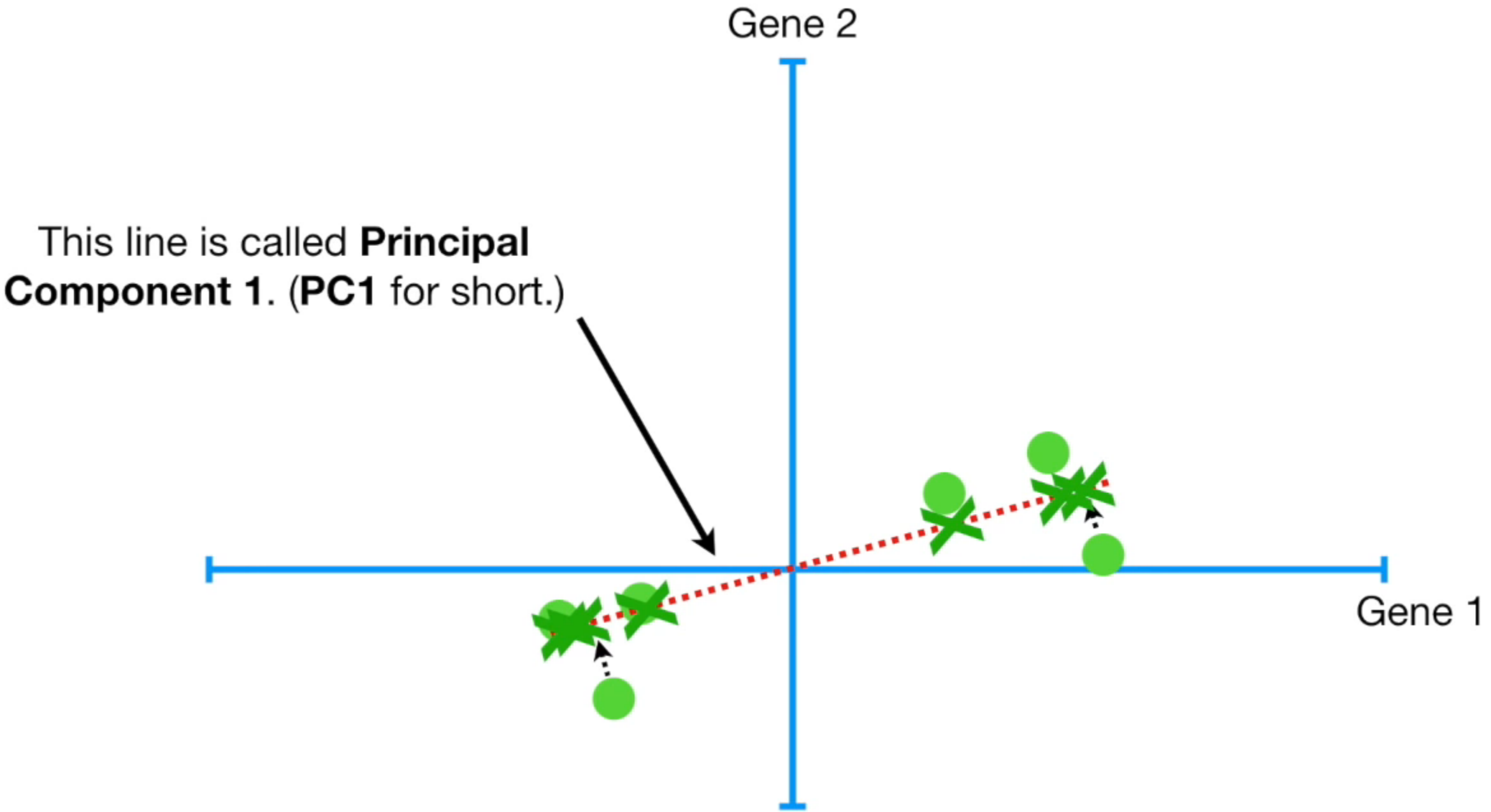
…and we repeat until we end up with the line with the largest sum of squared distances between the projected points and the origin.

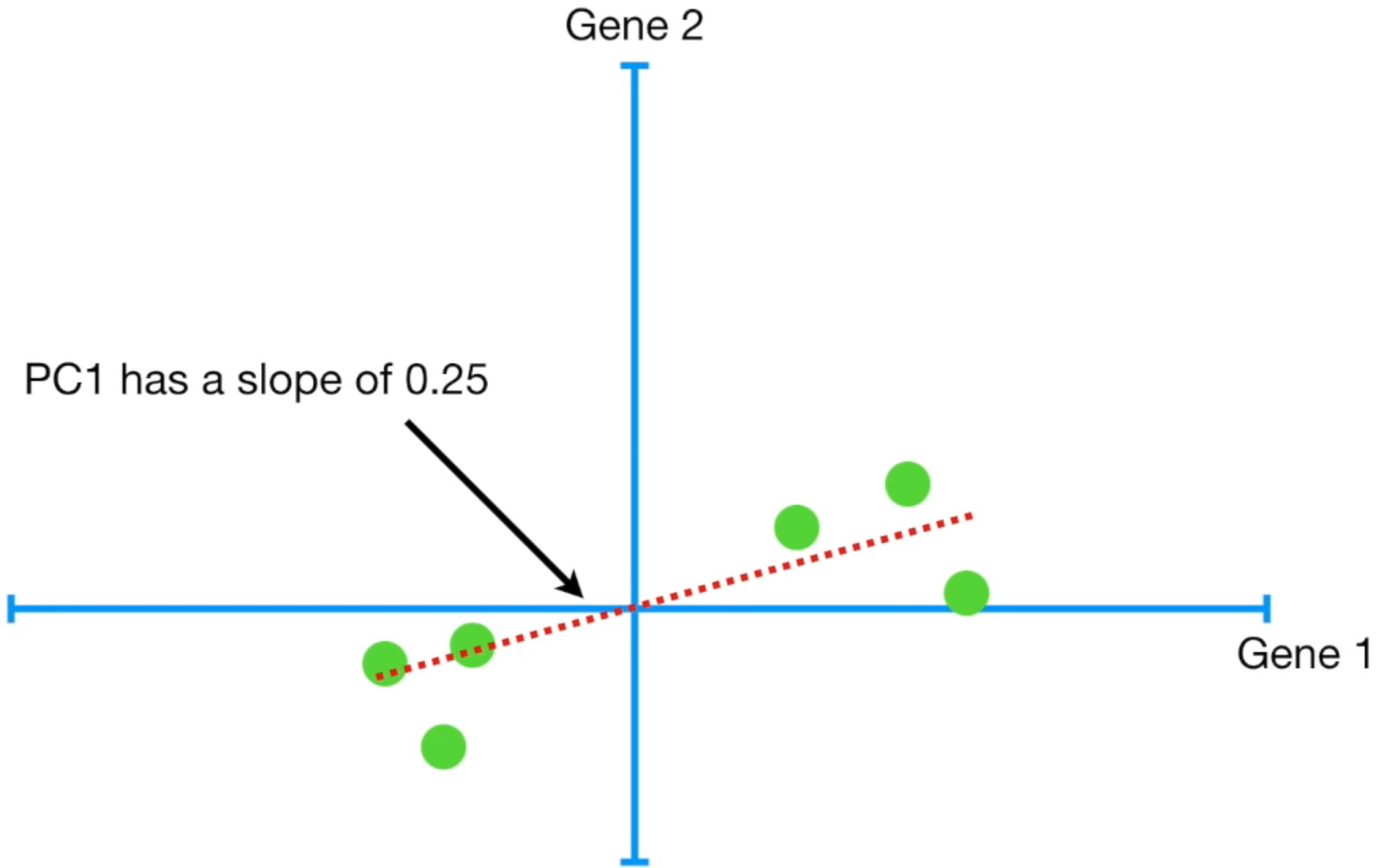


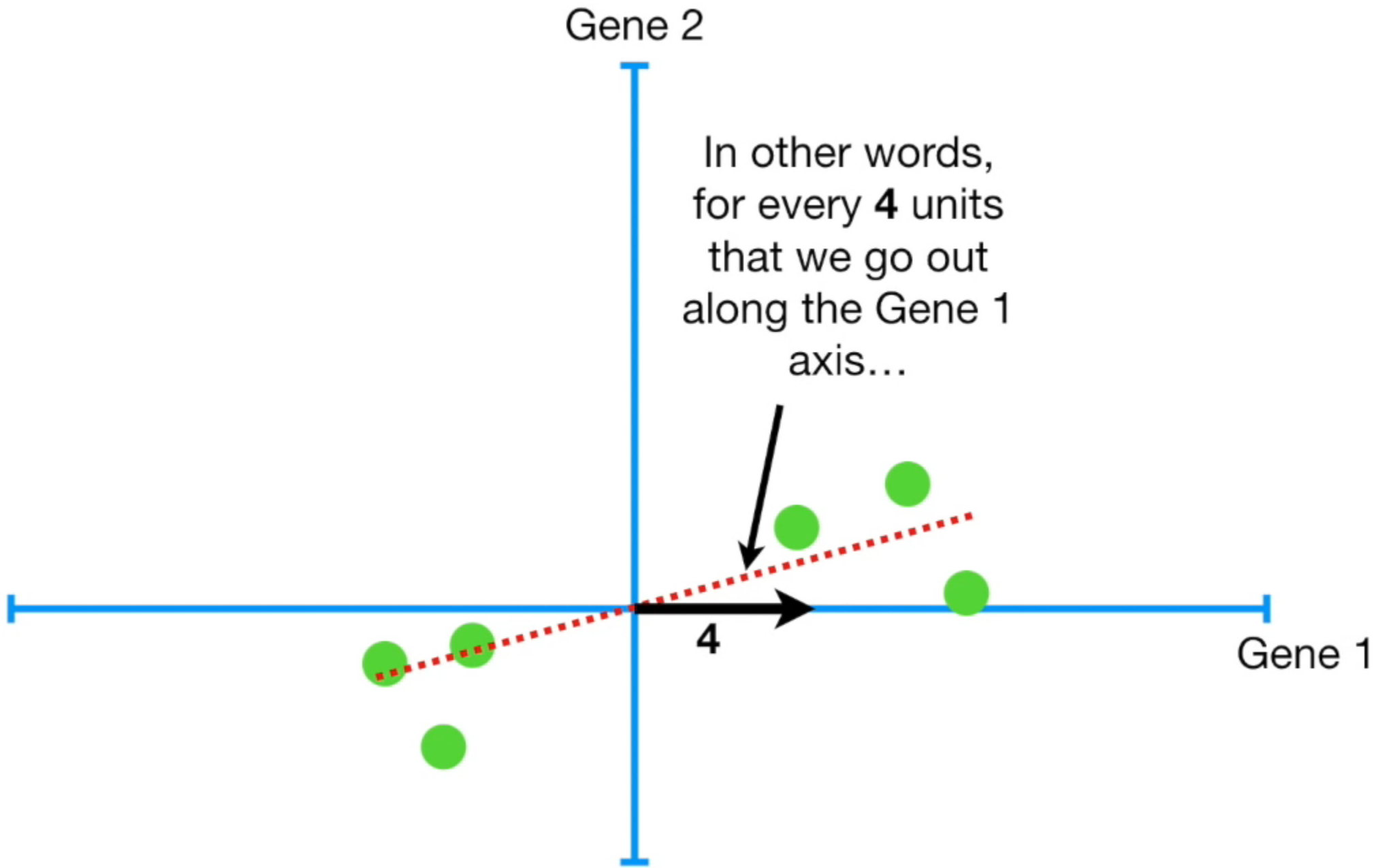
$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

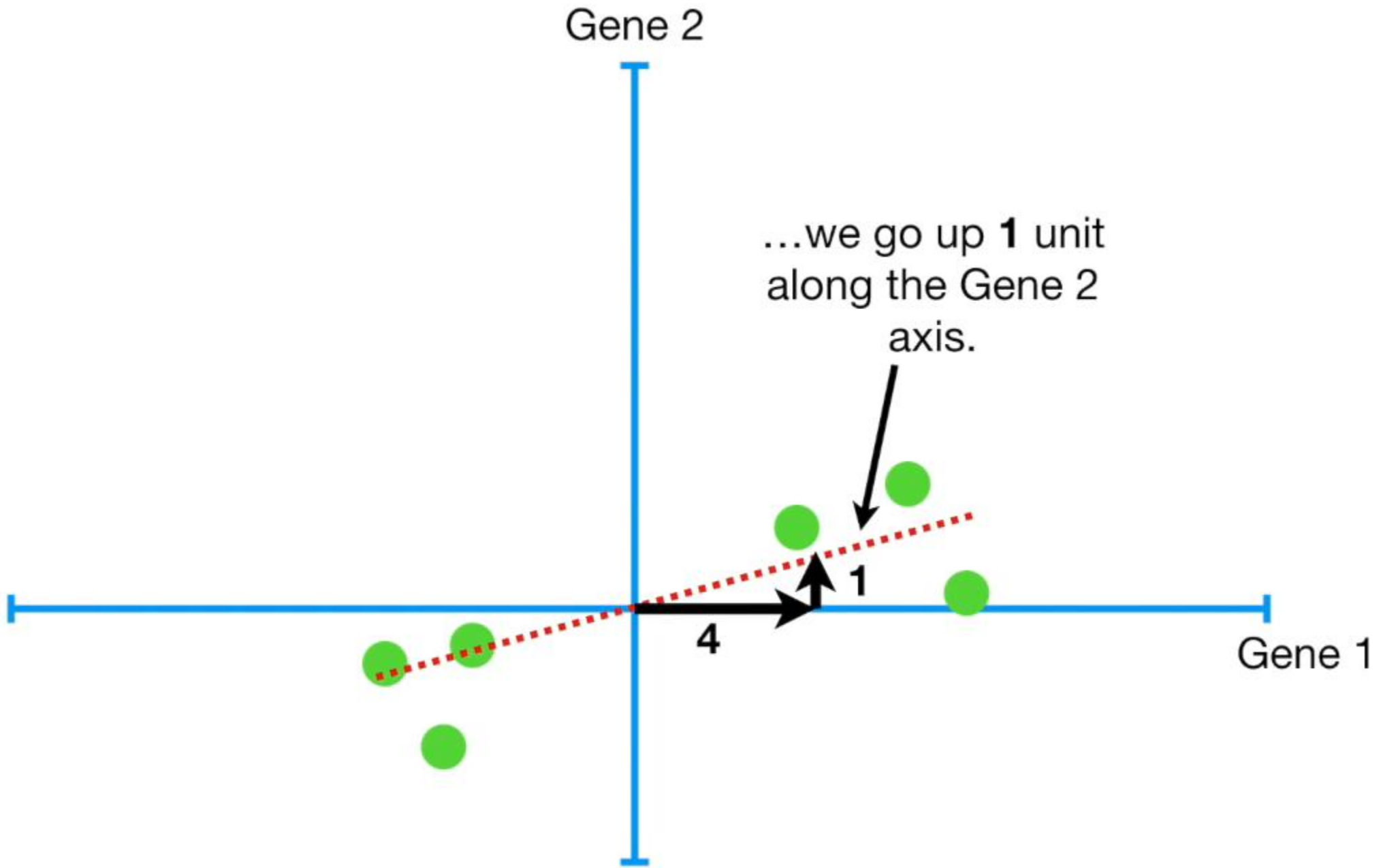
Ultimately, we end up with this line. It has the largest SS(distances).

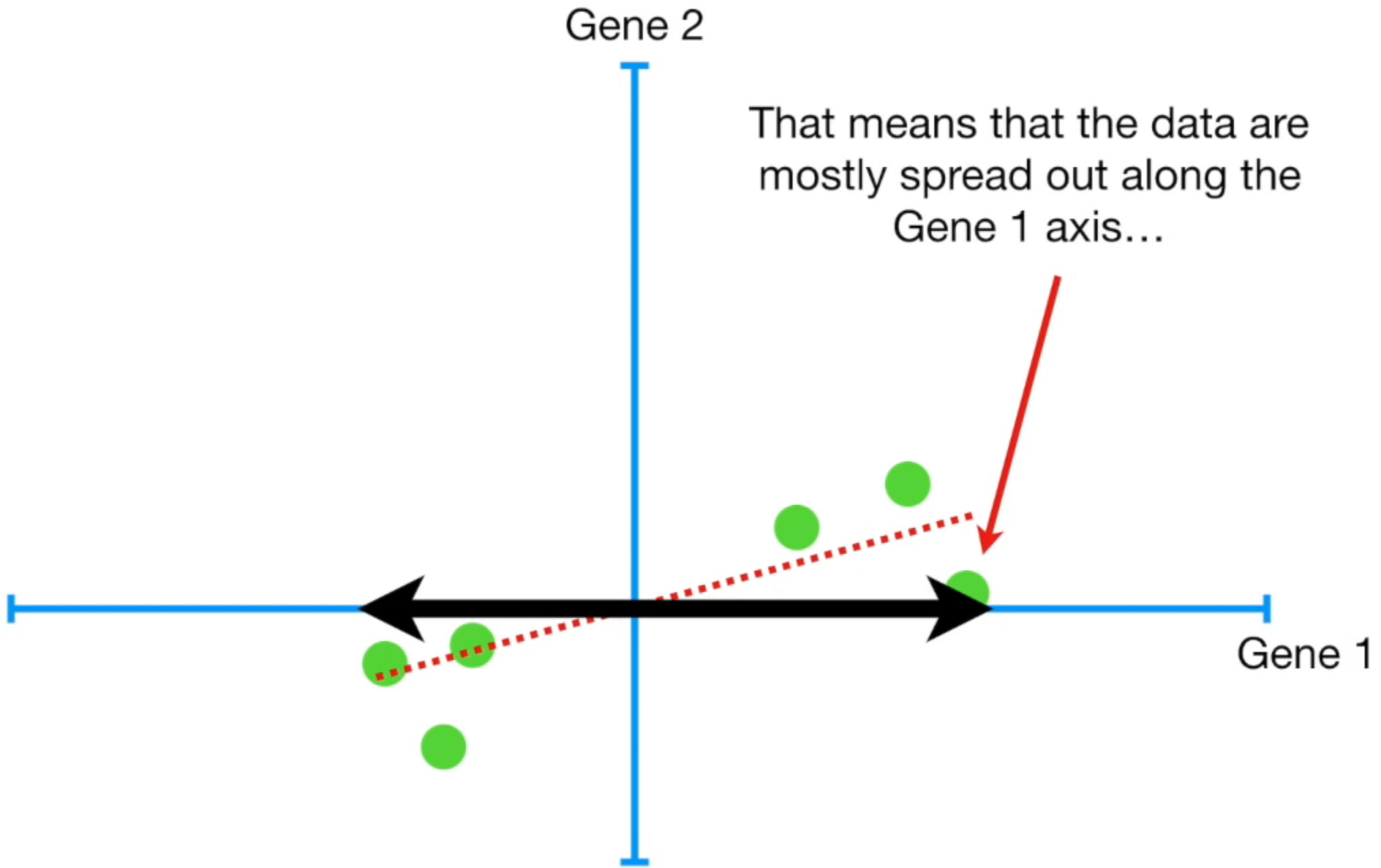


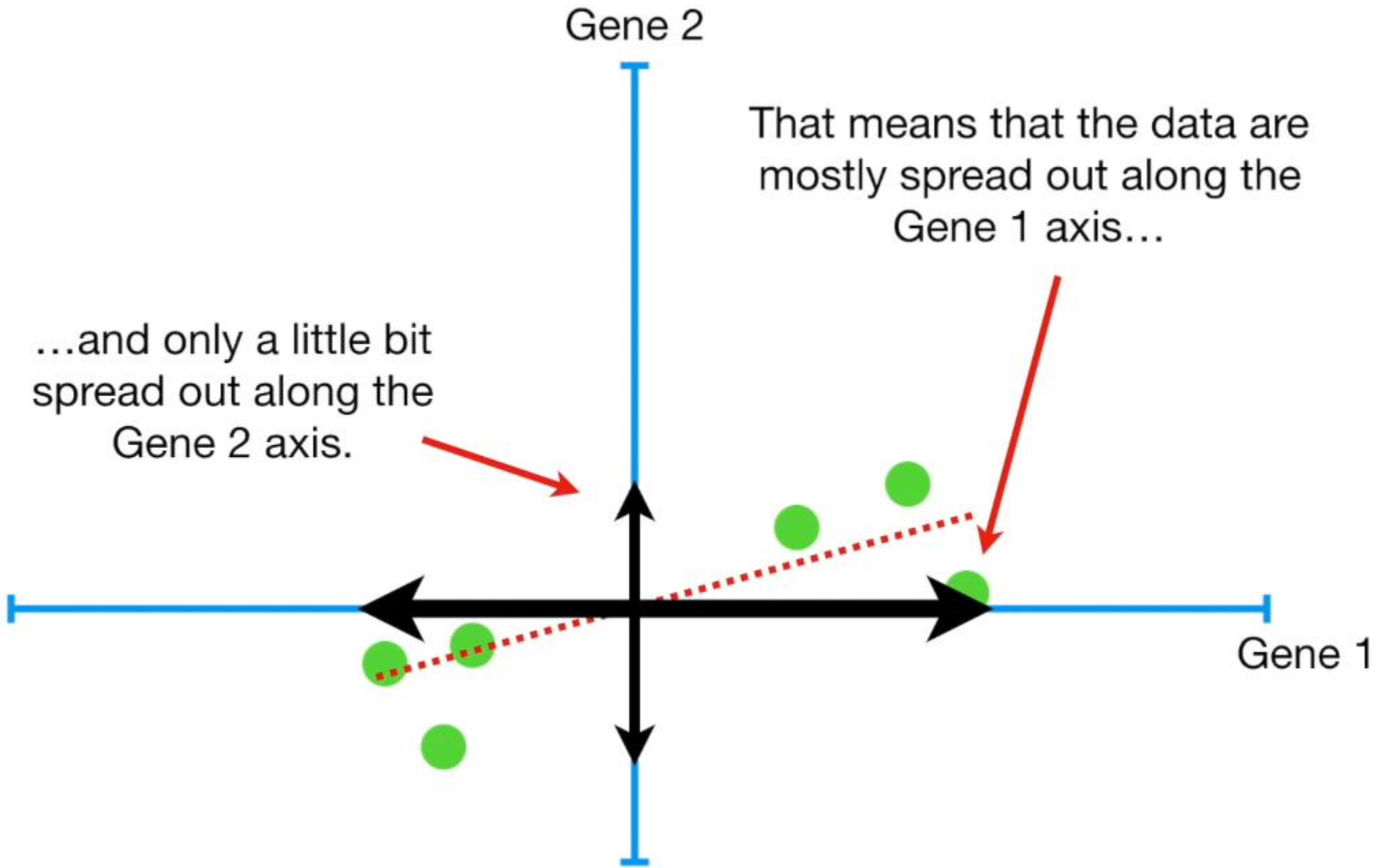

Gene 2
This line is called **Principal Component 1**. (**PC1** for short.)
Gene 1


Gene 2
PC1 has a slope of 0.25
Gene 1

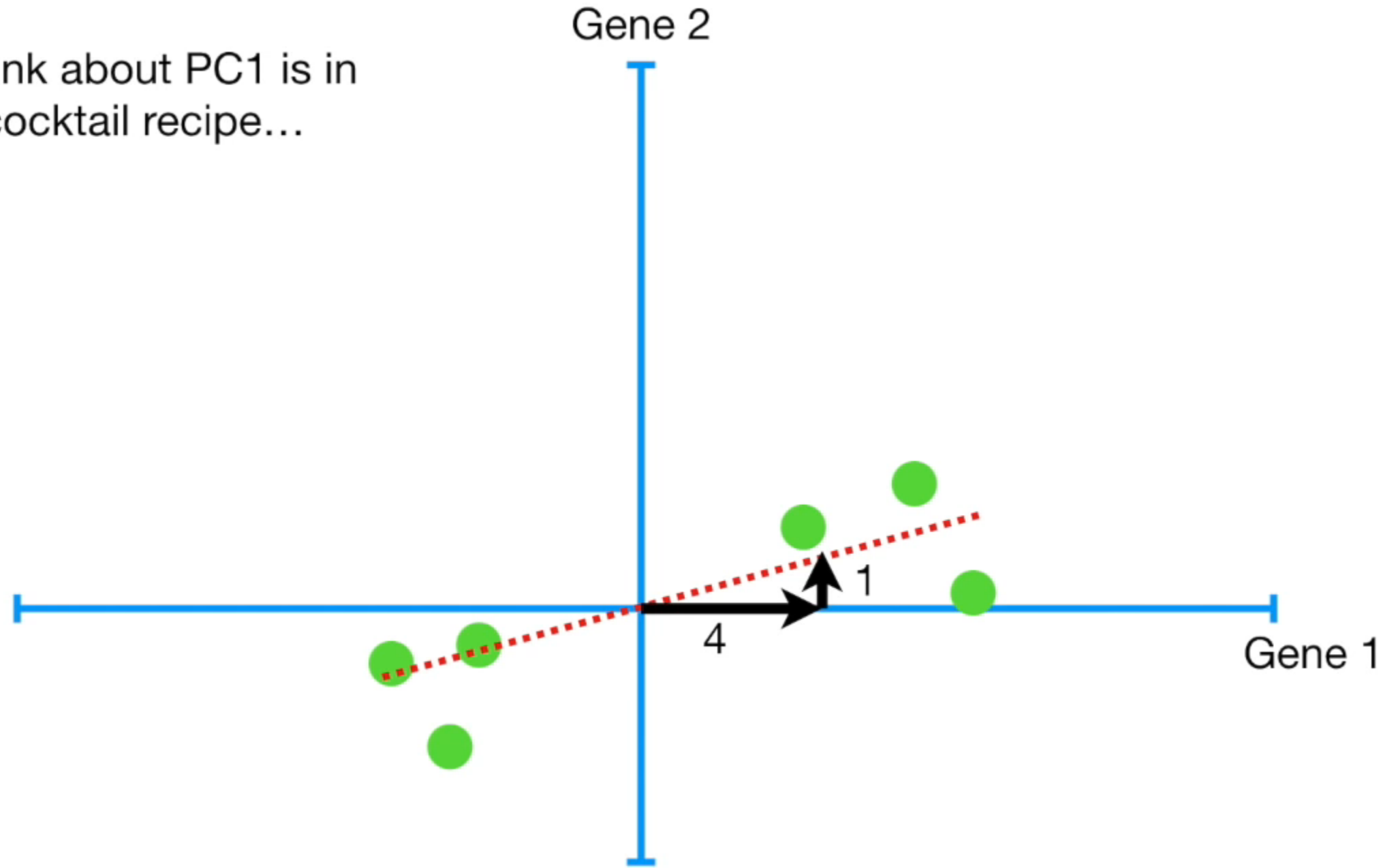

Gene 2
In other words,
for every 4 units
that we go out
along the Gene 1
axis…
4
Gene 1


Gene 2
…we go up **1** unit along the Gene 2 axis.
1
4
Gene 1


Gene 2
That means that the data are mostly spread out along the Gene 1 axis…
Gene 1

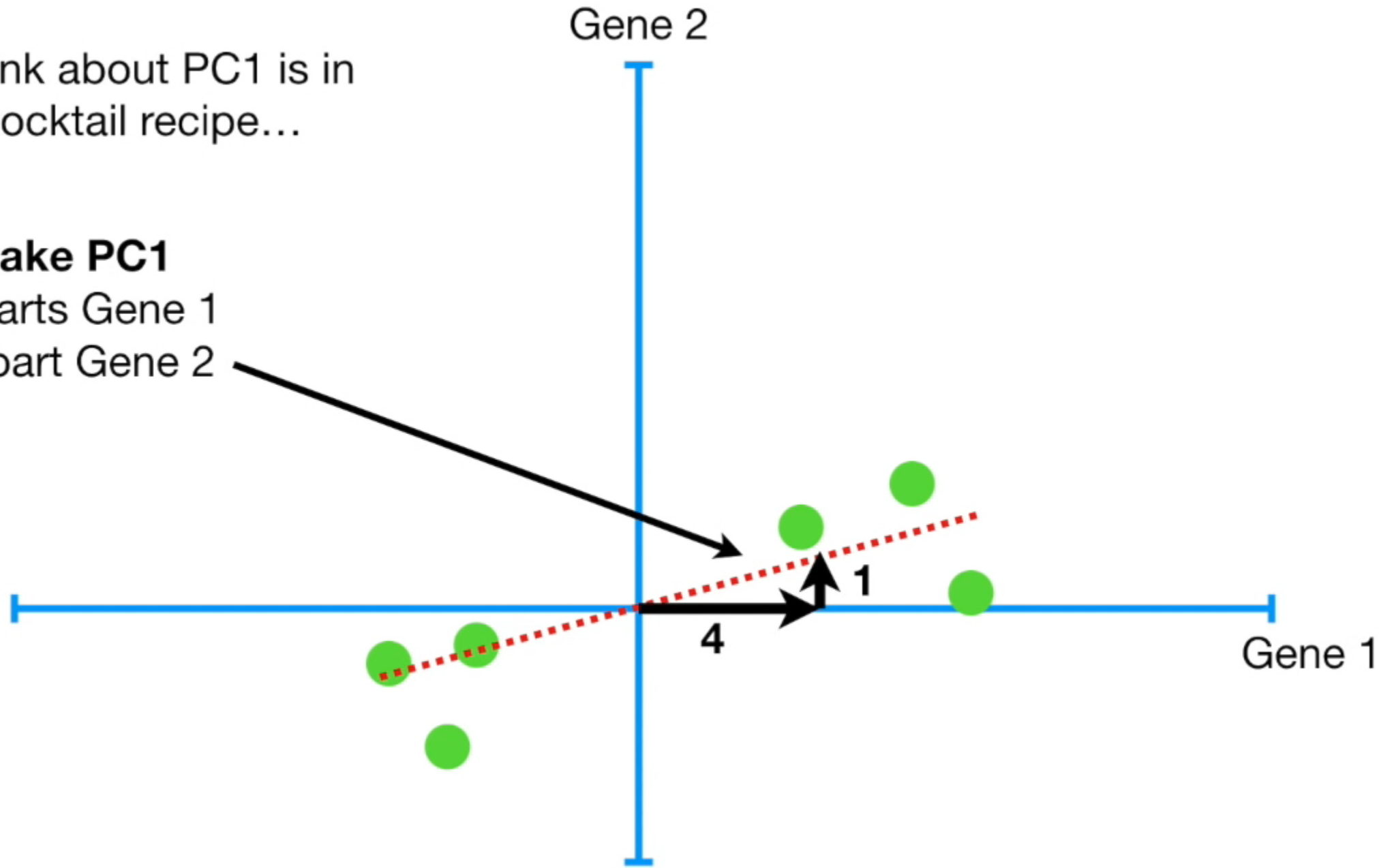

Gene 2
That means that the data are mostly spread out along the Gene 1 axis…
…and only a little bit spread out along the Gene 2 axis.
Gene 1

One way to think about PC1 is in terms of a cocktail recipe…



One way to think about PC1 is in terms of a cocktail recipe…

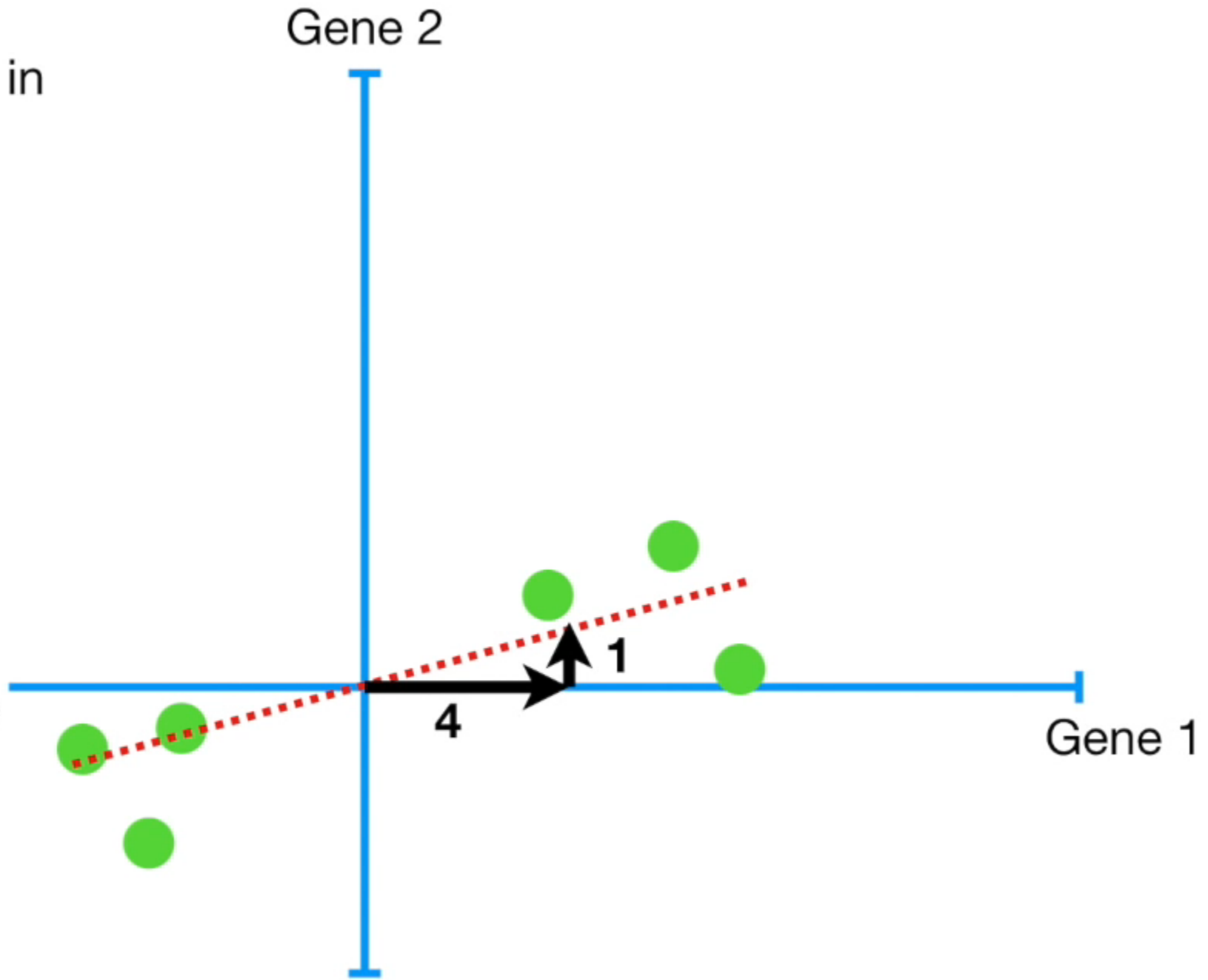
To make PC1
Mix **4** parts Gene 1
with **1** part Gene 2



One way to think about PC1 is in terms of a cocktail recipe…

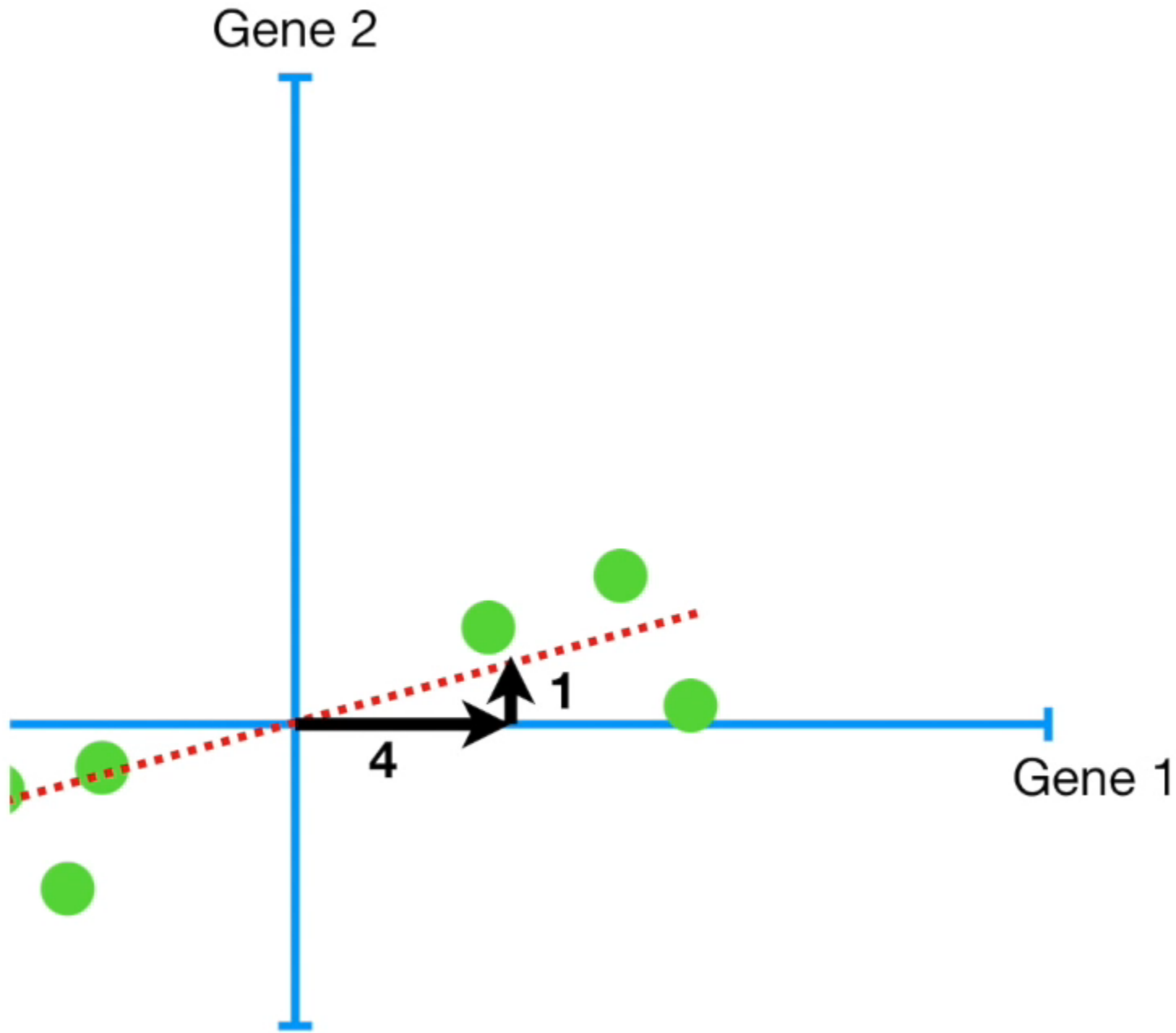
To make PC1
Mix **4** parts Gene 1
with **1** part Gene 2

The ratio of Gene 1 to Gene 2 tells you that Gene 1 is more important when it comes to describing how the data are spread out..



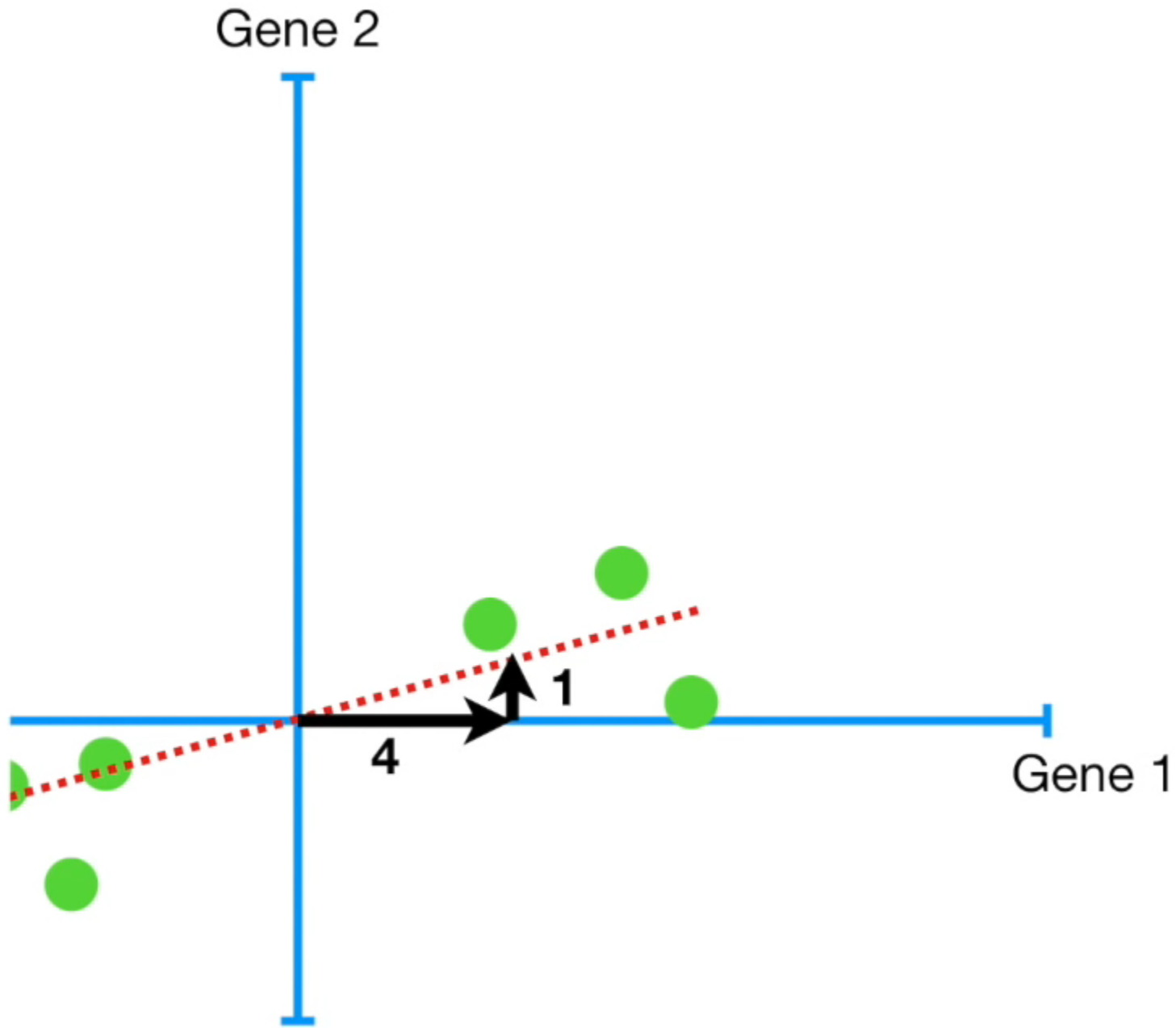
To make PC1
Mix **4** parts Gene 1
with **1** part Gene 2

Terminology Alert!!!!
Mathematicians call this cocktail recipe a "*linear combination*" of Genes 1 and 2.



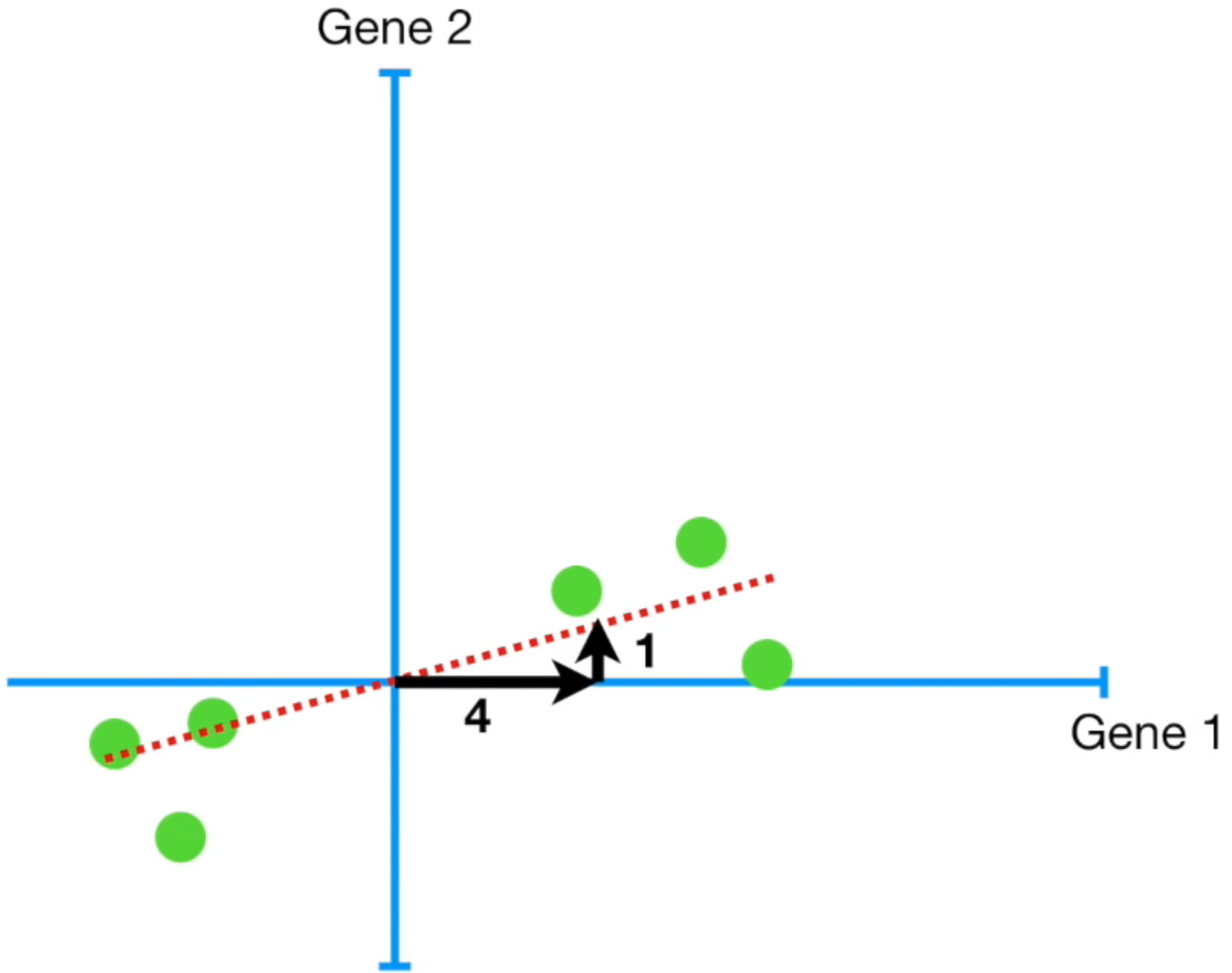
To make PC1
Mix **4** parts Gene 1
with **1** part Gene 2

I mention this because when someone says, “PC1 is a linear combination of variables…”

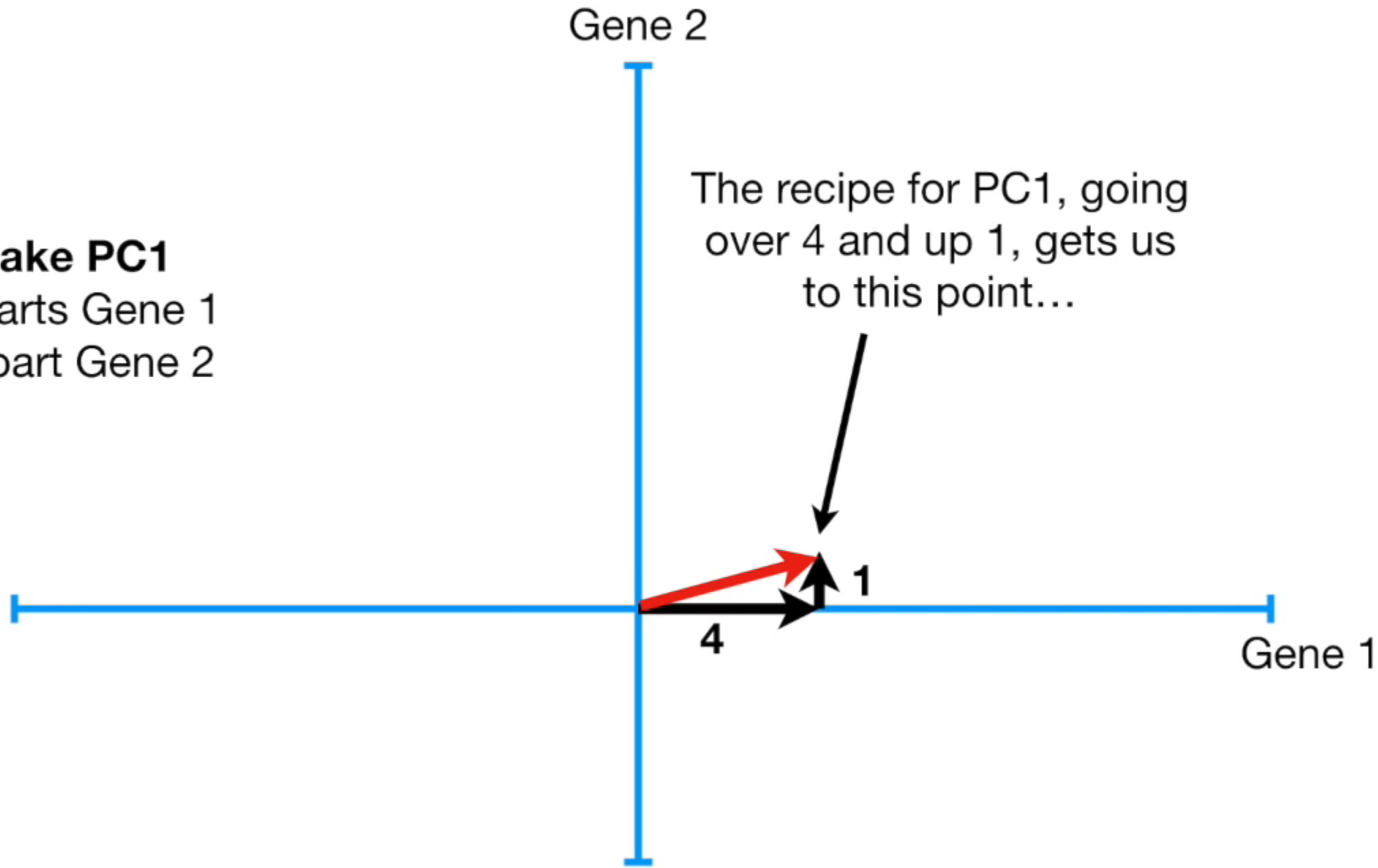


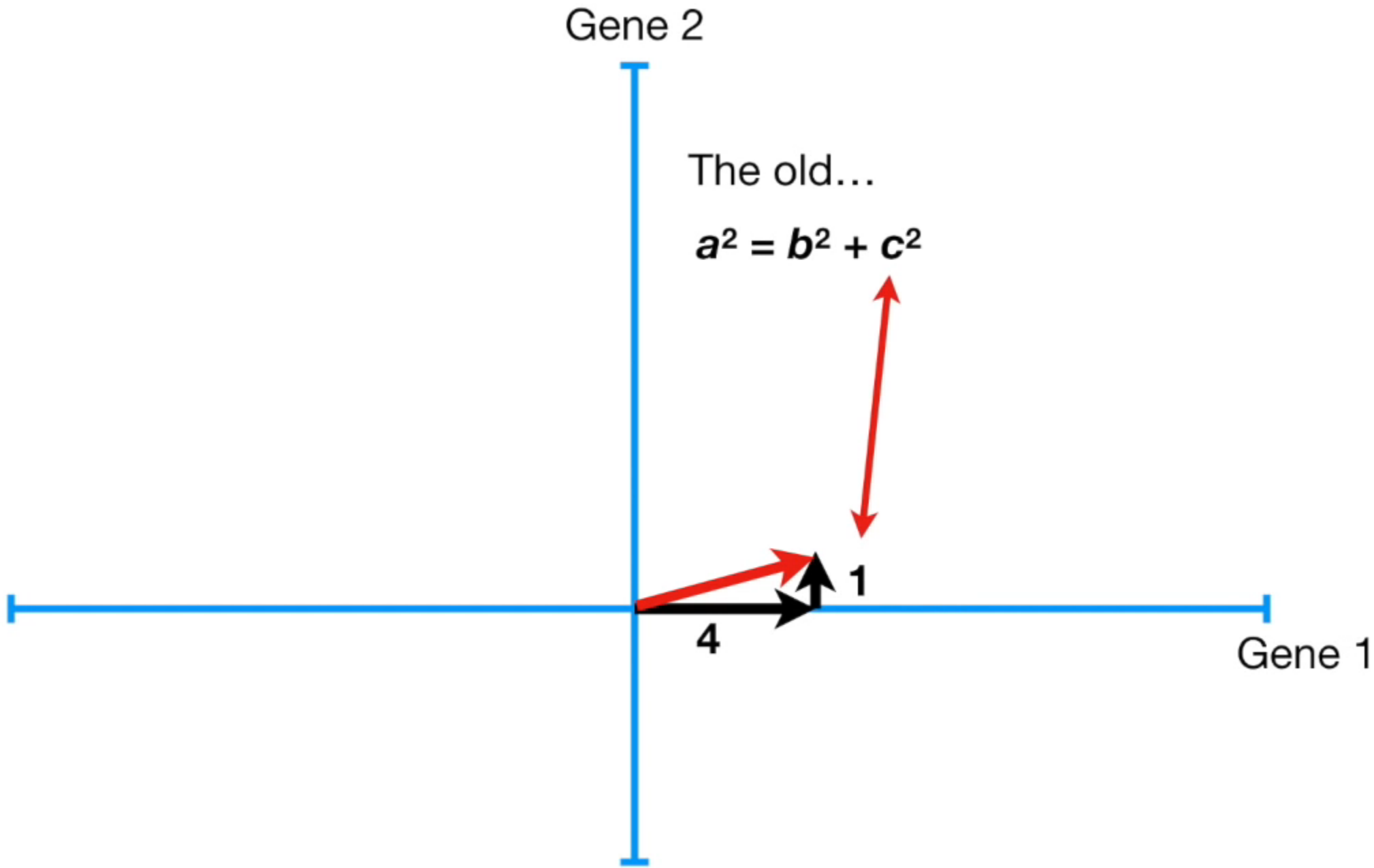
To make PC1
Mix **4** parts Gene 1
with **1** part Gene 2

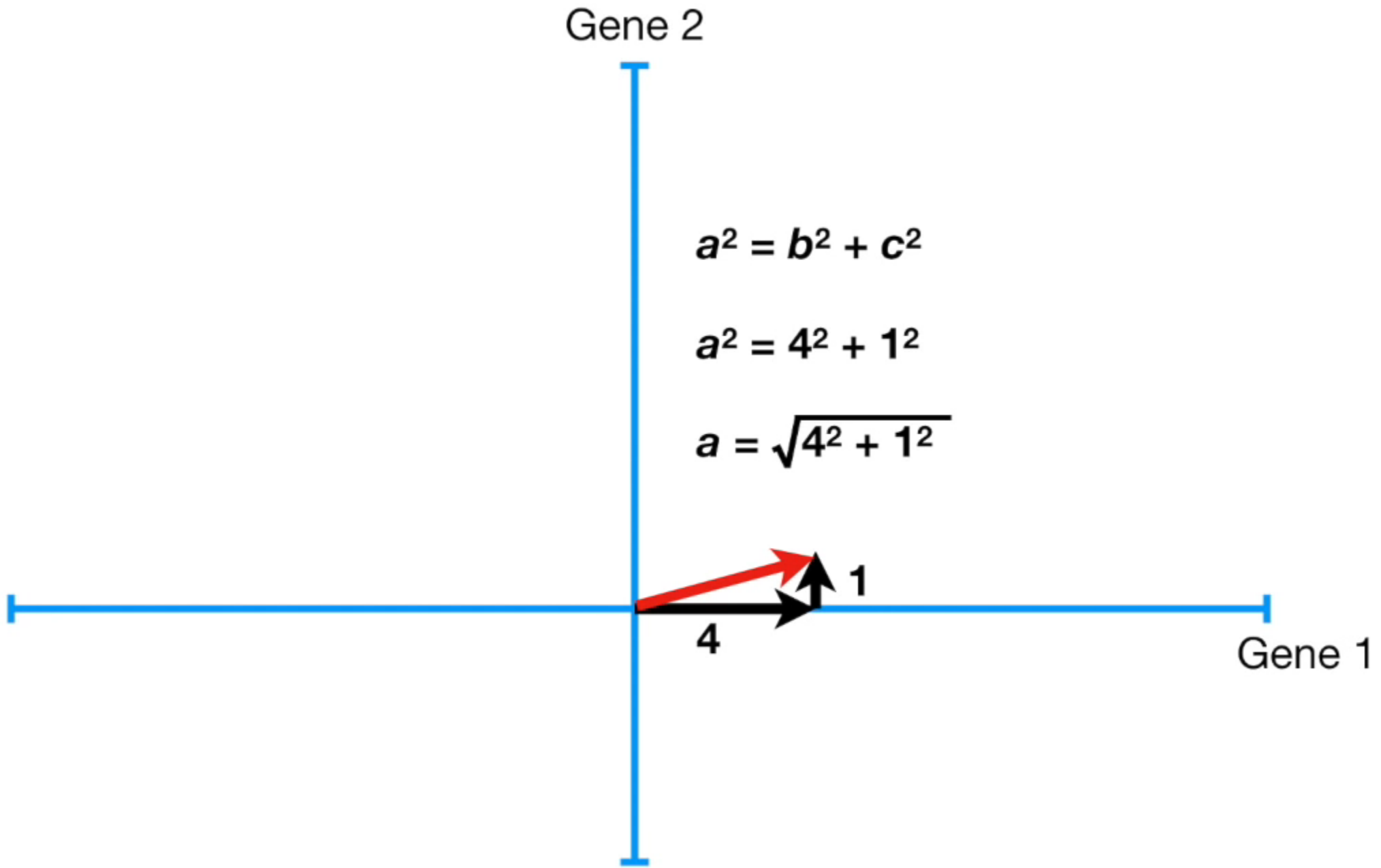
…this is what they are talking about.

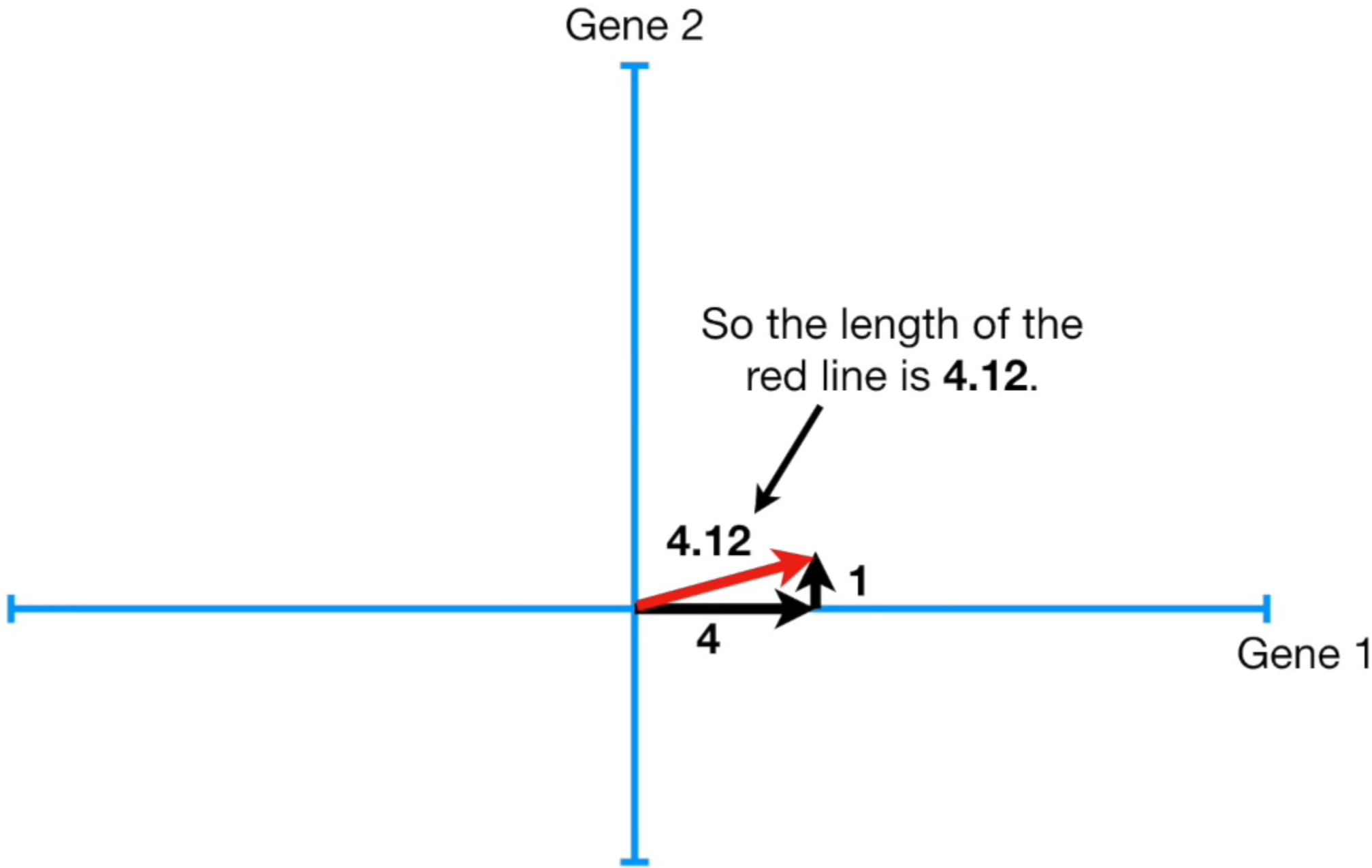


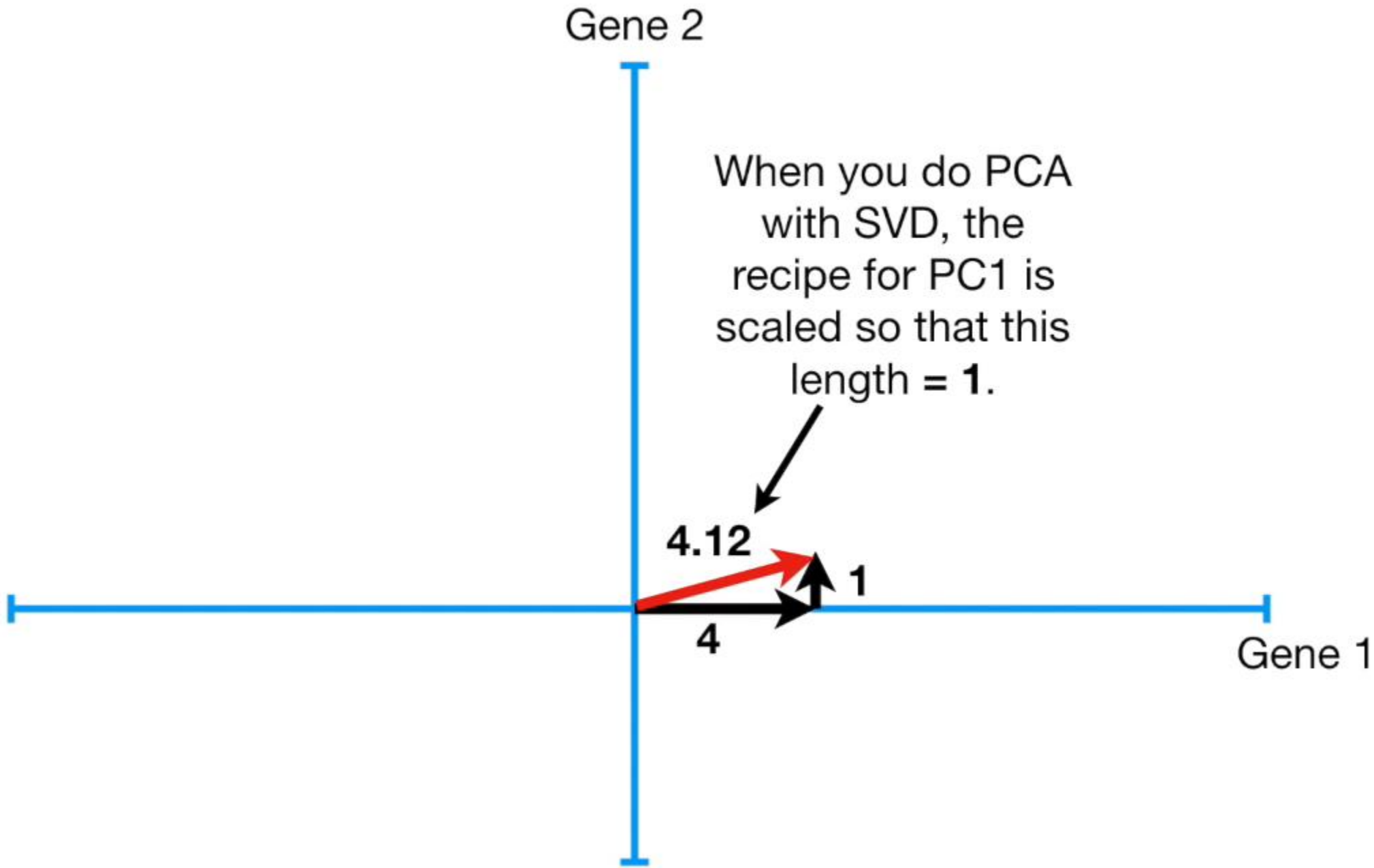
To make PC1
Mix **4** parts Gene 1
with **1** part Gene 2

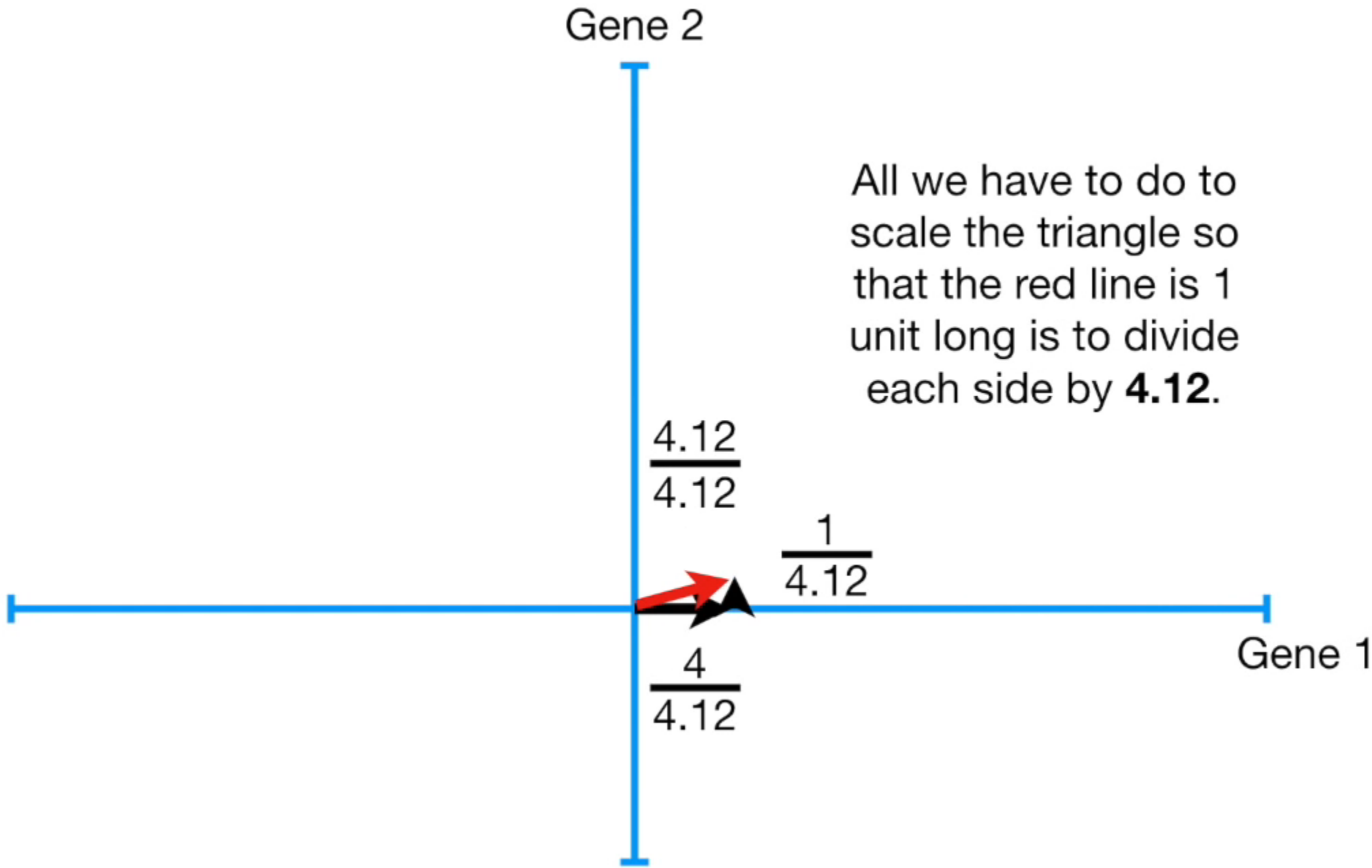


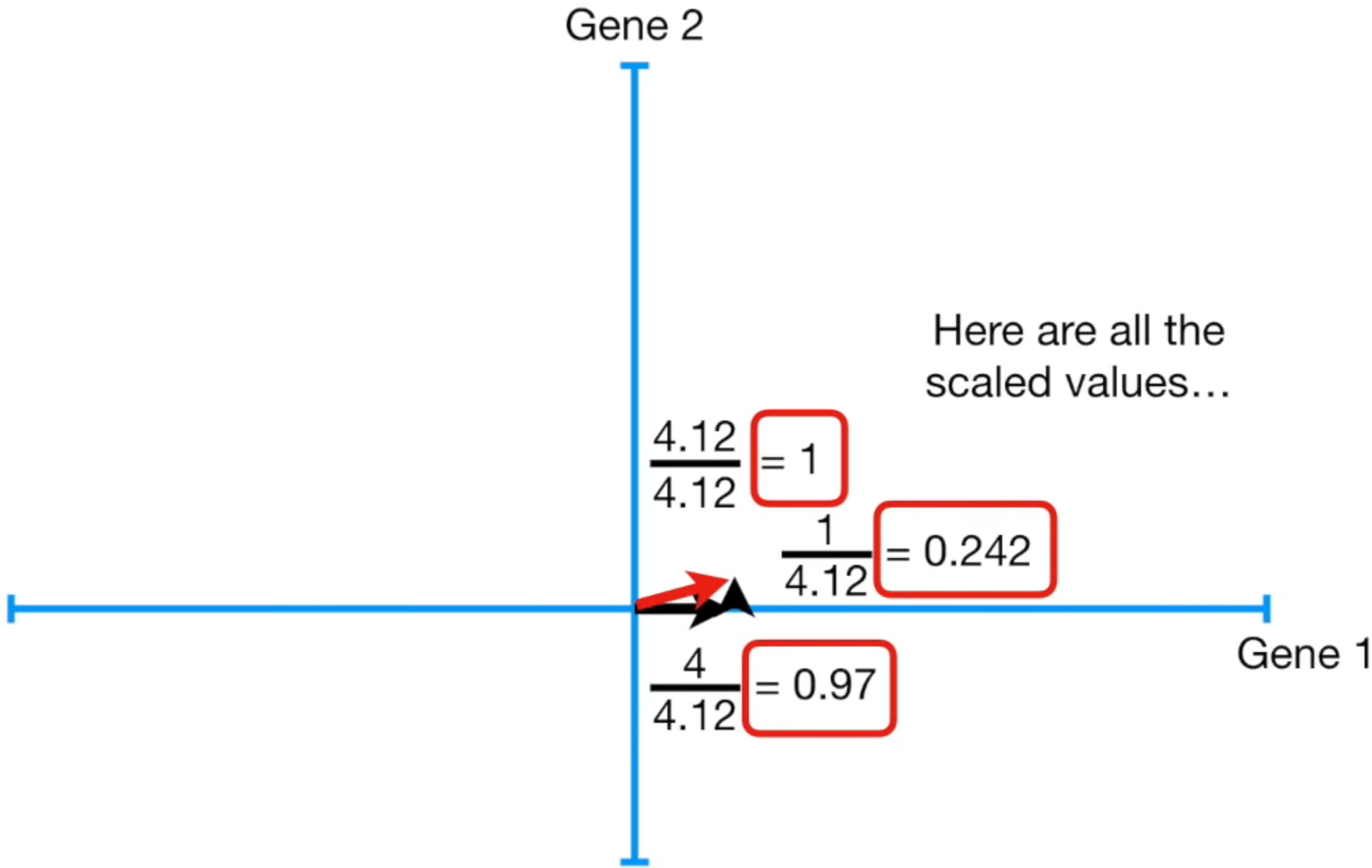

Gene 2
The old…
$a^2 = b^2 + c^2$
1
4
Gene 1


Gene 2
$a^2 = b^2 + c^2$
$a^2 = 4^2 + 1^2$
$a = \sqrt{4^2 + 1^2}$
1
4
Gene 1


Gene 2
So the length of the red line is **4.12**.
4.12
1
4
Gene 1

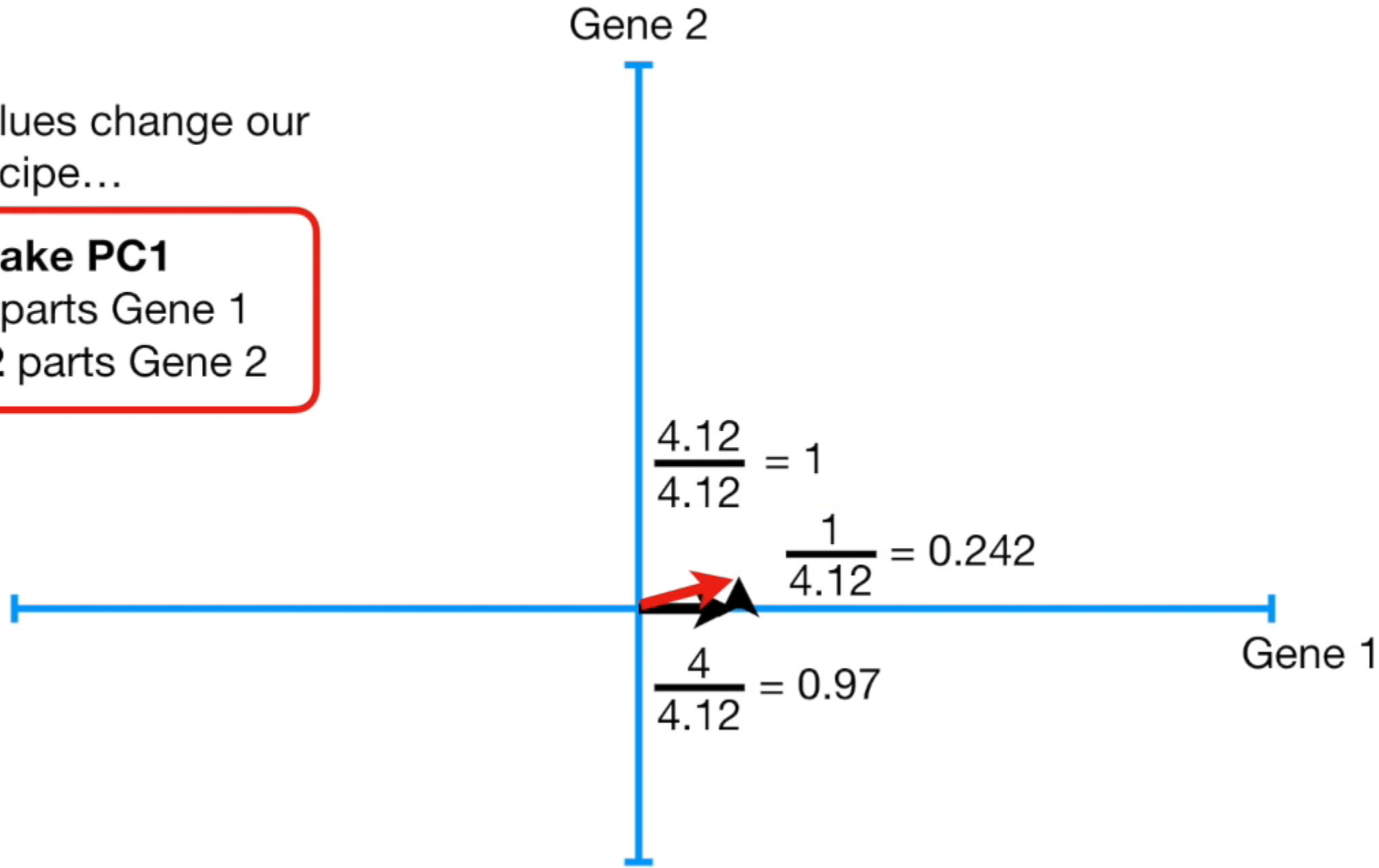

Gene 2
When you do PCA with SVD, the recipe for PC1 is scaled so that this length = 1.
4.12
1
4
Gene 1


Gene 2
All we have to do to scale the triangle so that the red line is 1 unit long is to divide each side by **4.12**.
4.12
4.12
1
4.12
4
4.12
Gene 1


Gene 2
Here are all the scaled values…
4.12 / 4.12 = 1
1 / 4.12 = 0.242
4 / 4.12 = 0.97
Gene 1

The new values change our recipe…

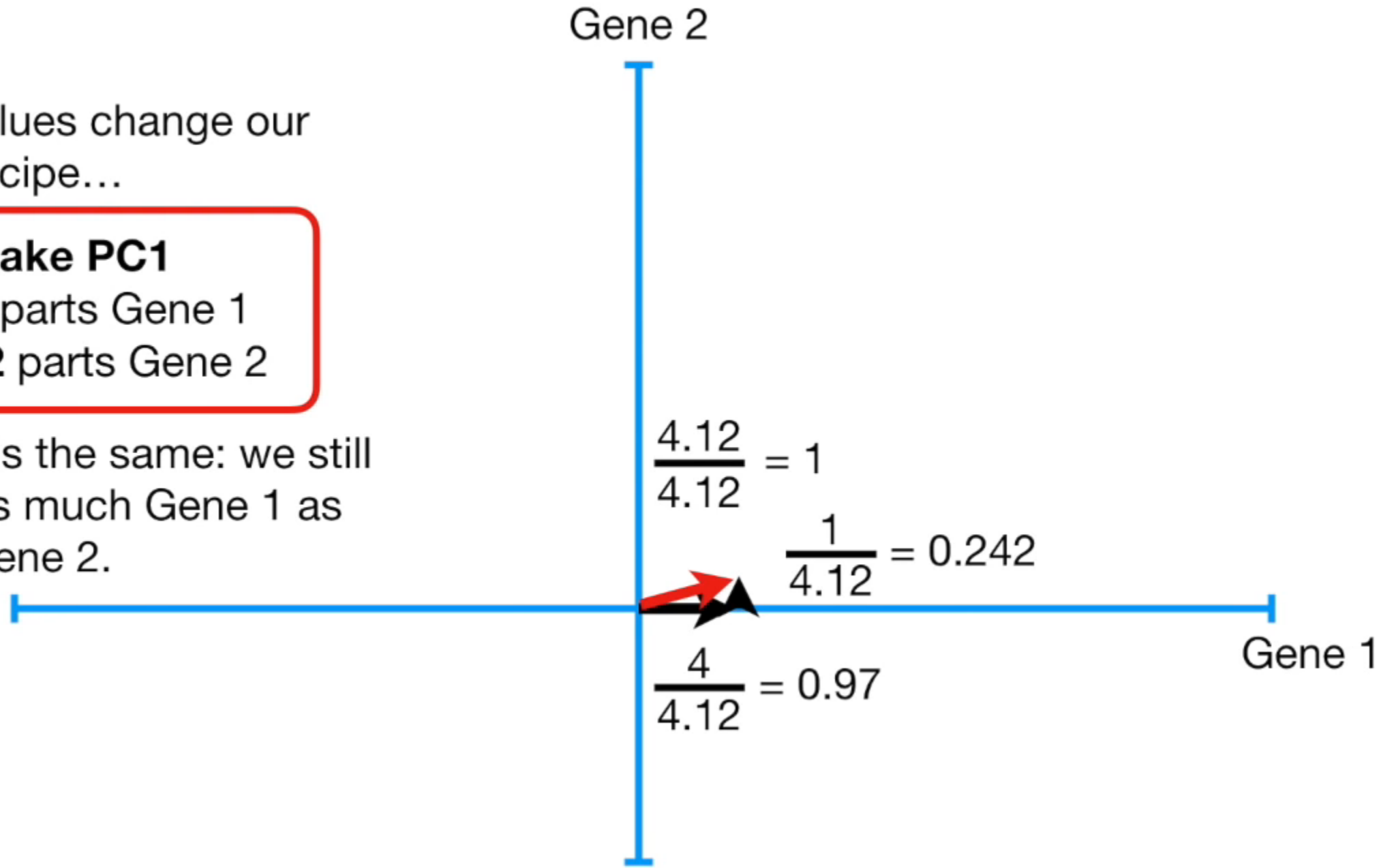
To make PC1
Mix **0.97** parts Gene 1
with **0.242** parts Gene 2



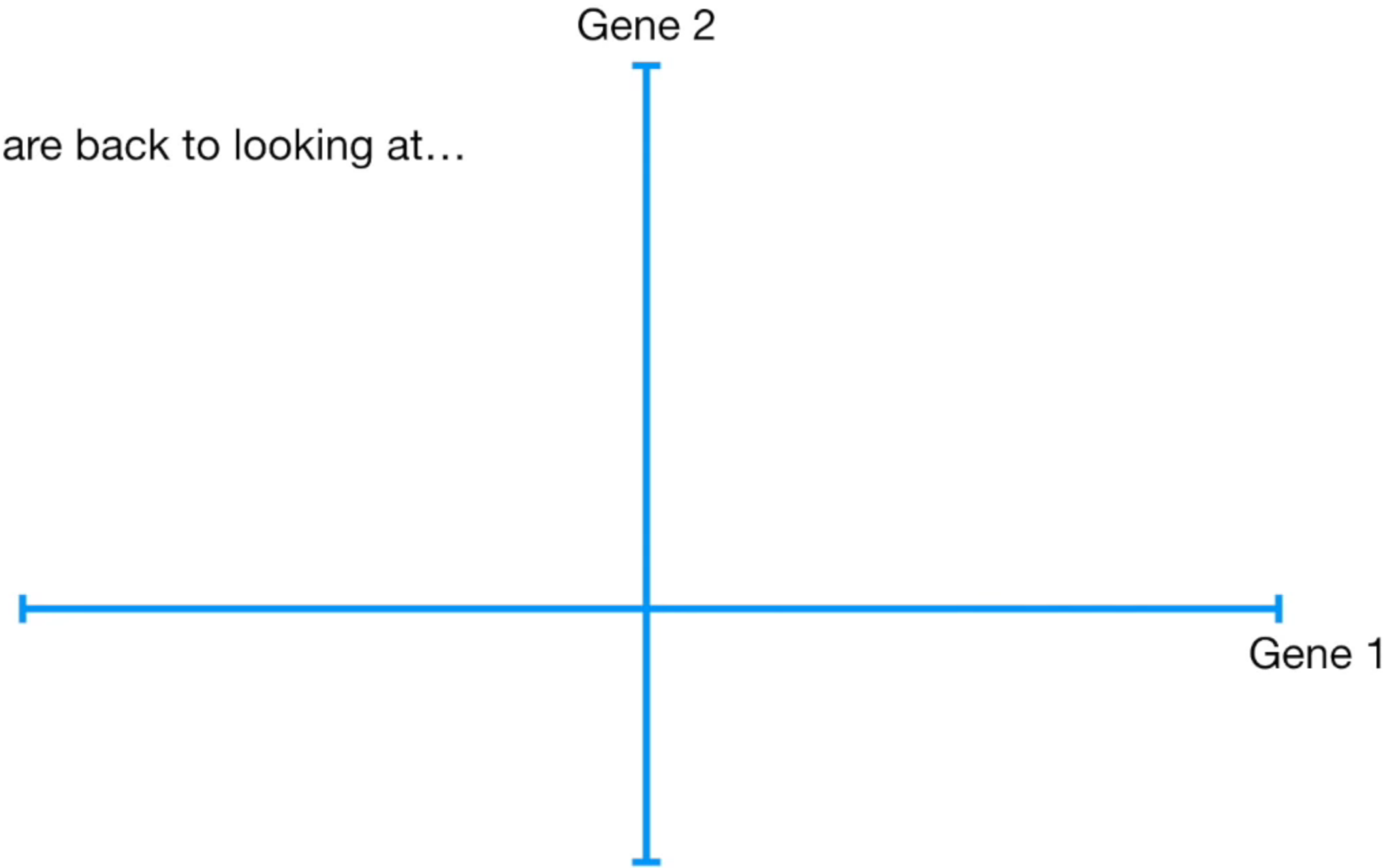
The new values change our recipe…

To make PC1
Mix **0.97** parts Gene 1
with **0.242** parts Gene 2

…but the ratio is the same: we still use 4 times as much Gene 1 as Gene 2.

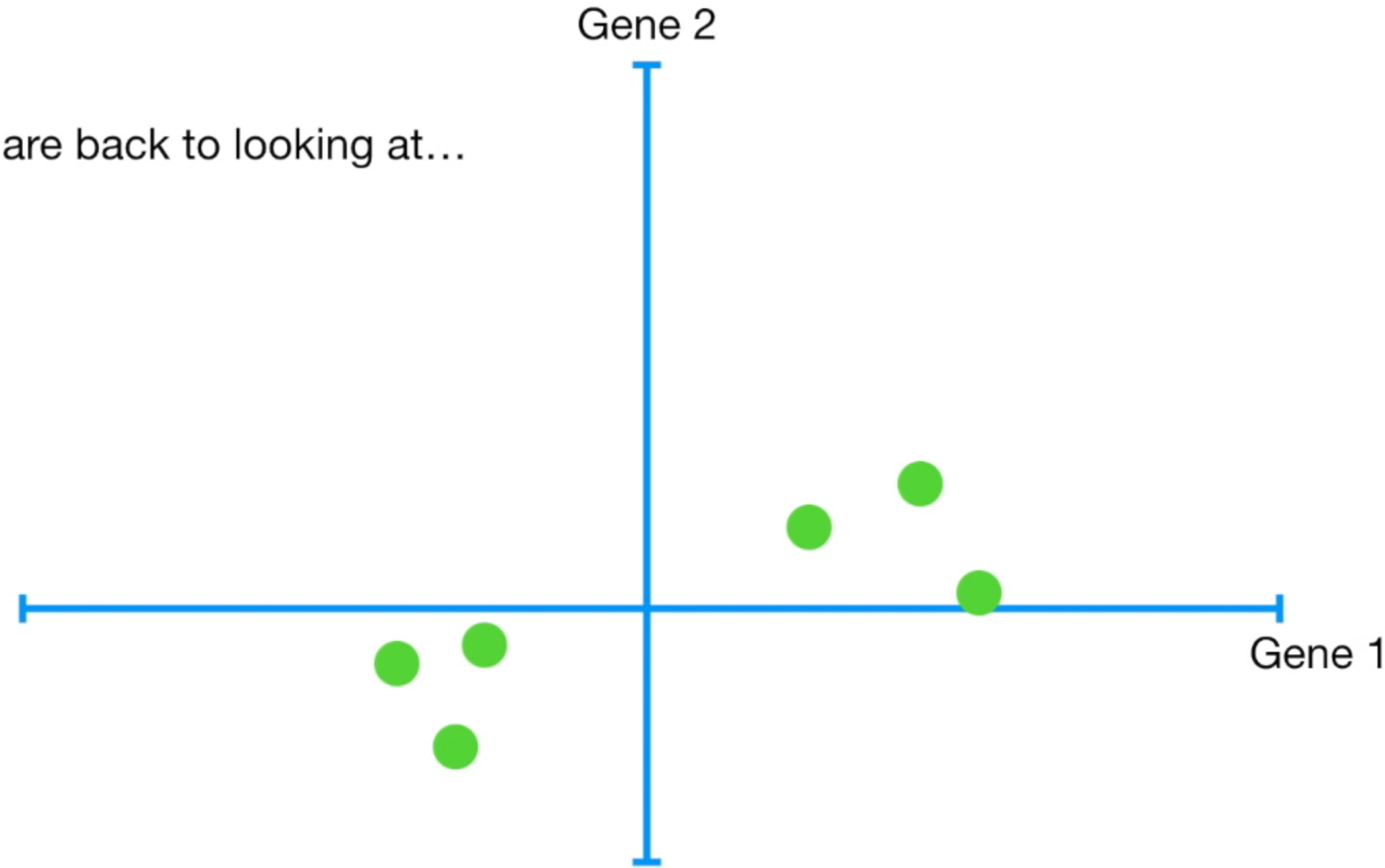


So now we are back to looking at…



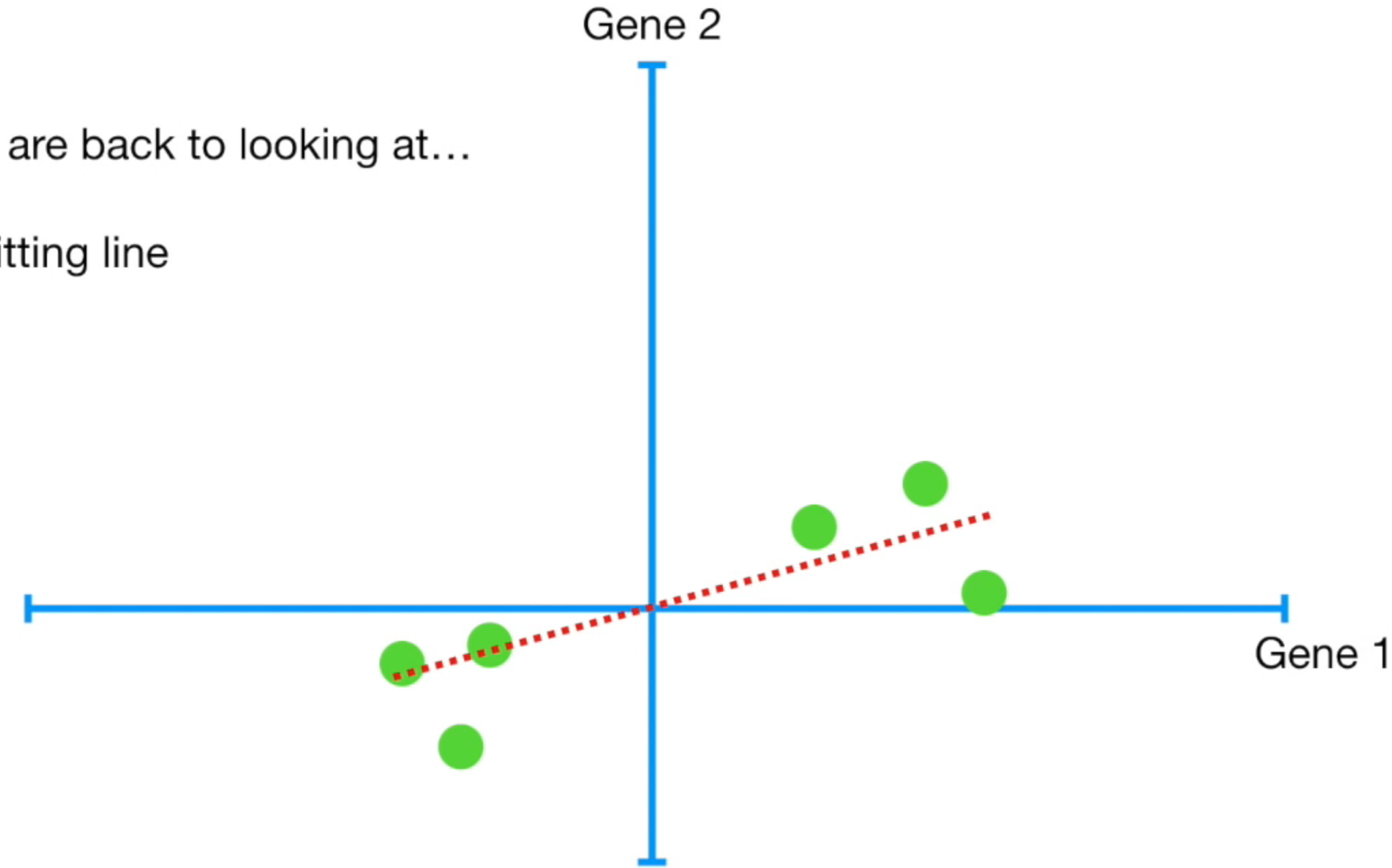
So now we are back to looking at…

- The data



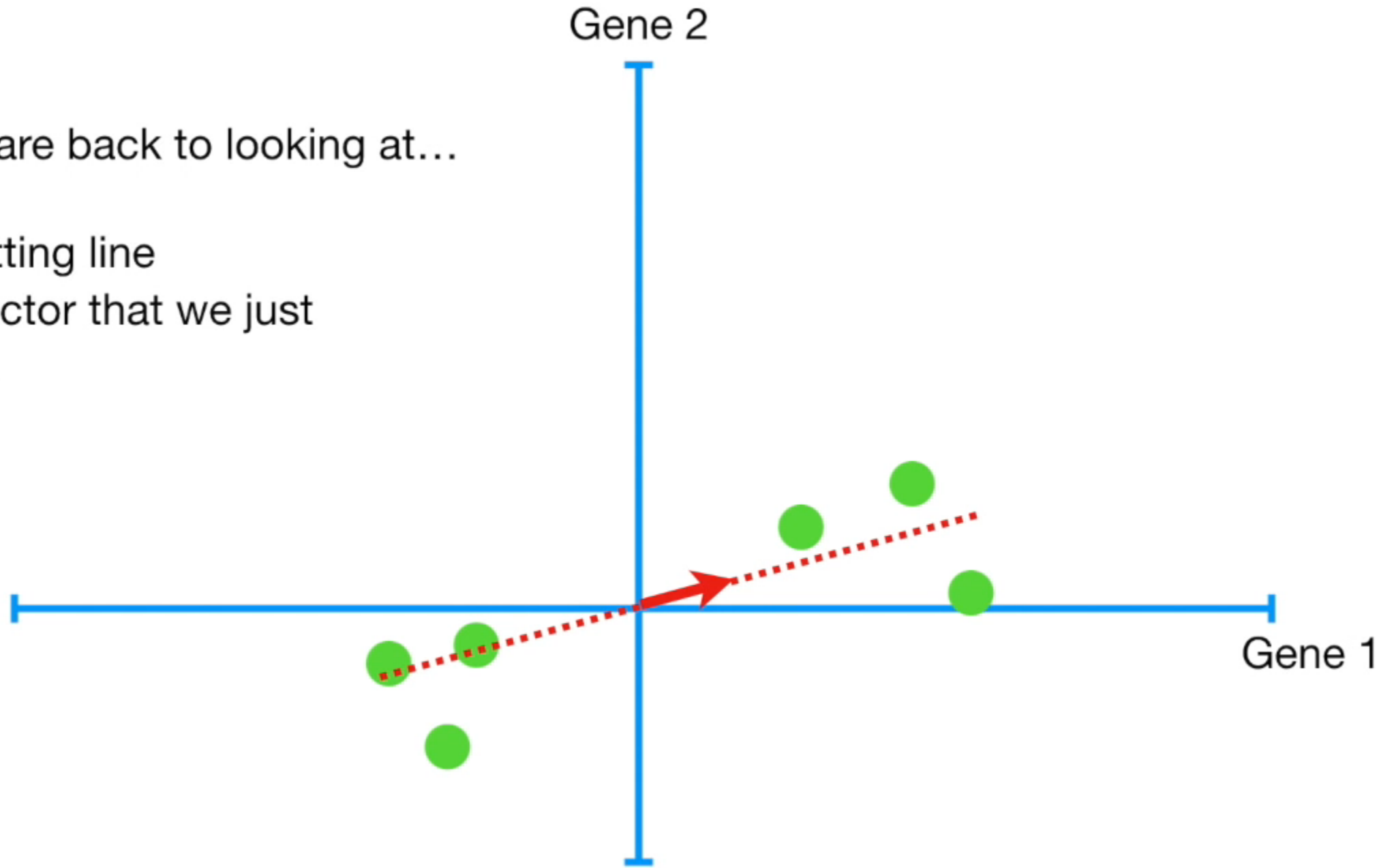
So now we are back to looking at…

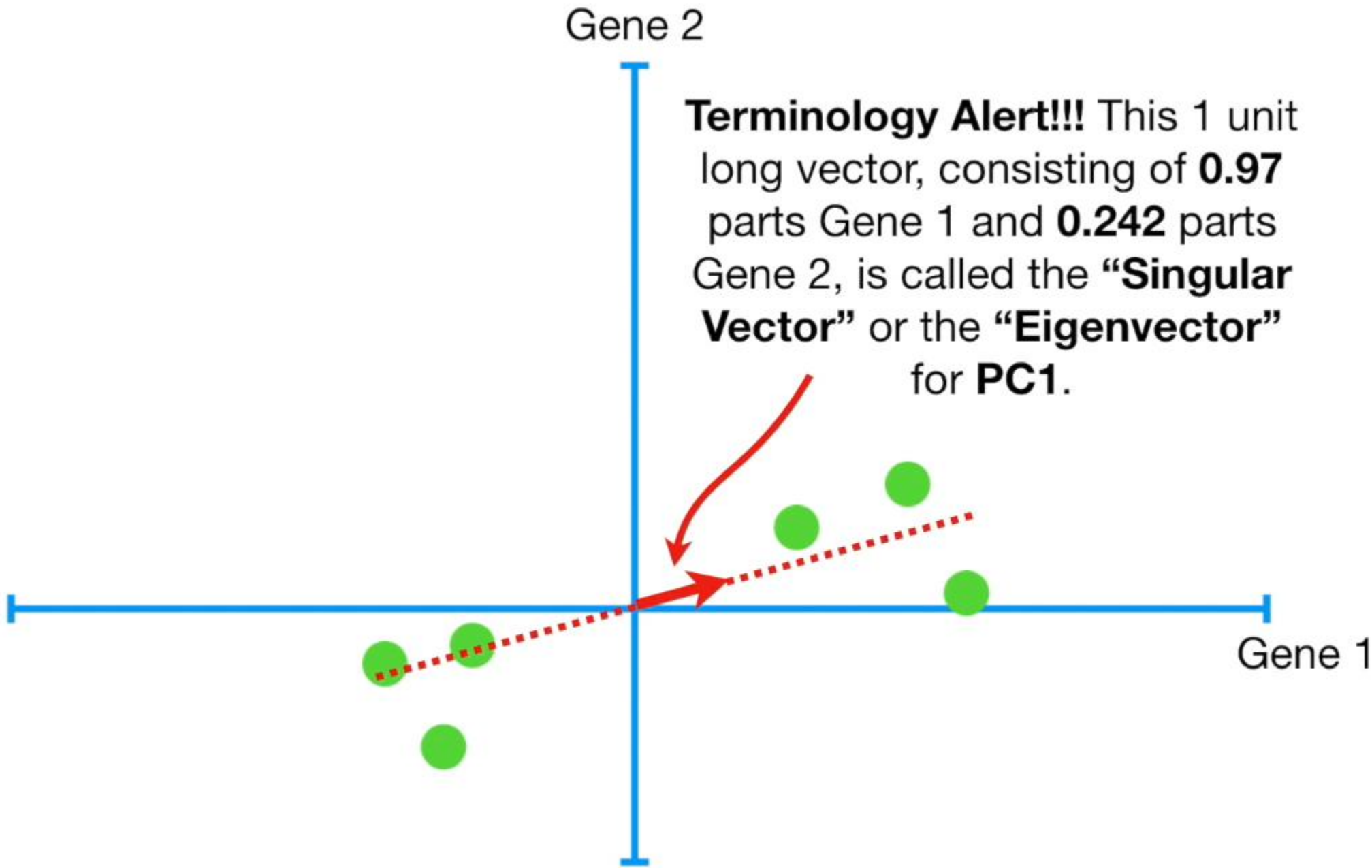
- The data
- The best fitting line

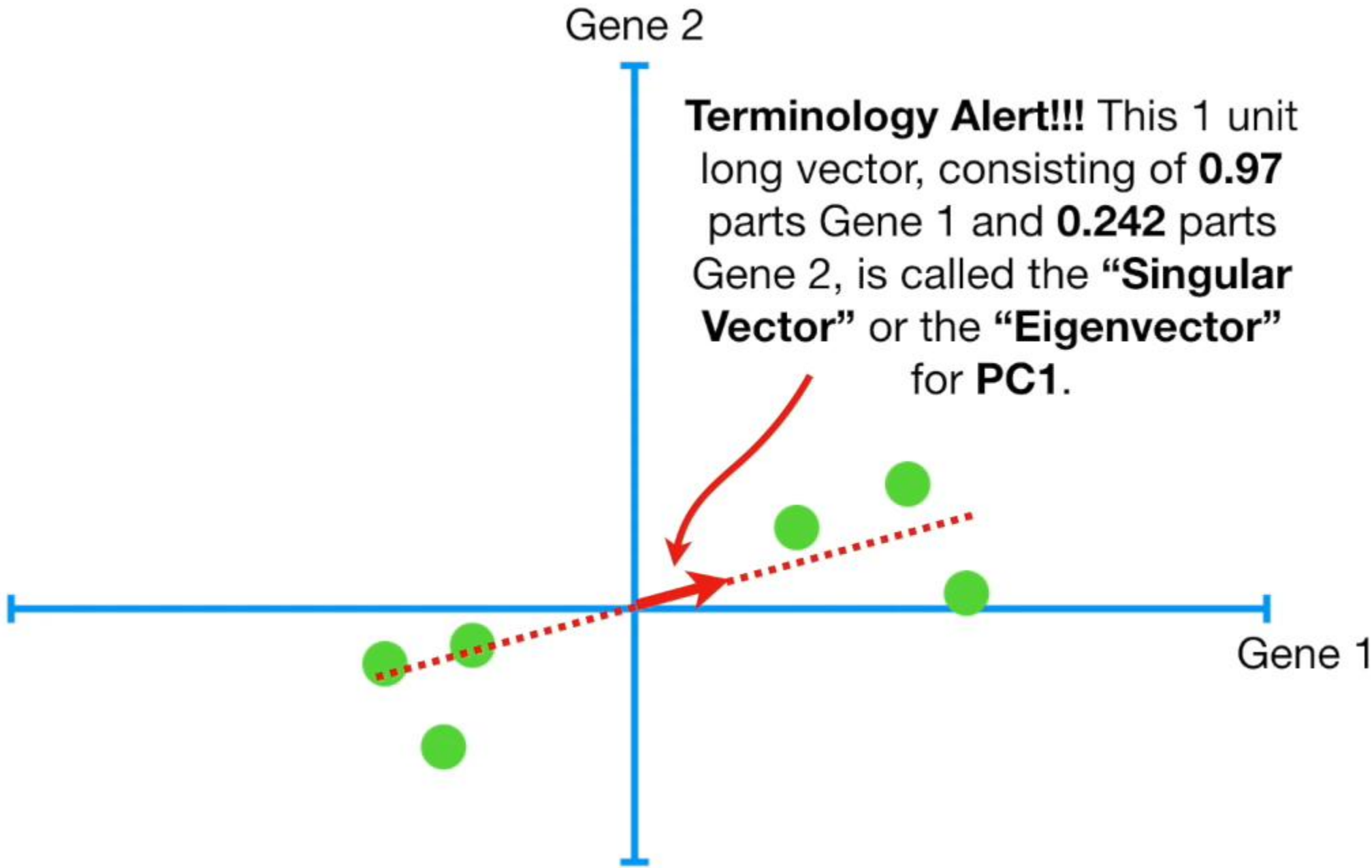


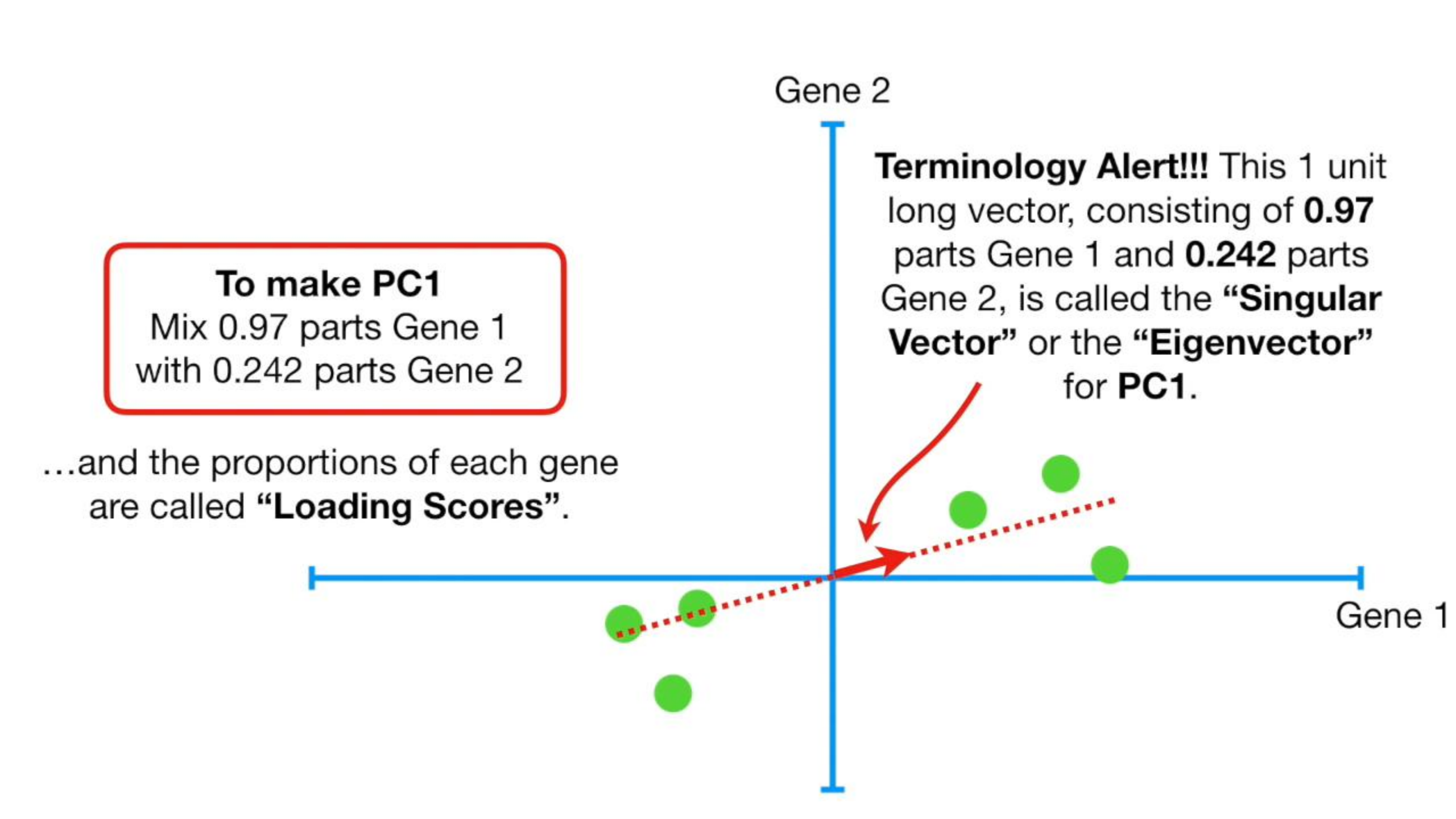
So now we are back to looking at…

- The data
- The best fitting line
- The unit vector that we just calculated.

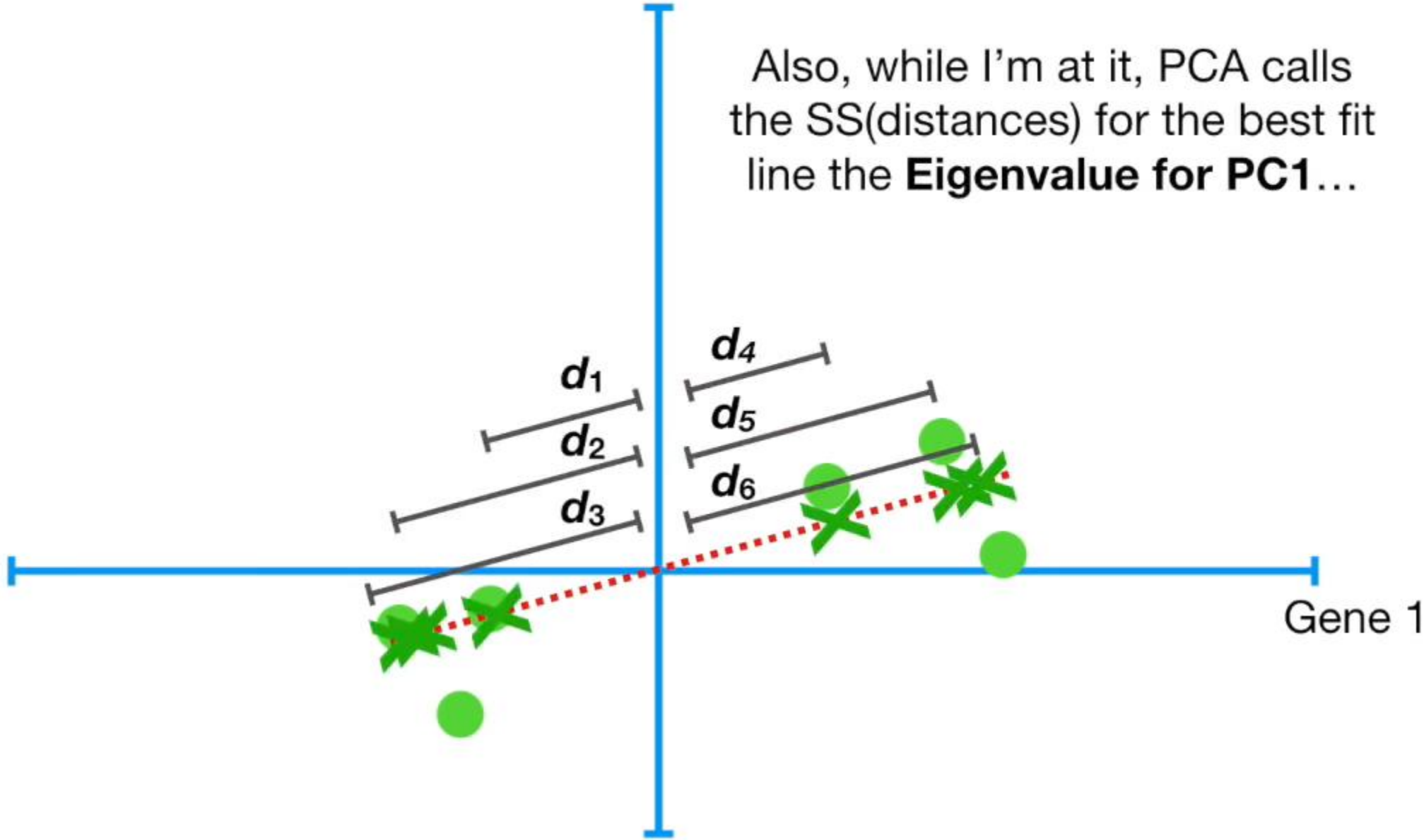



Gene 2
Terminology Alert!!! This 1 unit long vector, consisting of **0.97** parts Gene 1 and **0.242** parts Gene 2, is called the **"Singular Vector"** or the **"Eigenvector"** for **PC1**.
Gene 1

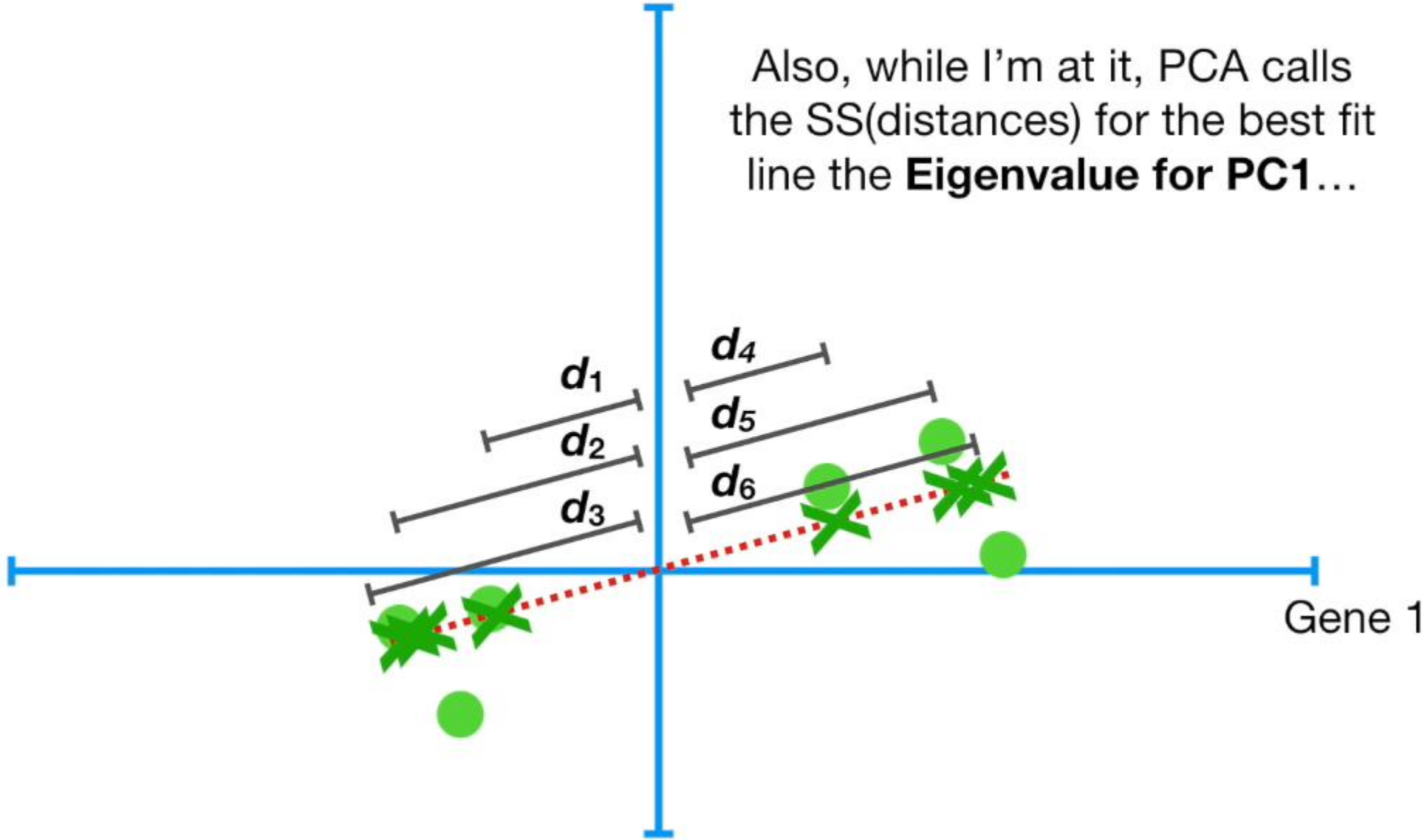

Gene 2
Terminology Alert!!! This 1 unit long vector, consisting of **0.97** parts Gene 1 and **0.242** parts Gene 2, is called the **"Singular Vector"** or the **"Eigenvector"** for **PC1**.
Gene 1


Gene 2
Terminology Alert!!! This 1 unit long vector, consisting of 0.97 parts Gene 1 and 0.242 parts Gene 2, is called the "Singular Vector" or the "Eigenvector" for PC1.
To make PC1
Mix 0.97 parts Gene 1 with 0.242 parts Gene 2
…and the proportions of each gene are called "Loading Scores".
Gene 1

$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)



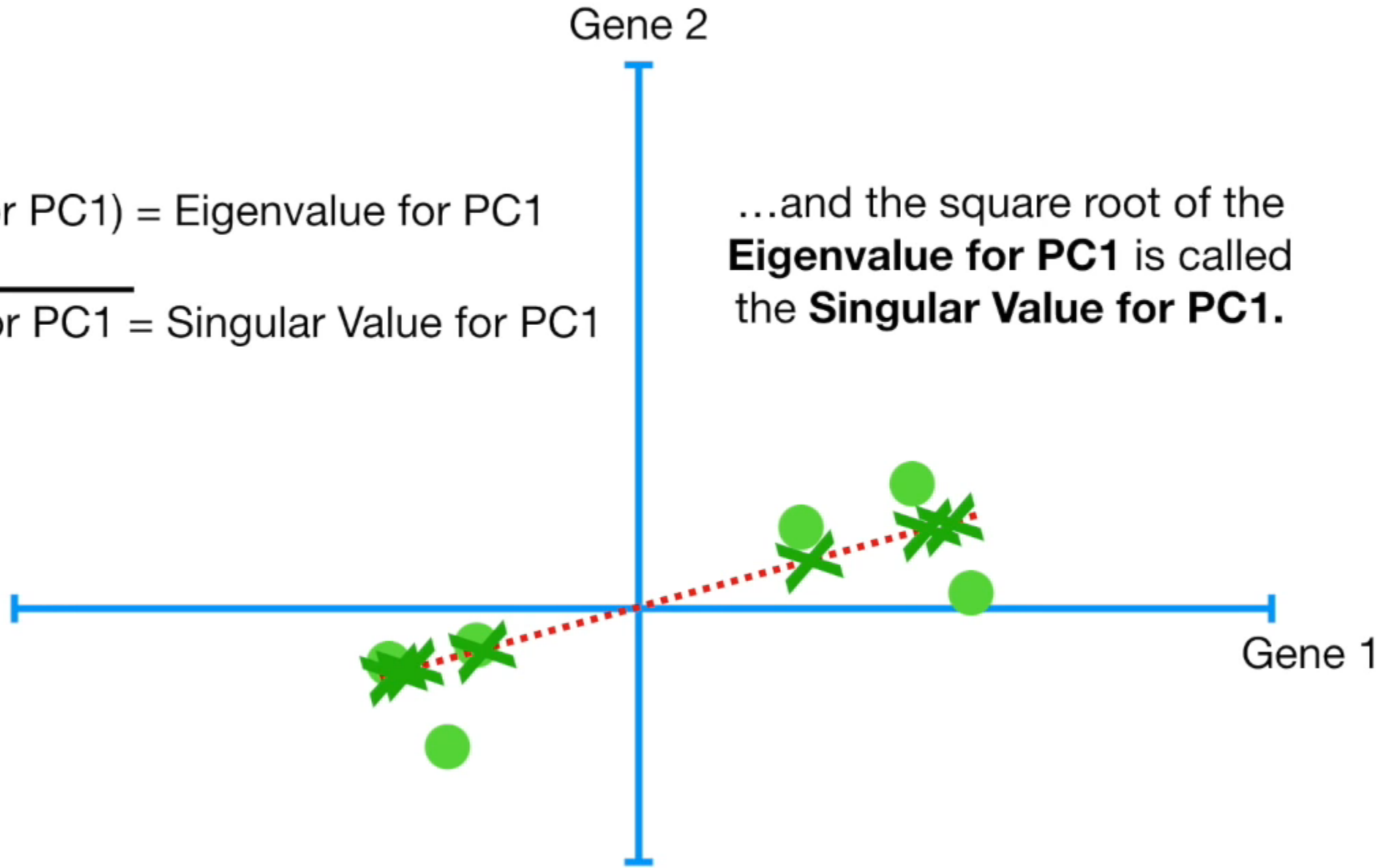
$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)



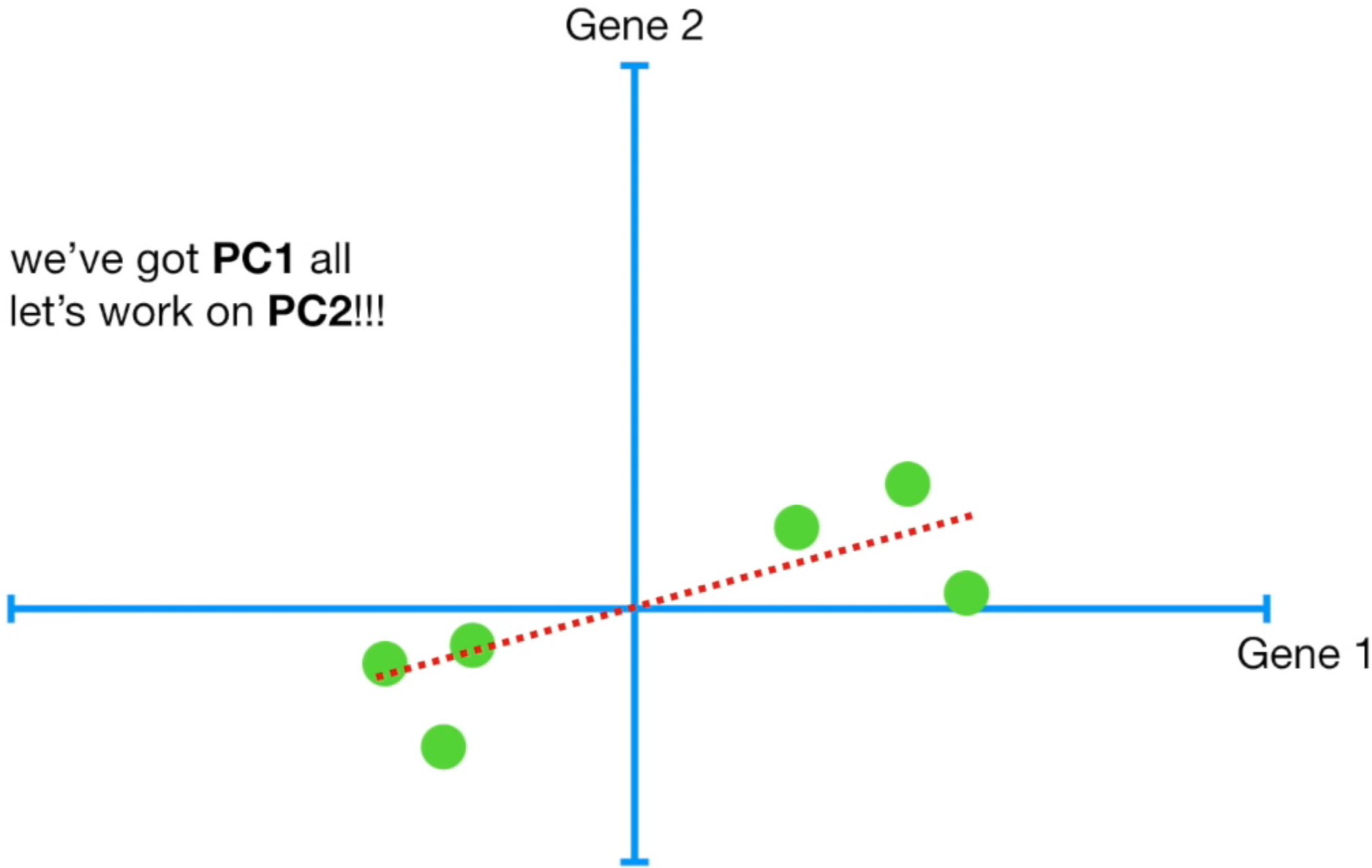
SS(distances for PC1) = Eigenvalue for PC1

$\sqrt{\text{Eigenvalue for PC1}}$ = Singular Value for PC1

…and the square root of the **Eigenvalue for PC1** is called the **Singular Value for PC1.**



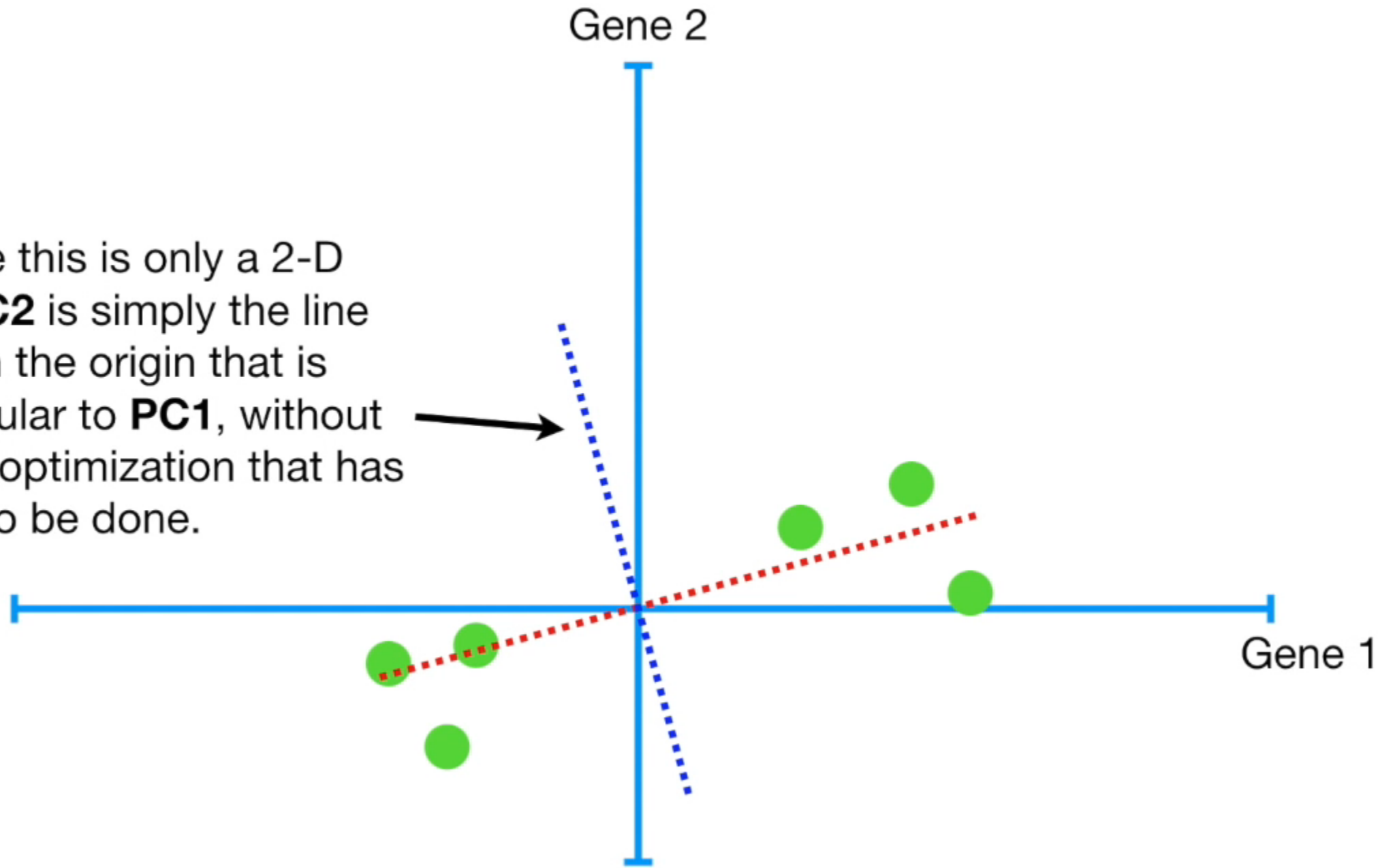
Now that we've got **PC1** all figured out let's work on **PC2**!!!



We have found the first principal component or PC 1.
Let’s find the PC 2, second principal component.

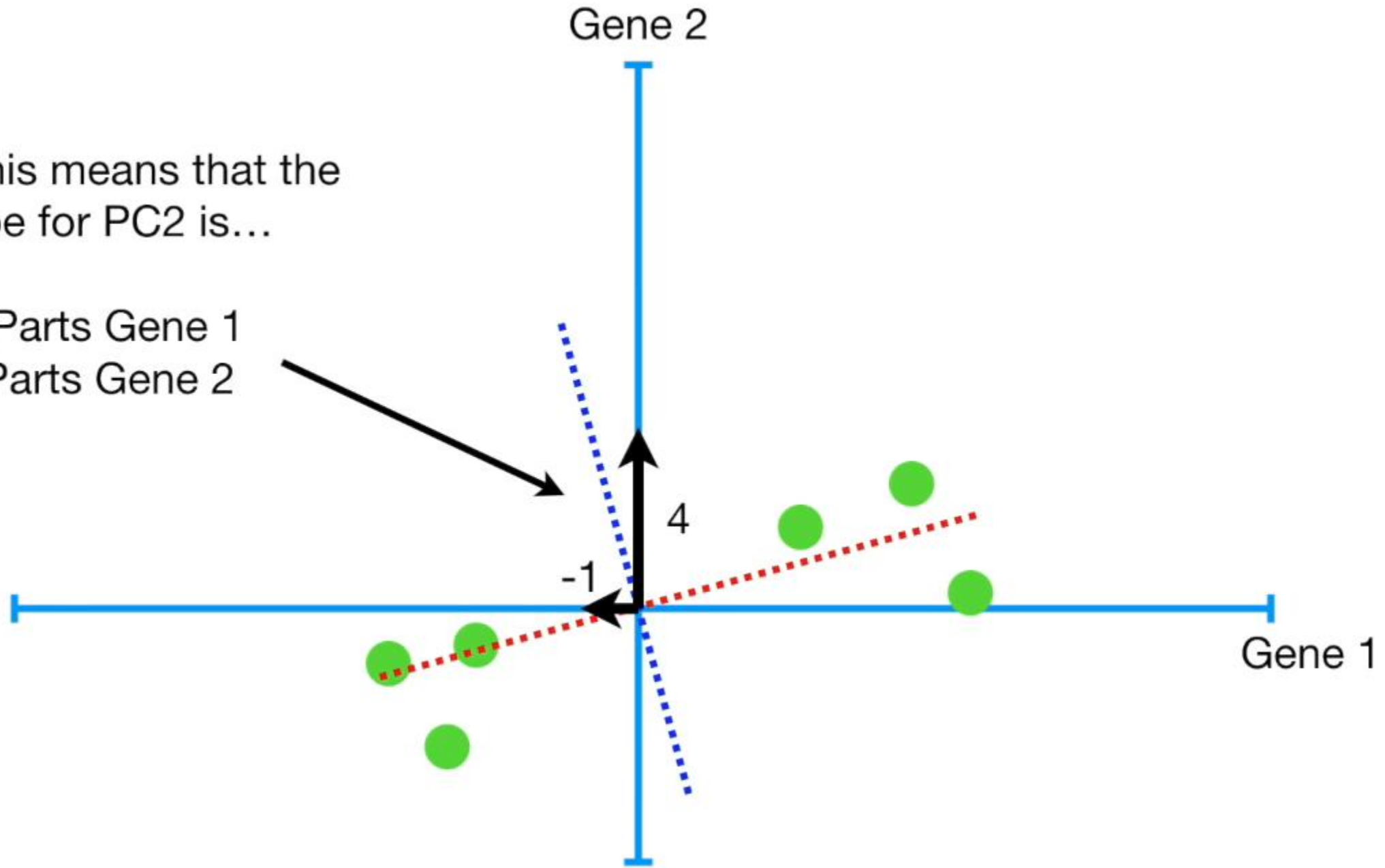
Remember, PC 1 got the highest sum of squared distances. We find the PC 2 such that it is perpendicular (orthogonal) to PC 1. Also it will have the second highest sum of squared distances.

Because this is only a 2-D graph, **PC2** is simply the line through the origin that is perpendicular to **PC1**, without any further optimization that has to be done.



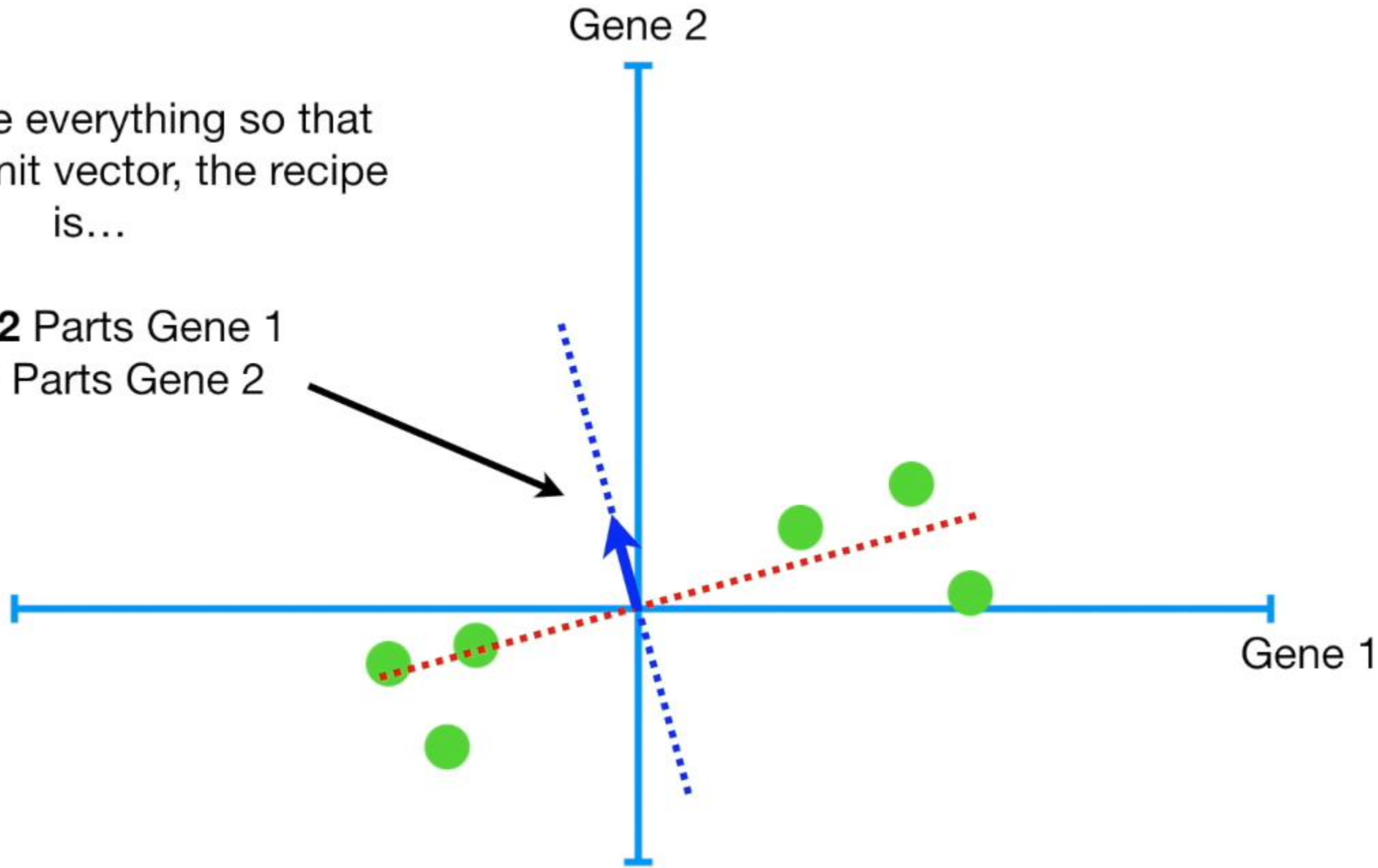
…and this means that the recipe for PC2 is…

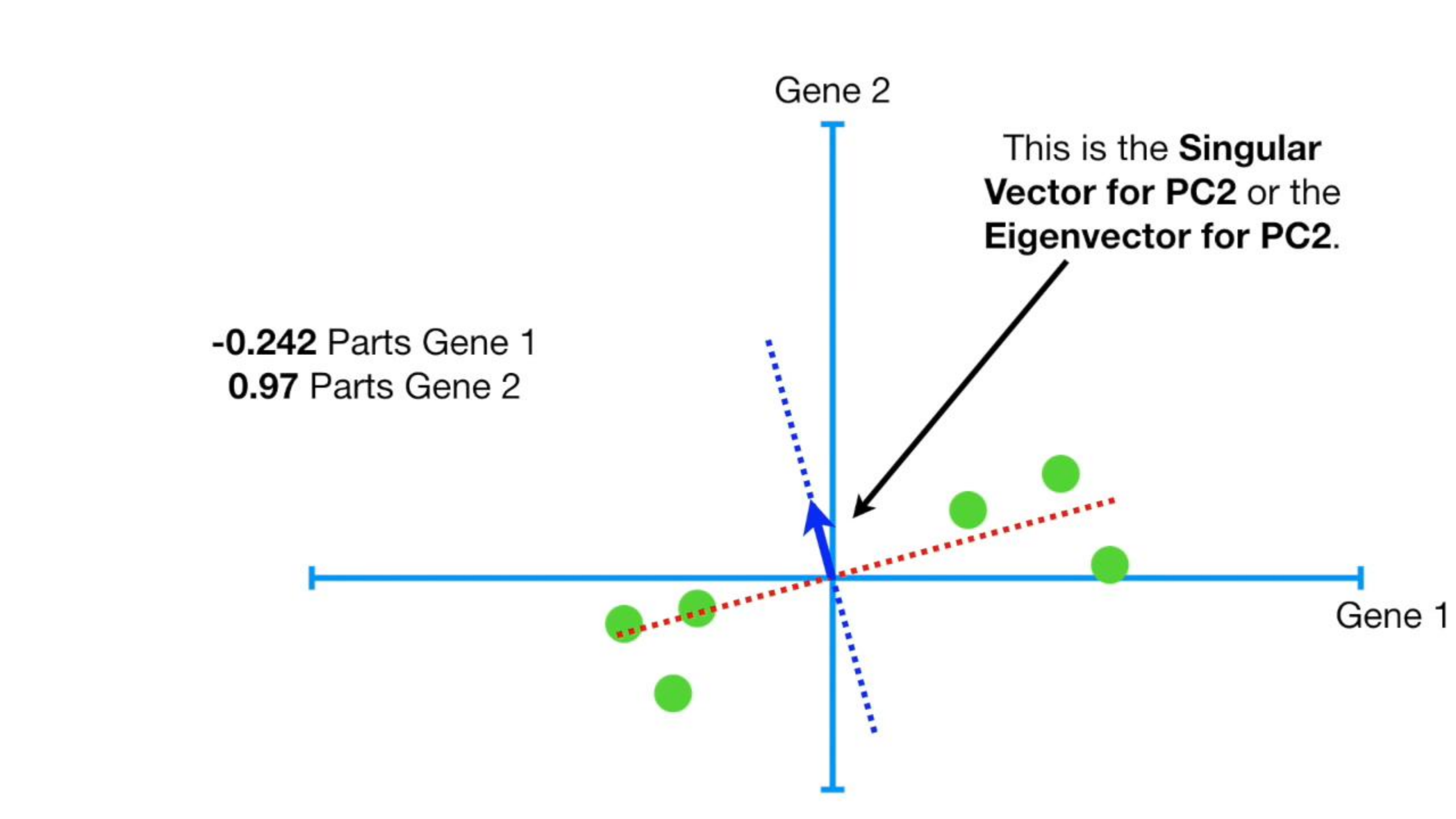
-1 Parts Gene 1
4 Parts Gene 2

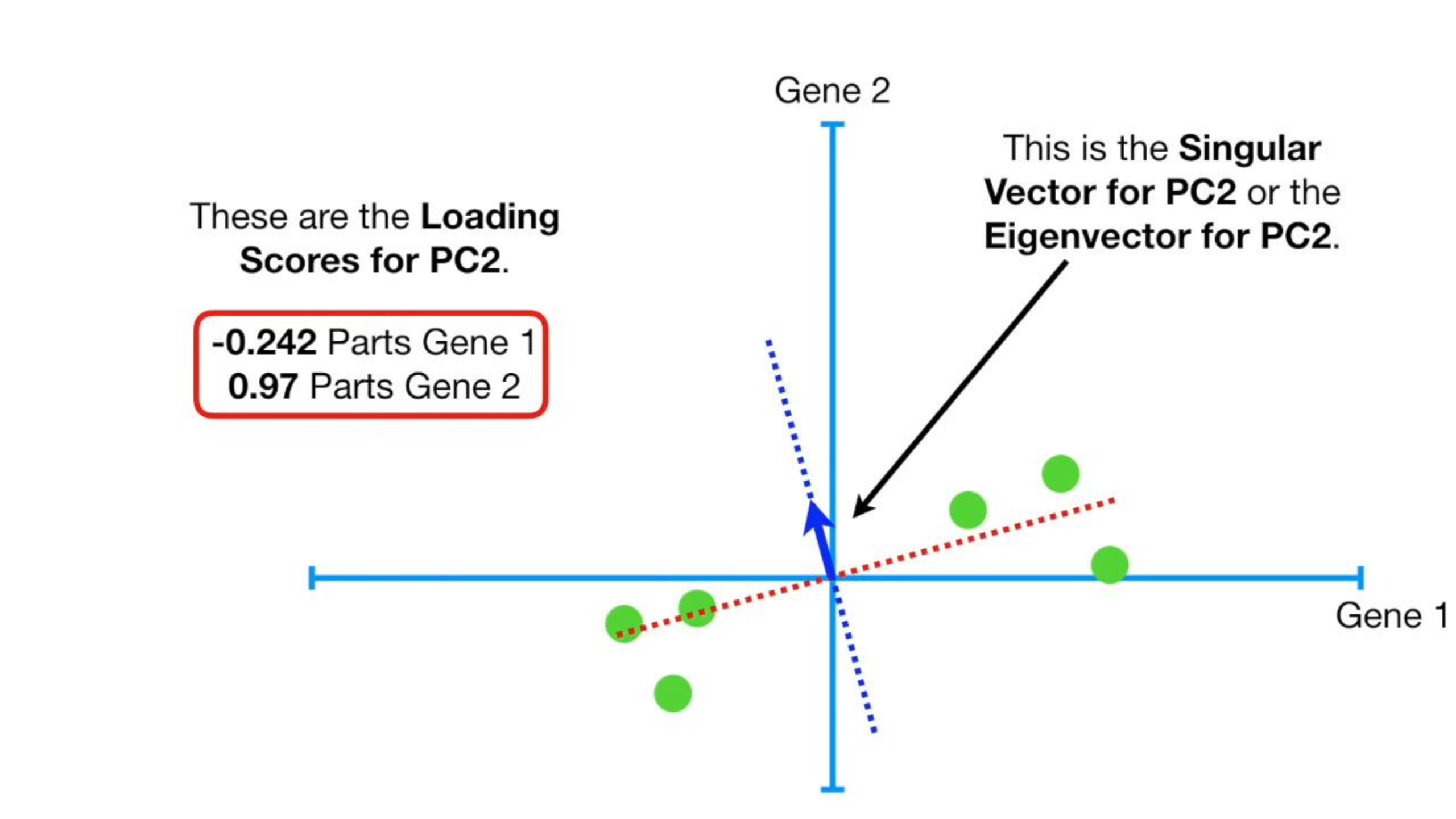


If we scale everything so that we get a unit vector, the recipe is…

-0.242 Parts Gene 1
0.97 Parts Gene 2



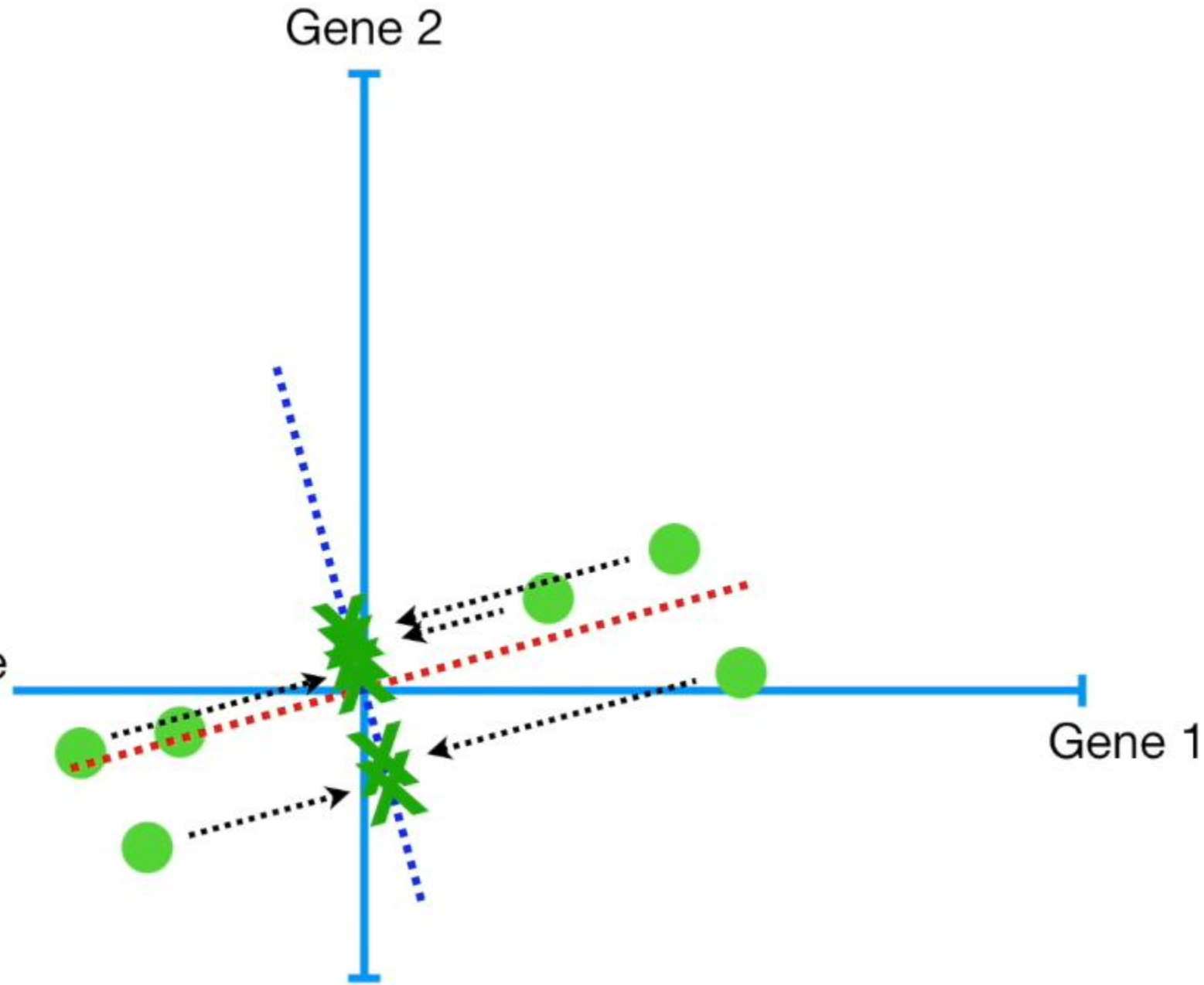

Gene 2
This is the **Singular Vector for PC2** or the **Eigenvector for PC2**.
-0.242 Parts Gene 1
0.97 Parts Gene 2
Gene 1


Gene 2
This is the **Singular Vector for PC2** or the **Eigenvector for PC2**.
These are the **Loading Scores for PC2**.
-0.242 Parts Gene 1
0.97 Parts Gene 2
Gene 1

These are the **Loading Scores for PC2**.

-0.242 Parts Gene 1
0.97 Parts Gene 2

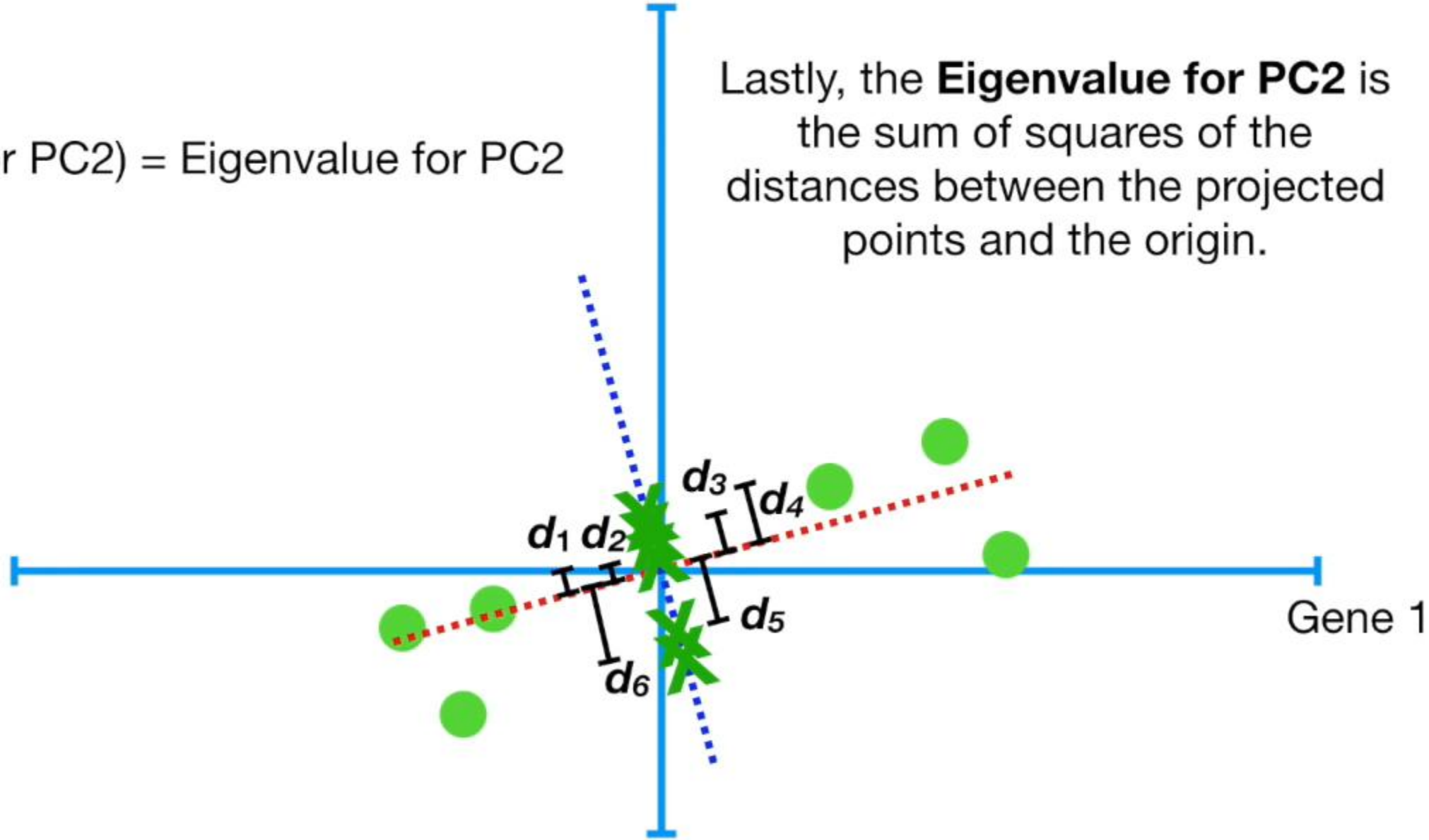
They tell us that, in terms of how the values are projected onto PC2, Gene 2 is 4 times as important as Gene 1.



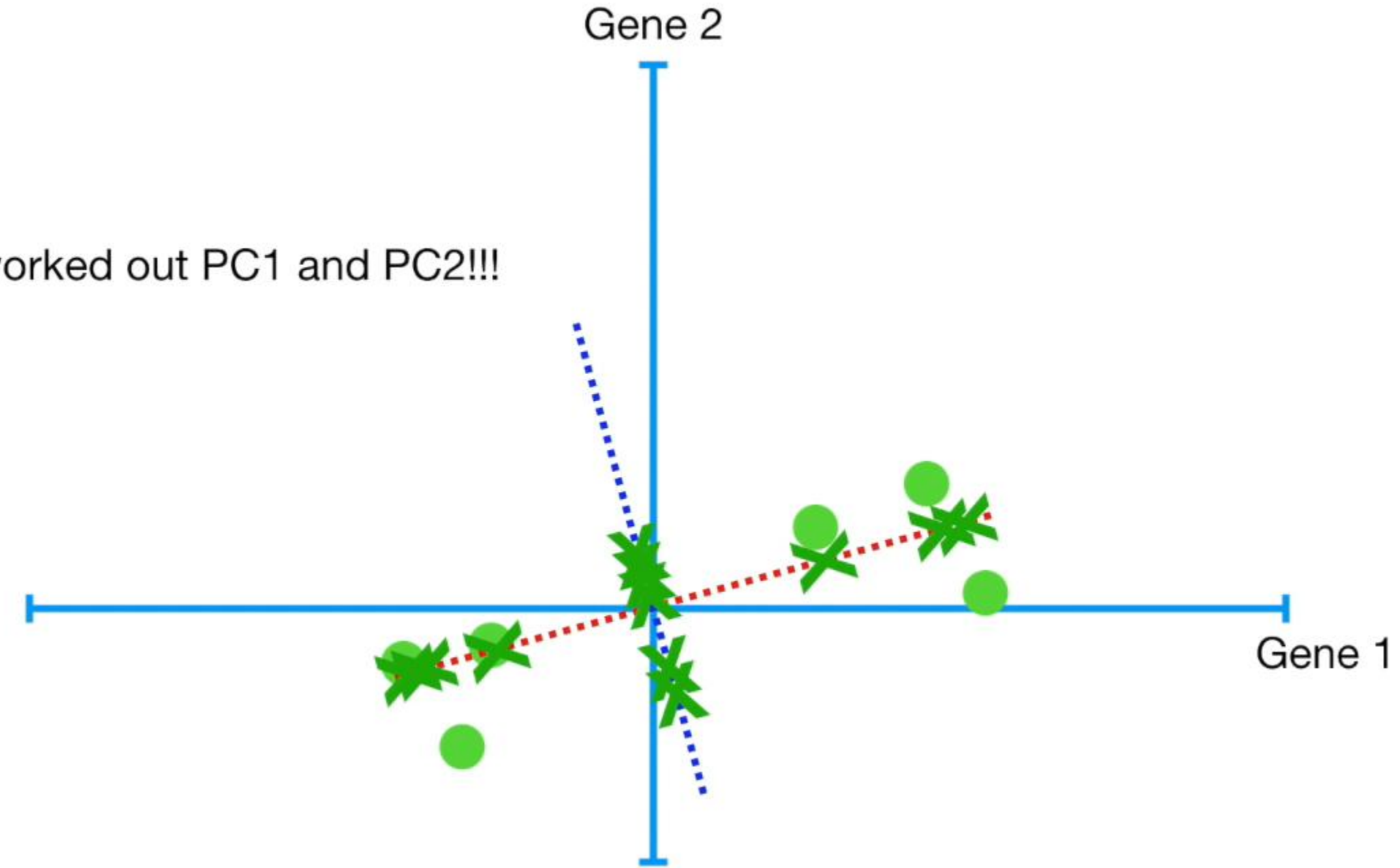
$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

SS(distances for PC2) = Eigenvalue for PC2

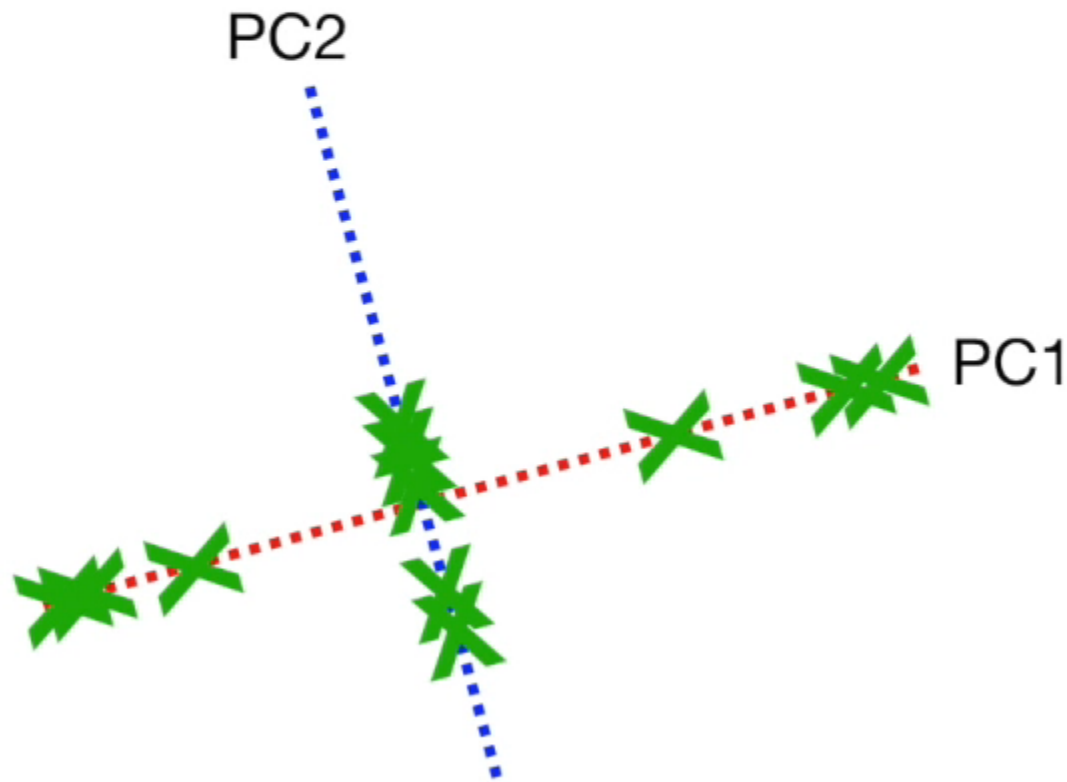
Lastly, the **Eigenvalue for PC2** is the sum of squares of the distances between the projected points and the origin.



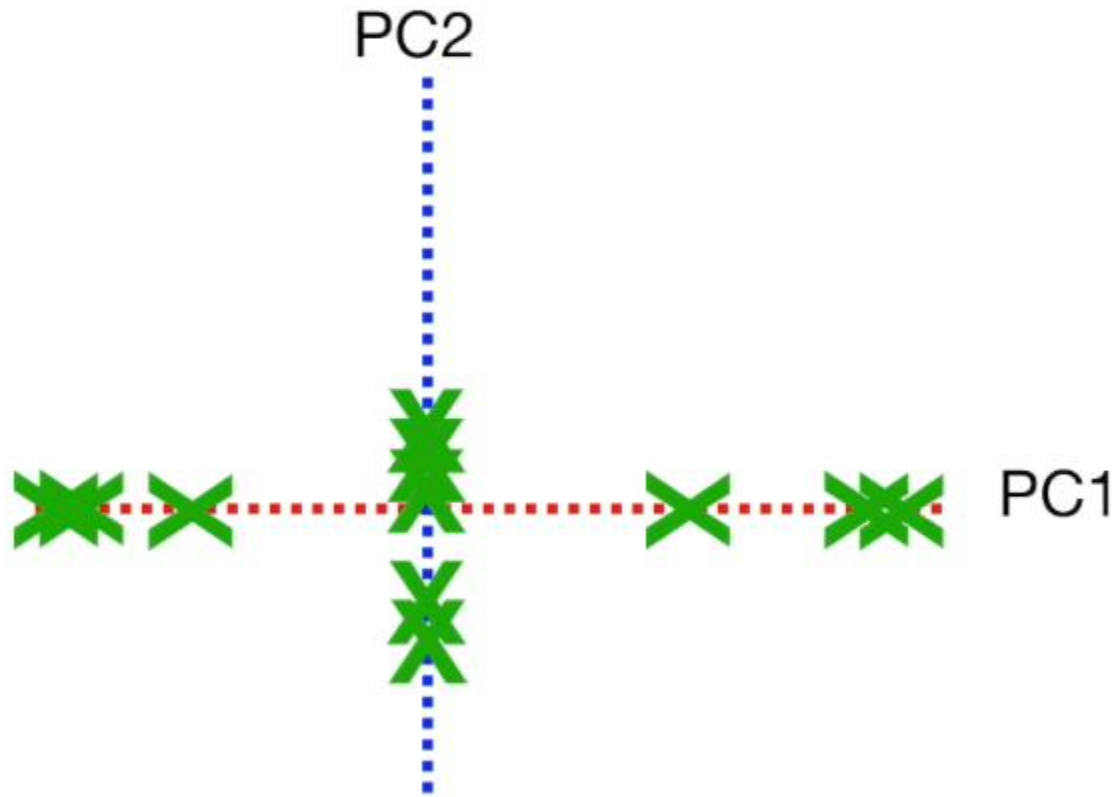
We've worked out PC1 and PC2!!!

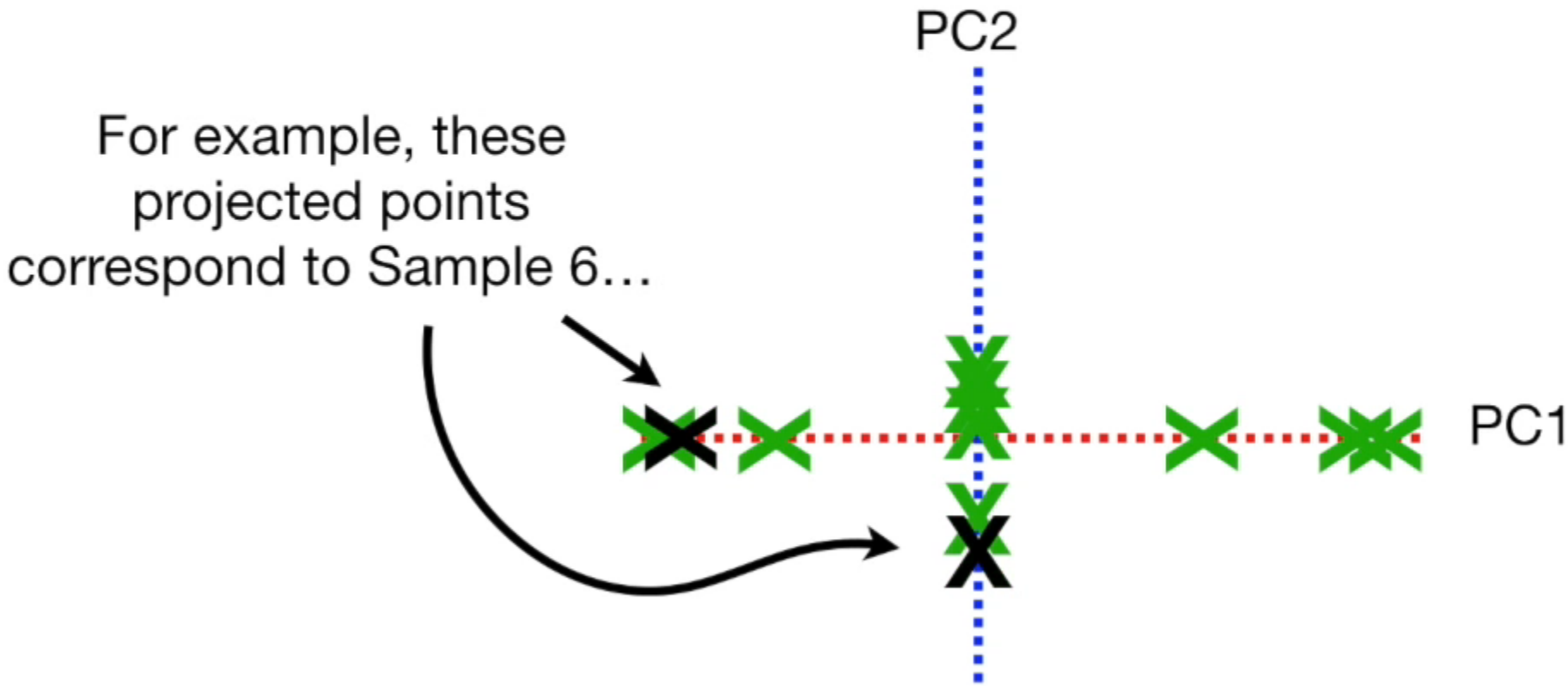


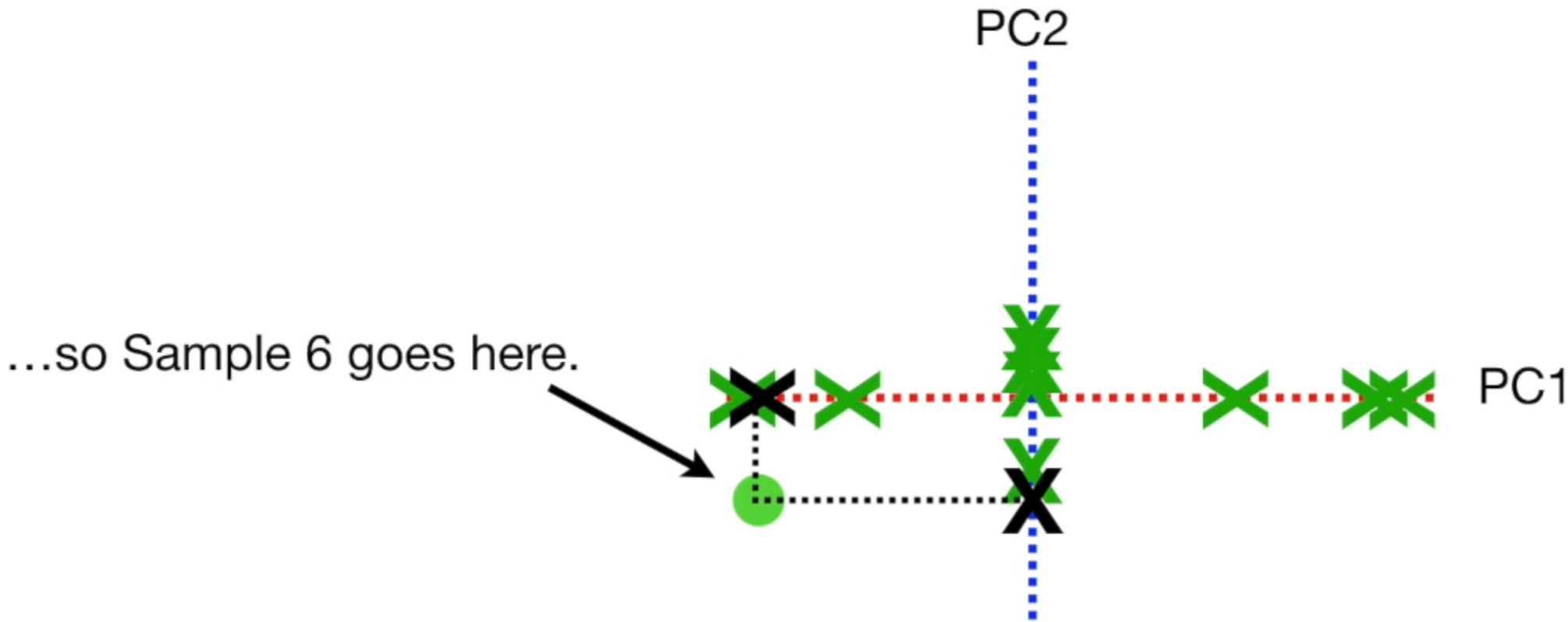
We simply rotate everything so that PC1 is horizontal…

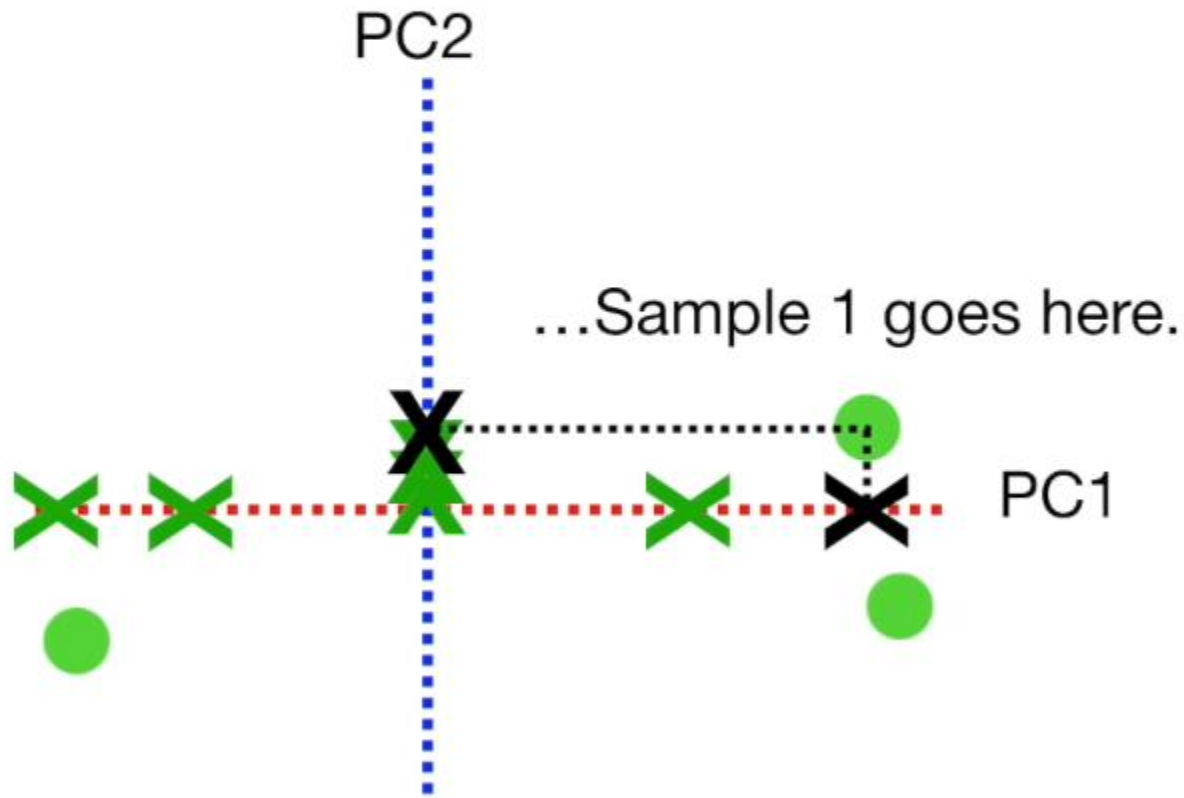


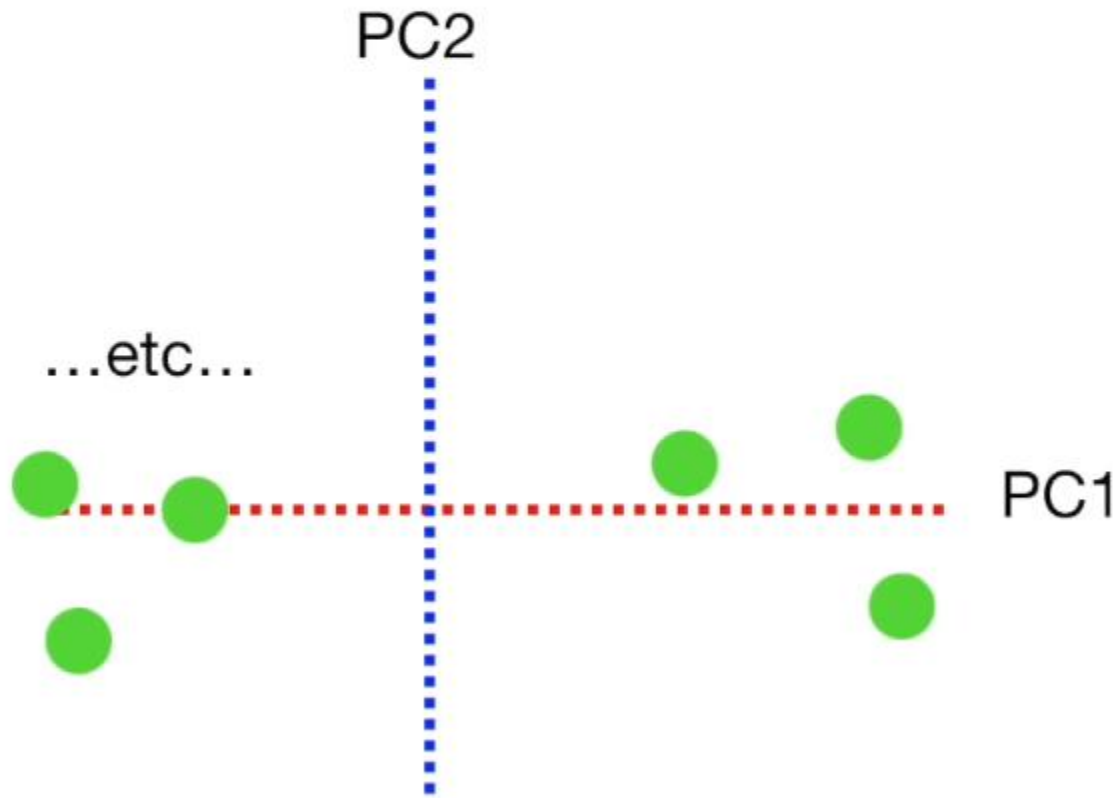
…then we use the projected points to find where the samples go in the PCA plot.




PC2
For example, these
projected points
correspond to Sample 6…
PC1


PC2
…so Sample 6 goes here.
PC1

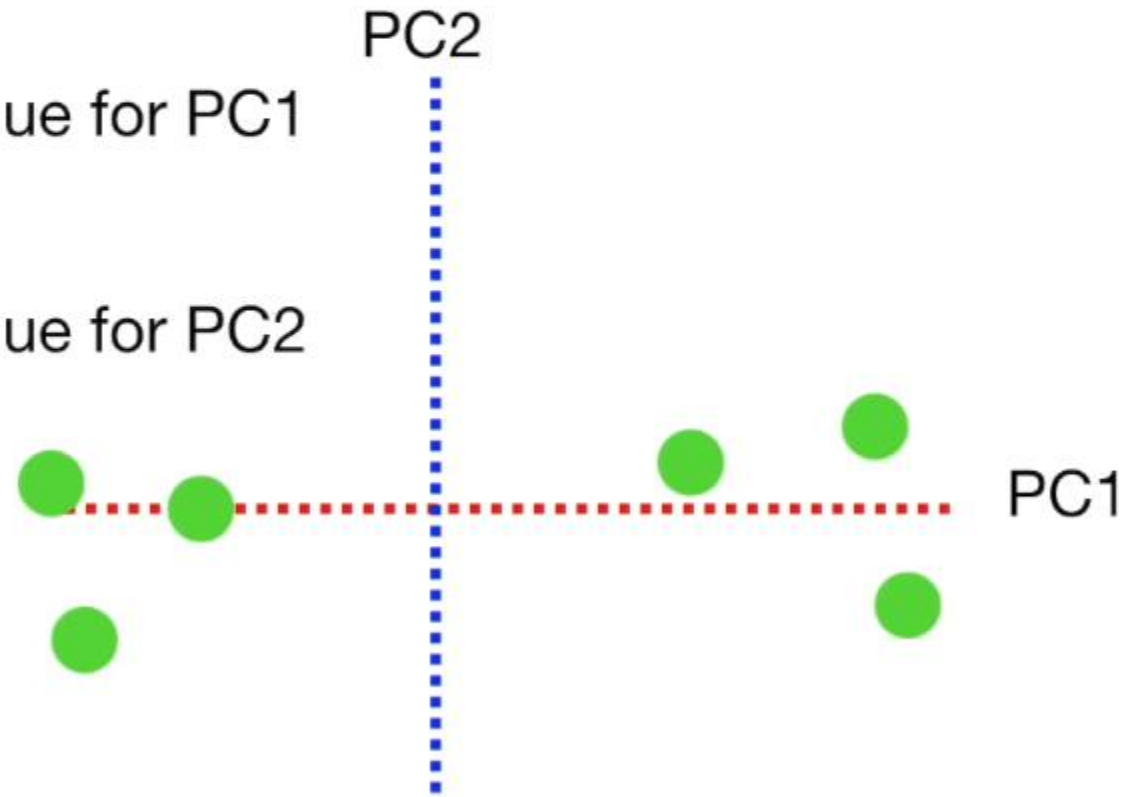

PC2
…Sample 1 goes here.
PC1


PC2
…etc…
PC1

Remember the eigenvalues?

SS(distances for PC1) = Eigenvalue for PC1

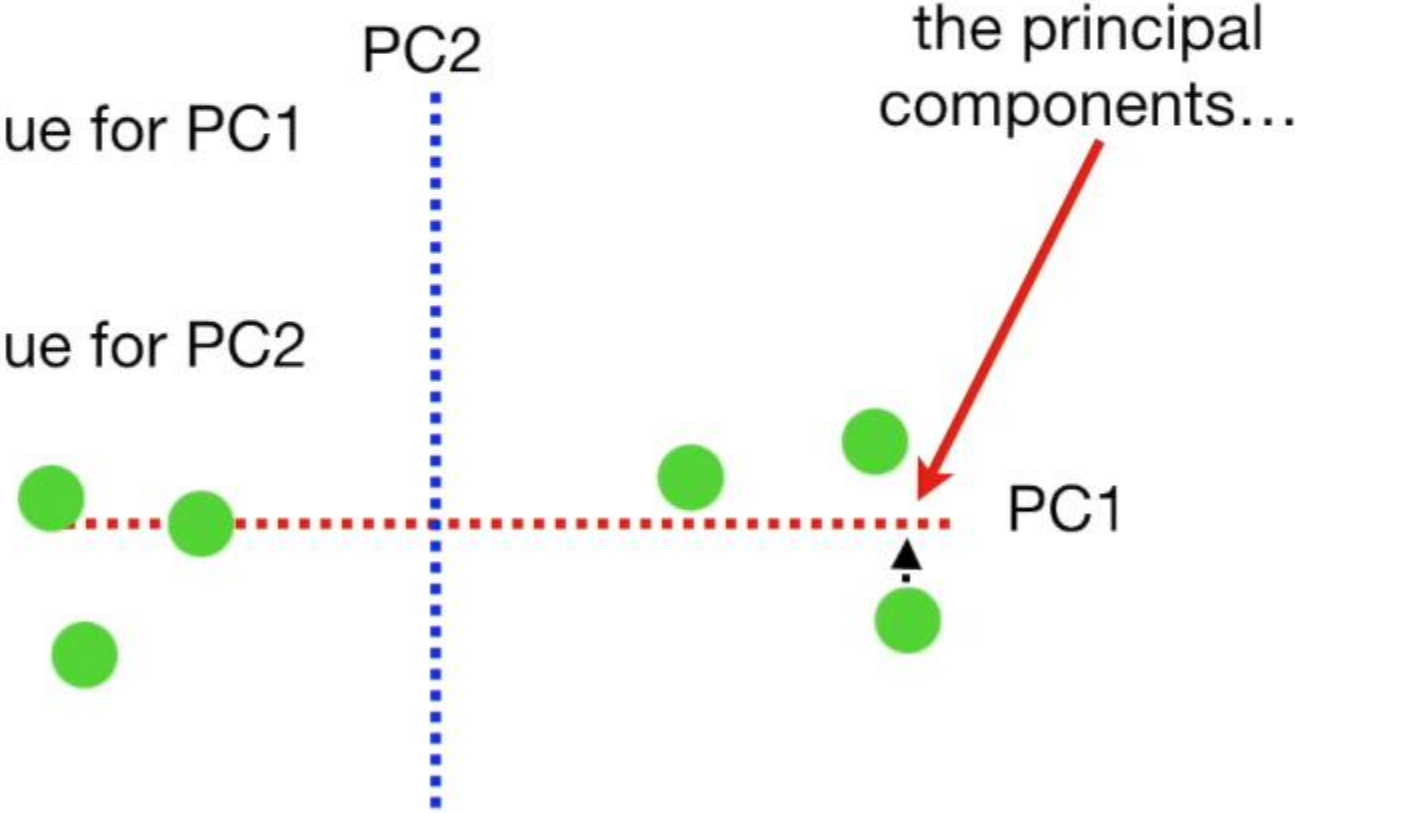
SS(distances for PC2) = Eigenvalue for PC2



Remember the eigenvalues?

SS(distances for PC1) = Eigenvalue for PC1

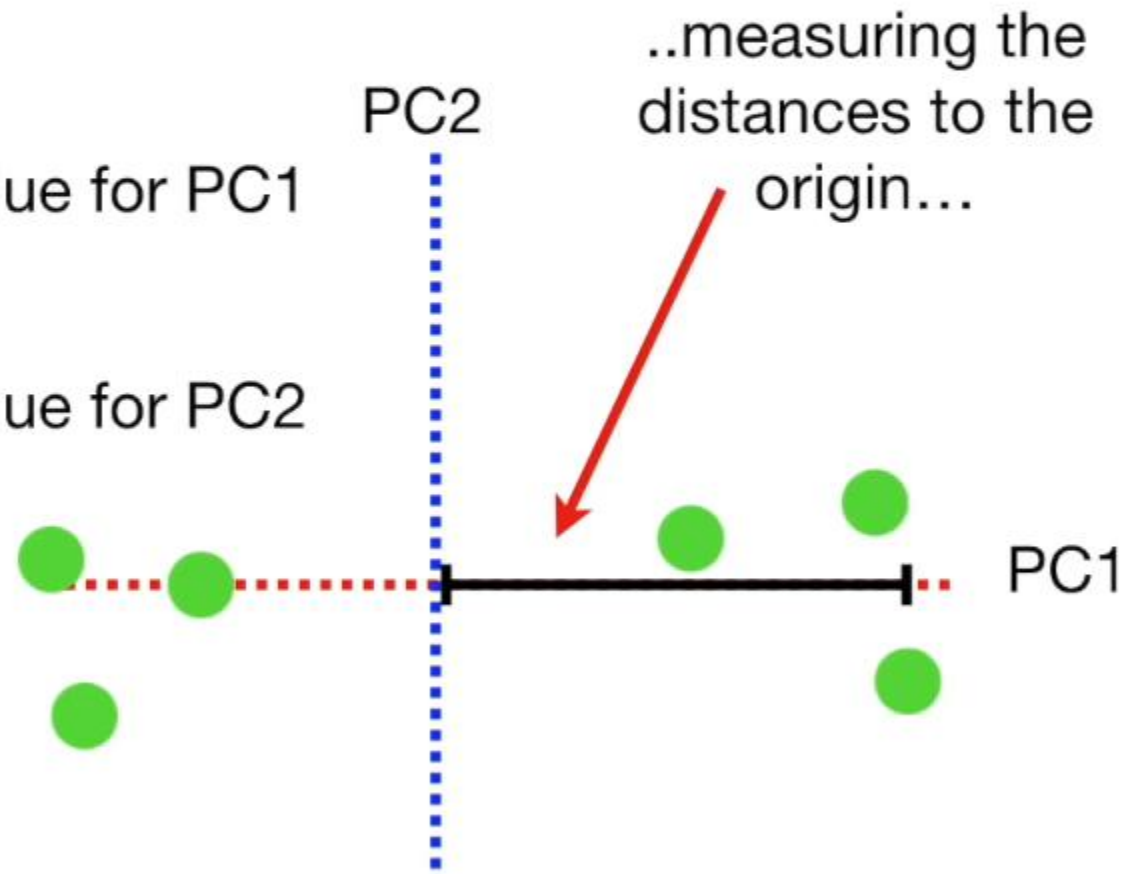
SS(distances for PC2) = Eigenvalue for PC2



Remember the eigenvalues?

SS(distances for PC1) = Eigenvalue for PC1

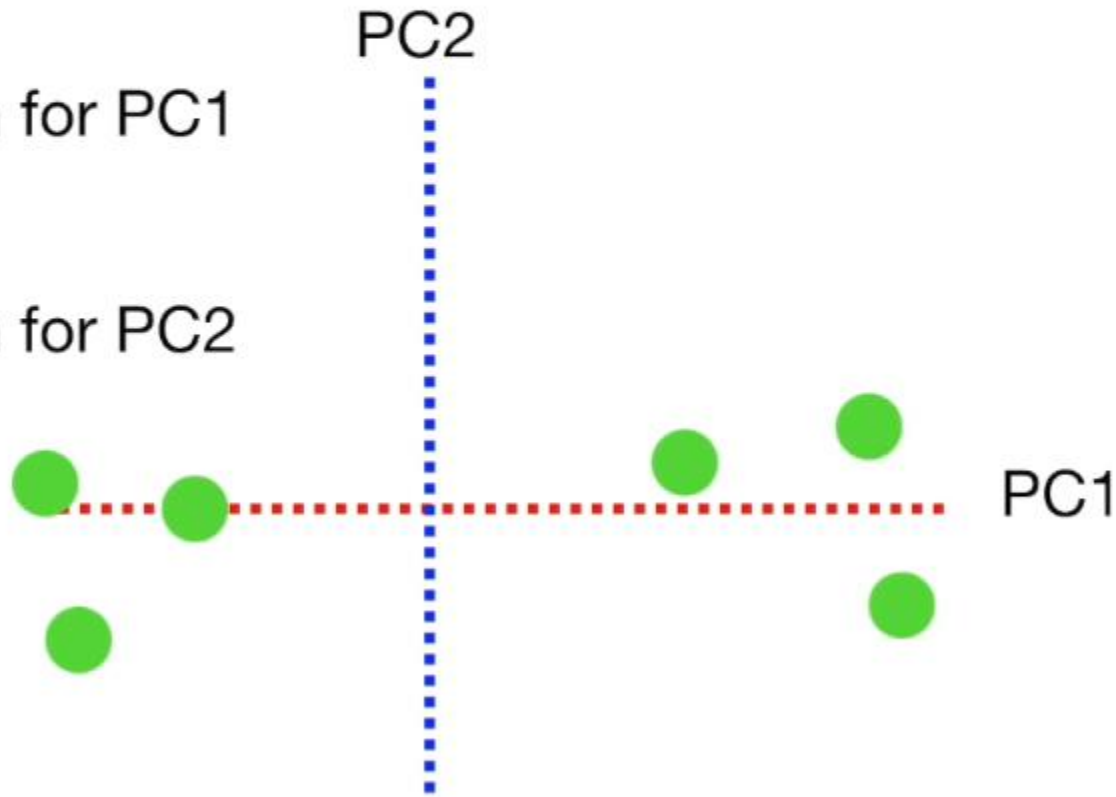
SS(distances for PC2) = Eigenvalue for PC2



We can convert them into variation around the origin (0, 0) by dividing by the sample size minus 1 (i.e. n - 1).

$$\frac{\text{SS(distances for PC1)}}{n - 1} = \text{Variation for PC1}$$

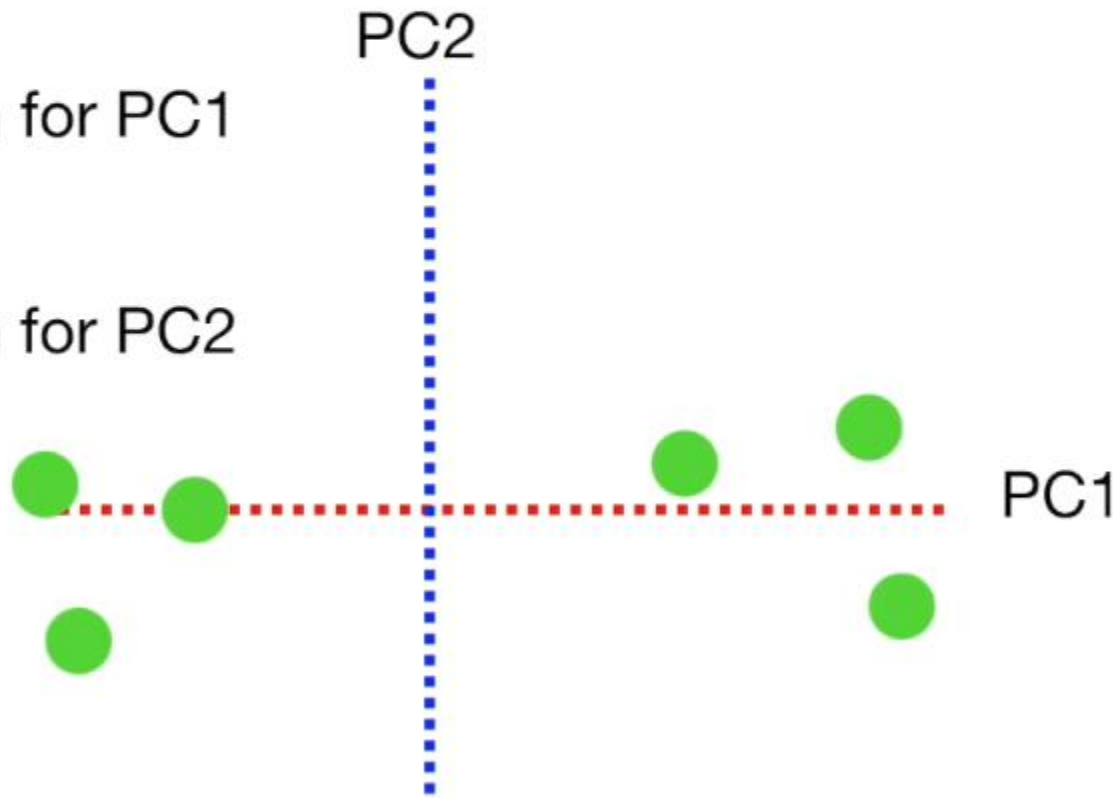
$$\frac{\text{SS(distances for PC2)}}{n - 1} = \text{Variation for PC2}$$



For the sake of the example, imagine that the Variation for **PC1 = 15**, and the variation for **PC2 = 3**.

$$\frac{\text{SS(distances for PC1)}}{n - 1} = \text{Variation for PC1}$$

$$\frac{\text{SS(distances for PC2)}}{n - 1} = \text{Variation for PC2}$$

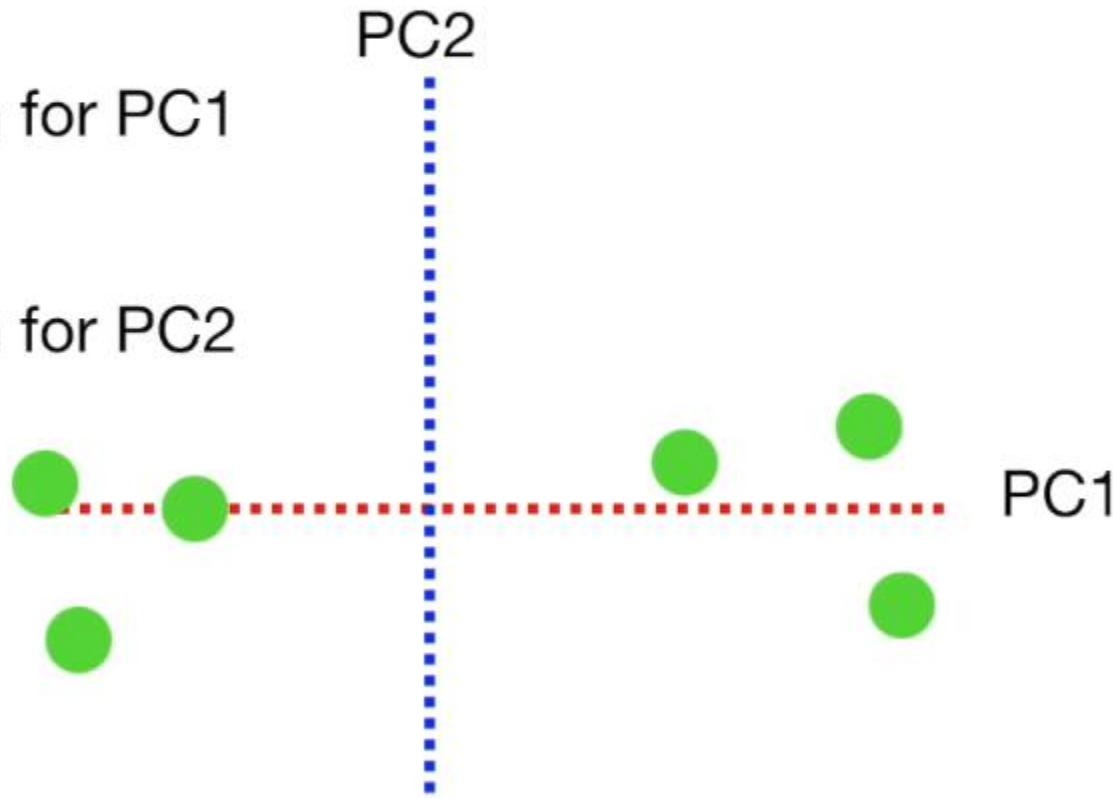


For the sake of the example, imagine that the Variation for **PC1 = 15**, and the variation for **PC2 = 3**.

That means that the total variation around both PCs is **15 + 3 = 18**...

$$\frac{SS(\text{distances for PC1})}{n - 1} = \text{Variation for PC1}$$

$$\frac{SS(\text{distances for PC2})}{n - 1} = \text{Variation for PC2}$$

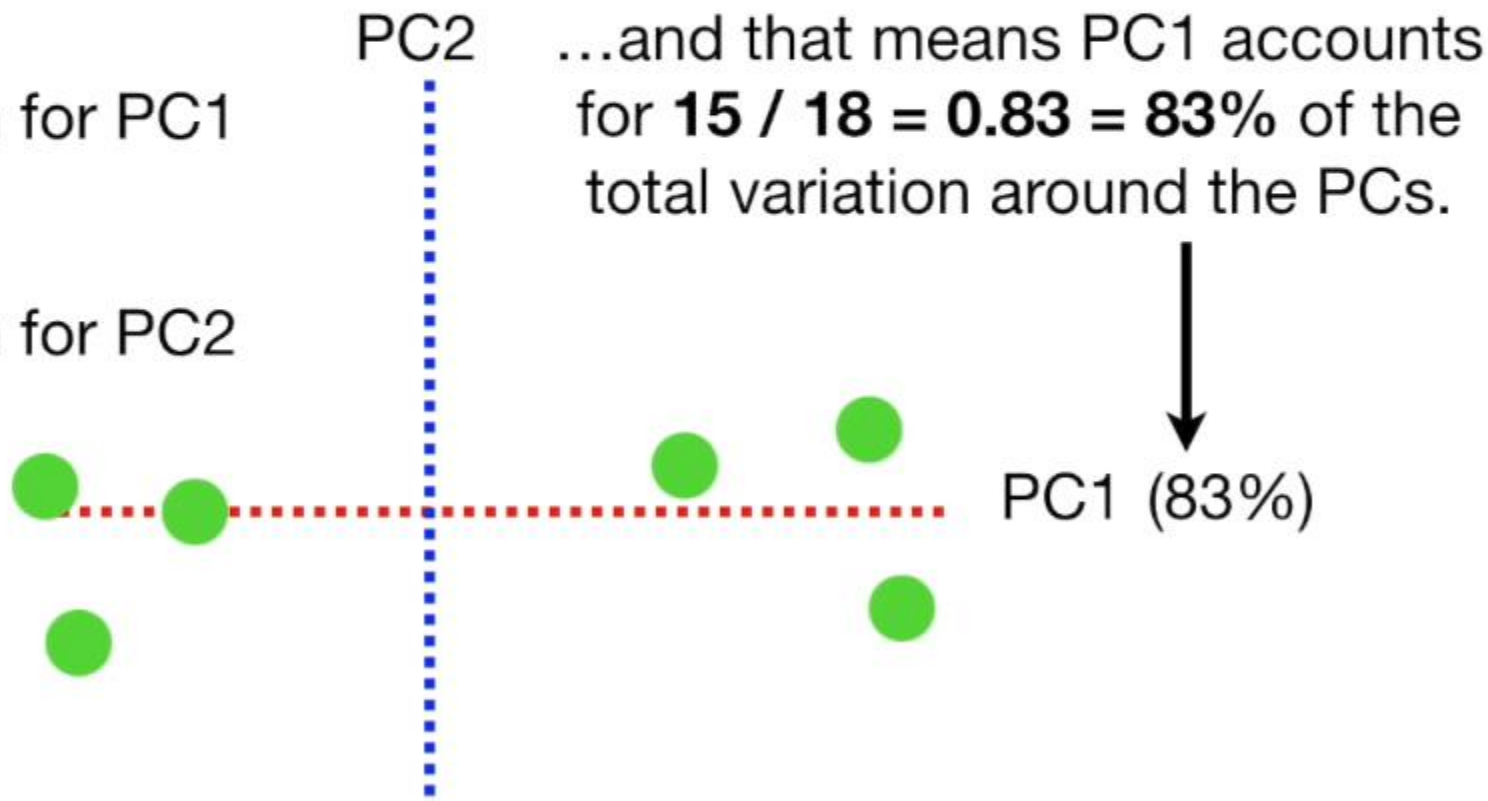


For the sake of the example, imagine that the Variation for **PC1 = 15**, and the variation for **PC2 = 3**.

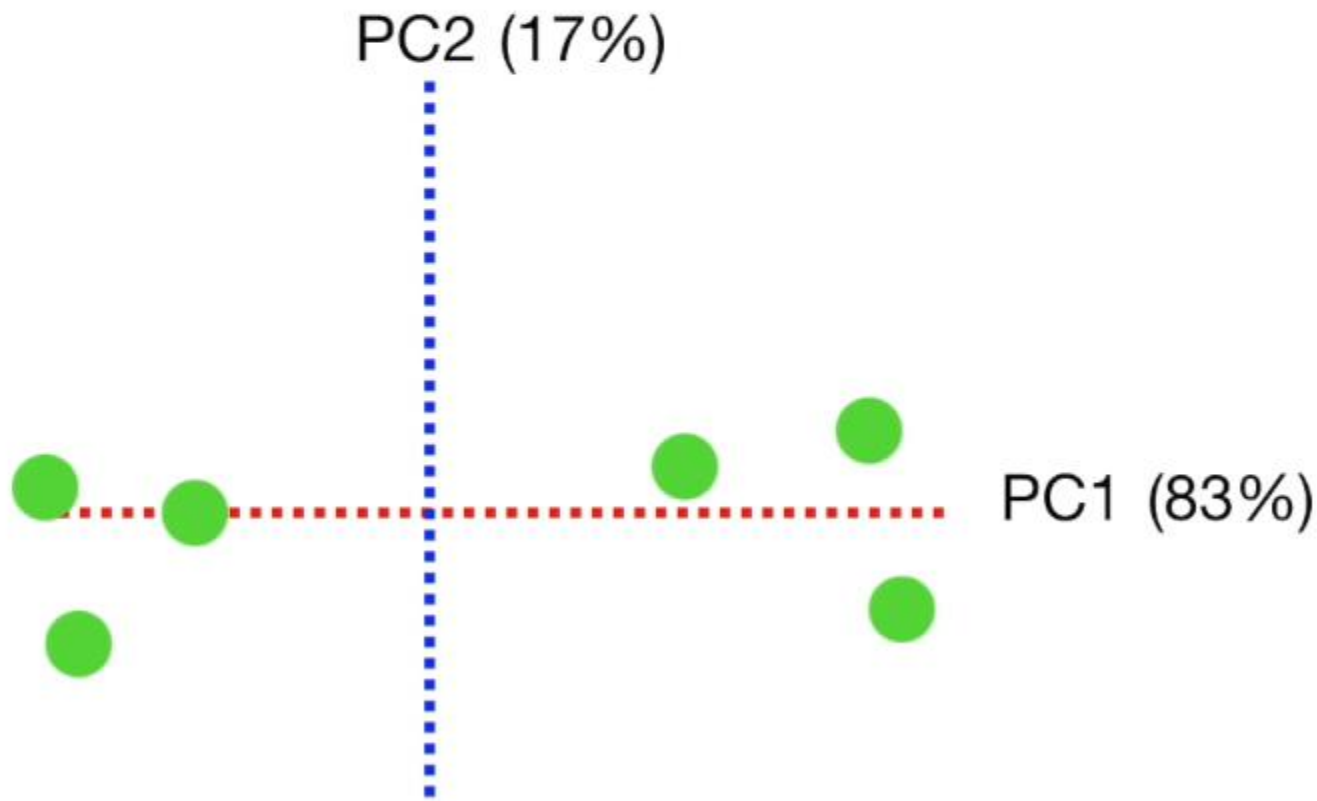
That means that the total variation around both PCs is **15 + 3 = 18**...

$$\frac{SS(\text{distances for PC1})}{n - 1} = \text{Variation for PC1}$$

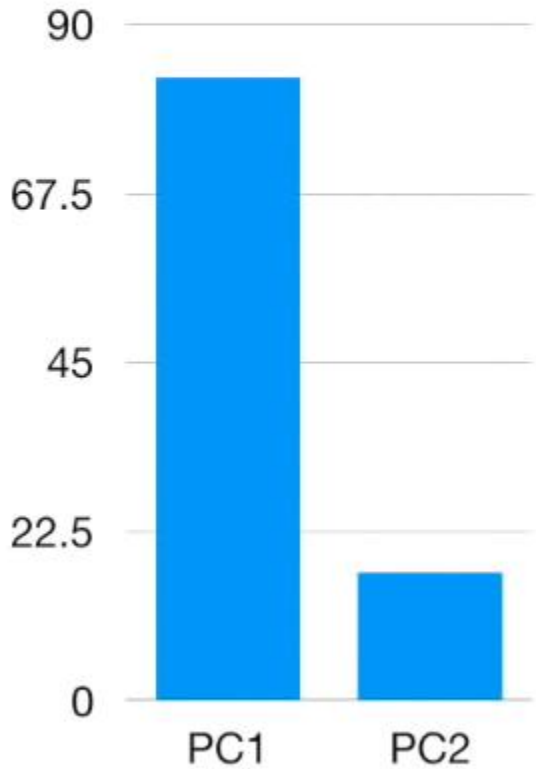
$$\frac{SS(\text{distances for PC2})}{n - 1} = \text{Variation for PC2}$$

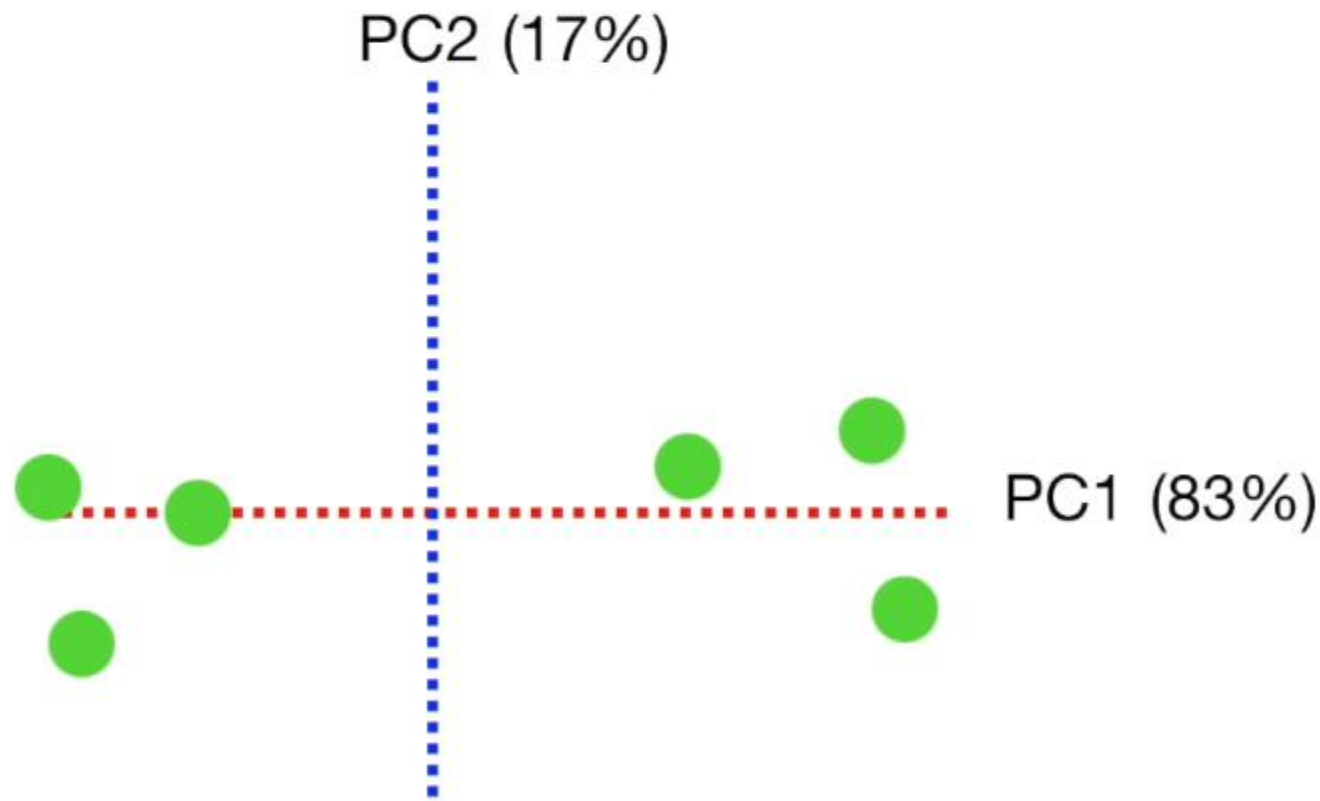


PC2 accounts for **3 / 18 = 0.17 = 17%** of the total variation around the PCs.



TERMINOLOGY ALERT!!!! A **Scree Plot** is a graphical representation of the percentages of variation that each PC accounts for.

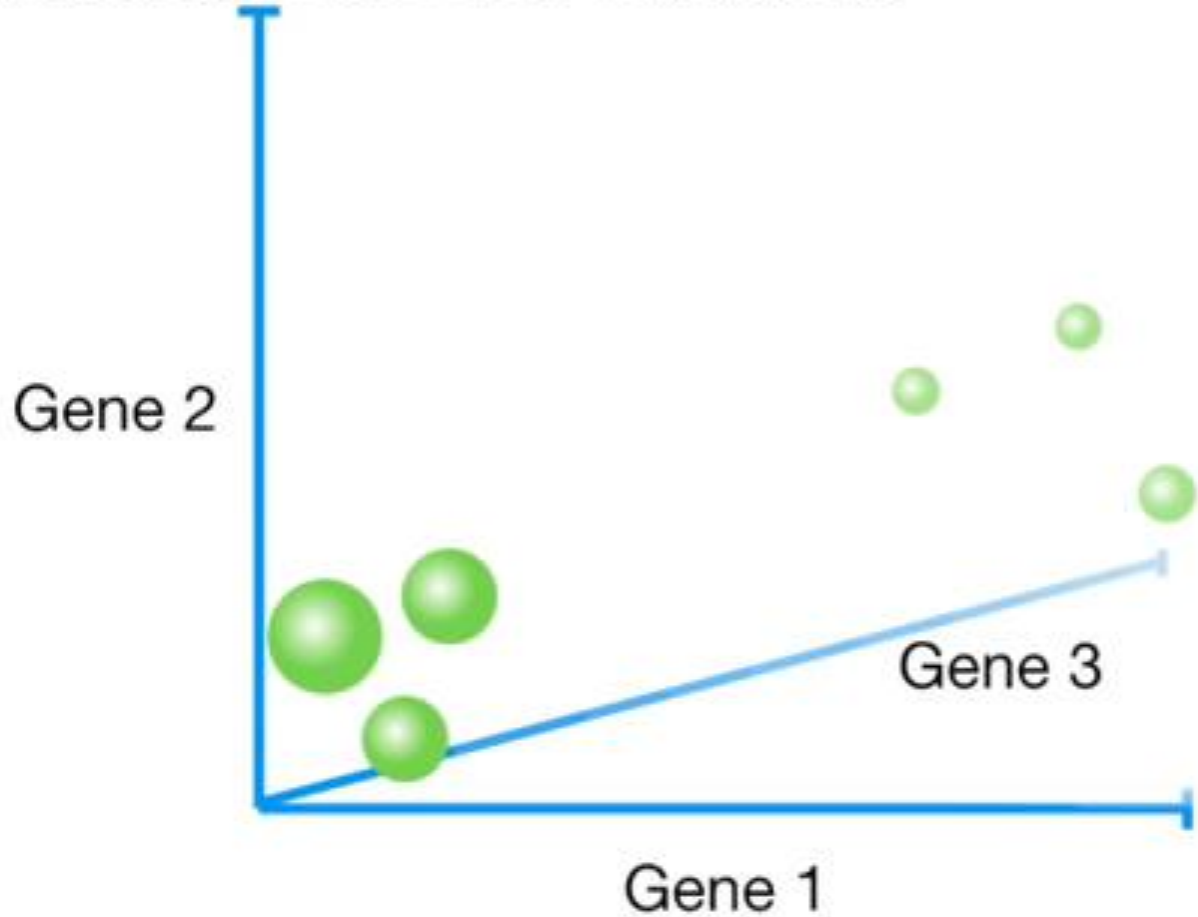




OK - Now let's quickly go through a slightly more complicated example

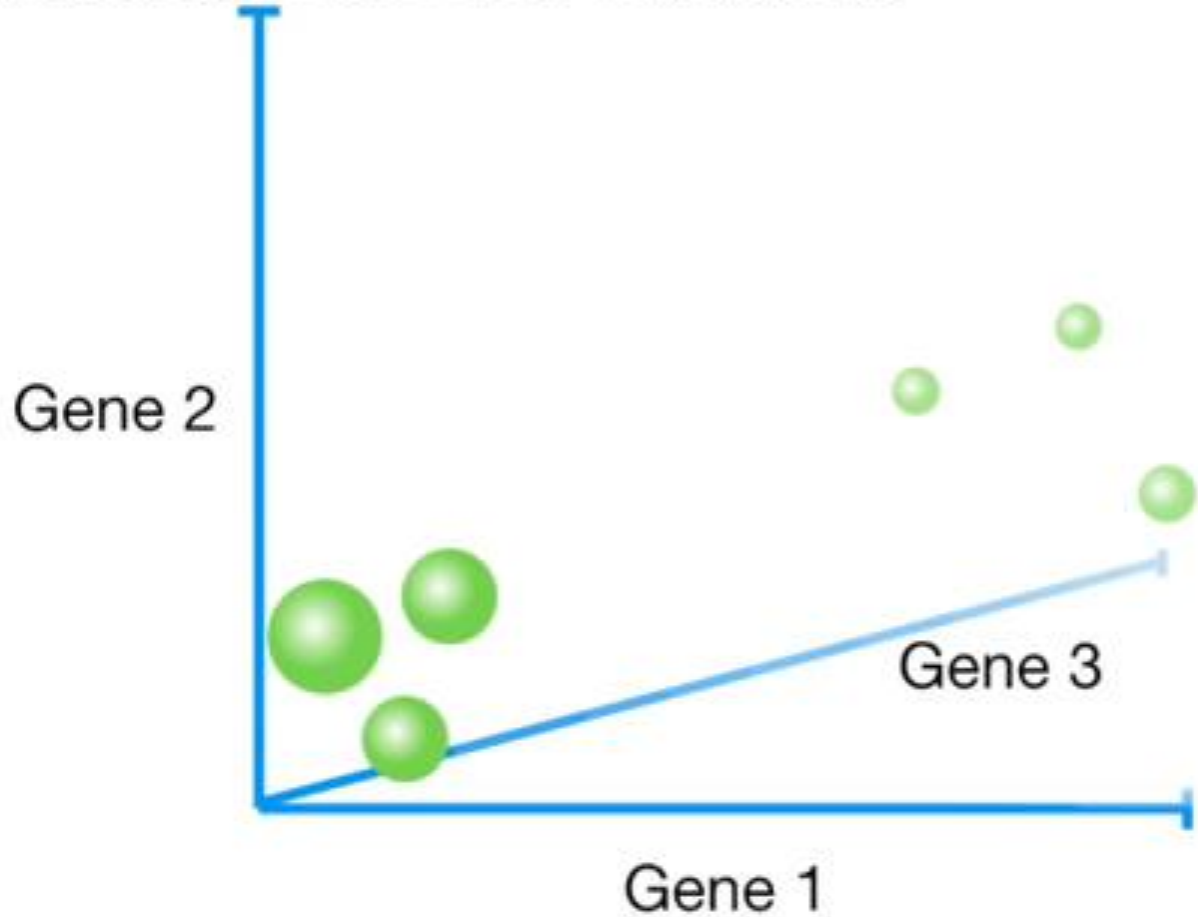
PCA with 3 variables (in this case, that means 3 genes) is pretty much the same as 2 variables…

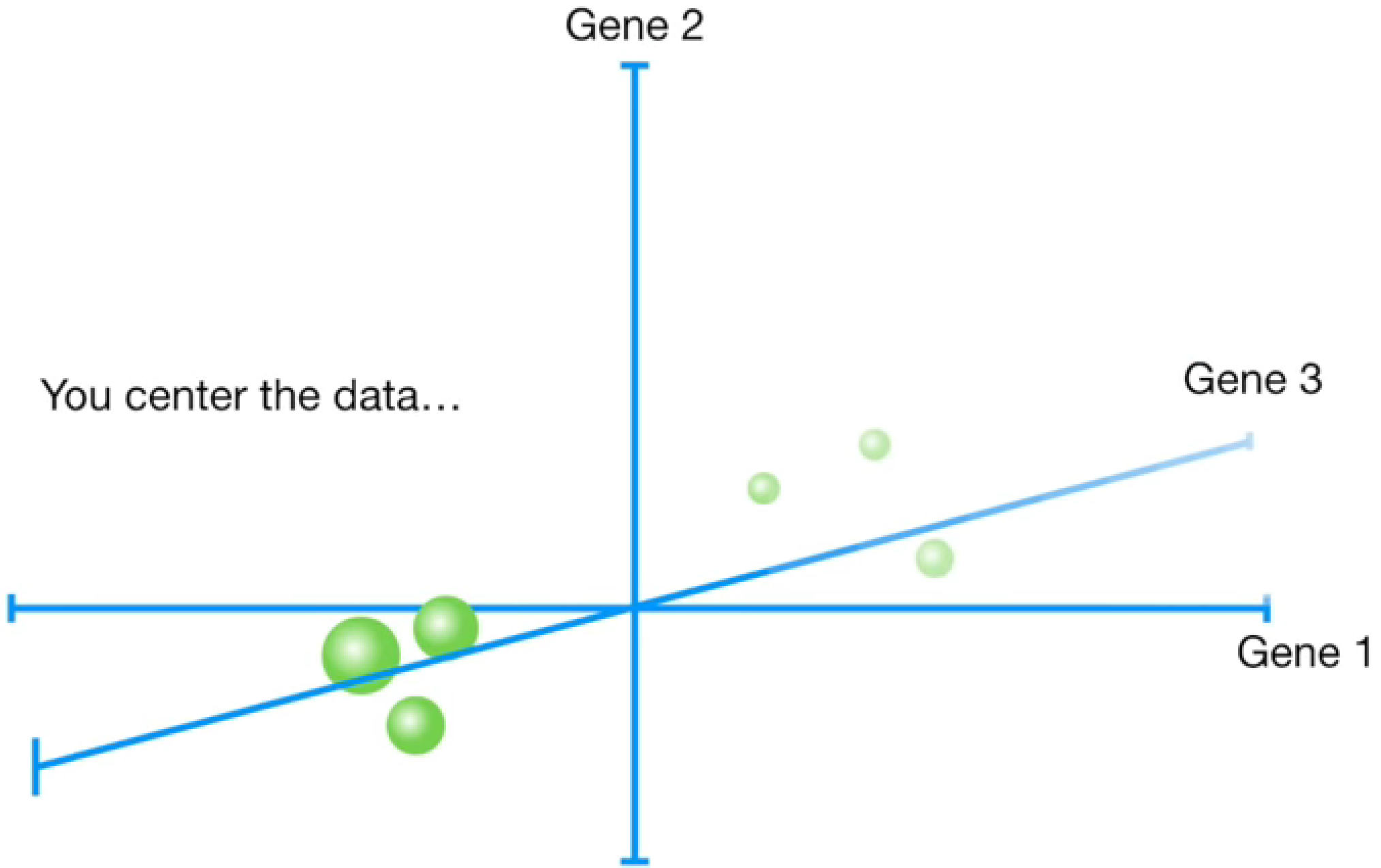
	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1
Gene 3	12	9	10	2.5	1.3	2



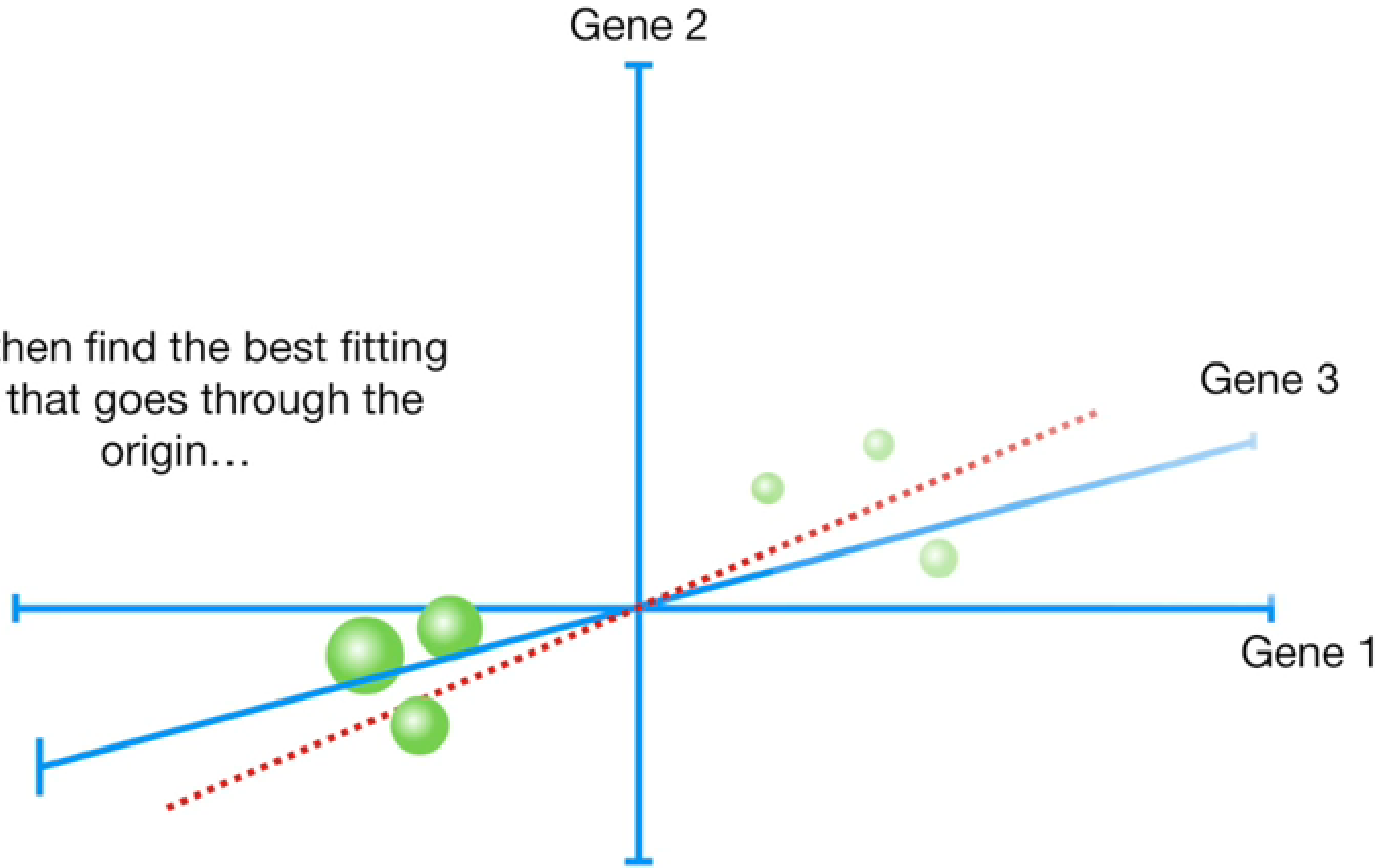
PCA with 3 variables (in this case, that means 3 genes) is pretty much the same as 2 variables…

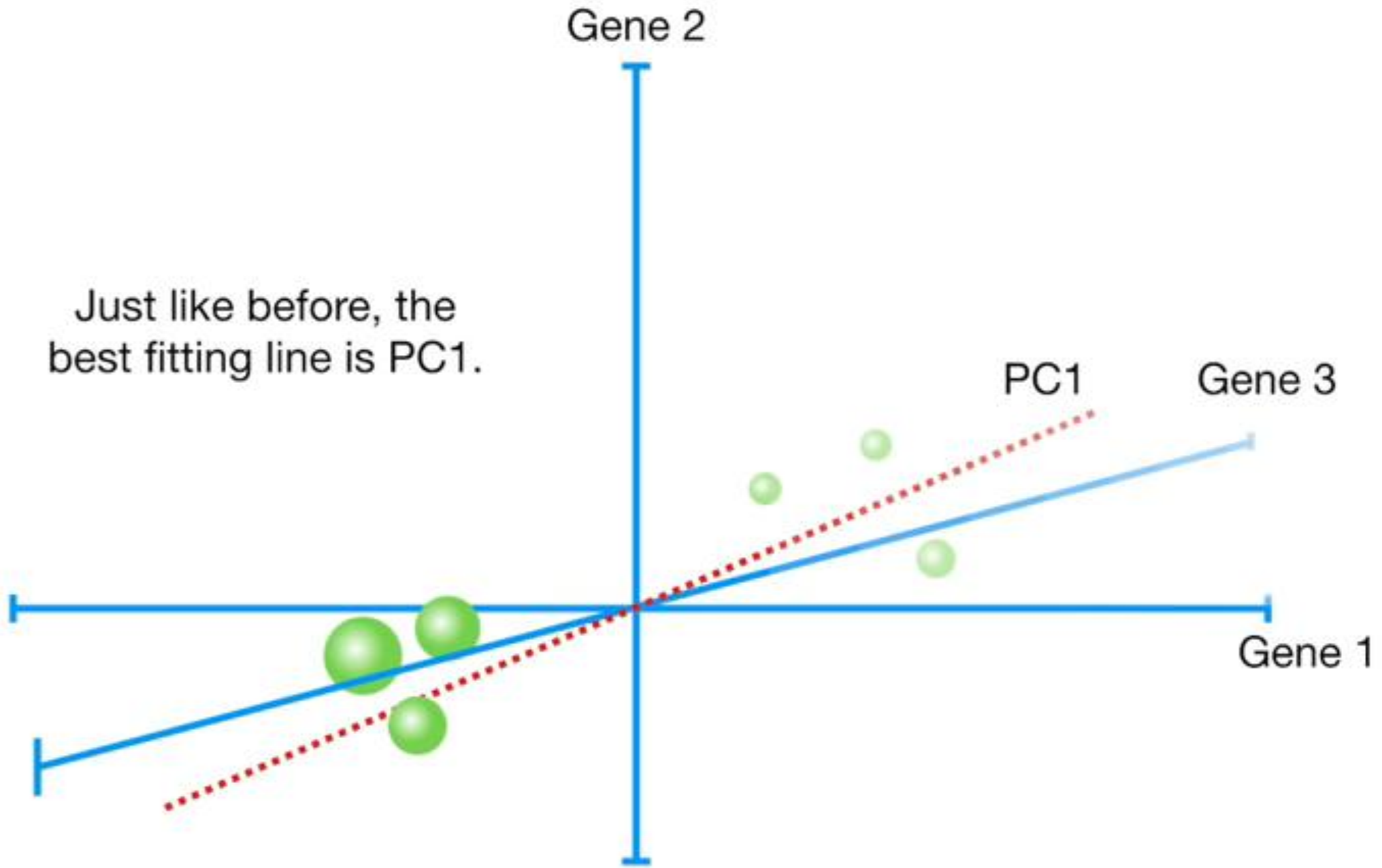
	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1
Gene 3	12	9	10	2.5	1.3	2




Gene 2
You center the data…
Gene 3
Gene 1

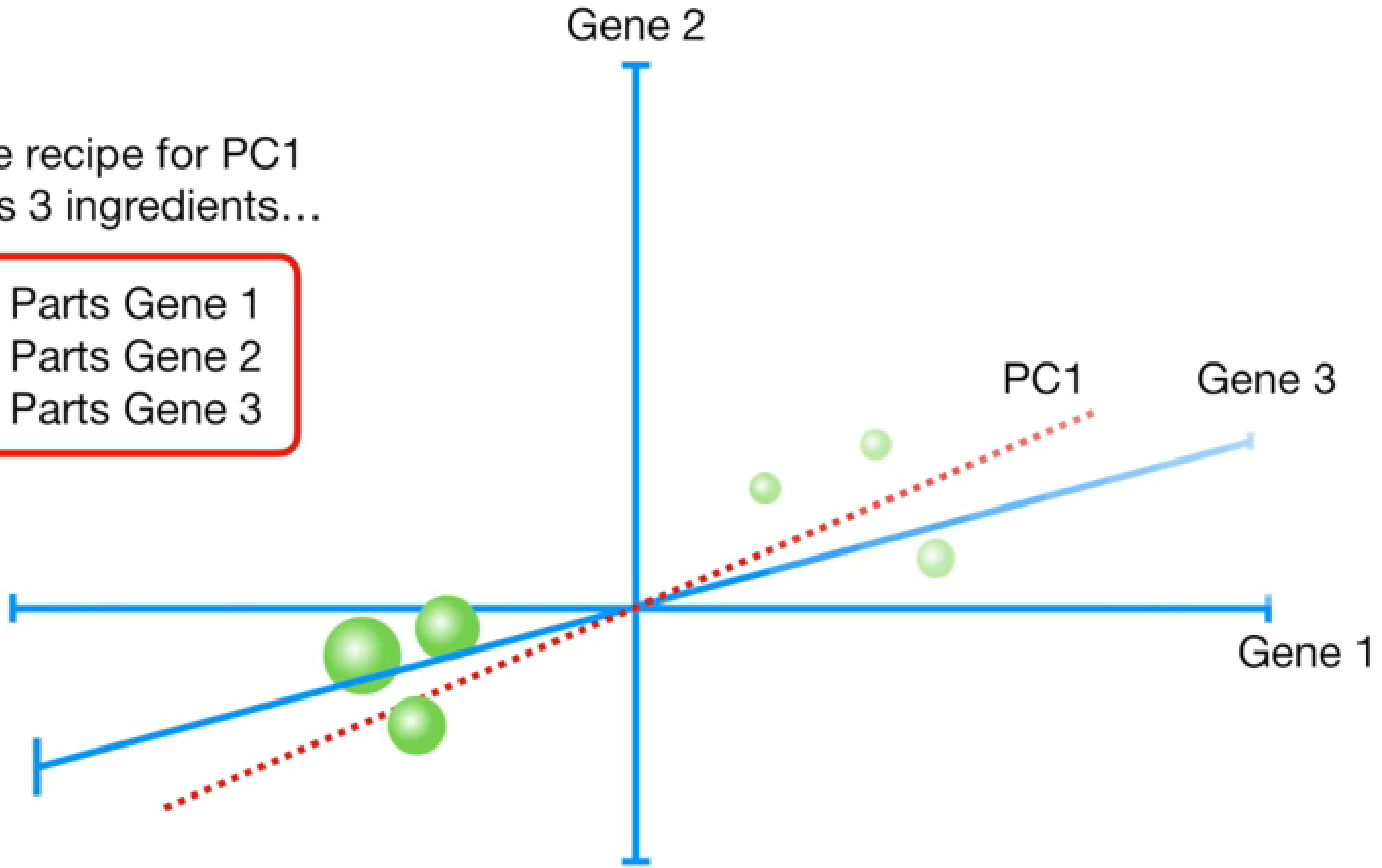
You then find the best fitting line that goes through the origin…

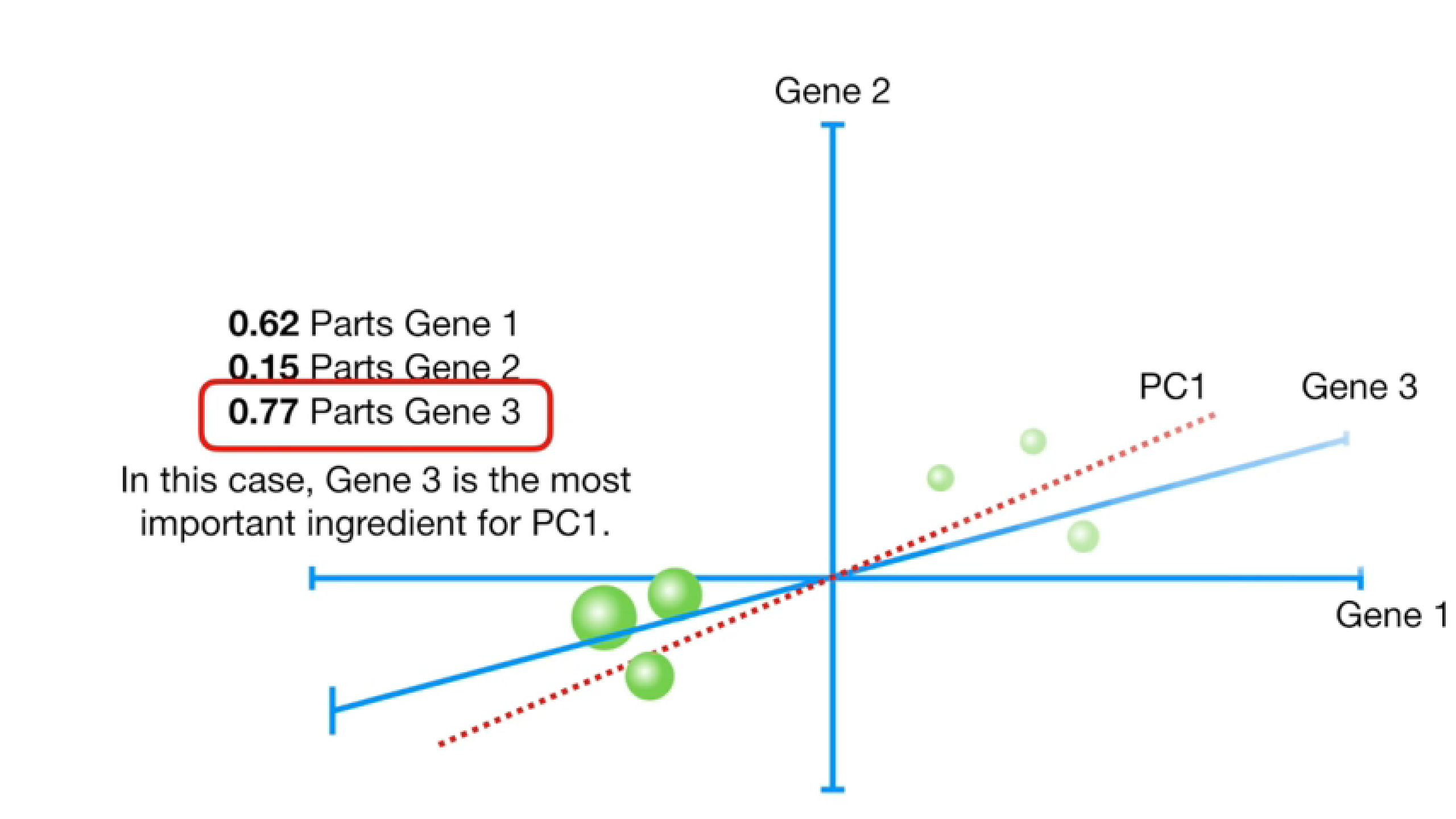


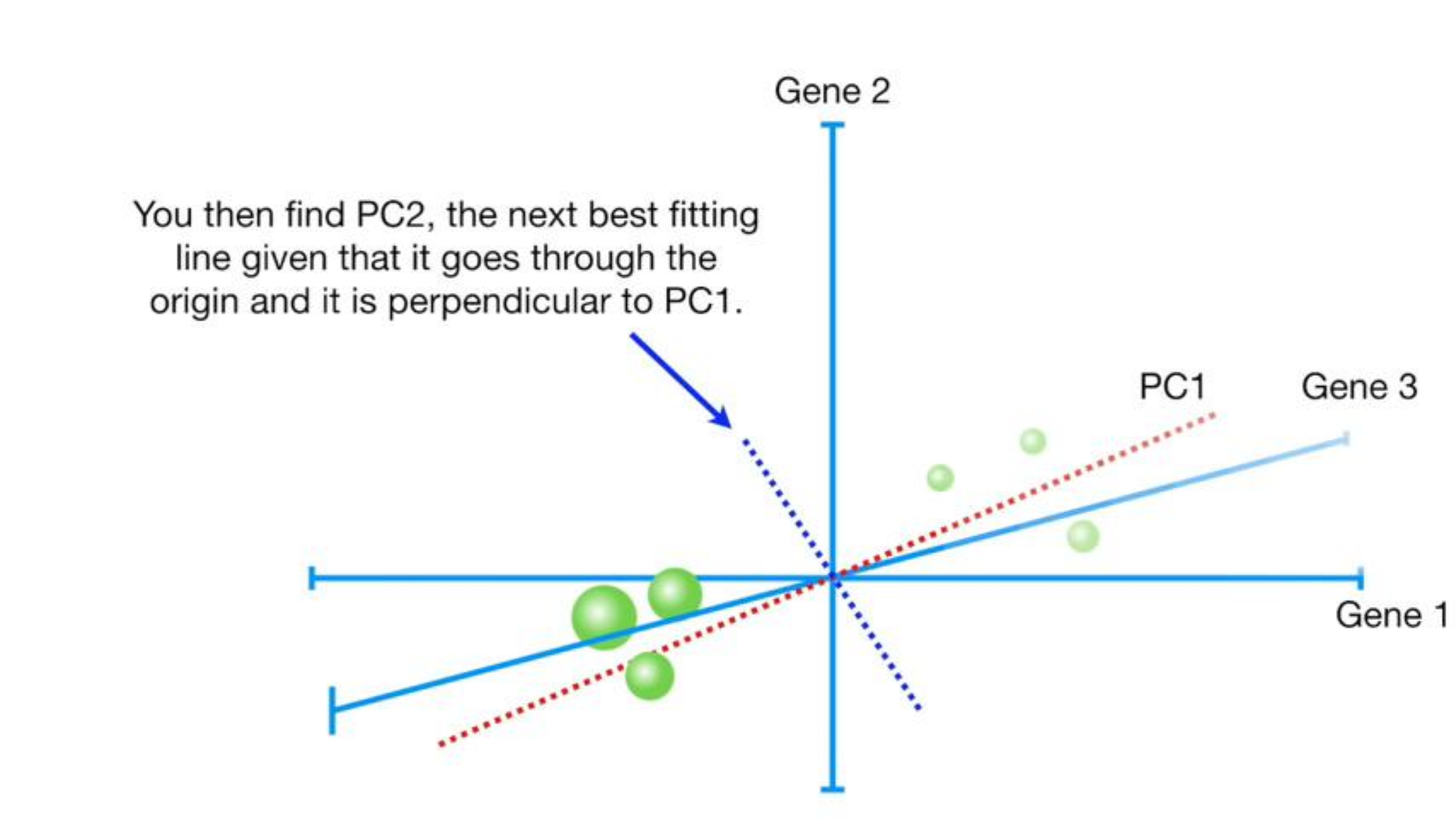

Gene 2
Just like before, the
best fitting line is PC1.
PC1
Gene 3
Gene 1

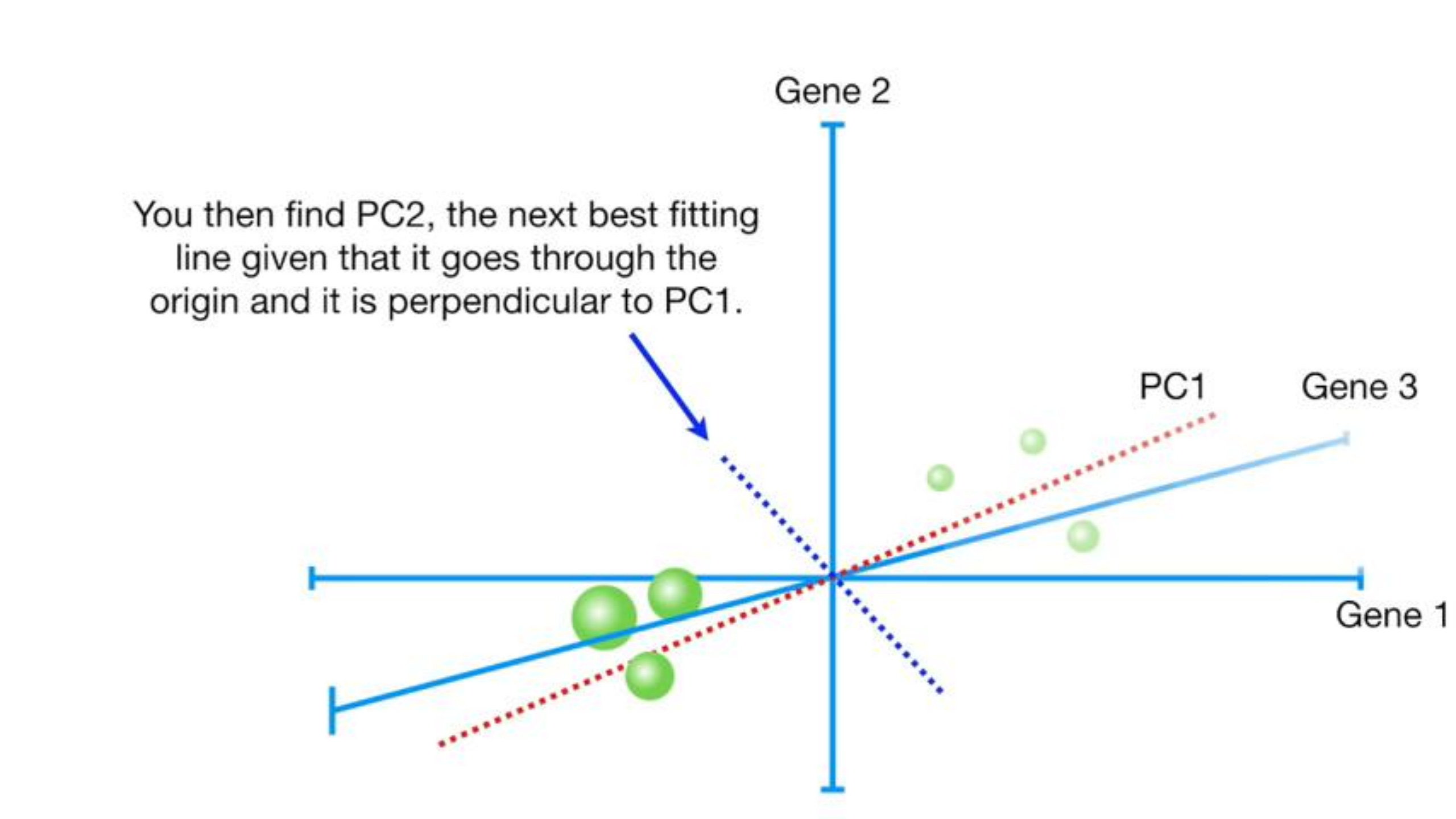
But the recipe for PC1
now has 3 ingredients…

0.62 Parts Gene 1
0.15 Parts Gene 2
0.77 Parts Gene 3



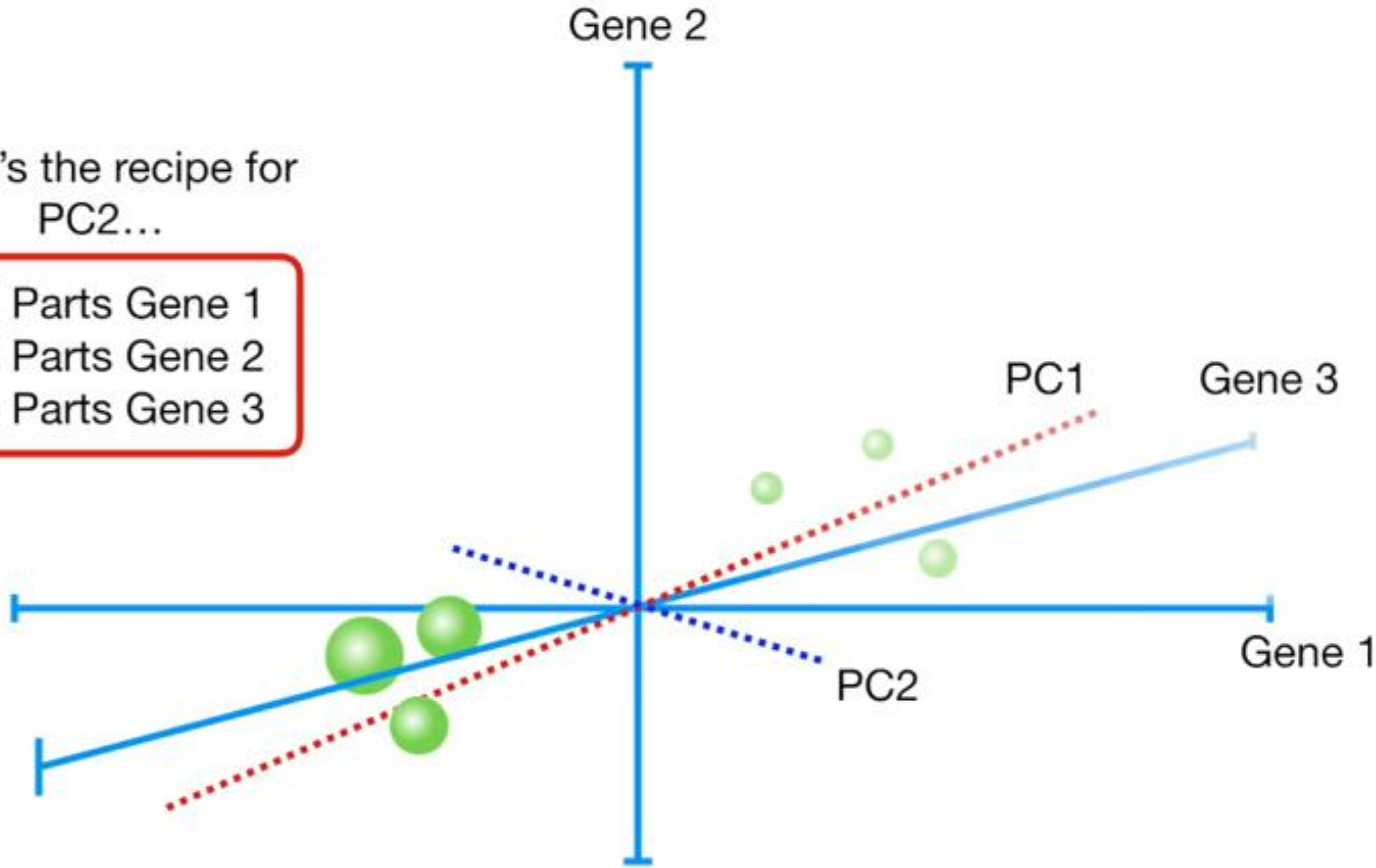

Gene 2
0.62 Parts Gene 1
0.15 Parts Gene 2
0.77 Parts Gene 3
In this case, Gene 3 is the most important ingredient for PC1.
PC1
Gene 3
Gene 1

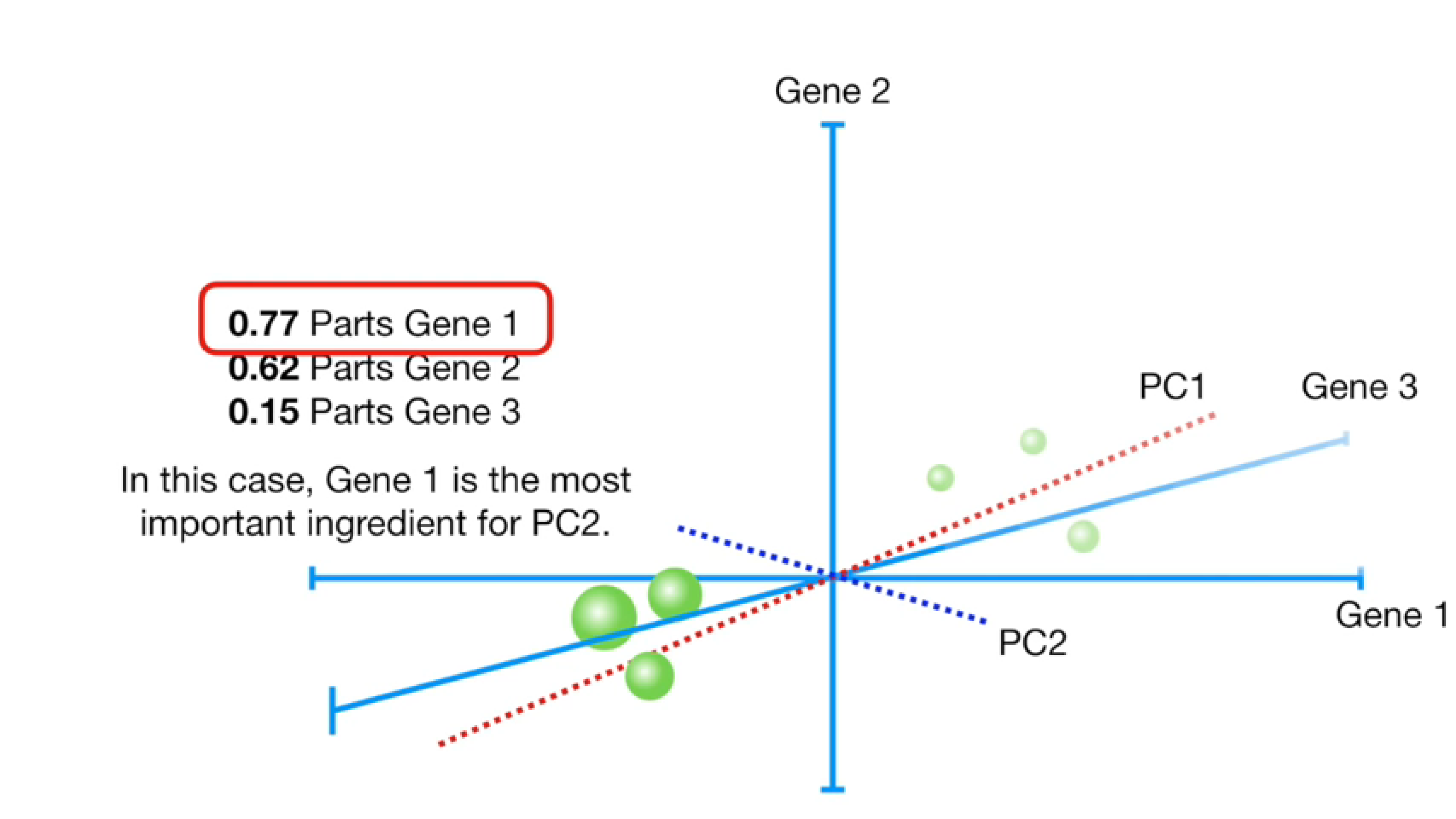

You then find PC2, the next best fitting line given that it goes through the origin and it is perpendicular to PC1.
Gene 2
PC1
Gene 3
Gene 1

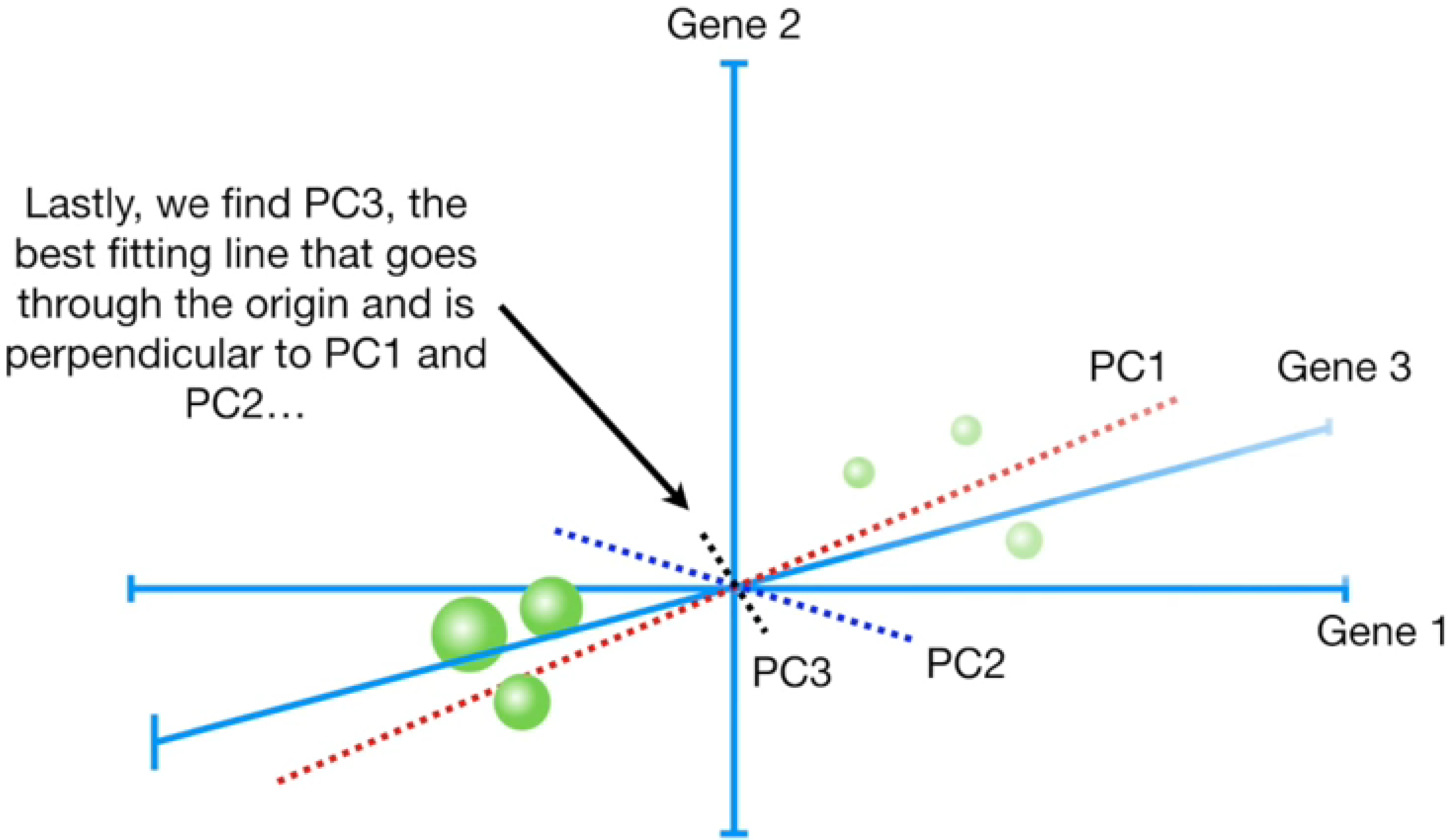

You then find PC2, the next best fitting line given that it goes through the origin and it is perpendicular to PC1.
Gene 2
PC1
Gene 3
Gene 1

Here's the recipe for PC2…

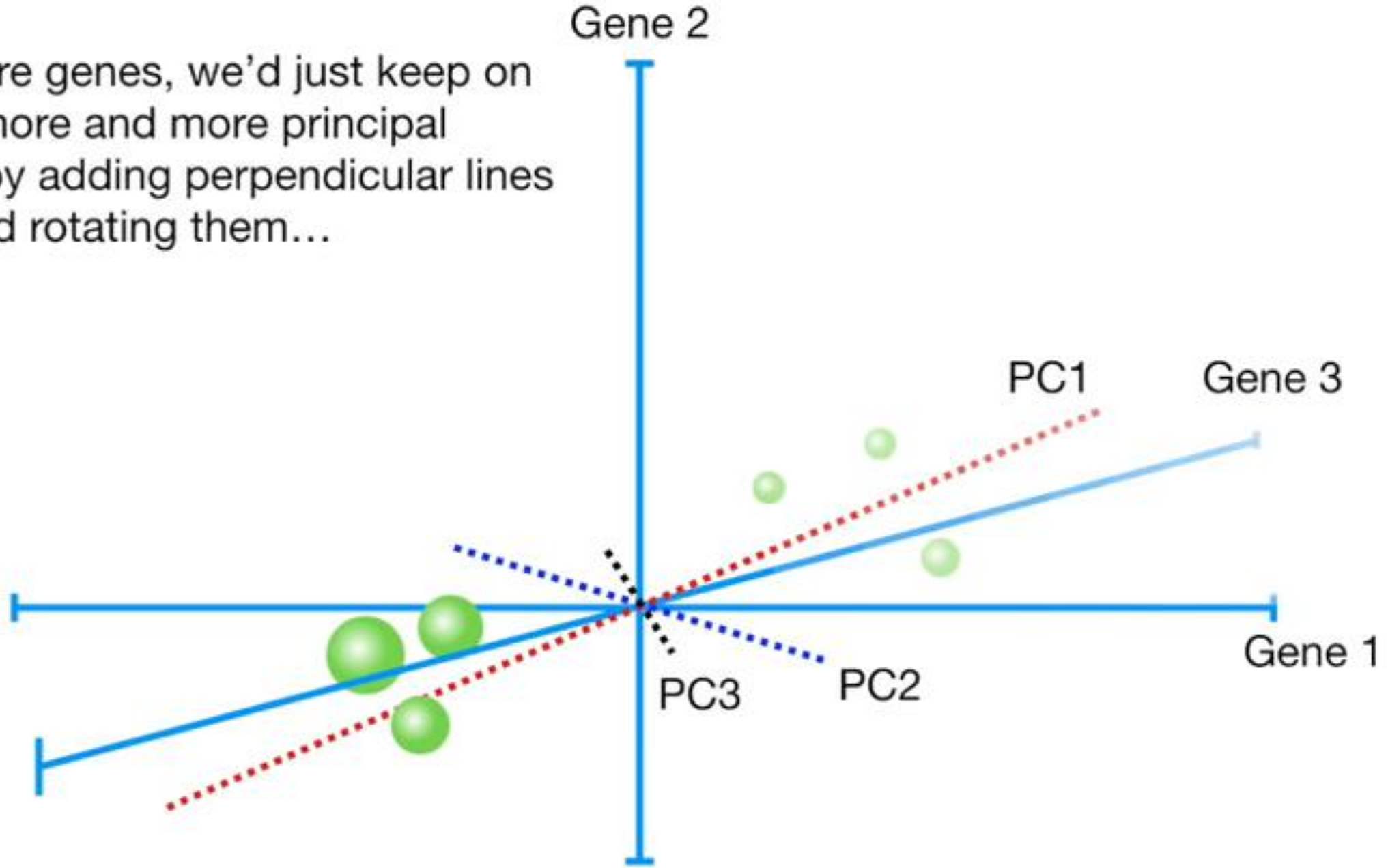
0.77 Parts Gene 1
0.62 Parts Gene 2
0.15 Parts Gene 3




Gene 2
0.77 Parts Gene 1
0.62 Parts Gene 2
0.15 Parts Gene 3
In this case, Gene 1 is the most important ingredient for PC2.
PC1
Gene 3
Gene 1
PC2

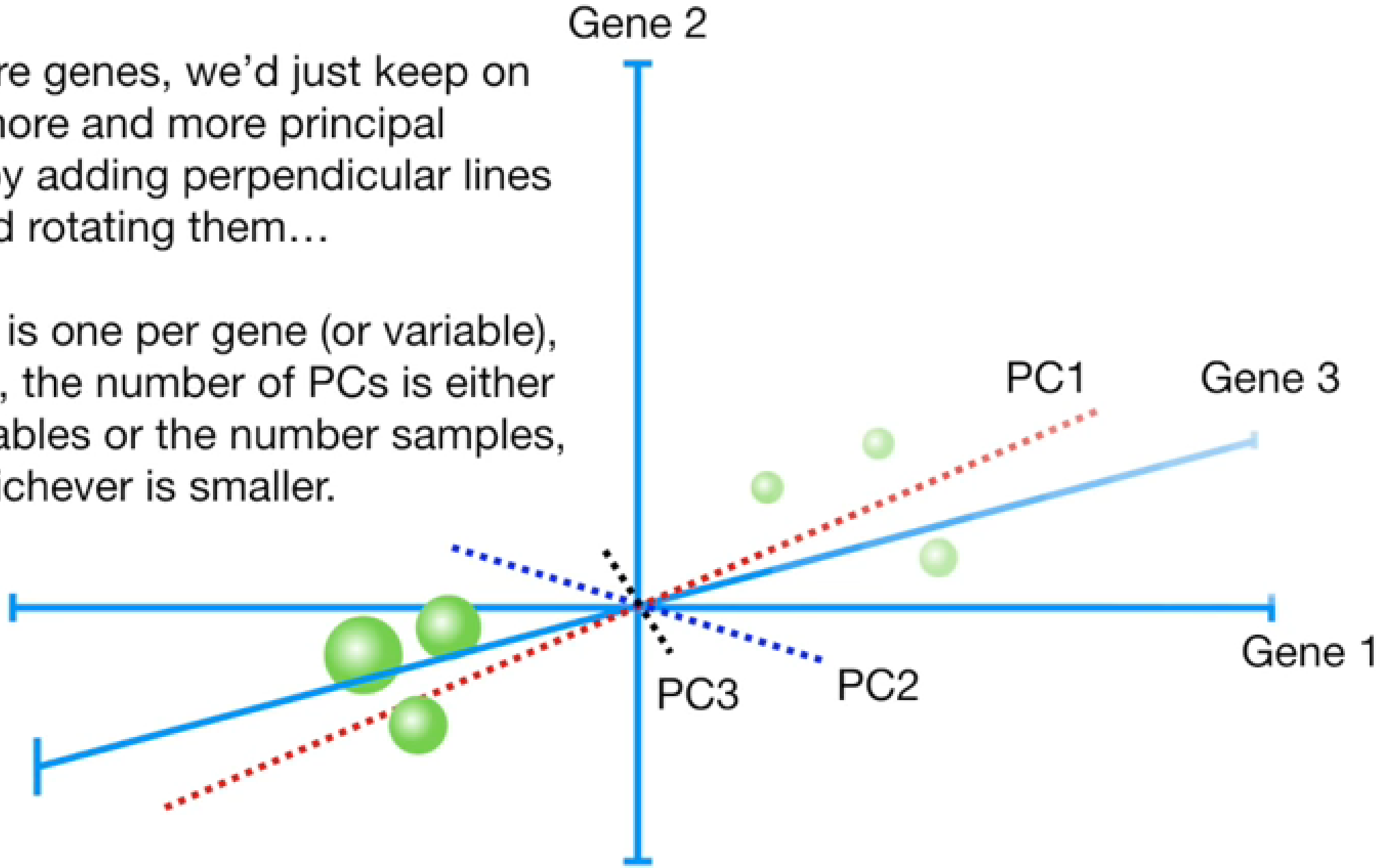

Gene 2
Lastly, we find PC3, the best fitting line that goes through the origin and is perpendicular to PC1 and PC2…
PC1
Gene 3
Gene 1
PC3
PC2

If we had more genes, we'd just keep on finding more and more principal components by adding perpendicular lines and rotating them…



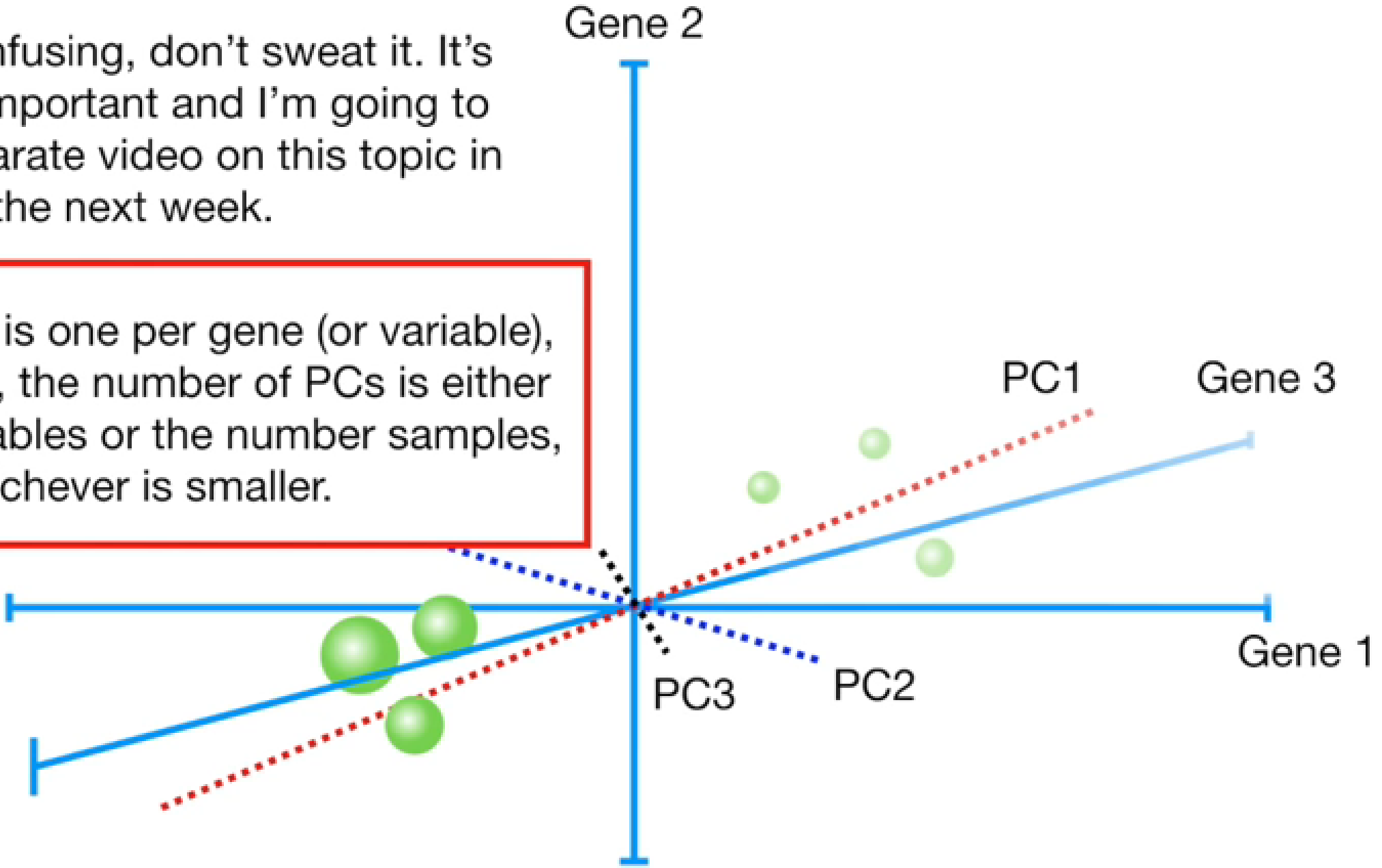
If we had more genes, we'd just keep on finding more and more principal components by adding perpendicular lines and rotating them…

In theory there is one per gene (or variable), but in practice, the number of PCs is either number of variables or the number samples, whichever is smaller.

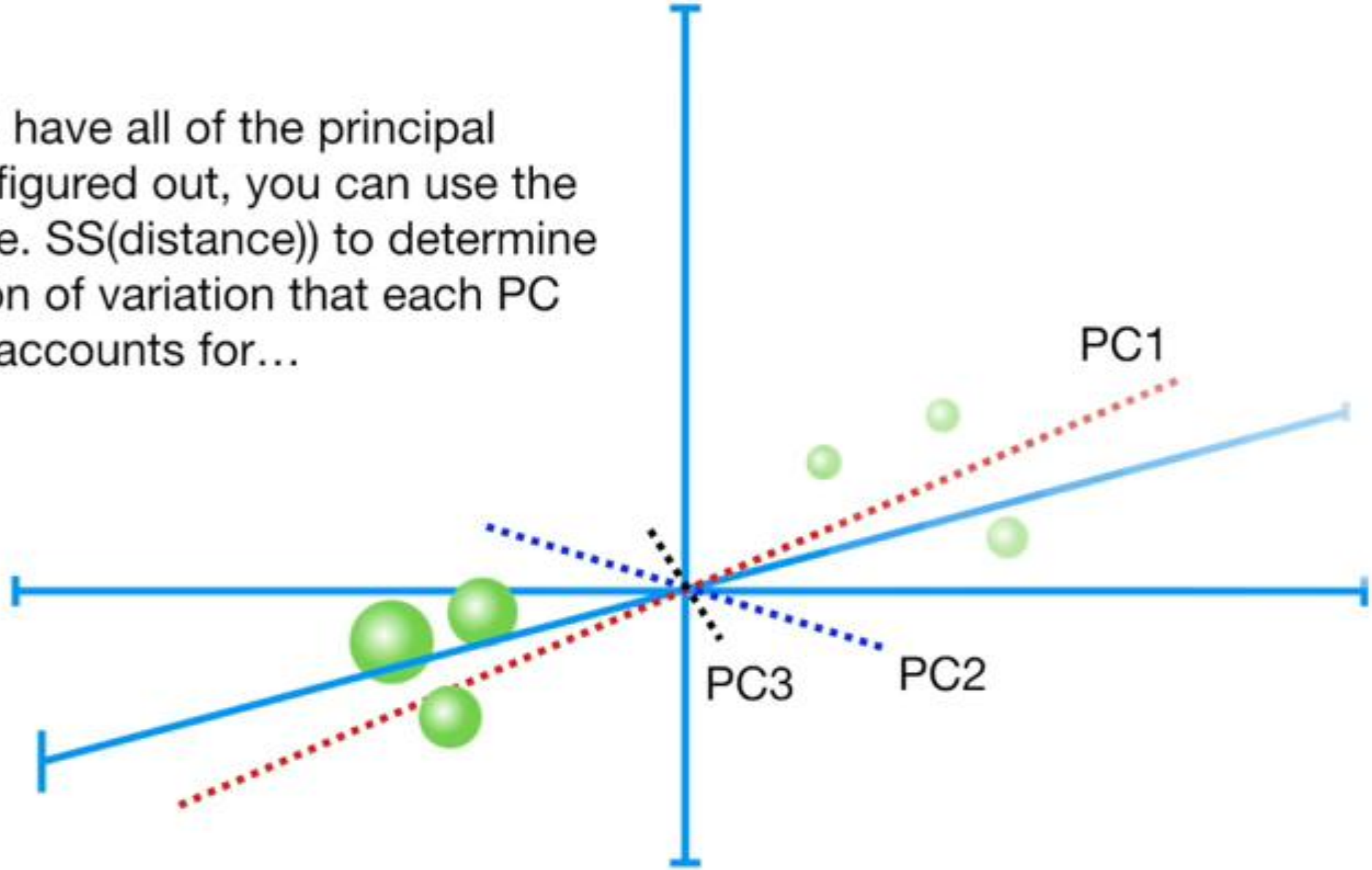


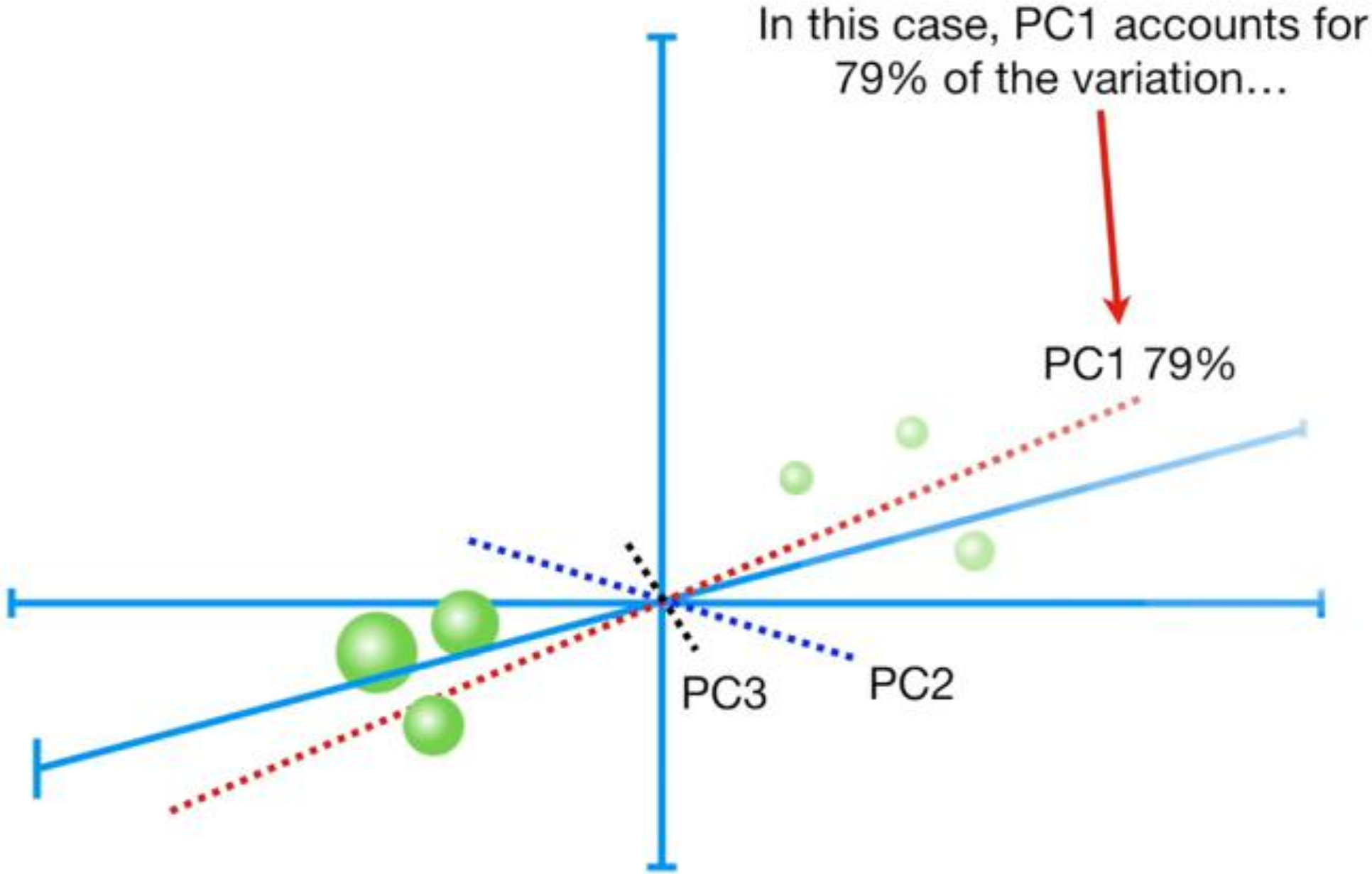
If this is confusing, don't sweat it. It's not super important and I'm going to make a separate video on this topic in the next week.

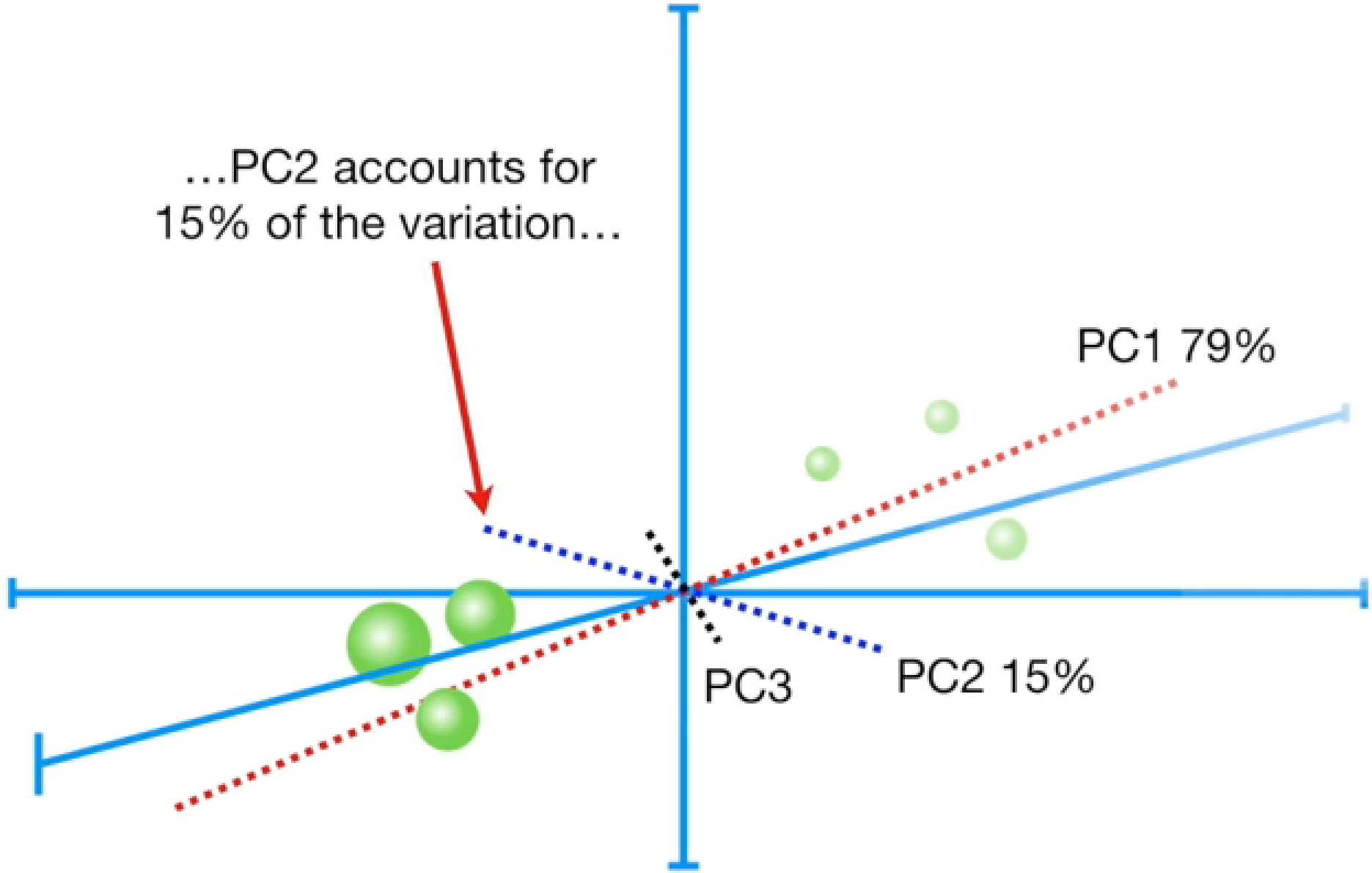
In theory there is one per gene (or variable), but in practice, the number of PCs is either number of variables or the number samples, whichever is smaller.

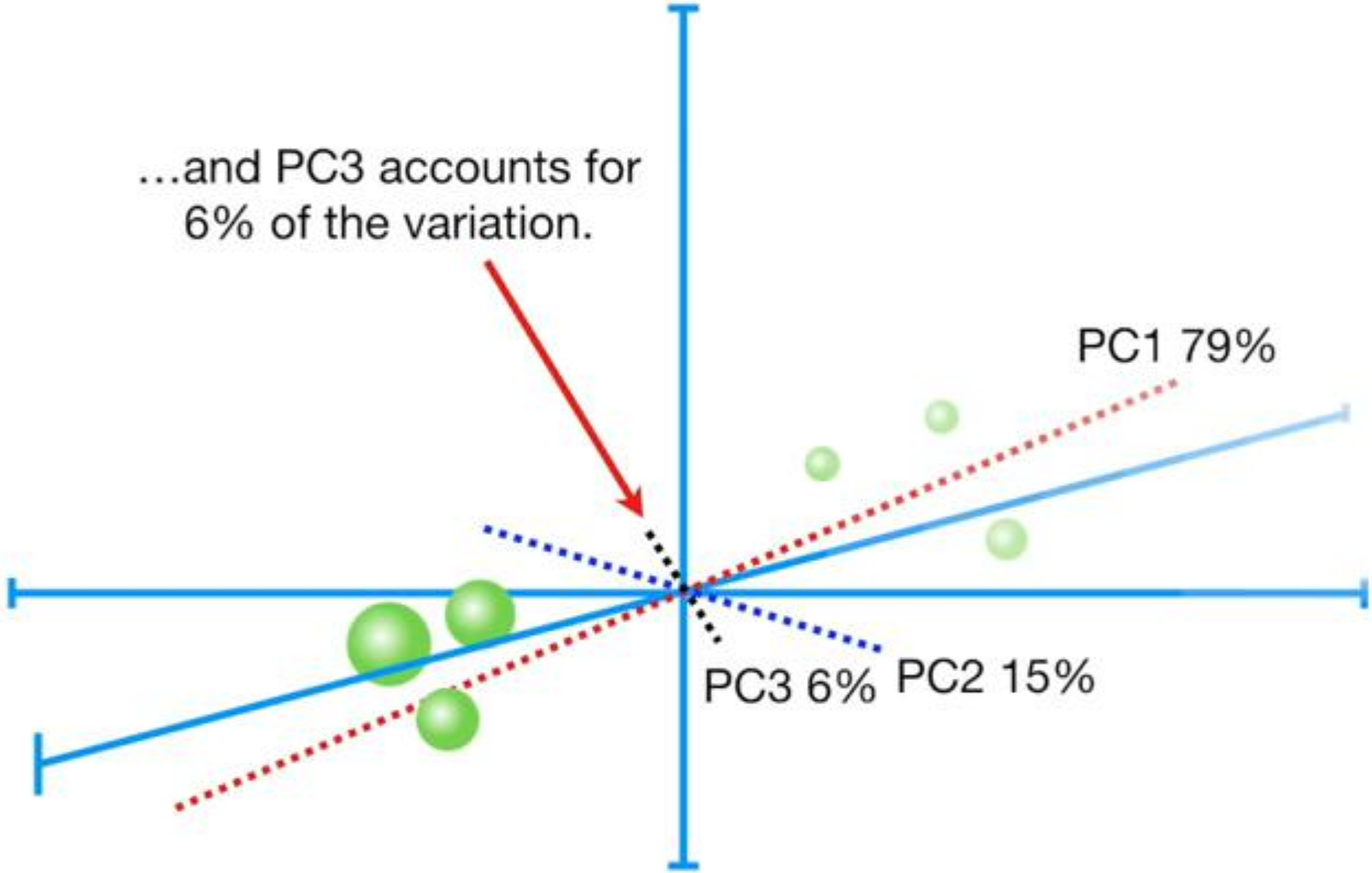


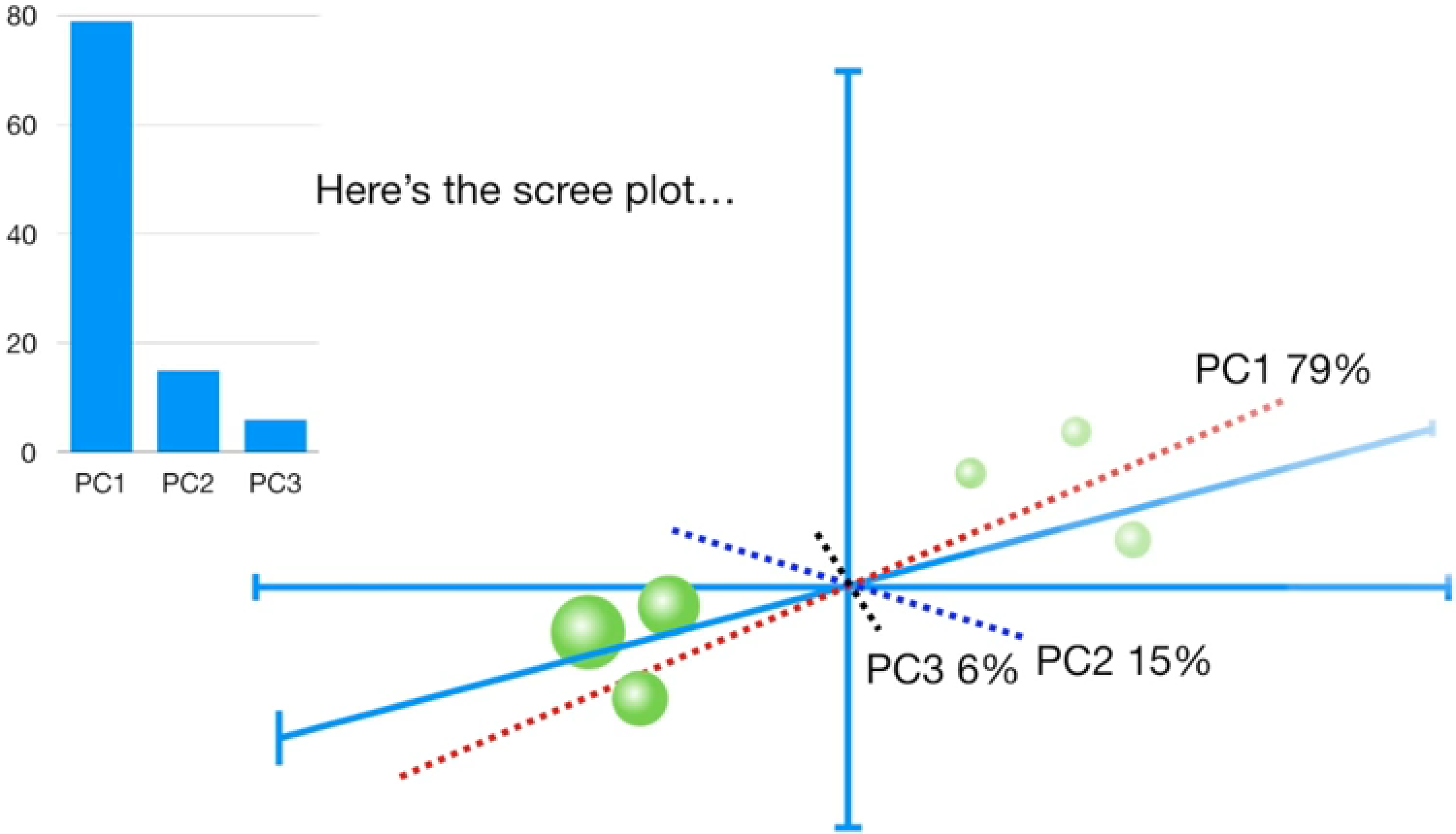
Once you have all of the principal components figured out, you can use the eigenvalues (i.e. SS(distance)) to determine the proportion of variation that each PC accounts for…

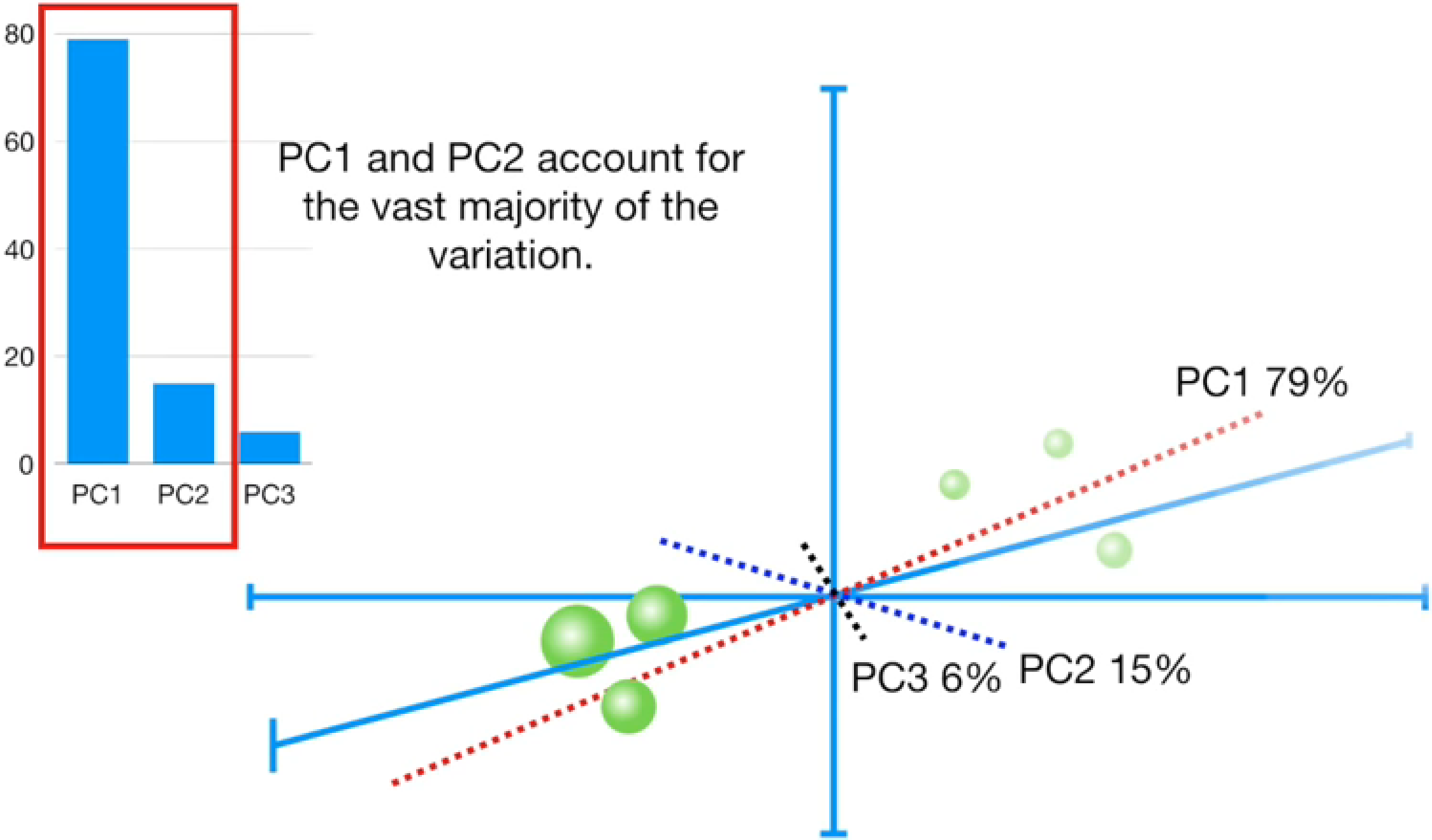


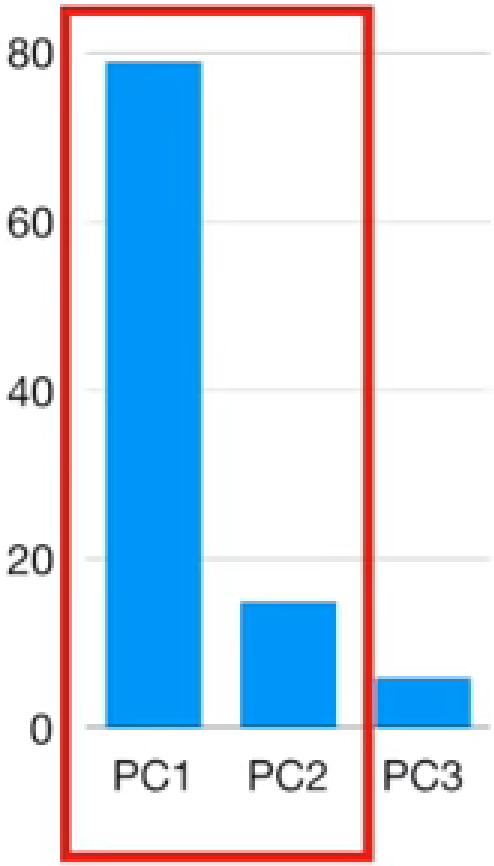

In this case, PC1 accounts for
79% of the variation…
PC1 79%
PC3
PC2


…PC2 accounts for
15% of the variation…
PC1 79%
PC3
PC2 15%

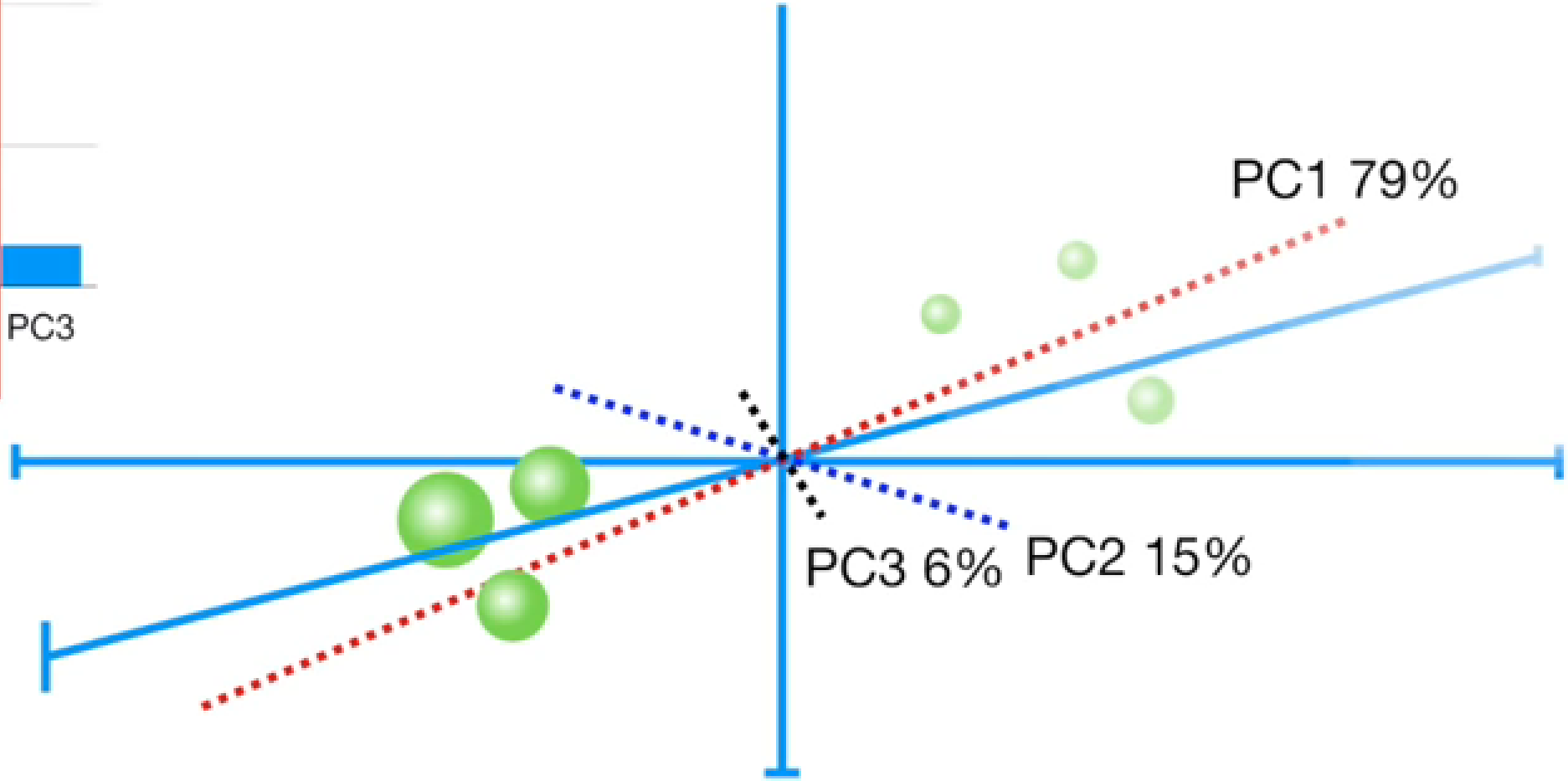

…and PC3 accounts for
6% of the variation.
PC1 79%
PC3 6%
PC2 15%


80
60
40
20
0
PC1
PC2
PC3
Here's the scree plot…
PC1 79%
PC3 6%
PC2 15%

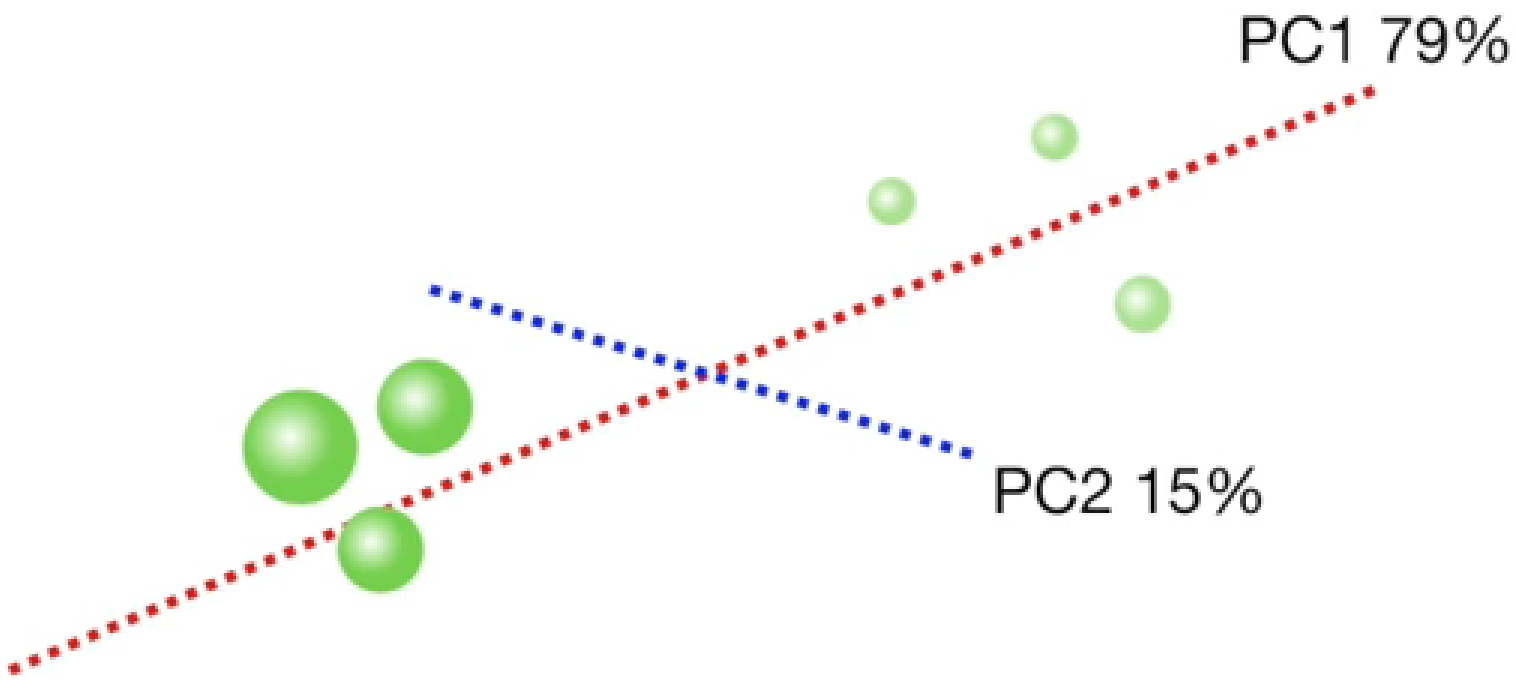

80
60
40
20
0
PC1
PC2
PC3
PC1 and PC2 account for the vast majority of the variation.
PC1 79%
PC3 6%
PC2 15%



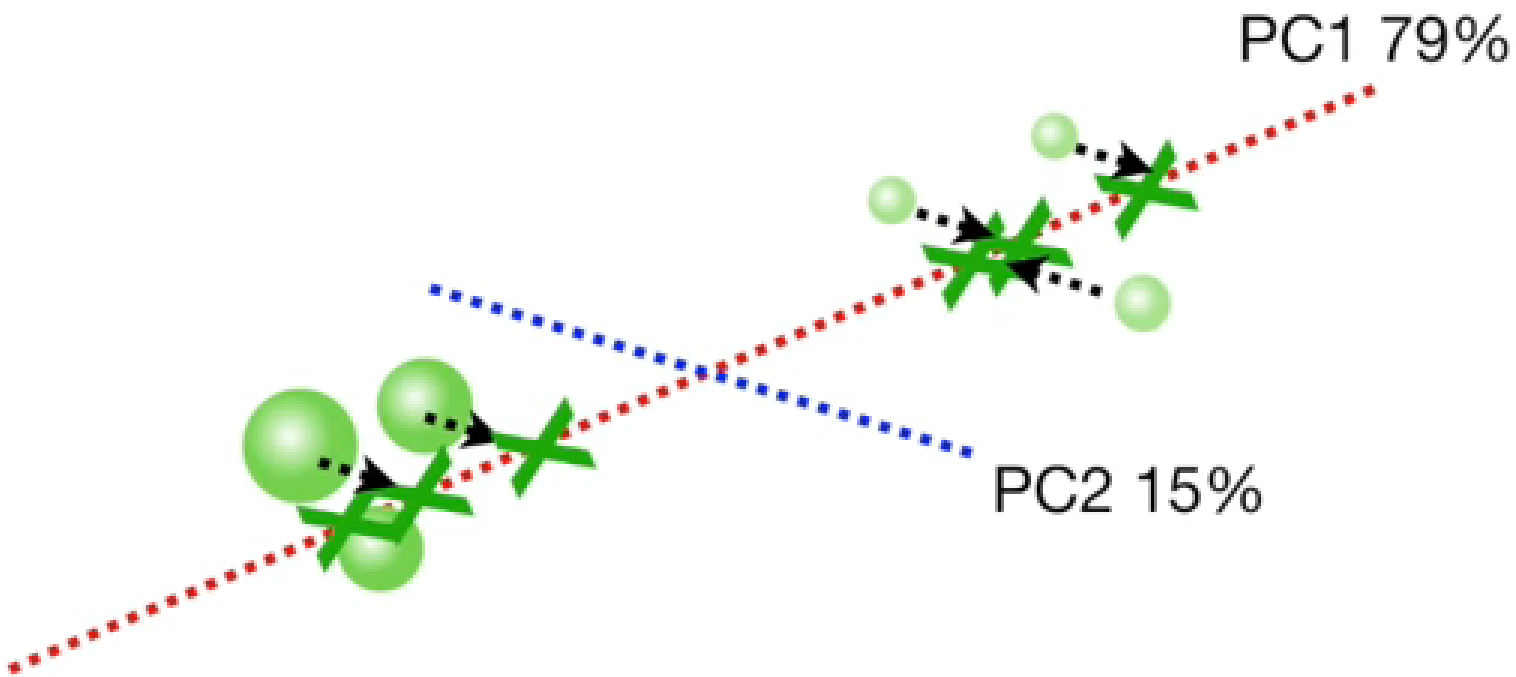
That means that a 2-D graph, using just PC1 and PC2, would be a good approximation of this 3-D graph since it would account for 94% of the variation in the data.



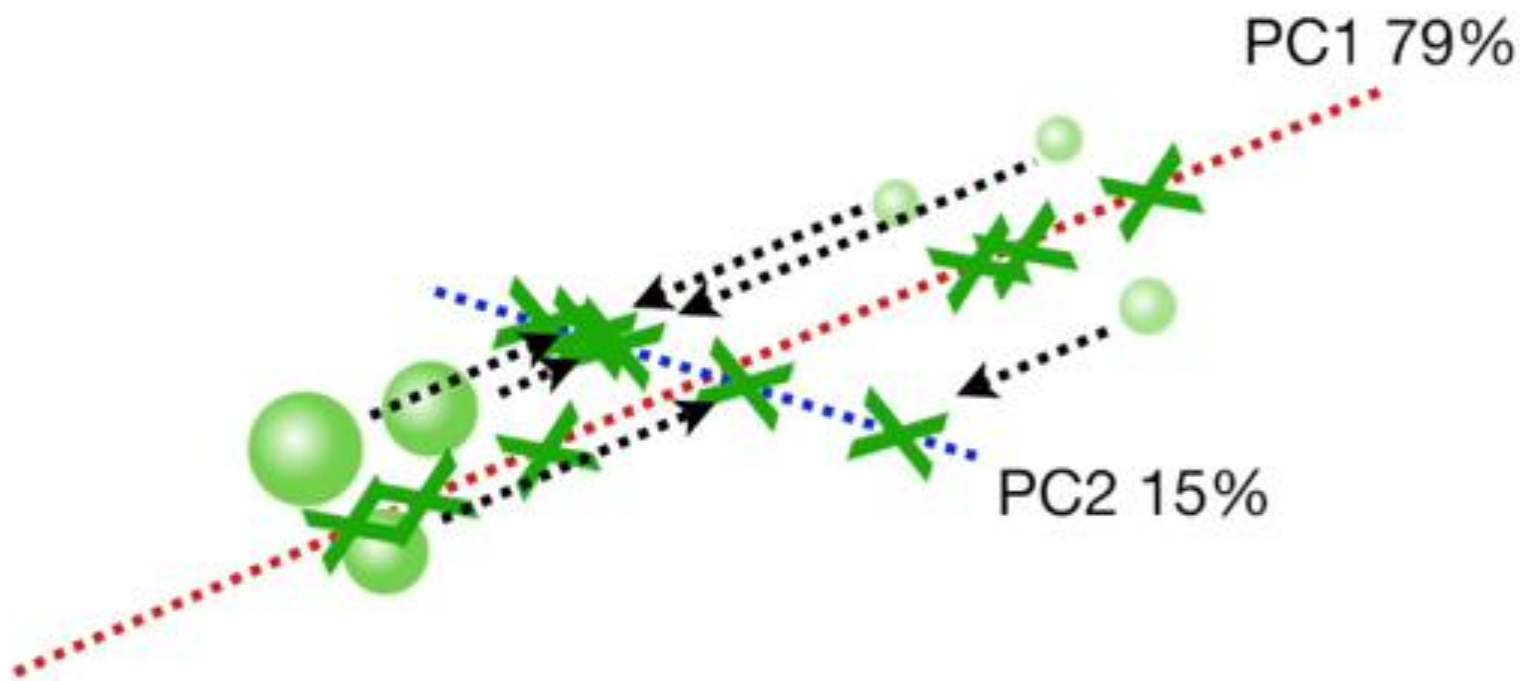
To convert the 3-D graph into a 2-D PCA graph, we just strip away everything but the data and PC1 and PC2…



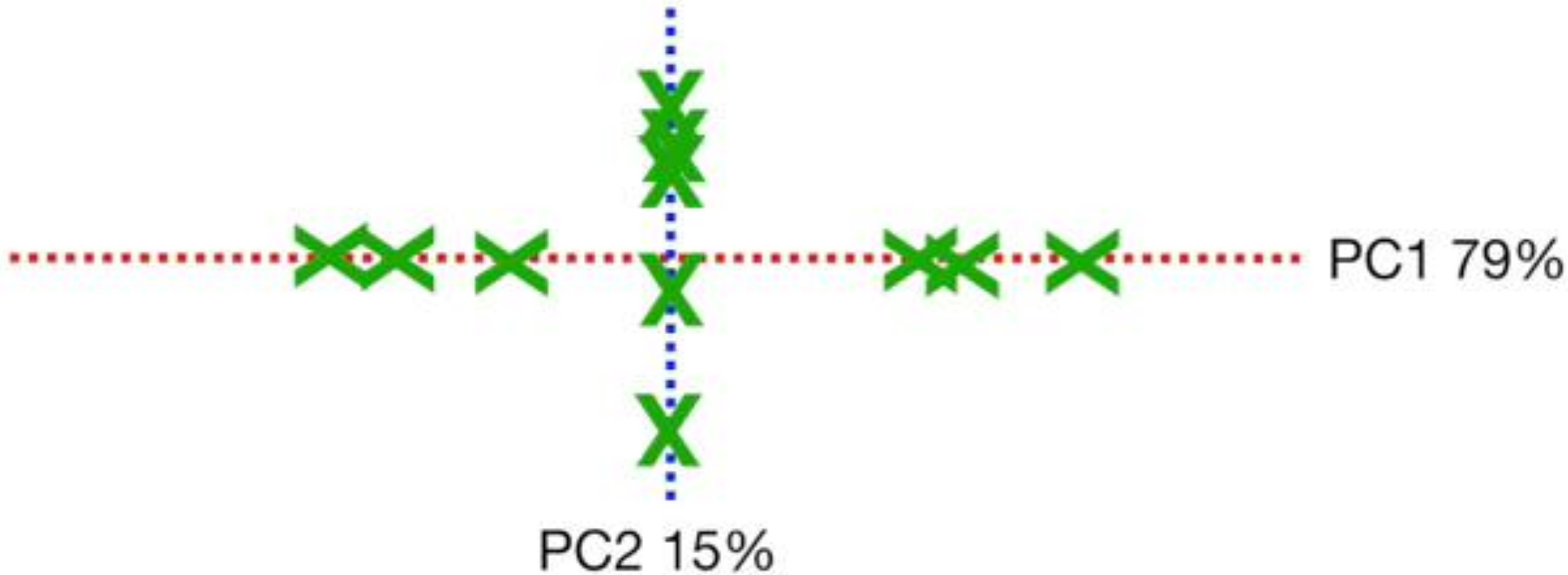
Then project the samples onto PC1…

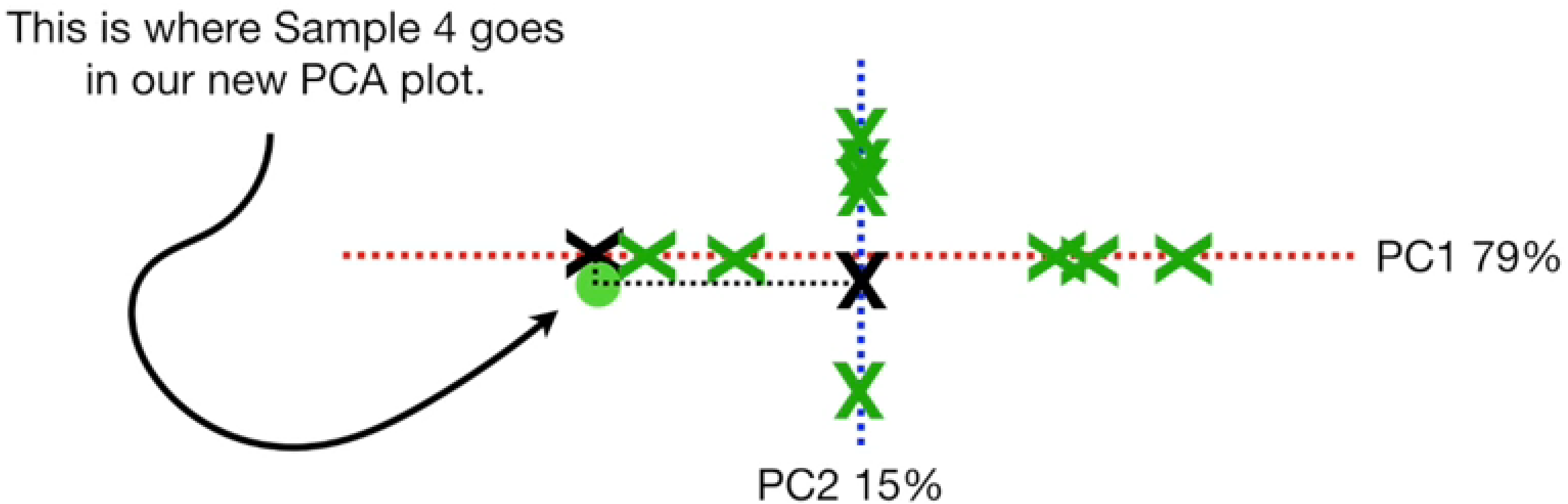


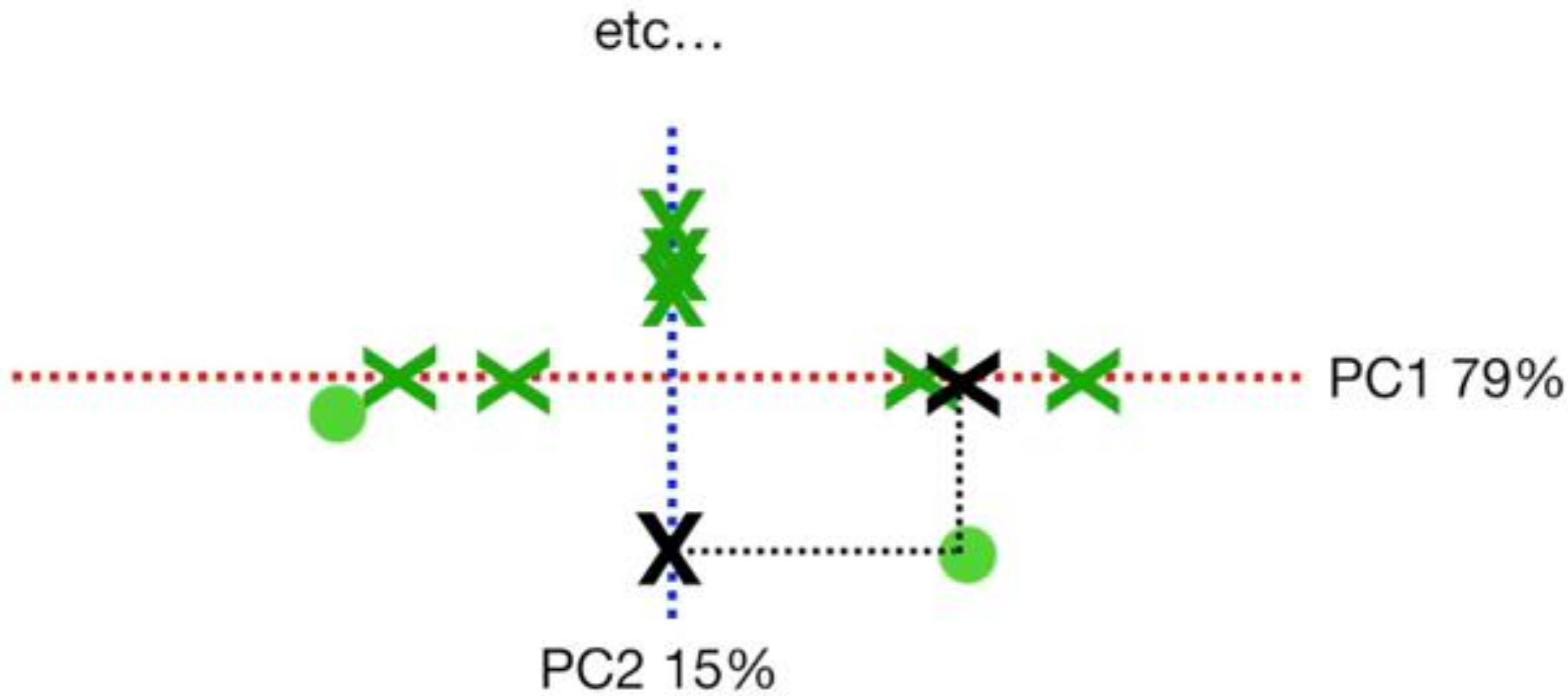
…and PC2.

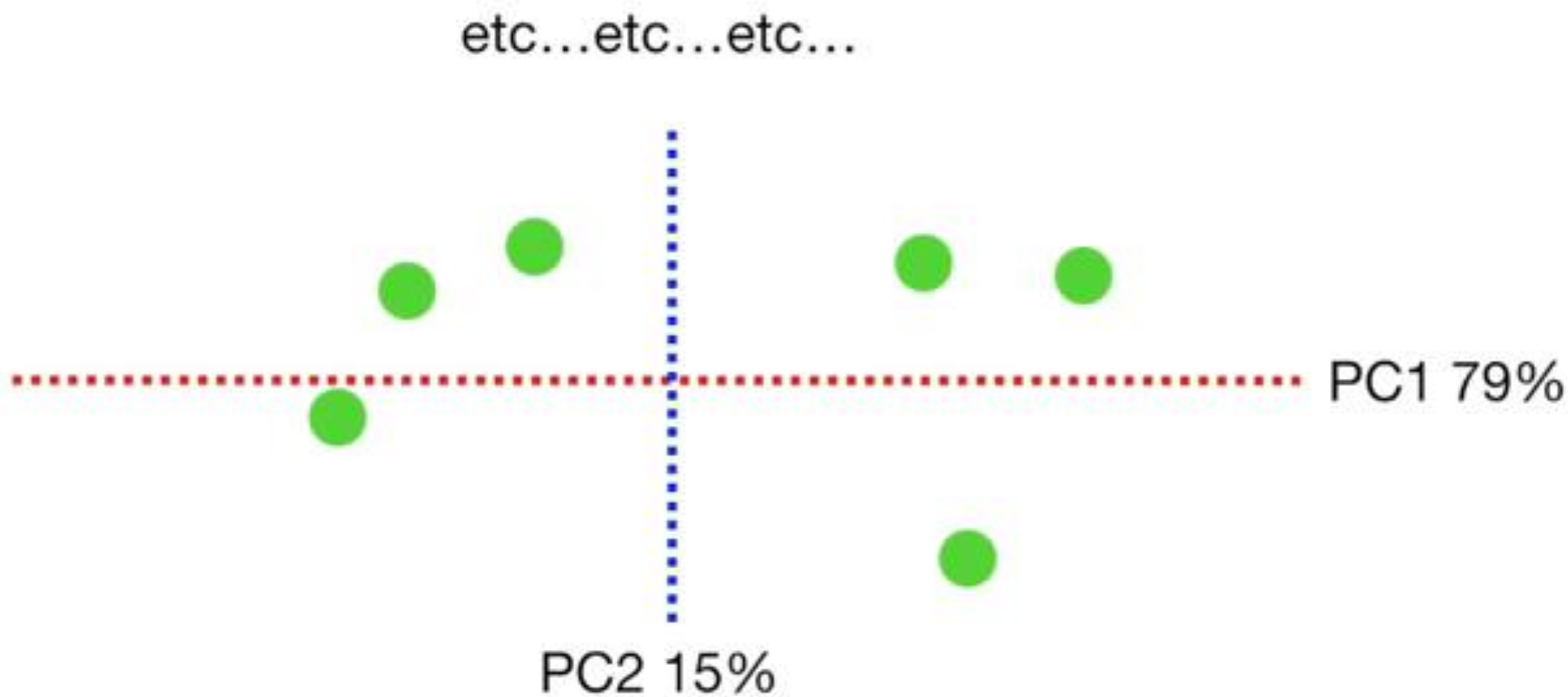


Then we rotate so that PC1 is horizontal and PC2 is vertical (this just makes it easier to look at).

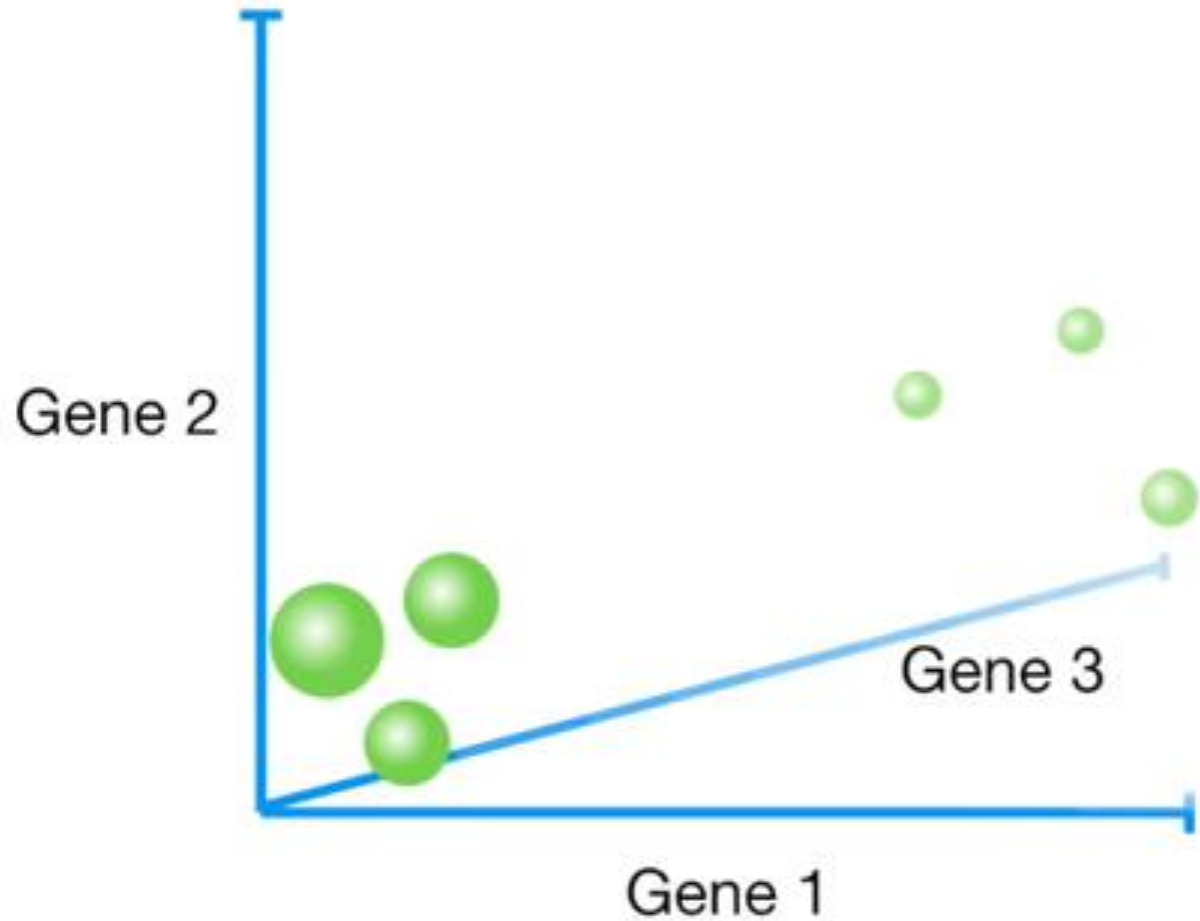



This is where Sample 4 goes
in our new PCA plot.
PC1 79%
PC2 15%

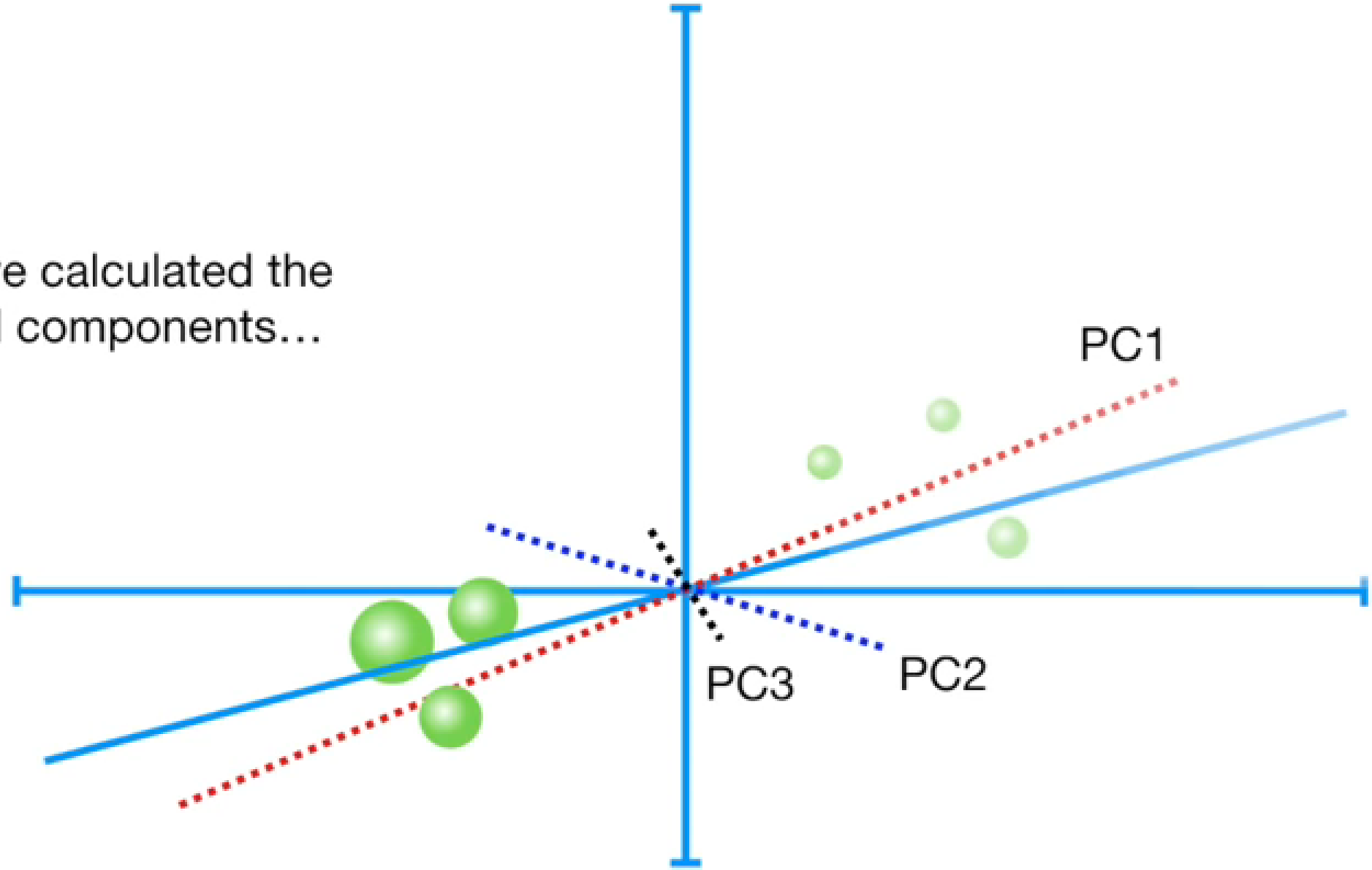

etc…
PC1 79%
PC2 15%

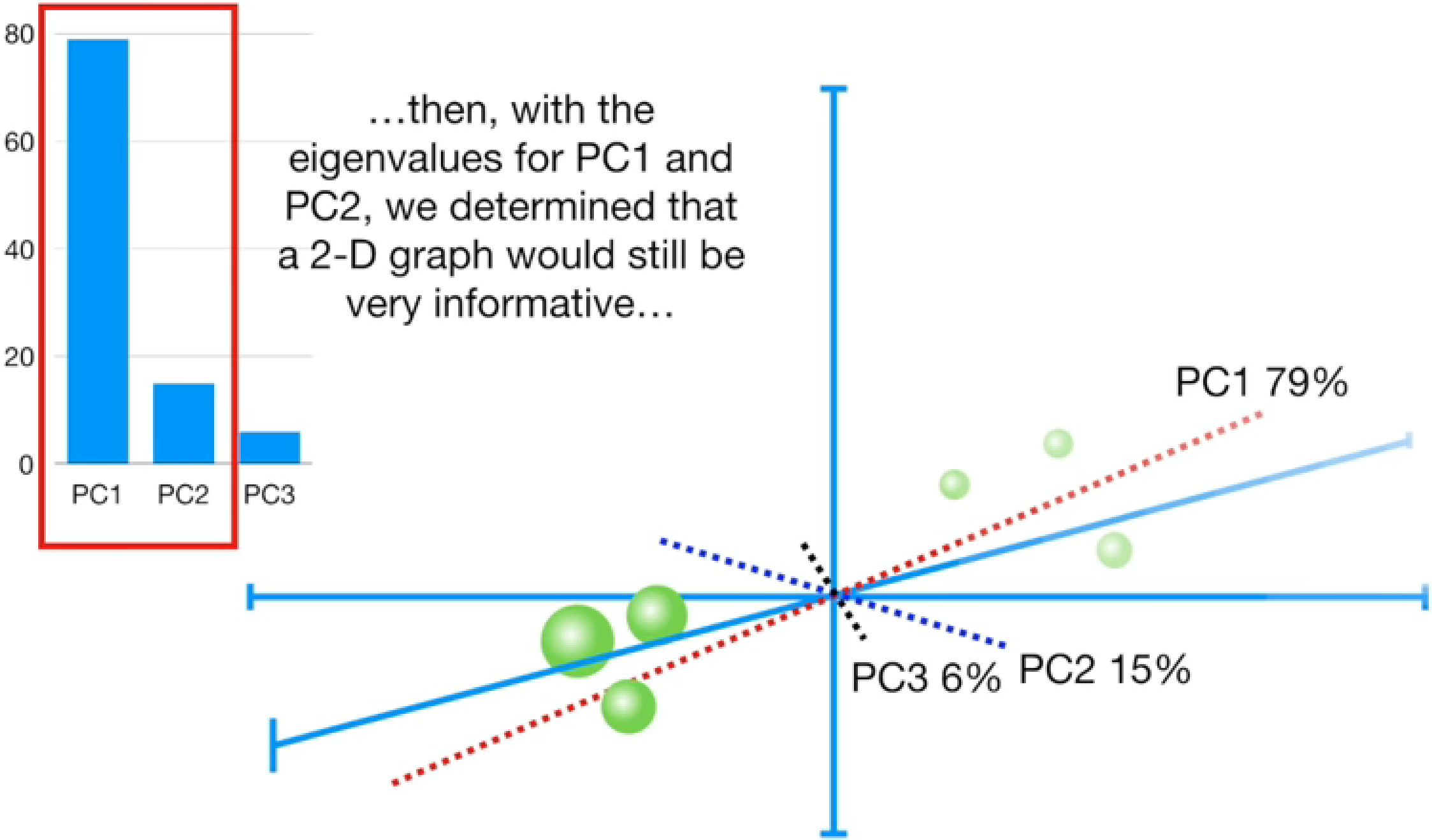

etc…etc…etc…
PC1 79%
PC2 15%

To review, we started with an awkward 3-D graph that was kind of hard to read…

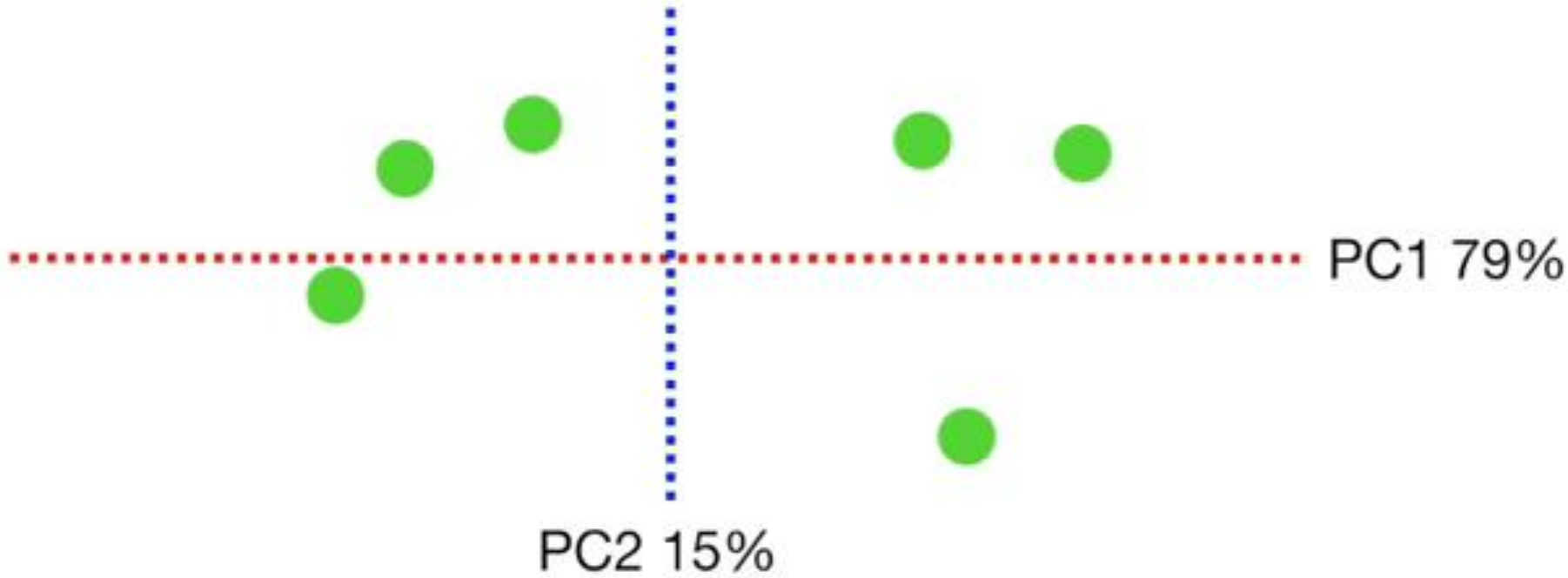


…then we calculated the principal components…



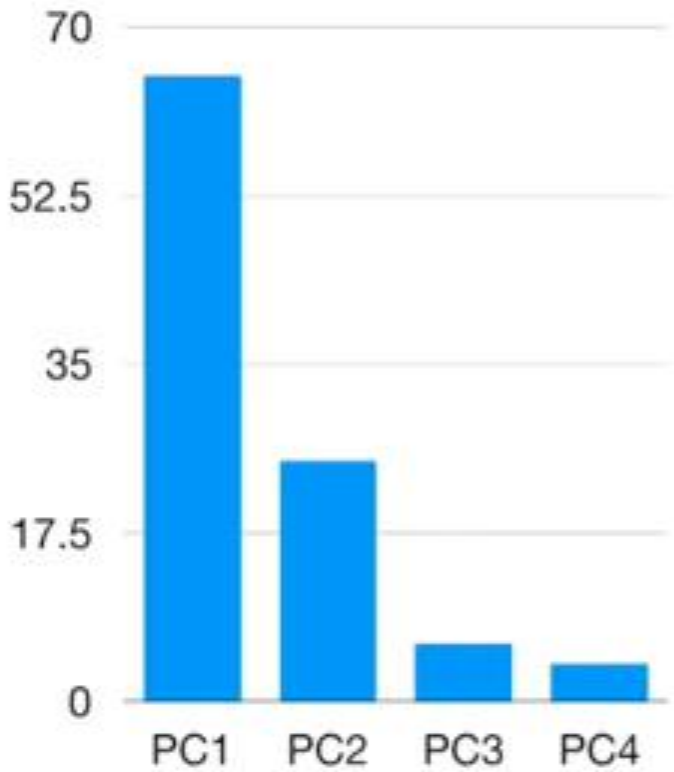

80
60
40
20
0
PC1
PC2
PC3
…then, with the eigenvalues for PC1 and PC2, we determined that a 2-D graph would still be very informative…
PC1 79%
PC3 6%
PC2 15%

…lastly, we used PC1 and PC2 to draw a 2-Dimensional graph with the data.

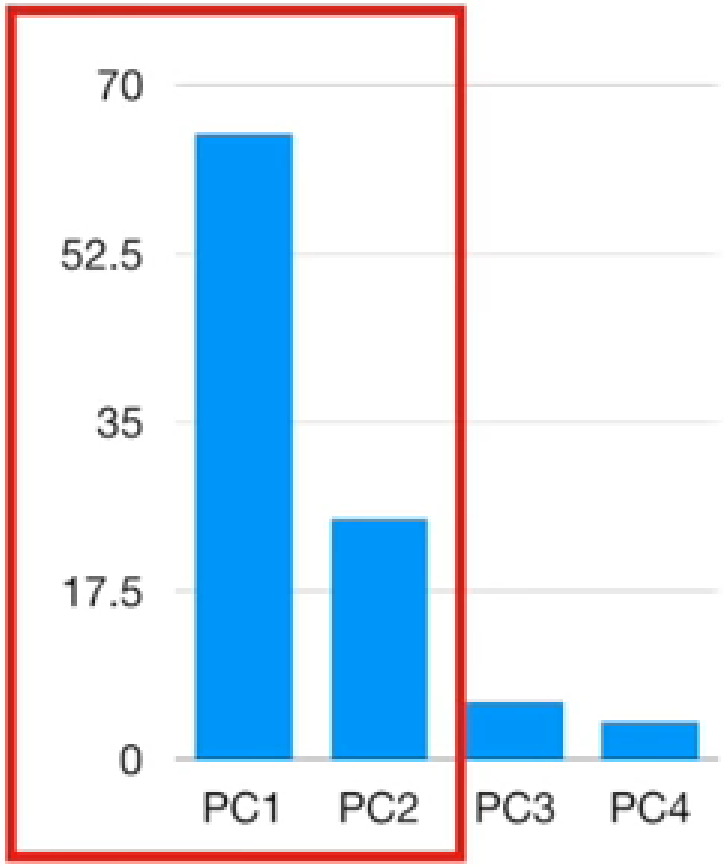


	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	2	1
Gene 2	6	4	5	3	2.8	1
Gene 3	12	9	10	2.5	1.3	2
Gene 4	5	20	6	2	18	19

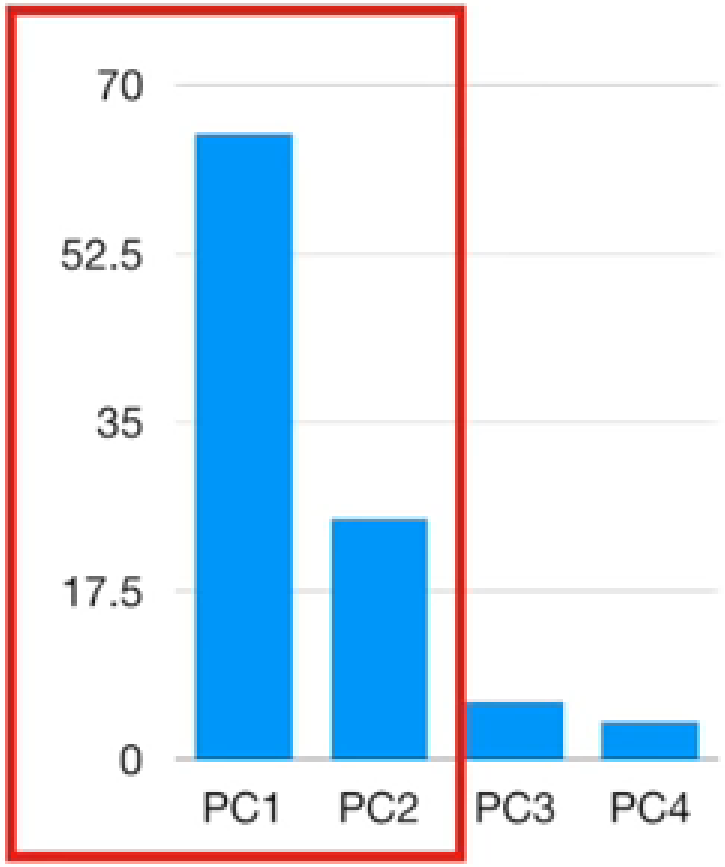
If we measured 4 genes per mouse, we would not be able to draw a 4-dimensional graph of the data…



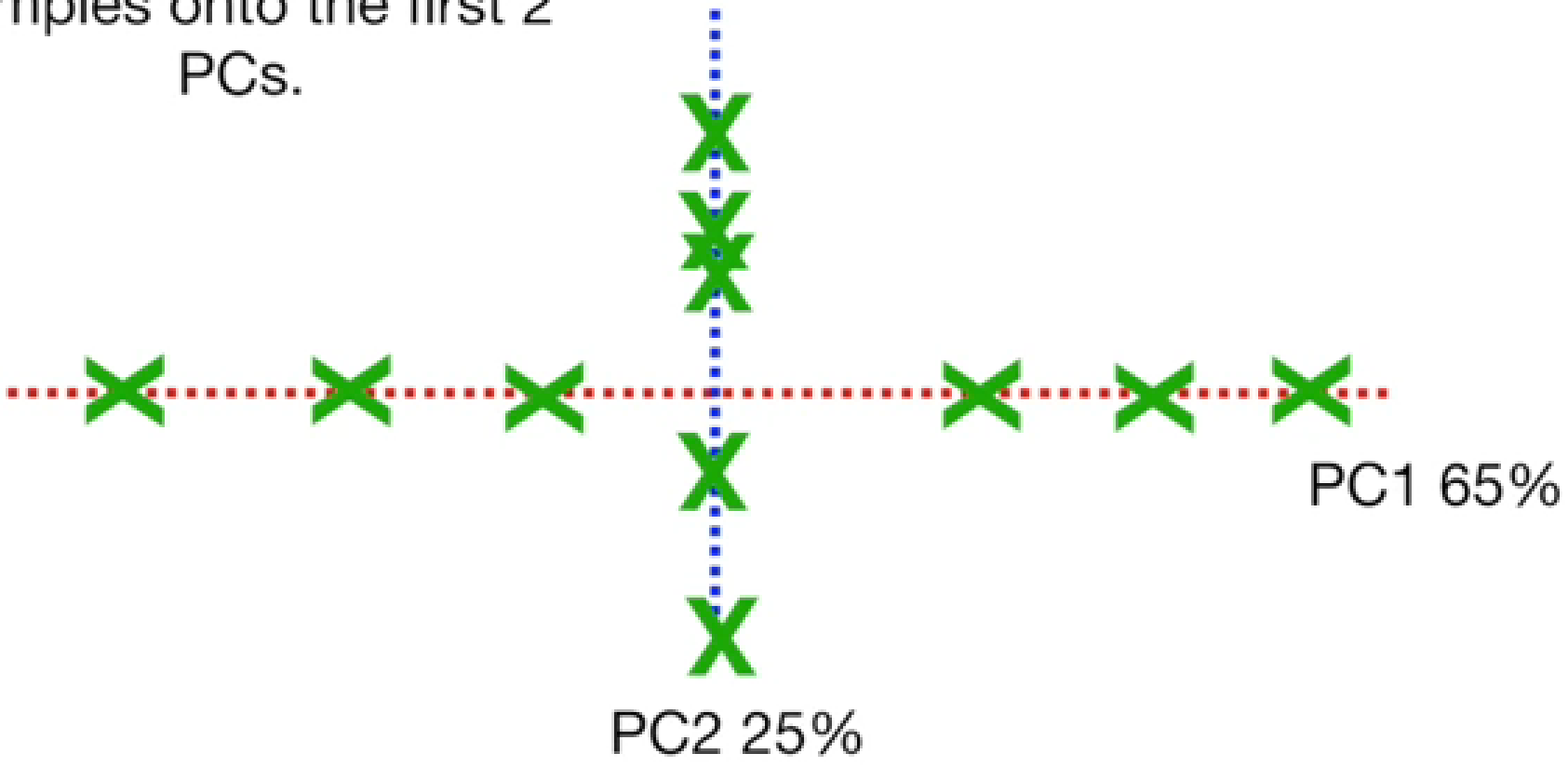
…but that doesn't stop us from doing the PCA math (which doesn't care if we can draw a picture of it or not) and looking at the scree plot…

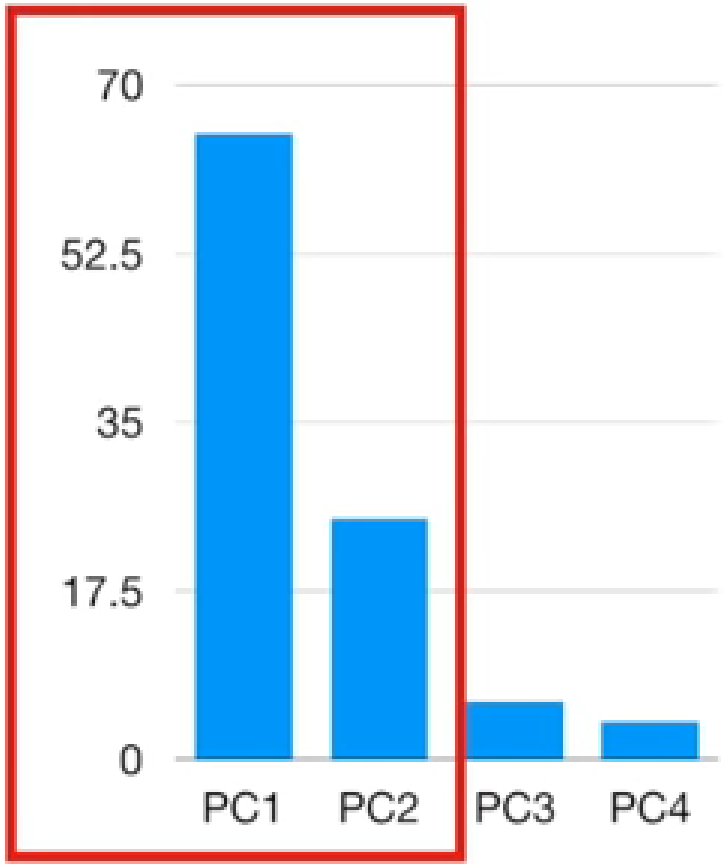


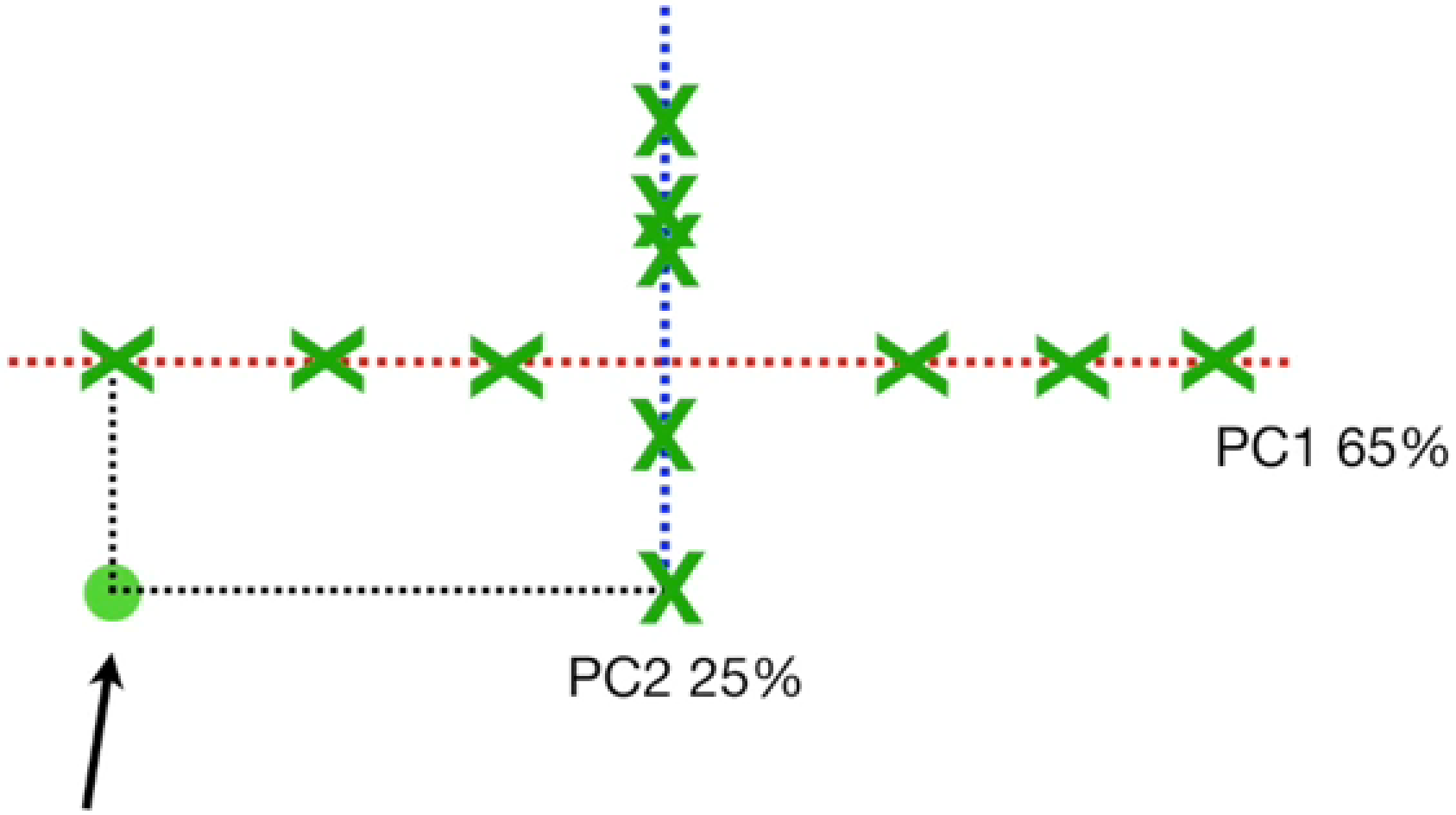
…in this case, PC1 and PC2 account for 90% of the variation, so we can just use those to draw a 2-dimensional PCA graph.



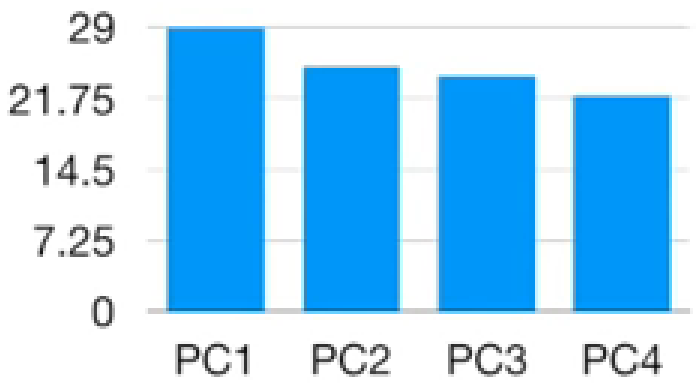
So we project the samples onto the first 2 PCs.





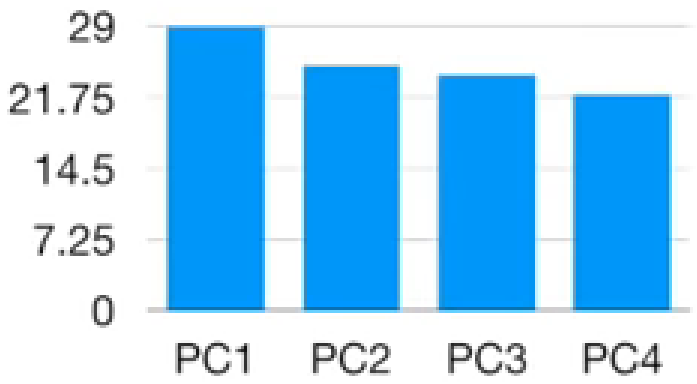


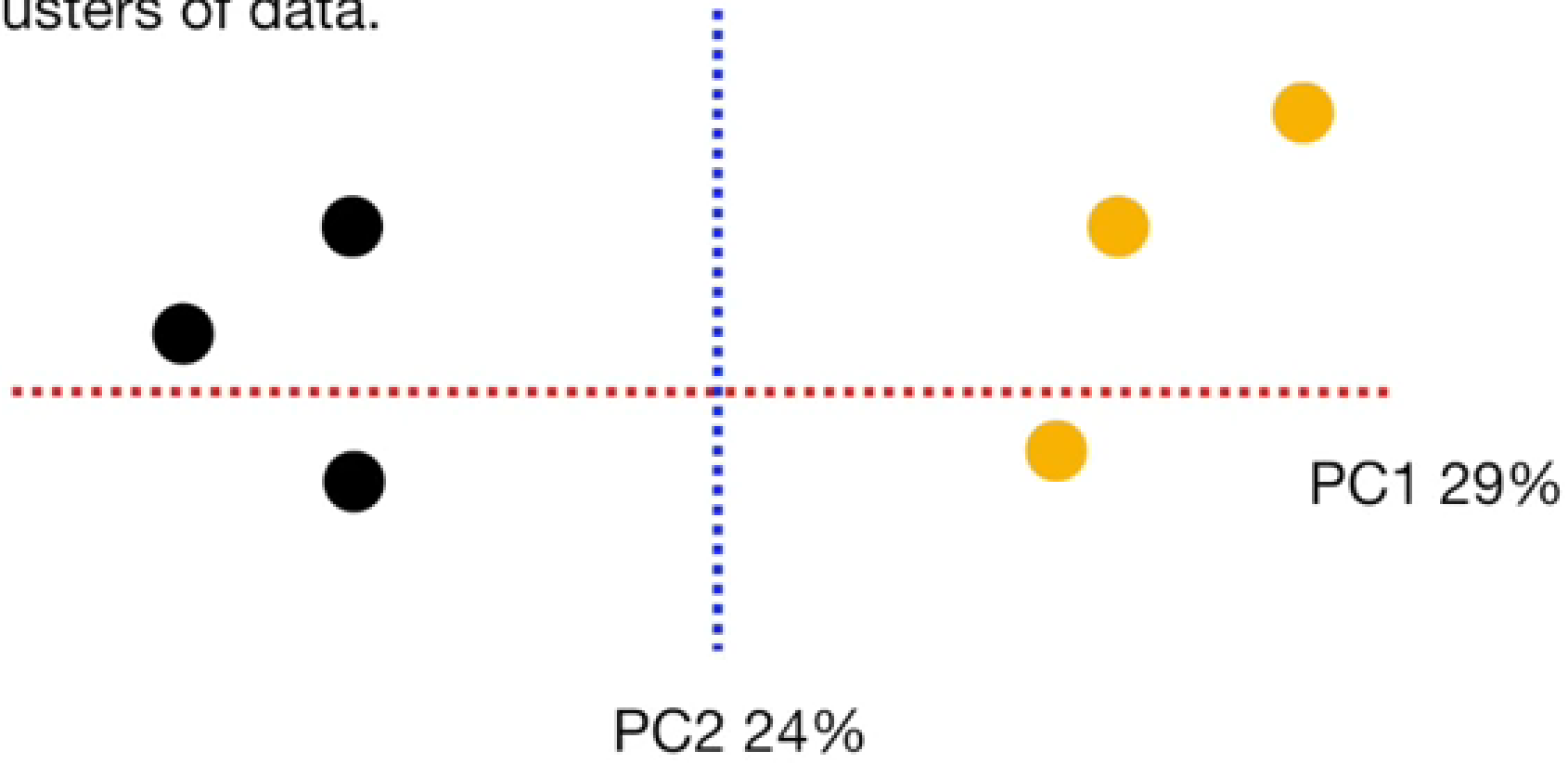
So Sample 2 goes here.



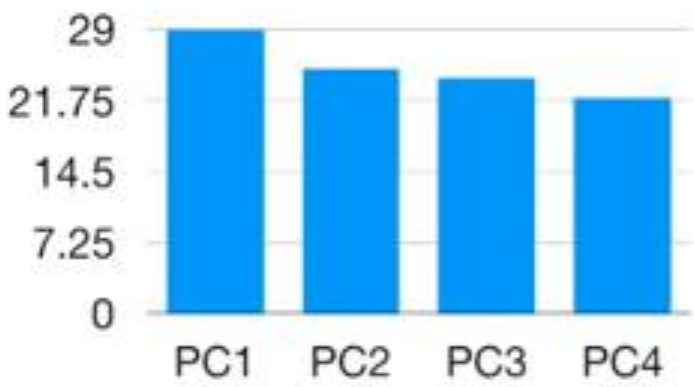
NOTE: If the scree plot looked like this, where PC3 and PC4 account for a substantial amount of variation, then just using the first 2 PCs would not create a very accurate representation of the data.

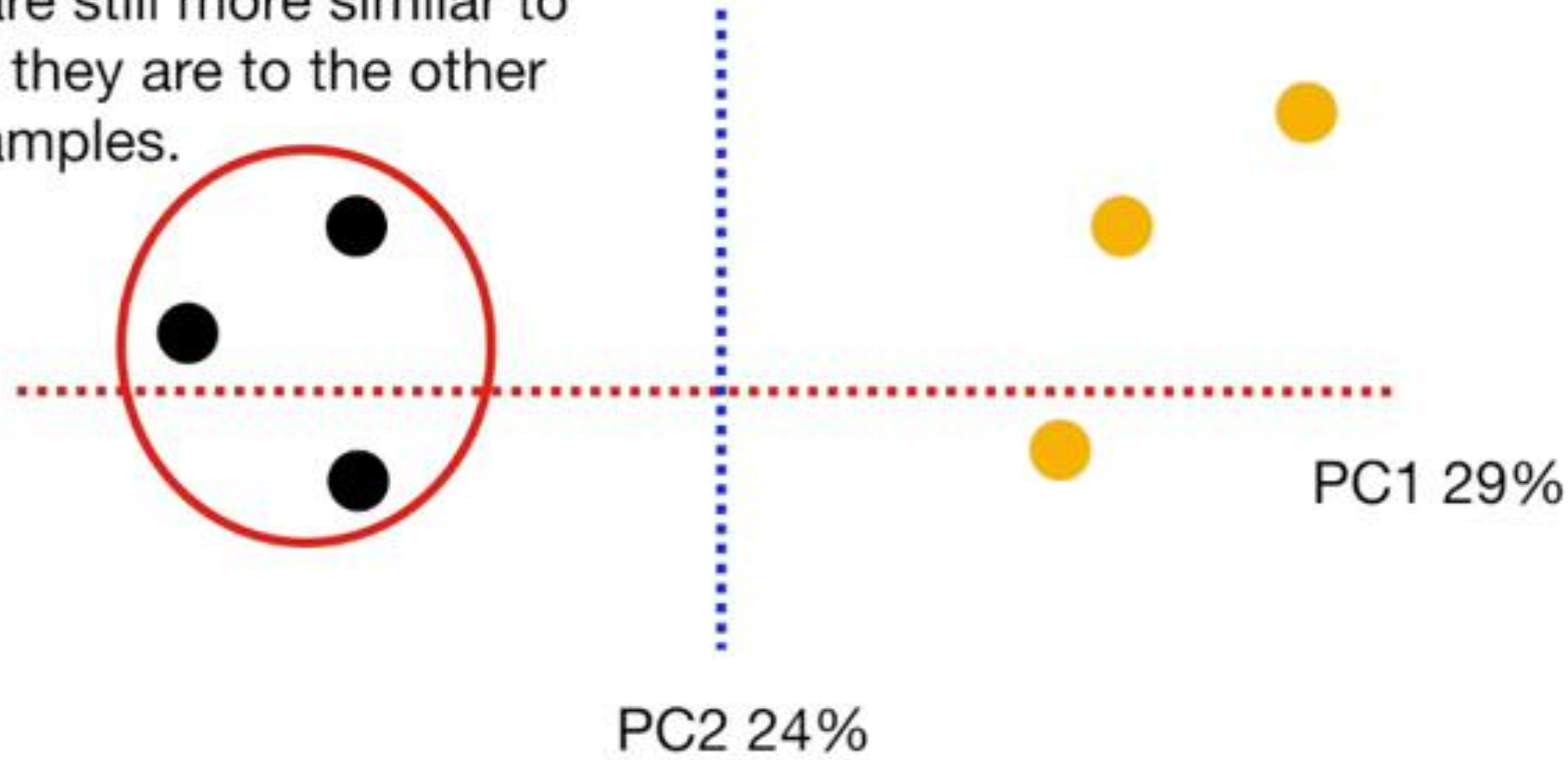
However, even a noisy PCA plot like this can be used to identify clusters of data.





These samples are still more similar to each other than they are to the other samples.





Slides courtesy

StatQuest (https://youtu.be/FgakZw6K1QQ)